



West Kowloon Cultural District Authority

for the

West Kowloon Cultural District

Technology Partnership Package 2

Ticketing and Admission System

and ancillary B2C applications

Requirement Document

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Table of Contents

1. Introduction	7
2. Requirement Catalogue	9
3. Business Requirement Definition	42
3.1. Business Pain Points and Market Opportunity	42
3.1.1. Business Pain Points	42
3.1.2. Market Opportunity.....	43
3.2. Business Background and Objectives	44
3.2.1. Business Background.....	44
3.2.2. Business Objectives.....	44
3.3. Box Office	46
3.3.1. Business Requirements	46
3.3.2. Business Requirement Analysis	49
3.3.3. Operating Model	53
3.3.4. Customer Journey.....	54
3.3.5. Business Scenarios	55
3.4. Website.....	81
3.4.1. Business Requirements	81
3.4.2. Business Requirement Analysis	84
3.4.3. Operating Model	88
3.4.4. Customer Journey.....	89
3.4.5. Business Scenarios	91
3.5. Kiosk	112
3.5.1. Business Requirements	112
3.5.2. Business Requirement Analysis	113
3.5.3. Operating Model	118
3.5.4. Customer Journey.....	118
3.5.5. Business Scenarios	120
3.6. OTA Integration	129
3.6.1. Business Requirements	130
3.6.2. Business Requirements Analysis.....	131
3.6.3. Operating Model	132
3.6.4. Customer Journey.....	133
3.6.5. Business Scenarios	135
3.7. Ticketing System	140
3.7.1. Business Requirements	140
3.7.2. Business Requirement Analysis	144
3.7.3. Customer Journey.....	148
3.7.4. Business Scenarios	149
3.8. Reporting.....	164
3.8.1. Business Requirements	164
3.8.2. Business Requirements Analysis.....	165
3.8.3. Operating Model	166
3.8.4. Customer Journey.....	167
3.8.5. Business Scenarios	168
3.9. Admission.....	173
3.9.1. Business Requirements	173

3.9.2. Business Requirement Analysis	174
3.9.3. Operating Model	179
3.9.4. Customer Journey	180
3.9.5. Business Scenarios	182
3.10. Membership and Membership Card	196
3.10.1. Business Requirements	196
3.10.2. Business Requirement Analysis	199
3.10.3. Customer Journey	204
3.10.4. Business Scenarios	205
3.11. Marketing, Promotion and RSVP	209
3.11.1. Business Requirements	209
3.11.2. Business Requirement Analysis	211
3.11.3. Customer Journey	215
3.11.4. Business Scenarios	216
3.12. Customer Data Platform	220
3.12.1. Business Requirements	220
3.12.2. Business Requirement Analysis	221
3.12.3. Operating Model	222
3.12.4. Customer Journey	222
3.12.5. Business Scenarios	223
3.13. CRM and ERP	229
3.13.1. Business Requirements	229
3.13.2. Business Requirements Analysis	231
3.13.3. Operating Model	232
3.13.4. Customer Journey	233
3.13.5. Business Scenario	234
3.14. Payment	236
3.14.1. Business Requirements	236
3.14.2. Business Requirement Analysis	237
3.14.3. Operating Model	239
3.14.4. Customer Journey	239
3.14.5. Business Scenarios	240
3.15. Ticketing Operation Centre	243
3.15.1. Business requirements	243
3.15.2. Business Requirements Analysis	243
3.15.3. Operation Model	244
3.15.4. Customer Journey	245
3.15.5. Business Scenarios	246
3.16. B2B Portal (Tour, School Tour, and Academy)	248
3.16.1. Business Requirements	248
3.16.2. Business Requirement Analysis	252
3.16.3. Operating Model	255
3.16.4. Customer Journey	256
3.16.5. Business Scenarios	259
3.17. WeChat Mini Program	278
3.17.1. Business Requirements	278
3.17.2. Business Requirements Analysis	282
3.17.3. Operating Model	286
3.17.4. Customer Journey	287
3.17.5. Business Scenarios	290

3.18. Douyin Mini Program.....	306
3.18.1. Business Requirements	306
3.18.2. Business Requirement Analysis	307
3.18.3. Operating Model	311
3.18.4. Customer Journey.....	312
3.18.5. Business Scenarios.....	314
3.19. Xiaohongshu Mini Program.....	333
3.19.1. Business Requirements	334
3.19.2. Business Requirement Analysis	335
3.19.3. Operating Model	338
3.19.4. Customer Journey.....	340
3.19.5. Business Scenarios.....	342
4. Technical Requirement Definition.....	363
4.1. Customer Data Platform	363
4.1.1. Technical Requirement	363
4.1.2. Technical Requirement Analysis	363
4.1.3. Technical Requirement List.....	365
4.2. Reporting.....	366
4.2.1. Technical Requirements	366
4.2.2. Technical Requirements Analysis	366
4.2.3. Technical Requirement List.....	367
4.3. Ticketing Operation Centre	367
4.3.1. Technical Requirements	367
4.3.2. Requirements Analysis	367
4.3.3. Technical Requirement List.....	368
4.4. Monitoring.....	369
4.4.1. Technical Requirement	369
4.4.2. Technical Requirement Analysis	370
4.4.3. Operating Model	374
4.4.4. Customer Journey.....	376
4.4.5. Business Scenarios	377
4.5. Permission.....	383
4.5.1. Technical Requirements	383
4.5.2. Technical Requirement Analysis	384
4.5.3. Business Scenario	385
4.6. Real-Name Authentication	392
4.6.1. Technical Requirements	392
4.6.2. Technical Requirement Analysis	393
4.6.3. Operating Model	394
4.6.4. Customer Journey.....	394
4.6.5. Business Scenarios	395
4.7. Technical Requirements.....	397
4.7.1. General	397
4.7.2. Log System.....	403
5. Organisation Chart.....	416
6. RACI Chart and Roles & Responsibilities	417
6.1. Roles & Responsibilities and Detail Job Descriptions.....	417
6.2. RACI Chart	418
Appendix.....	419

Appendix A – Abbreviations and Definitions.....	419
Appendix B – OTA Platform List	421
Appendix C – OTA Specific Requirements	421
Appendix D – Definitions of Technical Key Terms	423

1. Introduction

This document is expected to align all stakeholders' expectations to develop a joint understanding of all future business processes with clearly defined roles & activities, corresponding detailed business requirements and additional wishes West Kowloon Cultural District Authority (WKCD) would like to realise through this project.

This Requirement Document defines the complete business, functional, and technical scope for the Ticketing and Admission System (TAS) and its associated Business-to-Business (B2B) and Business-to-Consumer (B2C) channels to be implemented for WestK. It consolidates requirements gathered from multiple business units and channels, ensuring a unified understanding of future-state processes, system capabilities, operational roles, and integration points across the entire ticketing ecosystem.

Developed as formal business requirements specification, this document describes the target state Ticketing and Admission System (TAS) and its integration with a suite of supporting platforms and channels. These include visitor facing channels such as the Website, Kiosk, WeChat Mini Program, Douyin Mini Program, Xiaohongshu Mini Program, and Online Travel Agent (OTA) integrations, as well as enabling platforms such as the Membership System, B2B Portal, Marketing System, unified Payment Framework, and Customer Data Platform (CDP) event tracking. Together, these components form an integrated omnichannel ecosystem designed to support WestK's strategic objectives, operational efficiency, and end-to-end visitor experience.

Core areas addressed in this Requirement Document include:

- **Business Pain Points & Rationales**

A detailed assessment of current operational challenges, including fragmented systems, inconsistent workflows across venues, limited integration with digital channels, manual processes, and insufficient data visibility, establishing the need for a unified, scalable, and omnichannel solution.

- **Future Business Processes & Operating Models**

The document formalises standardised workflows across all sales and admission channels, such as box office operations, website ticketing, kiosk self-service flows, group bookings, and multi-terminal admission control. Each process is accompanied by role definitions and operating responsibilities for WestK and system providers.

- **Comprehensive Functional Requirements**

For each module, clear and traceable requirements (with unique Requirement IDs) describe expected system behaviour, including ticket management, seat management, inventory rules, pricing configuration, promotion mechanisms, membership entitlements, discount logic, order lifecycle management, payment handling, data synchronisation, logging, monitoring, security, and compliance.

- **Cross-Channel Customer Journeys**

Structured end-to-end journeys illustrate how visitors, members, institutions, and frontline staff interact with the system across different touchpoints—from browsing events, selecting tickets, completing payment, receiving e-tickets, and entering venues, to handling refunds, exchanges, and after-sales services.

- **Technical Architecture & Platform Expectations**

The document specifies technical requirements for stability, scalability, interoperability, permission management, real-name authentication, API standards, high-concurrency handling, monitoring, failover, disaster recovery, data governance, and integration with WestK's CDP and backend systems.

AnTank's platform is defined as the authoritative source for ticketing data and core transaction processing.

- **Governance, Roles, and Delivery Responsibilities**

Clear definitions of the responsibilities across WestK project teams, AnTank, and external platforms ensure accountability during solution design, implementation, testing, launch, and long-term operation.

By defining these requirements in a structured and traceable manner, this document serves as the foundational blueprint guiding system design, implementation, testing, and future enhancements. It ensures alignment among all stakeholders and provides WestK with a unified, future-ready ticketing and admission ecosystem that supports operational efficiency, cross-channel consistency, scalability, and enhanced visitor experience.

The intended audience for this document is the Project Steering Committee (PjSC), Project Management Team (PjMT) and other project team members involved in WKCD.A.

Appendix A provides the list of acronyms used in this document.

Appendix B provides the list of OTA platforms referenced in this project, outlining all third-party online travel agencies integrated or planned for integration within WestK ticketing ecosystem.

Appendix C provides the list of OTA-specific requirements, including platform-level constraints, naming rules, character limits, data formatting specifications, search requirements, and any special considerations needed to ensure compatibility with each OTA's technical and operational standards.

Appendix D provides the definitions of key terms used throughout this document, ensuring consistent understanding across all stakeholders by clarifying terminology related to ticketing, admission, membership, payment, promotions, technical components, and operational processes.

2. Requirement Catalogue

The requirements are prioritised into the following categories:

Priority	Description
High	The function requirements MUST be included in the proposed system and available to be used at Go-Live date. Assessment would be made on the feasibility to include the functional requirements in the proposed system or providing work around for the requirement.
Medium	The function requirements NEED to be included in the proposed system and to be supplied after Go-Live date. Assessment would be made on the feasibility to include the functional requirements in the proposed system or providing workaround for the requirement.
Low	The function requirements are NICE to have in the proposed system.

Table 2 (1): Requirement Priority Description

For each requirement, Requirement ID (RID), requirement description, and priority are documented.

The following table consolidates the requirements, enhancement requests, and clarification points raised during cross-functional requirement-gathering meetings for the project. Inputs were contributed by representatives from multiple WestK Business Units (BU) including M+ Museum(M+), Hong Kong Palace Museum (HKPM), Performing Arts (PA), as well as stakeholders from Marketing, Membership, Operations, Business Intelligence (BI), and other functional teams. Each entry captures the original requirement as raised, the module and category it relates to, and the corresponding response or preliminary position from the AnTank product and solution team.

This table below serves as a traceable record of all discussions held during the requirement-definition phase and provides transparency regarding stakeholder expectations, alignment status, and items requiring further design or evaluation. It also acts as a reference source for formalizing and validating the detailed requirements documented in subsequent sections of this Requirement Document.

Module	Category	Requirement	Raised By
Admission Tickets (Refer to Section 3.6)	OTA Integration	Admission ticket functions, such as admission rules and date extension, shall also apply to OTA channels.	Lydia Zou

Admission Tickets (Refer to Section 3.6)	Promotion/ OTA	Promotion strategies shall be synchronisable to OTA channels and updated in real time.	Lydia Zou
Admission Tickets (Refer to Section 3.7)	Product Bundling	The system shall support the purchase of mixed ticket types, such as standard and concession tickets, within the same transaction.	Lydia Zou/ Kamen Lam
Admission Tickets (Refer to Section 3.7)	Product Design	Where platforms do not support mixed ticket combinations, bundled ticket products such as 2 adults + 1 child shall be used as an alternative.	Kamen Lam/Lydia Zou
Admission Tickets (Refer to Section 3.7)	Admission Rules	The system shall support configurable admission time rules, such as how long before or after showtime admission is allowed.	Peggy Fung
Admission Tickets (Refer to Section 3.9)	Admission/ Exception Handling	The system shall support manual Box Office handling when a QR code cannot be scanned.	Peggy Fung
Admission Tickets (Refer to Section 3.9)	Admission/ Gate Control	Different entrances shall be able to apply restrictions to different tickets, such as by floor or area.	Peggy Fung
Admission Tickets (Refer to Section 3.9)	Admission/ User Experience (UX)	Scanning at the wrong entrance shall automatically trigger a prompt and skip forward without requiring manual staff action.	Peggy Fung
Admission Tickets (Refer to Section 3.9)	Admission/ Duplicate Ticket	The system shall detect duplicate admissions, such as scalped or reused tickets.	Peggy Fung
Admission Tickets (Refer to Section 3.9)	Admission/ Fraud Control	The system shall allow suspicious tickets to be flagged and prompt staff during admission.	Peggy Fung

Technology Partnership Package 2
Requirement Document

Admission Tickets (Refer to Section 3.9)	Admission/ Seat Issue	Seat issues shall trigger prompts for seat reassignment during admission.	Peggy Fung
Admission Tickets (Refer to Section 3.9)	Admission/ Real-time Monitoring	The system shall support real-time monitoring of admission headcount, sales, and validation rate.	Peggy Fung
Admission Tickets (Refer to Section 3.15)	Admission/ Operation	Real-time admission data shall be available to support operational decisions, such as whether to start the show.	Peggy Fung
Admission Tickets (Refer to Section 3.9)	Admission/ Gate Control	The admission scanning interface shall allow frontline staff to switch between different gates or scanning points within the same login, instead of being fixed to a single predefined location.	Jack Yeung
Admission Tickets (Refer to Section 3.7)	Inventory Sharing	Different ticket types shall be able to share quota.	Kamen Lam
Admission Tickets (Refer to Section 3.7)	Channel Management	The system shall support channel sales permissions and inventory allocation.	Kamen Lam/ Lydia Zou
Admission Tickets (Refer to Section 3.9)	Queue Management	The system shall support queue number management for admission.	Kamen Lam
Admission Tickets (Refer to Section 3.9)	Queue Management	The system shall support reserved queue numbers for Very Important Person (VIP) priority.	Kamen Lam
Admission Tickets (Refer to Section 3.7)	Ticket Template	The system shall support customization of physical ticket templates, including language and logo.	Lydia Zou

Technology Partnership Package 2
Requirement Document

Admission Tickets (Refer to Section 3.7)	Electronic Ticket (E-ticket) Template	The system shall support customization of e-ticket templates and styles.	Lydia Zou
Admission Tickets (Refer to Section 3.7)	Reservation	The system shall support internal reserved tickets.	Kamen Lam
Admission Tickets (Refer to Section 3.7)	Agent Sales	The system shall support an agent ticketing portal for hotels and travel agencies without requiring Application Programming Interface (API) integration.	Kamen Lam
Admission Tickets (Refer to Section 3.14)	Payment/ Settlement	The system shall support channel prepayment and credit limit management.	Kamen Lam
Admission Tickets (Refer to Section 3.11)	Promotion	The system shall support promotions such as vouchers, discounts, and bundles.	Lydia Zou
Admission Tickets (Refer to Section 3.11)	Promotion	The system shall support exclusive links, promotion codes, and credit card identification.	Lydia Zou
Admission Tickets (Refer to Section 3.10)	Membership	The system shall support member discounts and a points scheme.	Lydia Zou
Admission Tickets (Refer to Section 3.7)	Limit Control	The system shall support purchase limits by user, session, and ticket type.	Kamen Lam
Admission Tickets (Refer to Section 3.7)	Anti-scalping	The system shall support shared purchase limit rules across activities.	Kamen Lam
Seat Selection	Payment Flow	The system shall support a workflow in which seats are locked first, a payment link is	Kamen Lam

Technology Partnership Package 2
Requirement Document

(Refer to Section 3.11)		sent, and the customer completes payment online.	
Seat Selection (Refer to Section 3.16)	Group Booking	The system shall support batch seat reservation with payment assigned to designated customers, such as VIPs or guest-list users.	Susanna Yu
Seat Selection (Refer to Section 3.14)	Corporate Payment	The system shall support corporate payment by bank transfer or other non-online payment methods.	Susanna Yu
Seat Selection (Refer to Section 3.11)	Promotion Code	The system shall support different codes with different discounts for different partners.	Susanna Yu
Seat Selection (Refer to Section 3.11)	Promotion Code	The system shall support tracking the sales source through Promo Codes.	Susanna Yu
Seat Selection (Refer to Section 3.11)	Private Purchase Link	The system shall support a dedicated Private Link for exclusive ticket purchase access.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Private Purchase Link	The Private Link shall support user identity restrictions, such as email or membership status.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Private Purchase Link	The system shall support purchase limits per user, such as 2 to 4 tickets.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Private Purchase Link	The system shall support specified payment methods, such as designated credit cards.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Private Purchase Link	The system shall support credit card band identification number (BIN) verification or code verification.	Kamen Lam

Technology Partnership Package 2
Requirement Document

Seat Selection (Refer to Section 3.7)	Seat Plan	The system shall support multiple seat map versions for the same venue.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Seat Plan	The system shall support table seating, where 4 seats must be purchased together.	Susanna Yu/Peggy Fung
Seat Selection (Refer to Section 3.7)	Seat Plan	Accessible seats shall support automatic rules, such as companion seat binding.	Peggy Fung/ Carmen Lam
Seat Selection (Refer to Section 3.7)	Seat Plan	Accessible seats shall be capable of being automatically released as regular seats closer to showtime.	Carmen Lam
Seat Selection (Refer to Section 3.7)	Seat Plan	The system shall support tags such as restricted view and entrance reminders.	Peggy Fung
Seat Selection (Refer to Section 3.7)	Seat Plan	The system shall support automatic seat allocation strategies, such as from centre, left/right, or front/back.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Inventory	The system shall adopt a unified inventory pool shared across all channels.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Inventory	The system shall support allocating designated seat areas for OTA sales.	Kamen Lam/Carmen Lam
Seat Selection (Refer to Section 3.7)	Inventory	The system shall support real-time adjustment of inventory allocation.	Carmen Lam
Seat Selection	Inventory	The system shall support locking seats so that they are not available for public sale.	Susanna Yu/Peggy Fung

Technology Partnership Package 2
Requirement Document

(Refer to Section 3.7)			
Seat Selection (Refer to Section 3.7)	Inventory	The system shall support different seat availability by channel.	Kamen Lam
Seat Selection (Refer to Section 3.7)	Bundling	The system shall support bundled sales of tickets plus products such as beverages or merchandise.	Peggy Fung/ Susanna Yu
Seat Selection (Refer to Section 3.7)	Bundling	Bundled products shall support redemption by voucher or QR code.	Peggy Fung
Seat Selection (Refer to Section 3.7)	Add-on	Add-on products shall have separate QR codes.	Peggy Fung
Seat Selection (Refer to Section 3.7)	Sales	The system shall support agent ticket issuance for channels without API integration.	Susanna Yu
Seat Selection (Refer to Section 3.16)	Sales	The system shall support corporate account prepayment or credit arrangements.	Susanna Yu
Marketing (Refer to Section 3.11 and 3.10)	Voucher Rules	Promotional campaigns shall support gift vouchers, such as vouchers for redeeming gifts rather than discounts.	Carman Man
Marketing (Refer to Section 3.11 and 3.10)	Voucher Distribution	The system shall support voucher distribution through multiple channels, including Mini Program, Website, and Box Office.	Carman Man
Marketing (Refer to Section 3.11 and 3.10)	Voucher Distribution	Non-members shall be able to receive vouchers, for example after purchasing tickets as a guest.	Nicolas Leung

Technology Partnership Package 2
Requirement Document

Marketing (Refer to Section 3.11 and 3.10)	Voucher Usage	Vouchers shall support both QR code redemption and direct code entry.	Nicolas Leung
Marketing (Refer to Section 3.11 and 3.10)	Voucher Usage	Promotion Codes shall be supported across all channels, including Website, Kiosk, and Mini Program.	Nicolas Leung
Marketing (Refer to Section 3.11 and 3.10)	CRM & Voucher	The responsibility boundary for voucher issuance between the Customer Relationship Management (CRM) system and the Ticketing System shall be clearly defined.	Carman Man
Marketing (Refer to Section 3.11 and 3.10)	CRM Integration	The ownership of Electronic Direct Mail (EDM) sending and voucher distribution between the CRM system and the Ticketing System shall be clearly defined.	Carman Man
Marketing (Refer to Section 3.11 and 3.10)	Scope Definition	The scope of responsibilities among Marketing, Membership, and CRM under the TP2 project shall be clearly defined.	Carman Man
Marketing (Refer to Section 3.11 and 3.10)	Guest Mode	Guest users who are not logged in shall be able to redeem tickets using vouchers.	Billy Kwok
Marketing (Refer to Section 3.11 and 3.10)	Promotion Code	Promotion Codes shall support ticket redemption, including free tickets.	Billy Kwok
Marketing (Refer to Section 3.11 and 3.10)	Promotion Code	The system shall support corporate sponsors issuing independent redemption codes.	Billy Kwok
Marketing (Refer to Section 3.11 and 3.10)	Promotion Code	The system shall support online redemption without requiring on-site handling.	Billy Kwok
Marketing	Loyalty Mechanism	The membership points scheme may either be unified across BUs	Michael KH Lee

(Refer to Section 3.11 and 3.10)		(M+/HKPM/PA) or allow multiple schemes to coexist.	
Marketing (Refer to Section 3.11 and 3.10)	Loyalty Mechanism	The system shall support accumulation of member spending across BUs under a unified account.	Carman Man
Marketing (Refer to Section 3.11 and 3.10)	Points Mechanism	Whether points can be converted into cash or stored-value credit.	Michael KH Lee
Marketing (Refer to Section 3.11 and 3.10)	Ticket Upgrade	The system shall support membership upgrade after ticket purchase (Ticket → Membership).	June SK Wong
Marketing (Refer to Section 3.11 and 3.10)	Ticket Upgrade	Whether Ticket Upgrade is supported for users who purchase through OTA channels.	Angel Yu
Marketing (Refer to Section 3.11 and 3.10)	Functional Requirement	The system shall support membership purchase and gifting.	Billy Kwok
Marketing (Refer to Section 3.11 and 3.10)	Functional Requirement	The system shall support membership renewal.	Billy Kwok
Report (Refer to Section 3.8)	Report Presentation	Reports shall support multiple presentation formats, including in-system viewing via dashboard, export to Excel, and scheduled delivery by email.	Kamen Lam
Report (Refer to Section 3.8 and 4.5)	Report Presentation	Reports shall be viewable by external users, such as venue hirers.	Kamen Lam
Report (Refer to Section 3.8)	Dashboard	Dashboards shall present data in visual formats, such as trend charts and pie charts.	Nick Chan /Winnie Lau

Technology Partnership Package 2
Requirement Document

Report (Refer to Section 3.8 and 4.5)	Access Control	Different users, such as BUs and external users, shall only be able to view the data relevant to them.	Kamen Lam
Report (Refer to Section 3.8 and 4.5)	Access Control	Data across different BUs, such as HKPM, M+, and PA, shall be segregated.	Winnie Lau
Report (Refer to Section 3.8 and 4.5)	Access Control	The system shall support a multi-level access model for Group, BU, and external users.	Kamen Lam
Report (Refer to Section 3.8)	Data Consistency	The same metric, such as total tickets, shall be consistent across different reports.	Lydia Zou
Report (Refer to Section 3.8)	Data Consistency	Field naming and calculation logic shall be standardised.	Lydia Zou
Report (Refer to Section 3.8)	Data Definition	Different date dimensions, such as sales date and admission date, shall be clearly distinguished.	Lydia Zou
Report (Refer to Section 3.8)	Ticketing Report	The system shall support sales analysis by session, ticket price, and channel.	Nick Chan
Report (Refer to Section 3.8)	Ticketing Report	The system shall support sales analysis across different OTA channels.	Nick Chan
Report (Refer to Section 3.8)	Ticketing Report	A single dashboard shall cover all dimensions to avoid the need for multiple files.	Nick Chan
Report (Refer to Section 3.8)	Report Design	Each report shall provide a field list for confirmation.	Winnie Lau
Report (Refer to Section 3.8)	Report Design	Different BUs may have different report structures depending on their complexity.	Winnie Lau

Technology Partnership Package 2
Requirement Document

Report (Refer to Section 3.8)	Report Design	The calculation logic for all calculated indicators shall be clearly defined.	Lydia Zou
Report (Refer to Section 3.8)	Data Access	Users shall be able to select fields and export data themselves.	Winnie Lau
Report (Refer to Section 3.8)	Data Access	The system shall provide complete transaction-level data as a fallback report.	Lydia Zou
Report (Refer to Section 3.8)	Data Access	The system shall support flexible querying across dimensions such as time, ticket type, and price.	Winnie Lau
Report (Refer to Section 3.8)	Report Design	Sample reports shall be provided to validate the data structure.	Winnie Lau/ Naomi Chung
Report (Refer to Section 3.8)	Daily Operations	Daily visibility of sales, admissions, and inventory shall be provided.	Naomi Chung
Report (Refer to Section 3.8)	Daily Operations	Daily data shall support evaluation of promotional effectiveness.	Naomi Chung
Report (Refer to Section 3.8)	User Analysis	The system shall support identification of high-frequency ticket purchasers who are not members.	Naomi Chung
Report (Refer to Section 3.8)	User Analysis	The system shall support user identification by email or phone number.	Naomi Chung
Box Office (BO) (Refer to Section 3.3)	Sales Flow	The BO system shall support purchasing seat-selection and admission items in the same cart with a single checkout.	Kamen Lam
Box Office (Refer to Section 3.3)	Search	The BO system shall provide keyword search with a single input supporting multiple search types, including time, venue, and programme.	Kamen Lam

Technology Partnership Package 2
Requirement Document

Box Office (Refer to Section 3.3)	Ticketing	After selecting ticket type or base price, the system shall allow selection of price type, such as standard or concession.	Kamen Lam
Box Office (Refer to Section 3.3)	Cart	On the cart confirmation page, staff shall be able to change the price type of a ticket line without changing the seat or ticket type.	Kamen Lam
Box Office (Refer to Section 3.3)	Cart	On the cart confirmation page, staff shall be able to remove any ticket line item or remove an event-session sub-order.	Kamen Lam
Box Office (Refer to Section 3.3)	Cart	The system shall allow additional ticket lines to be added under a session sub-order by returning to the session for seat or ticket selection, rather than adding directly on the confirmation page.	Kamen Lam
Box Office (Refer to Section 3.3)	Discount Handling	The BO system shall allow manual cancellation of package discounts upon customer request.	Kamen Lam
Box Office (Refer to Section 3.3)	Seat Locking	Once a seat is selected on one BO screen, it shall be locked in real time across other BO screens.	Kamen Lam
Box Office (Refer to Section 3.3)	Cart	The cart confirmation page shall provide a quick button to show customers the seat location again.	Kamen Lam
Box Office (Refer to Section 3.3)	Ticket Exchange	The system shall support ticket exchange scenarios such as attendee type change, date change, and upgrade to higher-priced seats or tickets.	Kamen Lam
Box Office (Refer to Section 3.3)	Promotion Code	The system shall support promotion code discounts, typically with no more than two codes per event.	Kamen Lam
Box Office (Refer to Section 3.3)	Promotion Code	In cart mode, the system shall allow different promotion codes to be applied to different event or session sub-orders.	Kamen Lam
Box Office	Promotion Code	Reporting shall support sales statistics by promotion code, as	Kamen Lam

Technology Partnership Package 2

Requirement Document

(Refer to Section 3.3)		codes may correspond to different channels or audiences.	
Box Office (Refer to Section 3.3)	Package Discount	The system shall automatically apply package pricing when selected price types and quantities meet predefined thresholds.	Kamen Lam
Box Office (Refer to Section 3.3)	Package Discount	Package eligibility shall support dependency on combinations of price types, such as “2 adult + 2 child” qualifying while “4 adult” does not.	Kamen Lam
Box Office (Refer to Section 3.3)	Discount Handling	All discount policies shall be manually cancellable by staff.	Kamen Lam
Box Office (Refer to Section 3.3 and 3.14)	Payment	The system shall support direct payment methods such as WeChat Pay, Alipay, UnionPay, and Visa, as well as a “WestK” payment option.	Kamen Lam
Box Office (Refer to Section 3.3 and 3.14)	Payment	The feasibility of separate QR scanning and card-reading peripherals provided by Apollo shall be confirmed.	Kamen Lam
Box Office (Refer to Section 3.3)	Hotline/Call Centre	If a customer is willing to provide credit card details, staff shall be able to complete payment and ticket issuance directly in BO.	Kamen Lam
Box Office (Refer to Section 3.3)	Hotline/Call Centre	If a customer is unwilling to share card details, staff shall be able to send a payment link by email with a payment deadline, after which the order will be cancelled if unpaid.	Kamen Lam
Box Office (Refer to Section 3.3)	Service Fee	Service fees shall be configurable by channel, event, sales phase, and payment method.	Kamen Lam
Box Office (Refer to Section 3.3)	Service Fee	Service fees shall typically be calculated per ticket.	Kamen Lam

Technology Partnership Package 2
Requirement Document

Box Office (Refer to Section 3.3)	Service Fee	BO staff shall have the ability to adjust the service fee amount.	Kamen Lam
Box Office (Refer to Section 3.3)	Service Fee	The system shall support a per-order service fee cap.	Kamen Lam
Box Office (Refer to Section 3.3)	Service Fee	During refund processing, staff shall be able to choose whether the service fee is refundable.	Kamen Lam
Box Office (Refer to Section 3.3)	Refund	The system shall support scanning multiple printed ticket barcodes to process refunds together.	Kamen Lam
Box Office (Refer to Section 3.3)	Refund	Refunds shall be processed against a single original order.	Kamen Lam
Box Office (Refer to Section 3.3)	Refund	During refund, staff shall be able to choose whether the refund is booked to the current operator account or the original sales account.	Kamen Lam
Box Office (Refer to Section 3.3)	Refund	During refund or cancellation, staff shall be able to choose whether inventory is automatically released back to the public pool or manually released.	Kamen Lam
Box Office (Refer to Section 3.3)	Customer Display	During sales, the customer screen shall show a read-only view of staff selections. otherwise, it shall display poster images only.	Kamen Lam
Box Office (Refer to Section 3.3)	Customer Display	The customer screen shall display bilingual content in English and Chinese.	Kamen Lam
Box Office (Refer to Section 3.3)	Customer Display	Staff shall be able to toggle whether transaction content is shown on the customer screen and preview the customer-facing display in a floating window.	Kamen Lam
Box Office	Customer Display	The feasibility of allowing customers to select payment	Kamen Lam

(Refer to Section 3.3)		method on the customer screen and displaying QR payment codes there shall be explored.	
Box Office (Refer to Section 3.3)	Shift Handover /Settlement	On login and logout, users shall be able to choose whether to print a login or logout receipt.	Kamen Lam
Box Office (Refer to Section 3.3)	Shift Handover /Settlement	BO screens shall support query of staff sales, collections, and ticket paper usage during duty time.	Kamen Lam
Box Office (Refer to Section 3.3)	Shift Handover /Settlement	The settlement model for Apollo shall be further discussed, including the possibility of once-daily settlement while retaining accountability by system user.	Kamen Lam
Box Office (Refer to Section 3.3)	Language	The BO system shall support language switching after login.	Kamen Lam
Box Office (Refer to Section 3.3)	Operator Account Management	Staff should preferably use a single account to access BO systems of different BUs.	Kamen Lam
Box Office (Refer to Section 3.3)	Reporting	The system shall support sales statistics by terminal or device, since staff may sell across BUs using the same login.	Kamen Lam
Box Office (Refer to Section 3.3)	Anonymous Ticketing	For anonymous tickets, the system shall support a method for customers to obtain e-ticket or collection proof without providing mobile number or email.	Kamen Lam
Box Office (Refer to Section 3.3)	Anonymous Ticketing	The solution shall consider elderly customers who are not familiar with digital devices.	Kamen Lam
Box Office (Refer to Section 3.3)	Anonymous Ticketing	The final approach for e-code and collection code, whether the same or separate, shall be clarified.	Kamen Lam
Box Office (Refer to Section 3.3)	Delivery	The system shall support postal delivery fee handling by region, such as Hong Kong, Mainland, and Overseas, using simple flat-fee rules.	Kamen Lam

Technology Partnership Package 2
Requirement Document

Box Office (Refer to Section 3.3)	Reservation/ Seat Hold	Reservation or seat-hold orders shall support a configurable notification time to remind staff whether to keep or cancel the hold.	Kamen Lam
Box Office (Refer to Section 3.3)	Printing	Tickets shall be printed by ticket printer and receipts by Electronic Cash Register (ECR) device.	Kamen Lam
Box Office (Refer to Section 3.3)	Seat Sales Query	Unsold seats shall still display their base price rather than appearing fully greyed out.	Kamen Lam
Box Office (Refer to Section 3.3)	Seat Sales Query	Sold seats shall display the sold price type, such as standard or student.	Kamen Lam
Website (Refer to Section 3.4)	Content & UX	The ticket purchase flow shall provide clear step-by-step guidance and consistent copywriting to ensure users can clearly understand and complete the purchase process smoothly.	Naomi Chung
Website (Refer to Section 3.4)	Content & UX	The website shall adopt consistent terminology in both Chinese and English across all pages to ensure clarity and ease of use for users.	Cindy Chan
Website (Refer to Section 3.4)	Content & UX	All BUs (M+, HKPM, PA) shall participate in the definition of copywriting.	Cindy Chan
Website (Refer to Section 3.4)	Booking Flow	Users shall be able to click “Buy Tickets” from a BU event page and be directed straight to the corresponding purchase page on the AnTank Website without needing to reselect the event.	Nicolas Leung
Website (Refer to Section 3.4)	Login	Guest login shall be supported for ticket purchase.	Amelie Cheng
Website (Refer to Section 3.4)	Responsive	The mobile experience shall be consistent with the desktop experience.	Naomi Chung

Technology Partnership Package 2
Requirement Document

Website (Refer to Section 3.4)	Integration	Users expect to complete browsing and ticket purchasing within the same website without being redirected across multiple pages.	Lydia Zou
Website (Refer to Section 3.4)	Cart	The system shall support a shopping cart to allow the purchase of multiple products or sessions together.	Lydia Zou / Naomi Chung
Website (Refer to Section 3.4)	Cart	Users expect to purchase multiple dates or sessions in a single transaction.	Naomi Chung
Website (Refer to Section 3.4)	Session Display	The display of long-running sessions with dozens of occurrences shall be optimised, for example through a calendar view.	Kamen Lam
Website (Refer to Section 3.4)	Pricing & Seat Map	Ticket price distribution by seating zone shall be clearly displayed before purchase.	Kamen Lam / Kelvin Lam
Website (Refer to Section 3.4)	Seat Selection	The seat selection interface shall display detailed information such as seat number and row number.	Kamen Lam
Website (Refer to Section 3.4)	Accessibility	The website shall support accessibility features, including colour accessibility, display modes, and screen reader compatibility.	Kamen Lam / Amelie Cheng
Website (Refer to Section 3.4)	Ticket Type	Where there are many ticket types, such as student and senior tickets, the display method shall be optimised, for example by using dropdowns.	Kamen Lam
Website (Refer to Section 3.4)	Ticket Type	Ticket types shall not be selected by default, to avoid accidental purchase by users.	Kamen Lam
Website (Refer to Section 3.4)	Seat Selection	Special seats, such as restricted-view seats and wheelchair seats, shall be clearly indicated.	Kamen Lam

Technology Partnership Package 2
Requirement Document

Website (Refer to Section 3.4)	Session Display	Session information shall be displayed in full, such as different casts or different versions.	Kamen Lam
Website (Refer to Section 3.4)	Ticketing Model	The system shall support multi-day passes, such as 2-day passes or 3-day passes.	Kamen Lam
Website (Refer to Section 3.4)	Ticketing Model	Different ticket types, such as day pass and flex entry pass, shall not be split into separate products and shall instead be selected through a unified flow.	Kamen Lam
Website (Refer to Section 3.4)	Cart	The shopping cart shall display the seats already selected, so that users do not forget them.	Kamen Lam
Website (Refer to Section 3.4)	Seat Selection	Once a seat is selected, it shall be locked to prevent it from being taken during checkout.	Kamen Lam
Website (Refer to Section 3.4)	Venue & Seat Map	A venue zoning view, such as different floors or areas, shall be provided to assist user selection.	Kelvin Lam
Website Accessibility (Refer to Section 3.4)	Accessibility Standard	The ticketing website shall comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standards to support users with visual impairments, physical disabilities, and elderly users.	Cymie Yeung
Website Accessibility (Refer to Section 3.4)	Visual Accessibility	The website shall meet text contrast requirements of at least 4.5:1 for normal text and 3:1 for large text.	Cymie Yeung
Website Accessibility (Refer to Section 3.4)	Visual Accessibility	The website shall support 200% browser zoom without loss of content or functionality.	Cymie Yeung
Website Accessibility (Refer to Section 3.4)	Interaction Accessibility	The website shall provide clear focus states for all interactive elements.	Cymie Yeung

Technology Partnership Package 2
Requirement Document

Website Accessibility (Refer to Section 3.4)	Interaction Accessibility	The website shall support full keyboard navigation across the website.	Cymie Yeung
Website Accessibility (Refer to Section 3.4)	Interaction Accessibility	Status indicators shall not rely on colour alone and shall include alternative visual or textual cues.	Cymie Yeung
Website Accessibility (Refer to Section 3.4)	Seat Selection Accessibility	The seat map shall support both a graphical mode and a seat list mode to assist screen reader users.	Cymie Yeung
Website Accessibility (Refer to Section 3.4)	Accessibility Scope	Accessibility requirements shall cover the entire MOBILE WEBSITE site, including order lookup and member pages, rather than only the core ticketing journey.	Cymie Yeung, Kamen Lam
Website Accessibility (Refer to Section 3.4)	Special Ticketing Logic	The website shall display a disclaimer when wheelchair seats are selected to confirm the user's eligibility or need.	Kamen Lam
Website Accessibility (Refer to Section 3.4)	Special Ticketing Logic	The system shall support validation logic such that a Companion Ticket can only be purchased when a Wheelchair User Ticket is already in the cart.	Kamen Lam
Website Accessibility (Refer to Section 3.4)	Information Transparency	The website shall clearly state physical environment details relevant to accessibility, such as stairs and designated areas for guide dogs.	Cymie Yeung
Website Accessibility (Refer to Section 3.4)	User Testing	The project shall assess the use of third-party accessibility organizations for testing.	Cymie Yeung
Kiosk (Refer to Section 3.5)	Booking Flow	For M+ Cinema film programmes, the system shall provide an event detail page with text and images before ticket purchase to help users understand the programme content.	Lydia Zou

Kiosk (Refer to Section 3.5)	Booking Flow	The system shall support manual seat adjustment after automatic seat allocation.	Lydia Zou
Kiosk (Refer to Section 3.5)	Booking Flow	The ticket purchase flow shall be simplified as much as possible to improve on-site operational efficiency.	Naomi Chung
Kiosk (Refer to Section 3.5)	Cart	The system shall support cross-product checkout, such as Cinema and Exhibition, within a single shopping cart.	Lydia Zou /Michael KH Lee
Kiosk (Refer to Section 3.5)	Event Display	For PA events, the event list shall support categorization, such as Dance, Music, and Venue.	Jonas Chau
Kiosk (Refer to Section 3.5)	Event Display	The system shall support filtering by venue or programme type.	Jonas Chau
Kiosk (Refer to Section 3.5)	Event Display	The homepage shall provide separate entry points for different categories, such as Ticket, Cinema, and Workshop.	Michael KH Lee
Kiosk (Refer to Section 3.5)	Event Display	Different kiosks shall be configurable to display different content based on venue or location.	Jonas Chau
Kiosk (Refer to Section 3.5)	Session Display	The kiosk shall support guided tour products with multiple language sessions and clearly display the session language information during selection.	Jack Yeung
Kiosk (Refer to Section 3.5)	Session Display	The session list shall display the specific programme title, rather than only the time.	Kamen Lam
Kiosk (Refer to Section 3.5)	Session Display	The system shall support display of additional information, such as language and rating.	Willis Fong
Kiosk (Refer to Section 3.5)	Seat Map	The seat map shall display the full seating plan by default, rather than a partially zoomed-in view.	Lydia Zou

Technology Partnership Package 2
Requirement Document

Kiosk (Refer to Section 3.5)	Seat Map	The system shall support prevention of isolated single seats being left unsold.	Lydia Zou
Kiosk (Refer to Section 3.5)	Seat Map	Restricted-view seats shall display a warning prompt or pop-up.	Willis Fong
Kiosk (Refer to Section 3.5)	Seat Map	Sold, locked, or unavailable seats shall be displayed uniformly as “unavailable” to avoid exposing internal statuses.	Cheri Lo/ Willis Fong
Kiosk (Refer to Section 3.5)	Seat Map	The system shall support display of seat views, such as a 360-degree perspective, to assist seat selection.	Glades Sin
Kiosk (Refer to Section 3.5)	Membership & Discount	The system shall support member QR code scanning and automatic application of discounts.	Carman Man
Kiosk (Refer to Section 3.5)	Membership & Discount	The system shall support Promotion Code entry and instant discount calculation.	Carman Man
Kiosk (Refer to Section 3.5)	User Data	The system shall collect user contact details for notifications relating to programme changes.	Cheri Lo/ Willis Fong
Kiosk (Refer to Section 3.5)	User Data	The system shall allow non-member users to voluntarily provide contact information.	Cheri Lo
Kiosk (Refer to Section 3.5)	User Data	The system shall support newsletter subscription via email.	Jonas Chau
Kiosk (Refer to Section 3.5)	Survey	Whether the system should provide a questionnaire after ticket purchase to collect user feedback.	Lydia Zou
Kiosk (Refer to Section 3.5)	Survey	The system shall support guiding users to complete the questionnaire on their mobile phones via QR code.	Carman Man
Kiosk	Ticketing	The system shall support both e-ticket and physical ticket options.	Lydia Zou

Technology Partnership Package 2
Requirement Document

(Refer to Section 3.5)			
Kiosk (Refer to Section 3.5)	Ticketing	The system shall support QR code ticket collection for printing physical tickets.	Cheri Lo
Kiosk (Refer to Section 3.5)	Ticketing	The system shall support a more secure ticket collection method, such as a pickup password.	Cheri Lo
Kiosk (Refer to Section 3.5)	Accessibility	The system shall support automatic seat allocation under accessibility mode.	Michael KH Lee
Kiosk (Refer to Section 3.5)	Accessibility	The system shall support predefined logic for accessible seats and companion seats.	Glades Sin
Kiosk (Refer to Section 3.5)	UI/UX	The kiosk UI shall be aligned with the Website and Mini Program to ensure a consistent user experience across channels.	Jonas Chau
Kiosk (Refer to Section 3.5)	UI/UX	The homepage shall provide banner or recommendation slots for promotions and featured content.	Jonas Chau
Kiosk (Refer to Section 3.5)	UI/UX	Different BUs shall have distinct visual branding styles.	Jonas Chau
RSVP (Refer to Section 3.11)	RSVP Flow	The RSVP flow shall support a streamlined registration process with fewer steps.	Naomi Chung
RSVP (Refer to Section 3.11)	RSVP Flow	The system shall support registration for multiple activities within a single process.	Naomi Chung
RSVP (Refer to Section 3.11)	Quota Management	RSVP activities shall support quota limits and real-time quota deduction mechanisms.	Kamen Lam
RSVP (Refer to Section 3.11)	Quota Management	The system shall support a waitlist mechanism when quotas are full.	Kamen Lam

Technology Partnership Package 2
Requirement Document

RSVP (Refer to Section 3.11)	Quota Management	User quotas shall be locked after registration to prevent overbooking.	Kamen Lam
RSVP (Refer to Section 3.11)	Notification	Users shall receive a confirmation notification, such as by email or Short Message Service (SMS), after successful registration.	Cindy Chan
RSVP (Refer to Section 3.11)	Notification	The system shall support reminder notifications before an event starts.	Cindy Chan
RSVP (Refer to Section 3.11)	User Data	The system shall support the collection of additional user information for RSVP, such as questionnaires or forms.	Cindy Chan
RSVP (Refer to Section 3.11)	User Data	Different activities shall support different configurable information collection fields.	Cindy Chan
RSVP (Refer to Section 3.11)	Analytics	The system shall support tracking conversion from browsing to registration.	Nicolas Leung
RSVP (Refer to Section 3.11)	Analytics	The system shall support separation of registration data by different BUs.	Nicolas Leung
RSVP (Refer to Section 3.11)	Show	The system shall be able to support high-concurrency registration scenarios for popular activities.	Nicolas Leung
RSVP (Refer to Section 3.11)	Show	The system shall support a queue mechanism to handle peak registration traffic.	Nicolas Leung
RSVP (Refer to Section 3.11)	Security	The system shall support anti-bot registration measures, such as CAPTCHA and rate limiting.	Kelvin Lam
RSVP (Refer to Section 3.11)	Security	User data privacy and compliance requirements, such as General Data Protection Regulation (GDPR), shall be met.	Kelvin Lam

Technology Partnership Package 2
Requirement Document

Tour Booking & School Booking (Refer to Section 3.16)	Account/School Binding	The system shall allow users to select their school from a dropdown list during registration, rather than requiring a school-bound account.	Oscar Cheng
Tour Booking & School Booking (Refer to Section 3.16)	Account/ Verification	School identity verification may be completed after registration rather than being mandatory at the registration stage.	Bell Cheung
Tour Booking & School Booking (Refer to Section 3.16)	Booking/ Timeslot	The system shall support First Choice and Second Choice timeslot selection.	Bell Cheung
Tour Booking & School Booking (Refer to Section 3.16)	Booking/Group Scenario	The system shall support applications from cross-school joint groups, such as Scouts or inter-school groups.	Bell Cheung
Tour Booking & School Booking (Refer to Section 3.16)	Booking/NGO Flow	The system shall allow Non-Governmental Organizations (NGOs) to make bookings independently without going through the school-binding process.	Christy Wan
Tour Booking & School Booking (Refer to Section 3.16 and 3.14)	Payment	The system shall support post-visit payment within 30 to 45 days.	Oscar Cheng
Tour Booking & School Booking (Refer to Section 3.16 and 3.14)	Payment/Mixed Scenario	The system shall support mixed charging within the same order, such as 150 free places plus 50 paid places.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16)	Pricing/ Promotion	The system shall support dynamic promotions, for example temporary sponsorship resulting in changes between free and paid admission.	Oscar Cheng

Tour Booking & School Booking (Refer to Section 3.16)	Add-on/ Service	The system shall support add-ons, such as guided tours and learning materials, with independent charging.	Oscar Cheng
Tour Booking & School Booking (Refer to Section 3.16)	Add-on/ Headcount	Add-on quantities shall support an independent headcount rather than being required to match the main booking quantity.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16)	Booking Form	The system shall support dynamic forms, such as student list upload and recommendation letter upload.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16)	Booking Form	Form fields shall support both mandatory and optional settings, such as teacher information.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16)	Booking Modification	Back-office staff shall be able to modify bookings instead of requiring modification by users only.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16)	Booking Modification	The back office shall receive notifications when a user modifies a booking.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16)	Notification	The system shall support notifications, by email or system message, when a booking is created or updated.	Amelie Cheng
Tour Booking & School Booking (Refer to Section 3.16)	Booking Adjustment	The system shall support handling of on-site headcount increases, such as an additional two participants, with appropriate charge adjustment.	Phoebe Lai

Technology Partnership Package 2
Requirement Document

Tour Booking & School Booking (Refer to Section 3.16 and 3.7)	Event Config	Events shall support long-term validity, such as validity throughout the year.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16 and 3.7)	Event Config	The system shall support deep linking from external sources directly to a specific event booking page.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16)	UX/Message	The system shall support customised submission confirmation messages, allowing different activities to show different messages.	Phoebe Lai
Tour Booking & School Booking (Refer to Section 3.16 and 3.8)	Reporting	The system shall support automatic headcount statistics, Excel export, and conditional filtering.	Oscar Cheng
Tour Booking & School Booking (Refer to Section 3.16 and 3.8)	Reporting	The system shall support reporting across different dimensions, such as students, teachers, and school types.	Oscar Cheng
Tour Booking & School Booking (Refer to Section 3.16)	Quota Management	The back office shall be able to adjust quotas flexibly.	Oscar Cheng
Academy (Refer to Section 3.16)	Course Setup	Academy courses shall be configured in the AnTank system using a workshop-like setup model, with each course represented as a single course item without sessions, timeslots, ticket price categories, or ticket types.	Winsome Chow/ Angel Chan

Technology Partnership Package 2
Requirement Document

Academy (Refer to Section 3.16)	Enrolment Rules	The system shall allow users to enrol by selecting a single course item and submitting an application, with self-enrolment only, such that each applicant may enrol only for themselves.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Quota Management	Each course item shall have a capacity limit, and quota or inventory management shall be applied.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Application Workflow	Users shall be required to log in before submitting an application.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Application Workflow	The system shall support a questionnaire form as part of the course application process.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Document Upload	The system shall support upload of required supporting documents during application.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Application Workflow	Upon submission, the application shall enter a pending review status.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Approval Workflow	The enrolment process shall support an interview and approval workflow before payment is triggered.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Back-office Administration	Back-office staff shall be able to record interview outcomes in Admin as Approved or Rejected.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Back-office Administration	Admin shall support predefined rejection reasons for quick selection and tagging.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16 and 3.14)	Payment Flow	When an application is approved, the system shall send a payment notification email and allow the user to log in and complete online payment.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Notification	When an application is rejected, the system shall send a rejection notification email.	Winsome Chow/ Angel Chan

Academy (Refer to Section 3.16)	Notification	The rejection notification should include the rejection reason, if configured.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Quota Management	Quota or inventory shall only be deducted after successful payment.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Quota Management	When quota is fully consumed, waitlisted or subsequent applicants shall receive a quota-full notification.	Winsome Chow/ Angel Chan
Academy (Refer to Section 3.16)	Notification	The system shall support approval, rejection, and quota-full email notifications.	Winsome Chow/ Angel Chan
Ticketing Operation Centre (Refer to Section 3.15)	Access Control	The Operation Centre shall support BU-based access control, so that users from different BUs can only view data relevant to their own BU, while management-level users may view all BUs.	Kamen Lam
Ticketing Operation Centre (Refer to Section 3.15)	Admission Monitoring	For HKPM, the dashboard shall support viewing ticket sales and admissions by specific ticket products or exhibition combinations, rather than only showing overall museum-level totals.	Kamen Lam
Ticketing Operation Centre (Refer to Section 3.15)	Admission Monitoring	For HKPM, the dashboard shall support viewing sales and admission figures by time slot.	Kamen Lam
Ticketing Operation Centre (Refer to Section 3.15)	Admission Monitoring	For PA, the dashboard shall display the total expected attendance for each show, rather than requiring users to add up multiple figures manually.	Kamen Lam
Ticketing Operation Centre (Refer to Section 3.15)	Admission Monitoring	For PA, the dashboard shall show an attendance rate percentage in addition to raw admission and pending figures.	Kamen Lam

Technology Partnership Package 2
Requirement Document

Ticketing Operation Centre (Refer to Section 3.15)	Admission Monitoring	The dashboard shall cover workshop-type events in addition to standard shows and exhibitions.	Kamen Lam
Ticketing Operation Centre (Refer to Section 3.15)	Historical Data	The Operation Centre shall support viewing historical data, such as yesterday's or previous dates' figures, rather than only real-time data for the current day.	Kamen Lam
Ticketing Operation Centre (Refer to Section 3.15)	Session Display	For PA and similar show scenarios, the dashboard shall display session time clearly when multiple sessions of the same title occur on the same day.	Kamen Lam
BI (Refer to Section 3.15)	Data Definition	The data sources and definitions across Ticketing, CRM, and CDP are unclear and shall be clearly defined.	Lydia Zou
BI (Refer to Section 3.15)	Functional Design	Membership data from online channels, such as Mini Program, and offline channels, such as CRM or counter, shall be capable of being displayed in a unified manner.	Lydia Zou /Billy Kwok
BI (Refer to Section 3.15)	Functional Design	The system shall support analysis by channel.	Lydia Zou
BI (Refer to Section 3.15)	Functional Design	The system shall support analysis by user attributes such as gender, region, and age.	Charles Lau/ Michael KH Lee
BI (Refer to Section 3.15)	Membership	New Member statistics shall distinguish between new members and renewal members.	Janice Ho
BI (Refer to Section 3.15)	Membership	The system shall support reporting of membership income by membership type.	Janice Ho
BI (Refer to Section 3.15)	Membership	The system shall support analysis of member visit frequency.	Janice Ho

Technology Partnership Package 2
Requirement Document

BI (Refer to Section 3.15)	Membership	The system shall support analysis of membership benefit usage, such as coupons and discounts.	Angel Yu
BI (Refer to Section 3.15)	Membership	The calculation logic for benefit utilisation rate, including numerator and denominator, shall be clearly defined.	Janice Ho
BI (Refer to Section 3.15)	Functional Design	The system shall support separate analysis of membership data by campaign, such as roadshows.	Angel Yu
BI (Refer to Section 3.15)	Functional Design	Different BUs, such as M+, HKPM and PA, shall have separate dashboards.	Jackie Chan
BI (Refer to Section 3.15)	Data Consistency	Data across different dashboards shall remain consistent to avoid management seeing different figures.	Jackie Chan
BI (Refer to Section 3.15)	Access Control	Sensitive data shall be access controlled.	Jackie Chan
BI (Refer to Section 3.15)	Access Control	Different users, such as BUs and vendors, shall have different levels of data access.	Jackie Chan
BI (Refer to Section 3.15)	Data Definition	Sales date and admission date shall be clearly distinguished.	Lydia Zou /Nicole Mok
BI (Refer to Section 3.15)	Functional Design	The system shall support analysis by product type, such as standard tickets and family tickets.	Lydia Zou
BI (Refer to Section 3.15)	Functional Design	The system shall support filtering by event category, such as Exhibition and Event.	Michael KH Lee/ Nicole Mok
BI (Refer to Section 3.15)	Functional Design	The system shall support a festival dimension for aggregated analysis across multiple events.	Nick Chan

Technology Partnership Package 2
Requirement Document

BI (Refer to Section 3.15)	Functional Design	The system shall support distinguishing between free tickets and paid tickets.	Charles Lau
BI (Refer to Section 3.15)	Functional Design	The system shall support sales analysis by future admission date.	Nicole Mok
BI (Refer to Section 3.15)	Functional Design	The system shall support event-to-event comparison.	Michael KH Lee
BI (Refer to Section 3.15)	Functional Design	The system shall support time-based comparison, such as month-over-month and year-over-year.	Michael KH Lee
BI (Refer to Section 3.15)	Functional Design	The system shall support AI-generated analysis and recommendations based on data.	Michael KH Lee
Financial Report (Refer to Section 3.8)	Report	For an annual membership card, the finance team needs to recognise the revenue on a monthly basis according to the activation date, breaking it down into monthly income.	Samanda Cheung
Financial Report (Refer to Section 3.8)	Report	If an OTA involves a package scenario (for example, two independent AnTank events combined and sold as a package), the orders are counted separately in the AnTank order management system.	Finance
Financial Report (Refer to Section 3.8)	Report	Only the finance team is allowed to access the raw data (reports); the BU team cannot access any (Personally Identifiable Information) PII-related data.	Finance
Financial Report (Refer to Section 3.8)	Report	The Daily Fund Settlement Report to provide a daily fund verification to identity outstanding items, support T+N settlement monitoring for Faster Payment System (FPS)/Octopus.	Finance
Financial Report	Report	The Order and Payment Reconciliation Discrepancy Report to captures abnormal	Finance

Technology Partnership Package 2
Requirement Document

(Refer to Section 3.8)		transactions, reducing financial overages/shortages.	
Financial Report (Refer to Section 3.8)	Report	The Membership Revenue Allocation Detail Report to provide the monthly straight-line basis revenue over the membership period.	Finance
Financial Report (Refer to Section 3.8)	Report	The Ticketing Contract Liability Report to support revenue recognition judgment based on "transfer of control"; long-unredeemed tickets (e.g., beyond validity period).	Finance
Financial Report (Refer to Section 3.8)	Report	The Coupon/Gift Card Liability Report to support revenue recognition judgment based on "transfer of control"; long-unredeemed tickets (e.g., beyond validity period).	Finance
Financial Report (Refer to Section 3.8)	Report	The BU Profit and Loss Summary Report to provide the operating show for each BU including the profit and loss.	Finance
Financial Report (Refer to Section 3.8)	Report	The Product Line Revenue Detail Report to provide net revenue and associated channel costs by detailed product line to support profitability analysis.	Finance
Financial Report (Refer to Section 3.8)	Report	The Exceptional Ticket Issuance Report to display orders classified as "non-normal ticket issuance" (exceptional ticketing cases)	Finance
Financial Report (Refer to Section 3.8)	Report	The Event Discounted Sales Summary Report to provide a detailed analysis of discounted ticket sales by event, tracking the value and impact of different discount types.	Finance
Financial Report (Refer to Section 3.8)	Report	The Event Settlement Summary Report to provide statistics for each event, covering sales show across different ticket; issuance types; different sales channels	Finance

		and different ticket tiers (price categories).	
Financial Report (Refer to Section 3.8)	Report	The Tour/School/Group Booking Business Reports to provide statistical summaries of sales show related to Tour bookings, School bookings and Group bookings.	Finance
Financial Report (Refer to Section 3.8)	Report	The Key Account Prepayment Balance Summary Report to provide a summary of the current prepayment balance for each key account.	Finance
Financial Report (Refer to Section 3.8)	Report	The Key Account Outstanding Payment and Order Detail Summary Report to provide a summary of outstanding (unpaid) amounts for each key account, along with detailed information on all unreconciled/unsettled orders.	Finance

Table 2 (2): Requirement Catalogue

3. Business Requirement Definition

3.1. Business Pain Points and Market Opportunity

3.1.1. Business Pain Points

1) Fragmented Channel Management and Insufficient Real-Time Inventory Allocation Capability

The current ticketing landscape operates on two separate systems, without a unified real-time inventory linkage mechanism across online B2C, B2B (including OTAs/travel agents), and onsite box offices. This results in fragmented quota management and limits the ability to flexibly adjust allocations based on actual sales show. In addition, the current B2C website purchase journey requires users to be redirected to a third-party platform to complete transactions, preventing user behaviour data from being fully captured within the official ecosystem and affecting sales analysis, customer insight, and subsequent targeted operations.

2) Insufficient Adaptation to Mainland User Purchasing Habits and Gaps in Membership Conversion

For visitors from the Greater Bay Area and Mainland China, there is currently a lack of official native ticketing entry points aligned with their digital habits, particularly in terms of direct sales capabilities on mainstream platforms such as WeChat, Douyin, and Xiaohongshu. As a result, these user groups rely heavily on OTA platforms such as Meituan and Trip.com for ticket purchases. Given the relative data opacity of OTA platforms, the Authority is unable to directly obtain complete behavioural data and identity information for these users, resulting in a large number of high-value visitors not being effectively converted into the official membership ecosystem and limiting subsequent membership operations, precision marketing, and long-term customer relationship building.

3) Low Level of Automation in Group and School Booking Processes

At present, scenarios such as guided tours, school bookings, and corporate group purchases still rely heavily on manual handling or semi-automated processes. From online application, eligibility verification, and quota reservation to payment and settlement of outstanding balances, there is not yet a complete end-to-end business support capability in place. This leads to constrained operational efficiency, higher manpower costs, and longer customer response and confirmation cycles, which in turn affects the overall service experience.

4) Limited Cross-Selling Capability Across Business Lines

The current ticketing system lacks deep integration with related business systems such as Food & Beverage (F&B) and Retail and has not yet established a unified transaction journey or recommendation mechanism. During the ticket purchase process, the system is not yet able to intelligently recommend guided tours, dining offers, cultural merchandise, or co-branded products based on ticket type, venue, time slot, or customer preferences. As a result, it misses important opportunities to drive secondary conversion and increase average spending at key purchase decision points.

5) Insufficient Unified Multi-Venue Operations and Data Transparency

As multiple venues and business formats within the West Kowloon Cultural District continue to grow, the existing ticketing and operational support capabilities still face limitations in cross-venue, cross-channel, and cross-business management. Sales show, inventory status, channel show, refund and exchange status, and reconciliation data have not yet been sufficiently centralised or made transparent, affecting the ability of management and operations teams to monitor the overall business in real time and make timely decisions.

3.1.2. Market Opportunity

1) Building a Direct Sales Closed Loop from Content Engagement to Official Ticket Purchase

By establishing a Mini Program matrix across mainstream platforms such as WeChat, Douyin, and Xiaohongshu, and integrating it with the official website, OTAs, offline Box Office, and Kiosk channels, ticketing capabilities can be embedded into users' high-frequency digital touchpoints. This creates an official direct sales journey that is better aligned with the habits of Mainland China and Greater Bay Area users. Such an approach will not only help increase the proportion of official direct sales but also enable user traffic to be retained within WestK's owned platforms, forming a complete closed loop from content engagement and ticket conversion to membership acquisition.

2) Establishing Data-Driven Operational Capability

Through a unified TAS platform that integrates ticketing, Admission, Membership, Marketing & Promotion, CDP, Payment, and transaction data from all channels, user behaviour, consumption records, and preferences can be progressively structured and consolidated. This will provide the data foundation for future precision marketing, segmentation-based operations, membership lifecycle management, and brand communication, while reducing reliance on third-party platform data.

3) Driving Multi-Business Integration to Enhance Overall Revenue Contribution

Leveraging the integration capability of a unified ticketing platform with related business systems, the organisation can expand beyond ticket sales into guided tours, school and group bookings, retail products, dining offers, membership benefits, and RSVP-related activities. This enables bundled product offerings and cross-business synergies, thereby improving customer experience and increasing the contribution of non-ticketing revenue.

4) Establishing a Digital Operations Benchmark for a Smart Cultural District

Through the establishment of a Ticketing Operation Center (TOC), unified monitoring, alerting, operational analysis, and reporting support can be achieved across venues, channels, and business lines. This will significantly enhance management transparency and operational efficiency, positioning the West Kowloon Cultural District as a benchmark for digital transformation in the cultural and tourism industry.

3.2. Business Background and Objectives

3.2.1. Business Background

1) West Kowloon Cultural District Has Entered a New Stage of Multi-Venue Integrated Operations

With M+, the Hong Kong Palace Museum, and various performing arts venues entering full operation, ticketing management within the West Kowloon Cultural District has evolved from supporting single-venue, single-channel operations to meeting the needs of integrated cross-venue, cross-channel, and cross-business operations. There is now an urgent need to establish a unified, flexible, and scalable Ticketing and Admission System (TAS) to support growing visitor volume, increasingly complex business scenarios, and diversified user engagement channels.

2) TAS Is One of the Core Business Foundations of the TP2 Project

Within the overall TP2 project blueprint, TAS is not only responsible for core ticket sales and admission management but also serves as a key business hub connecting membership, marketing, payment, CDP, and various online and offline channels. The system must evolve from a traditional transaction-focused ticketing tool into an integrated business platform with capabilities in sales conversion, membership acquisition, data capture, precision marketing, and operational management, thereby supporting WestK's long-term development.

3) Changes in Customer Composition Are Driving the Need for Stronger Direct Sales and Digital Capabilities

The West Kowloon Cultural District serves a diverse customer base, including local audiences, Mainland visitors, overseas tourists, school groups, corporate customers, and travel agents. These different customer segments demonstrate clear differences in ticketing channels, payment preferences, booking methods, and service expectations. In particular, to better serve the Greater Bay Area and Mainland China markets, WestK needs to establish digital direct sales channels that align more closely with local user habits, while enhancing the end-to-end experience from ticket purchase and admission to membership conversion and ongoing engagement. This will strengthen the attractiveness of the official platform and improve WestK's operational control over customer relationships.

3.2.2. Business Objectives

1) Establish a Unified Official Direct Sales and Service Platform

Build an integrated ticketing platform covering Website, Box Office, Kiosk, OTA, Tour Booking/School Booking, WeChat/Douyin/Xiaohongshu Mini Program, and RSVP touchpoints, with unified management of inventory, orders, payments, memberships, and data, thereby enhancing the official platform's ability to serve different user groups.

2) Strengthen Direct Sales Conversion for Mainland China and Greater Bay Area Users

Develop native ticketing entry points on mainstream platforms such as WeChat, Douyin, and Xiaohongshu that are aligned with Mainland user habits, supported by local mainstream payment methods and more seamless membership onboarding journeys, in order to increase the proportion of Mainland visitors purchasing directly through official channels and improve conversion into the official membership ecosystem.

3) Enable Real-Time Inventory Sharing and Dynamic Allocation Across All Channels

Establish a centralised inventory and quota management mechanism to support real-time inventory sharing and unified control across B2C, B2B, OTA, Box Office, Kiosk, and other sales channels, thereby improving inventory utilisation, optimising sales strategies, and maximising the sales show of key programmes and high-demand ticket types.

4) Improve the Level of Automation for Group and School Booking Processes

For Tour Booking, School Booking, and related group business scenarios, establish end-to-end mechanisms covering online application, review, approval, inventory control, payment, and notifications, in order to reduce manual intervention, improve processing efficiency, ensure service consistency, and enhance the overall user experience.

5) Strengthen Membership Conversion, Marketing Integration, and Data-Driven Operations

By integrating Membership, Marketing and Promotion, CDP, and data from all sales channels, achieve unified management of purchase behaviour, membership identity, marketing campaigns, and customer preferences, thereby supporting precision marketing, benefit management, re-engagement initiatives, and customer lifecycle management, and progressively establishing a data-driven operating model.

6) Drive Growth in Non-Ticketing Revenue and Cross-Business Conversion

Through integration with retail, dining, guided tours, and other value-added service scenarios, enable product bundling, promotional offers, recommended selling, and campaign linkage within the ticket purchase journey and membership engagement scenarios, thereby increasing customer spend per transaction and the contribution of non-ticketing revenue, in support of WestK's broader commercial objectives.

7) Establish a Transparent and Efficient Ticketing Operations Management Framework

Establish a Ticketing Operation Center (TOC) to support centralised management and visual monitoring of sales show, inventory status, payment status, and on-site admission flow across multiple venues, channels, and business types, thereby enhancing real-time monitoring and decision-making capabilities for both operations teams and management.

8) Ensure High Availability, Scalability, and Compliance of the System

The system should be capable of supporting high-concurrency access, elastic scaling, multi-channel integration, and continuous evolution. It shall also comply with applicable privacy and data protection requirements, including but not limited to Personal Data (Privacy) Ordinance (PDPO), Personal Information Protection Law (PIPL), and GDPR, in areas such as data collection, payment processing, user identification, and marketing engagement, so as to ensure the platform's long-term stable operation and compliant governance.

3.3. Box Office

3.3.1. Business Requirements

From a business perspective, WestK requires the Box Office system to support onsite ticketing operations across multiple venues while ensuring operational efficiency, transaction accuracy, and consistent service workflows in high-traffic box office environments.

The following business requirements have been identified.

3.3.1.1. Onsite Ticket Sales Operations

WestK requires the Box Office system to support complete onsite ticket sales workflows, enabling staff to search events, search ticket type, select sessions, and complete ticket transactions through a unified operational interface. The system should provide clear operational flows to help staff efficiently complete transactions in high-traffic box office environments.

3.3.1.2. Event Search and Information Access

WestK requires the system to provide keyword-based search capabilities for events and sessions, along with clear event information displays. This allows staff to quickly retrieve relevant information when responding to customer enquiries and assisting with ticket purchases.

3.3.1.3. Seat Selection and Ticket Allocation

WestK hosts both exhibitions and shows. Therefore, the system is required to support both seat selection and admission ticketing models to accommodate the ticketing needs of different programme types. For the shows, staff should be able to select specific seats based on customer preferences. The system should also allow tickets for multiple sessions or different events to be purchased within a single transaction, supporting more complex onsite purchasing scenarios.

3.3.1.4. Real-time Seat Availability Management

For the shows, WestK requires the system to provide clear and reliable seat availability information so that box office staff can determine whether seats are available during the ticket selling process. Effective seat availability management helps reduce the risk of multiple terminals selecting the same seat at the same time, enabling staff to complete seat selection more efficiently while minimizing transaction conflicts.

3.3.1.5. Ticket Exchange Handling

WestK requires the system to support common ticket exchange scenarios at the box office, enabling staff to adjust issued tickets based on customer needs. For example, customers may request to change to another session if they are unable to attend the originally scheduled event, or upgrade to a higher-priced seat or ticket category if inventory is available.

In certain situations, such as when a student ticket was originally purchased but the attendee no longer qualifies for the discounted category, the system must support ticket exchange with additional price adjustments to ensure ticketing policies are correctly applied.

3.3.1.6. Onsite Refund Processing

WestK requires the system to support onsite refund operations, enabling staff to process customer refund requests directly at the box office. The system should allow staff to quickly locate relevant orders using ticket codes or order information and perform refund operations efficiently, including scenarios such as partial refunds for corporate purchases or group bookings.

During the refund process, the system should clearly record refund operations and related transaction information to support subsequent enquiries and operational reporting. Where permitted by business rules, refunded tickets should be returned to available inventory so that they may be resold.

3.3.1.7. Promotion and Discount Management

WestK requires the system to support various promotional mechanisms, such as promotional codes and bundled ticket pricing. The system should automatically apply eligible discounts when conditions are met, while also allowing staff to manually adjust or remove promotions when necessary.

3.3.1.8. Payment and Transaction Processing

The System shall integrate with the **Apollo payment terminal** to support payment processing for both onsite and remote payment scenarios, enabling staff to complete transactions under different operational requirements.

The System shall support the following payment methods through the Apollo payment terminal:

- Visa
- Mastercard
- UnionPay
- Octopus
- WeChat Pay
- Alipay

The System shall ensure that payment transactions are processed through the Apollo payment terminal in a consistent and controlled manner across the applicable operational scenarios.

3.3.1.9. Service Fee Management

WestK requires the system to support flexible service fee configuration to accommodate different fee policies across sales channels, event types, and payment methods. The system should automatically calculate applicable fees while allowing staff adjustments when required.

3.3.1.10. Customer Display Screen

WestK requires the system to support a dual screen display mode at the box office so that customers can view relevant ticketing information during the transaction process, improving transparency and overall customer experience. The customer-facing screen is primarily used for information display, allowing customers to clearly view selected tickets, quantities, and order amounts during the purchase process, and does not provide direct interaction functionality.

3.3.1.11. Shift Handover and Operational Settlement

WestK requires the system to support shift handover management for box office staff, ensuring that transactions performed by different operators on the same terminal can be clearly distinguished, and supporting onsite operational management and reconciliation needs. The system should allow shift credentials to be generated when staff log in or log out of the system, with logout credentials summarizing the operator's sales and settlement information for the shift.

In addition, the system should support enquiries of sales records, payment information, and related operational data during an operator's working session to support daily box office management and operational analysis. The system should also display operational notifications upon login and allow transaction records to be queried by both operator account and box office terminal, supporting operational tracking and reporting across multiple venues.

3.3.1.12. Lost Ticket Management

WestK requires the system to support lost ticket handling and management, enabling box office staff to verify and reissue tickets when customers are unable to present valid ticket proof. This ensures a smooth admission experience while maintaining operational control. The system should support ticket lookup and verification based on order information, purchaser identity, or other supporting credentials, and allow ticket reprinting or e-ticket reissuance in accordance with defined business rules, with complete audit logs maintained for traceability.

In addition, the system should support marking the original ticket status (e.g. invalid) during the lost ticket handling process and record the reason for reissuance and operator details, preventing the risk of both original and reissued tickets being used simultaneously. The system should also support enquiry of lost ticket handling records, allowing traceability by order or operator to support daily operations and audit requirements.

3.3.1.13. Membership Sales and Enrolment

WestK requires the Box Office system to support onsite membership sales and enrolment operations, enabling staff to introduce, register, and sell applicable membership products at the box office. The system should support the capture of required member information and the completion of membership transactions through a clear and efficient operational workflow.

The system shall support the sale of membership products for M+, HKPM, and PA as standalone transactions, subject to the applicable business rules and membership policies of each venue or programme. This helps support membership growth while providing customers with a convenient onsite membership purchase channel.

3.3.1.14. Membership Search and Information Access

WestK requires the system to provide membership search and enquiry capabilities so that box office staff can quickly retrieve relevant member information when assisting customers onsite. The system should support member lookup using identifiers such as membership code, registered mobile number, or email address, subject to access control and privacy requirements.

The system should present key membership information required for frontline operations, including membership status, validity period, membership type, and applicable benefits or discount eligibility, enabling staff to provide accurate customer service and support membership-related enquiries.

3.3.2. Business Requirement Analysis

This section provides further analysis and clarification of the business requirements described in Section 3.3.1. Each subsection elaborates on the business intent, operational considerations, and functional expectations for the Box Office system supporting onsite ticketing operations.

3.3.2.1. Onsite Ticket Sales Operations

Map to Business Requirement 3.3.1.1, 3.3.1.2, 3.3.1.3.

The Box Office system shall support a unified onsite ticket sales workflow enabling staff to efficiently perform ticket sales operations.

Key capabilities include:

- Search the exhibition and show using filters such as venue, date, event name, or category.
- Display of event information including session schedules, venue information, and ticket availability.
- Session selection and initiation of ticket sales.
- Support for both admission ticket and seat selection ticket.
- Ability to purchase tickets for multiple sessions or events within a single transaction.

3.3.2.2. Seat Selection and Seat Management

Map to Business Requirement 3.3.1.3 and 3.3.1.4.

The system shall provide seat-based ticketing capabilities allowing staff to select seats through a seat map while maintaining accurate seat availability management.

Key capabilities include:

- Display of seat maps including zones, rows, seats, and pricing tiers.
- Visual indicators for seat status such as available, sold, or unavailable.
- Selection of ticket price types for each seat.
- Seat inventory validation and seat locking mechanisms.
- Automatic release of locked seats upon timeout or order cancellation.

3.3.2.3. Admission Ticket Selection

Map to Business Requirement 3.3.1.3.

The system shall support ticket sales for exhibition events, allowing staff to select ticket categories and quantities.

Key capabilities include:

- Display of available ticket categories and prices.
- Selection of ticket types such as standard or student tickets.
- Quantity input with inventory validation.
- Adding selected ticket categories and quantities to the shopping cart.

3.3.2.4. Cart, Promotion and Order Confirmation

Map to Business Requirement 3.3.1.3 and 3.3.1.7.

The system shall provide a unified shopping cart mechanism supporting order management and promotion handling.

The unified shopping card shall support the following capabilities:

- Display of selected ticket items grouped by event and session.
- Allow users to modify or remove cart items.
- Automatically evaluate and apply eligible promotions and bundles.
- Allow manual application or removal of discounts, where applicable.
- Present an order confirmation step for users to review and confirm the order before proceeding to payment.

3.3.2.5. Payment and Transaction Handling

Map to Business Requirement 3.3.1.8 and 3.3.1.9.

The system shall support multiple payment methods and ensure reliable transaction processing.

Key capabilities include:

- Display of order summary and total payable amount.
- Support for multiple payment methods such as cash or card.
- Handling of integrated and non-integrated payment modes.
- Management of payment success, failure, or timeout scenarios.
- Support for assisted remote payment initiated by staff.

3.3.2.6. Ticket Fulfilment and Customer Display

Map to Business Requirement 3.3.1.1 and 3.3.1.10.

The system shall support ticket fulfilment and customer-facing display functions to enhance transaction transparency.

Key capabilities include:

- Physical ticket printing and electronic ticket delivery.
- Customer-facing display showing transaction information.
- Ticket collection and reprint functions.
- Logging of ticket issuance and printing activities.

3.3.2.7. Ticket Exchange and Refund Handling

Map to Business Requirement 3.3.1.5 and 3.3.1.6.

The system shall support ticket exchange and refund operations for onsite ticketing adjustments.

Key capabilities include:

- Retrieval of original order or ticket records.
- Validation of exchange or refund eligibility.
- Calculation of price differences or refund amounts.
- Update of order status and inventory after processing.

3.3.2.8. Operational Management and System Administration

Map to Business Requirement 3.3.1.11.

The system shall support operational management and administrative functions to ensure stable box office operations across venues.

Key capabilities include:

- User account and permission management.
- Shift handover and settlement management.
- Operational logging and audit traceability.
- Configuration and administrative functions required to support daily box office operations.

3.3.2.9. Lost Ticket Handling and Reissuance

Map to Business Requirement 3.3.1.12.

The system shall support lost ticket handling and reissuance to ensure service continuity in cases of missing tickets while maintaining ticket usage control and traceability.

Key capabilities include:

- Ticket lookup and verification based on order details, purchaser identity, or supporting credentials.
- Reprinting of physical tickets and re-delivery of electronic tickets.
- Invalidating original tickets to prevent duplicate usage.
- Logging of lost ticket handling and reissuance activities.

3.3.2.10. Membership Sales and Enrolment

Map to Business Requirement 3.3.1.13.

The system shall support onsite membership sales and enrolment operations, enabling box office staff to register customers and complete membership purchases through the Box Office system.

Key capabilities include:

- Capture of required customer information for membership registration.
- Support for the sale of membership products for M+, HKPM, and PA as standalone transactions.
- Display of available membership types, plans, or packages, where applicable.
- Validation of required information before membership purchase is completed.
- Creation of membership records and processing of membership transactions in accordance with applicable business rules and membership policies.

3.3.2.11. Membership Search and Information Access

Map to Business Requirement 3.3.1.14.

The system shall support membership search and enquiry functions, enabling box office staff to retrieve relevant membership information when assisting customers onsite.

Key capabilities include:

- Lookup of member records using membership code, registered mobile number, or email address.
- Display of key membership information including membership status, validity period, and membership type.
- Display of applicable membership benefits or discount eligibility for frontline reference.
- Support for membership-related enquiries to assist box office customer service operations.

3.3.3. Operating Model

This initiative adopts a collaborative operating model to support onsite ticket sales operations across WestK venues through the Box Office system.

The operating model defines the respective roles and responsibilities of the key parties involved in Box Office operations, with AnTank providing and supporting the ticketing platform, and WestK Box Office staff carrying out day-to-day ticketing operations at the venue, including applicable membership-related services.

AnTank – Ticketing Technology Provider

AnTank is responsible for providing and supporting the ticketing technology platform used for Box Office operations.

Key responsibilities include:

- Providing the Box Office ticketing system and related functional capabilities required to support onsite sales operations.
- Supporting ticketing configuration, including event setup, session setup, pricing, seating and inventory controls, in accordance with approved business requirements.
- Supporting membership-related system functions, including membership product setup, membership transaction processing, and membership enquiry capabilities, in accordance with approved business requirements.
- Supporting integration with applicable onsite payment devices and payment platforms.
- Maintaining system availability, operational monitoring, logging, and technical support in accordance with agreed service levels.

WestK – Box Office Operations

WestK Box Office staff are responsible for operating the system to deliver onsite ticket sales and customer service at the venue.

Key responsibilities include:

- Conducting onsite ticket sales operations, including ticket selection, seat allocation where applicable, order processing, and payment handling.
- Conducting onsite membership-related operations, including membership sales, customer registration, and membership information enquiry in accordance with applicable business rules and membership policies.
- Providing customer service functions such as ticket enquiries, ticket exchange, and refund handling in accordance with business policies.
- Managing ticket fulfilment activities, including ticket printing and ticket collection at the box office.
- Performing daily operational activities, including shift handover, transaction review, and ticket stock monitoring.

3.3.4. Customer Journey

This section describes the user story for Box Office staff when using the Box Office system to complete onsite ticket sales and customer service activities.

The table below illustrates how staff interact with the system to complete ticket sales, payment processing, ticket fulfilment, and post-transaction services at the venue.

No.	Category	Role	Operational Scenario
1	System Access	As a Box Office Staff	I log into the Box Office system at the beginning of my shift so that I can access ticketing functions and start daily operations.
2	Event Search	As a Box Office Staff	I search for events or sessions in the system so that I can quickly locate the event requested by the customer.
3	Ticket Selection (Seat Selection Ticket)	As a Box Office Staff	I select seats through the seat map so that customers can choose their preferred seats for seat selection events.
4	Ticket Selection (Admission Ticket)	As a Box Office Staff	I select ticket types and quantities for general admission events so that the required number of tickets can be added to the order.
5	Cart Management	As a Box Office Staff	I review the selected tickets in the shopping cart so that I can verify ticket details before proceeding with the transaction.
6	Promotion Adjustment	As a Box Office Staff	I adjust the system-applied bundle promotions, or apply other eligible discounts such as membership discounts, redemption vouchers, and promotional codes, so that the most appropriate offers can be applied before order confirmation.
7	Order Confirmation	As a Box Office Staff	I confirm the cart contents so that the order can be created and promotions can be applied before payment.
8	Payment Processing	As a Box Office Staff	I select the appropriate payment method and process the payment so that the ticket transaction can be completed.
9	Ticket Fulfilment	As a Box Office Staff	I print and deliver tickets to the customer so that the customer receives their purchased tickets immediately.

10	Customer Display	As a Box Office Staff	I use the customer-facing display to show order details so that customers can verify their ticket information during the transaction.
11	After-Sales Service	As a Box Office Staff	I process ticket exchanges or refunds when requested by customers in accordance with business policies.
12	Operational Management	As a Box Office Staff	I complete shift handover procedures and review transaction records so that daily box office operations can be properly tracked and reconciled.

Table 3.3.4: Description for the Box Office User Story

3.3.5. Business Scenarios

3.3.5.1. Ticket Sales

The Box Office governs the end-to-end workflow of onsite ticket sales within the Box Office system. It is designed to support high-volume, time-sensitive transactions while ensuring pricing accuracy, seat availability control, and consistent execution of sales policies across multiple venues.

This module enables box office staff to efficiently search and select events, manage seat selection and admission ticket sales, apply various pricing types and promotional rules, and complete transactions through a unified shopping cart and payment workflow. By centralising ticket selection, pricing logic, discount application, and payment processing, the system reduces operational complexity and minimises the risk of errors during onsite sales.

The Ticket Sales Process Management module also incorporates seat locking mechanisms, shopping cart management, and customer display screen support to enhance transaction transparency and operational reliability. It serves as the core foundation for subsequent processes such as ticket exchange, refund handling, and ticket issuance.

3.3.5.1.1. User Process – General Process

The following process describes the user process for ticket sales.

No.	Component	Description
1	Box Office Login	Box office staff logs into the Box Office system using authorised accounts. System validates user roles and related venue access rights.
2	Language Selection	Staff selects system display language. The selected language applies to both operator screen and customer display screen.

3	Exhibition and Show Search	Staff searches available exhibitions or shows by venue, date, event name, category, or availability status.
4	Exhibition and Show Selection	Staff selects the target event to initiate ticket sales.
5	Session Selection	Staff selects a specific session, show date, and time. System displays real-time availability and sales status.
6	Seat Mode Determination	System determines ticket type for the selected event: seat-selection, general admission, or mixed seating (seat + standing).
7	Seat Selection (if applicable)	Staff selects seat(s) from the seat map. System performs seat availability validation and temporarily locks selected seats.
8	Quantity Selection (if applicable)	For general admission tickets, staff inputs ticket quantity within allowed limits.
9	Ticket Price Type Selection	System displays available ticket price types for the select seats or admission tickets. Staff selects the applicable price types.
10	Special Ticket Validation	System validates special ticket rules (e.g. Accessible Ticket eligibility, Companion Seat, Special Price Availability).
11	Add to Shopping Cart	Selected tickets are added to the shopping cart. System confirms seat lock and price calculation.
12	Shopping Cart Review	Staff reviews shopping cart contents, including ticket details, pricing, discounts, and seat information.
13	Modify Cart (Optional)	Staff modifies cart contents by adding/removing tickets or adjusting price types. System revalidates availability and pricing rules.
14	Promotion / Bundle Application (Optional)	Staff applies applicable promotions, discount codes, or bundle offers. System recalculates total amount.
15	Order Confirmation	Staff confirms the order details with the customer before proceeding to payment.
16	Payment Method Selection	Staff selects payment method (e.g. cash, card, Octopus, Apollo-integrated payment, assisted remote payment).

17	Payment Processing	System processes payment and handles success, failure, or timeout scenarios.
18	Payment Confirmation	System confirms successful payment and generates order record.
19	Ticket Issuance	System issues tickets and triggers ticket printing or digital ticket according to configuration.
20	Receipt Printing (Optional)	System prints or provides a transaction receipt if required.
21	Transaction Completion	Transaction is completed. Seat status is finalised as sold and inventory is updated in real time.
22	Post-Transaction Actions	Staff may perform post-transaction actions such as order lookup, reprint, or proceed to next sale.

Table 3.3.5.1.1: Description of the Ticket Sales User Process

3.3.5.1.2. User Process - Seat Selection & Locking

No.	Component	Description
1	Enter Seat Selection Process	Staff enters the seat selection interface after selecting a seated event session.
2	Seat Map Initialisation	System loads the seat map for the selected session, including zone, row, seat layout, and pricing tiers.
3	Seat Status Rendering	System displays seat status using visual indicators (e.g. available, sold, locked, unavailable).
4	Seat Selection	Staff selects one or multiple available seats from the seat map based on customer requirements.
5	Validate Seat Availability	System validates seat eligibility based on session rules, ticket type constraints, and seat dependencies (e.g. companion seats).
6	Confirm Seat Selection and Add to Cart	When seats are confirmed, they are added to the shopping cart while maintaining lock status.
7	Seat Lock Execution	System locks selected seats for a defined lock duration to prevent concurrent selection by other users.
8	Seat Lock Status Synchronisation	Locked seats are shown as unavailable (greyed out) on other frontends and channels.

9	Seat Lock Failure Handling	If seat locking fails (e.g. seat already locked or sold), system notifies staff and prompts reselection.
10	Seat Deselection and Release	Staff may manually deselect seats, triggering immediate seat unlock.
11	Seat Lock Timeout and Release	If lock timer expires before order confirmation, system automatically releases the seats back to inventory.
12	Seat Lock Finalisation	Upon successful order completion, seat status is updated to sold and synchronised across all channels.

Table 3.3.5.1.2: Description of Seat Selection & Locking User Process

3.3.5.1.3. User Process - Admission Ticket Selection

The following process describes the Admission Ticket Selection.

No.	Component	Description
1	Enter Admission Ticket Selection Process	Staff enters the ticket selection interface after selecting an event session.
2	Load Available Ticket Categories	System loads available ticket categories and corresponding prices for the selected event session.
3	Display Ticket Information	System displays ticket category information including ticket name, price, and applicable conditions.
4	Select Ticket Category	Staff selects the appropriate ticket category according to customer requirements (e.g. standard, student, senior).
5	Input Ticket Quantity	Staff inputs the number of tickets requested by the customer within the allowed purchase limits.
6	Validate Ticket Availability	System validates the requested quantity against the available inventory and ticket purchase rules.
7	Add Tickets to Cart	Selected ticket category and quantity are added to the shopping cart.
8	Modify Ticket Quantity in Cart	Staff may adjust ticket quantities in the shopping cart, triggering automatic validation and price recalculation.

Table 3.3.5.1.3: Description of Admission Ticket Selection User Process

3.3.5.1.4. User Process - Shopping Cart Management

The following process describes the Shopping Cart Management.

No.	Component	Description
1	Enter Shopping Cart	Staff enters the shopping cart interface after adding tickets or products.
2	Display Shopping Cart Contents	System displays cart items grouped by event, session, and sub-order structure.
3	View Seat Location (Optional)	Staff may preview seat location to assist customer confirmation.
4	Modify Price Type in Cart (Optional)	Staff modifies price type of ticket items without changing seat.
5	Remove Ticket Item (Optional)	Staff removes individual ticket items from the cart.
6	Remove Session Sub-Order (Optional)	Staff removes all ticket items under a specific session sub-order.
7	Add Tickets to Existing Session (Optional)	Staff adds more tickets to an existing session by returning to seat or ticket type selection flow via sub-order.
8	Update Shopping Cart Contents	Any change to the shopping cart triggers a recalculation of discounts and related pricing information.
9	Proceed to Payment	Staff confirms cart contents and proceeds to the payment workflow.
10	Cancel Shopping Cart Order	Staff can cancel the shopping cart order; the system releases all locks and clears the cart contents.

Table 3.3.5.1.4: Description of Shopping Cart Management User Process

3.3.5.1.5. User Process - Discount & Bundle Application

The following process describes the Discount & Bundle Application.

No.	Component	Description
1	Enter Shopping Cart	Staff enters the shopping cart interface after adding tickets or products.
2	System Automatically Calculates Eligible Bundle Discounts	Upon entering the shopping cart, the system automatically triggers the discount and bundle calculation process, including event bundles and cross-event bundles.

3	Display the Discounted Amount after Bundle Discounts are Applied	System displays applied bundle details and price differences to staff.
4	Manual Bundle Cancellation	Staff may manually cancel applied bundle or cross-session bundle upon customer request.
5	Manual Discount Entry	Staff manually inputs discount information or discount code as required.
6	Manual Discount Validation	System validates manually entered discount against applicable rules and scope.
7	Manual Discount Application	Valid promotion codes and manual discounts are applied to eligible ticket items in the shopping cart.
8	Confirm Discounts and Initiate Payment	When entering the payment flow, all applied discounts are confirmed and locked and can no longer be modified.

Table 3.3.5.1.5: Description of Discount & Bundle Application User Process

3.3.5.1.6. User Process - Payment

The following process describes the Payment.

No.	Component	Description
1	Confirm Order Details and Proceed to the Payment Flow	After cart confirmation, staff initiates the payment process.
2	Display Order Summary	The system displays the order payable amount, including the total order amount, applied discounts, and service fees.
3	Select Payment Method	Staff selects the preferred payment method according to customer preference (or choose remote payment).
4	Display Payment Instructions or the Payment Interface for Customer	System prepares the payment interface or instructions for the customer (e.g. scan, tap, or link).
5	Customer Completes the Payment	Customer completes the payment action through the selected payment method.
6	Payment Status Monitoring	For integrated payment methods, system obtains payment status. For non-integrated payments, staff records payment result.

7	Payment Successful	Upon successful payment, system confirms transaction and proceeds to order completion.
8	Payment Failed	If payment fails or is cancelled, system notifies staff and provides retry or cancel options.
9	Payment Timeout	If payment exceeds allowed time, system handles timeout according to rules.
10	Display Payment Confirmation	System displays payment confirmation result to staff and customer.
11	Print Paper Tickets and Receipts	System triggers downstream actions such as ticket issuance and receipt printing.

Table 3.3.5.1.6: Description of Payment User Process

3.3.5.1.7. User Process - Assisted Remote Payment (BO Assisted Customer Payment)

The following process describes the Assisted Remote Payment.

No.	Component	Description
1	Initiate Assisted Remote Payment	Staff selects “Assisted Remote Payment” as the payment mode for the order.
2	Confirm Customer Contact Details	Staff confirms customer contact details for remote payment communication.
3	Confirm Order Details with Customer (Online/Phone)	Staff reviews order details with the customer before sending payment request.
4	Generate Remote Payment Request	System generates a remote payment request for the order.
5	Send Payment Link	System sends the payment link or instruction to the customer through the selected channel.
6	Wait for Customer Payment	Order enters a pending payment state while awaiting customer action.
7	Monitor Payment Time Window	System monitors the allowed payment time window for remote payment.
8	Payment Status Update	System updates payment status upon receiving payment result.
9	Payment Success Confirmation	System confirms successful payment and notifies staff.

10	Payment Failure Notification	If payment fails or is cancelled, system notifies staff.
11	Handle Payment Timeout	If customer does not pay within time limit, payment request expires.
12	Complete Order after Payment	After successful payment, the order is completed.
13	Deliver Electronic Tickets or Print Paper Tickets Onsite	The customer receives an electronic ticket or has a paper ticket printed on site.

Table 3.3.5.1.7: Description of Assisted Remote Payment User Process

3.3.5.1.8. User Process - Customer Display Screen

The following process describes the Customer Display Screen.

No.	Component	Description
1	Customer Display Screen Activation	Customer display screen is activated when a sales session starts.
2	Welcome/Idle Screen	Customer display screen displays welcome information or idle information when no active order exists.
3	Display Current Transaction Information	The customer display screen stays synchronised with the active transaction, showing only confirmed selections and not the operator's edits on operator screen.
4	Staff Can Turn Off the Synchronized Display on the Customer Screen at Any Time	The customer screen is enabled by default, but staff can turn off customer screen synchronization at any time.
5	Event/Session Information Display	Selected event, session, date and venue information are displayed to the customer.
6	Seat Selection Visualisation	Seat selection results are visually shown on the customer display screen.
7	Ticket Type and Quantity Display	Selected ticket types and quantities are displayed in real time.
8	Shopping Cart Summary Display	Customer screen shows a clear summary of items (products, tickets, etc) in the shopping cart.
9	Price Breakdown Display	Ticket prices, discounts, bundles and total amount are displayed transparently.

10	Promotion & Discount Visibility	Applied discounts or bundles are clearly indicated to the customer.
11	Payment Method Display	The selected payment method and payment instructions are shown.
12	Payment Progress Display	Customer screen displays payment status during payment processing.
13	Payment Result Display	Payment success or failure result is displayed to the customer.
14	Post-Payment Information Display	Post-payment messages such as ticket issuance or next steps are displayed.
15	Order Completes and Screen Resets	Customer screen resets to idle state after transaction ends.

Table 3.3.5.1.8: Description of Customer Display Screen**3.3.5.1.9. User Process - Ticket Issuance & Printing**

The following process describes the Ticket Issuance & Printing.

No.	Component	Description
1	Order Completed; Start Ticket Issuance	Upon payment success confirmation, the system triggers the ticket issuance process.
2	Printing Instruction Initiation	System initiates the ticket printing action.
3	Printing Status Monitoring	System tracks printing progress and status.
4	Successful Printing Confirmation	System confirms tickets are printed successfully.
5	Printing Exception Handling	In case of printing failure, staff is notified to take action.
6	Reprint Decision	Staff decides whether to reprint tickets if issues occur.
7	Customer Ticket Handover	Staff hands over printed tickets to the customer.
8	Post-Issuance Information Update	The ticket issuance record is updated, and the number of print attempts is recorded on the ticket information.
9	End of Ticket Issuance Process	Ticket issuance and printing workflow is completed.

Table 3.3.5.1.9: *Description of Ticket Issuance & Printing User Process*

3.3.5.1.10. Requirement List

RID	Description	Priority	Related Process
REQ-BO-TS-001	The system shall allow box office staff to log into the Box Office system using authorised accounts with role-based access control.	High	3.3.5.1.1: #1
REQ-BO-TS-002	The system shall allow staff to search events or programmes by venue, date, name, category, or availability status.	High	3.3.5.1.1: #3
REQ-BO-TS-003	The system shall allow staff to select event sessions including show date and time for ticket sales.	High	3.3.5.1.1: #5
REQ-BO-TS-004	The system shall support seat selection ticket through an interactive seat map.	High	3.3.5.1.2: #4
REQ-BO-TS-005	The system shall validate seat availability and seat rules before confirming seat selection.	High	3.3.5.1.2: #5
REQ-BO-TS-006	The system shall lock selected seats for a defined duration to prevent concurrent selection.	High	3.3.5.1.2: #7
REQ-BO-TS-007	The system shall automatically release locked seats if the lock duration expires before order confirmation.	High	3.3.5.1.2: #12
REQ-BO-TS-008	The system shall support ticket selection for admission events by ticket category and quantity.	High	3.3.5.1.3: #4
REQ-BO-TS-009	The system shall validate ticket quantity against inventory and purchase rules before adding tickets to the cart.	High	3.3.5.1.3: #6
REQ-BO-TS-010	The system shall allow staff to add selected tickets to a unified shopping cart.	High	3.3.5.1.3: #7
REQ-BO-TS-011	The system shall display shopping cart contents grouped by event, session, and sub-order structure.	High	3.3.5.1.4: #2
REQ-BO-TS-012	Staff shall be able to modify ticket items in the shopping cart including ticket quantity or price type where applicable.	Medium	3.3.5.1.4: #4

REQ-BO-TS-013	The system shall automatically recalculate pricing and validate inventory when shopping cart contents are modified.	High	3.3.5.1.4: #8
REQ-BO-TS-014	The system shall automatically calculate eligible bundle promotions and discounts.	High	3.3.5.1.5: #2
REQ-BO-TS-015	Staff shall be able to manually apply or remove discounts and promotion codes were permitted by business rules.	Medium	3.3.5.1.5: #5
REQ-BO-TS-016	The system shall confirm order details and create an order before initiating payment.	High	3.3.5.1.6: #1
REQ-BO-TS-017	The system shall support multiple payment methods including cash, card, Octopus, and integrated payment devices.	High	3.3.5.1.6: #3
REQ-BO-TS-018	For integrated payment methods, the system shall obtain payment status from the payment service.	High	3.3.5.1.6: #6
REQ-BO-TS-019	For non-integrated payment methods, the system shall allow staff to record payment results manually.	High	3.3.5.1.7: #7
REQ-BO-TS-020	The system shall support assisted remote payment where staff can send a payment link to customers.	Medium	3.3.5.1.7: #4
REQ-BO-TS-021	The system shall manage payment time windows for remote payments and update payment status accordingly.	Medium	3.3.5.1.7: #7
REQ-BO-TS-022	The system shall provide a customer-facing display screen showing transaction details to customers.	Medium	3.3.5.1.8: #5
REQ-BO-TS-023	The system shall display ticket information, pricing, discounts, and order summary on the customer display screen.	Medium	3.3.5.1.8: #9
REQ-BO-TS-024	The system shall issue tickets after successful payment and trigger ticket printing or digital ticket fulfilment based on configuration.	High	3.3.5.1.9: #1
REQ-BO-TS-025	The system shall support ticket printing through integrated ticket printing devices.	High	3.3.5.1.9: #2
REQ-BO-TS-026	The system shall record ticket issuance information including printing status and print attempts for operational tracking.	Medium	3.3.5.1.9: #8

REQ-BO-TS-027	The system shall update ticket inventory and order status after transaction completion.	High	3.3.5.1.6: #7
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Table 3.3.5.1.10: Requirement List of Ticket Sales Process

3.3.5.2. Ticket Exchange Process Management

This workflow covers **post-sale ticket modification scenarios** in BO operations, including ticket exchanges such as fare differences, date changes, and seat exchanges. The module shall enable BO staff to efficiently complete complex ticket adjustments at the counter while ensuring **ticket pool consistency, financial accuracy, and full operational traceability**. All changes must be **synchronised in real time with inventory and payment status**.

The system shall support multiple ticket exchange modes, and be tightly integrated with the **original order, payment method, service fee rules, and seat locking mechanisms**, to prevent inventory discrepancies or financial risks caused by manual handling.

3.3.5.2.1. User Process

The following process describes the general.

No.	Component	Description
1	Original Ticket Lookup	The operator queries the original order or ticket information to confirm the exchangeability of the ticket.
2	Validate Exchange Eligibility	The operator checks whether the ticket type, date, session, and user identity comply with the ticket exchange rules.
3	Select Exchange Type	The operator selects the ticket exchange type: exchange with price difference/change date/change seat.
4	Input New Ticket Details	The operator selects the seat, date, or fare of the new ticket based on the ticket exchange type.
5	Price Difference Calculation	If a price difference is involved, the system automatically calculates the difference for staff confirmation.
6	Confirm Exchange & Lock Seat	Upon confirmation of the ticket exchange, System locks the new seats and updates inventory.
7	Update Ticket Status	Original ticket status is updated to "Exchanged". Generates the new ticket, and links it to the original order.

8	Release Original Seat	The original seats from the exchanged ticket are released back to inventory according to exchange rules.
9	Complete Exchange	Print or generate a new ticket and notify the customer that the ticket exchange is completed.

Table 3.3.5.2.1: Description of the User Process of Ticket Exchange Process

3.3.5.2.2. Requirement List

RID	Description	Priority	Related Process
REQ-BO-TE-001	The system shall allow staff to retrieve original ticket information and initiate a ticket exchange process.	High	Refer to Section 3.3.5.2.1: #1
REQ-BO-TE-002	The system shall validate ticket exchange eligibility according to configured ticketing rules.	High	3.3.5.2.1: #2
REQ-BO-TE-003	The system shall support different ticket exchange scenarios including session change, seat change, and price difference exchange.	High	3.3.5.2.1: #3
REQ-BO-TE-004	The system shall manage inventory updates and ticket status changes during the exchange process.	High	3.3.5.2.1: #4, #5, #6
REQ-BO-TE-005	The system shall generate a new ticket and maintain linkage with the original ticket to ensure operational traceability.	Medium	3.3.5.2.1: #7

Table 3.3.5.2.2: Requirement List of Ticket Exchange Process

3.3.5.3. Onsite Refund Process Management

The refund process is used to support box office staff in handling operations when customers apply for a refund, including:

- **Single or batch ticket refund:** Supports scanning paper tickets or selecting multiple tickets via orders to initiate a refund in one go.
- **Refund processing:** You can choose to have the refund amount returned to the original ticketing account or the current operator's account.
- **Inventory restoration:** After the ticket refund is completed, the system can release the return the ticket inventory to the public ticket pool.
- **Operator friendliness:** Provides convenient interface operations, supports the generation of refund records, printing of receipts, and synchronous updating of sales data.

The core objective is to enable operators to process ticket refunds quickly and safely while ensuring the accuracy of system inventory and financial records.

3.3.5.3.1. User Process

The following process describes the general user process.

No.	Component	Description
1	Refund Entry	Operators can enter the refund interface and initiate the refund process by scanning the ticket code or querying the order.
2	Ticket/Order Selection	Select the tickets to be refunded, either individually or in batches, and the system will display the ticket information and refund amount.
3	Refund Amount Determination	The system calculates the refund amount based on the ticket price and applicable policies, including rules for discounts, package tickets, and handling fees. Verify whether the conditions for refund without review are met.
4	Choose whether to Refund the Handling Fee	If a handling fee was charged at the time of order payment, you can choose whether to refund the handling fee when refunding the ticket.
5	Record Refund Operator	System records the operator responsible for the refund transaction.
6	Determine whether the Conditions for "Refund without Approval" are Met	Determine whether the conditions for "refund without approval" are met.
6.1	Refund without Approval	If the conditions for a "refund without approval" are met, the operator shall confirm the refund information.
6.2	Submit a refund request for approval	If there are no conditions for "refund without approval", the operator shall submit a refund request and wait for approval by the administrator.
7	Refund Confirmation	After the refund request is successfully submitted or approved, the system conducts a secondary verification, including ticket status and inventory status.
8	Refund Execution	The system executes a refund, releases the ticket back to the public ticket pool (if applicable),

		updates order status and related operational reports.
9	Receipt/Confirmation Printing	The system provides the function of printing refund or transaction receipts.

Table 3.3.5.3.1: Description of the User Process of Onsite Refund Process Management

3.3.5.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-BO-RF-001	The system shall allow staff to initiate the refund process by scanning ticket codes or retrieving the original order.	High	3.3.5.3.1: #1
REQ-BO-RF-002	The system shall support refunding single or multiple tickets within one operation.	High	3.3.5.3.1: #2
REQ-BO-RF-003	The system shall calculate refund amounts based on ticket pricing rules, applied promotions, and refund policies.	High	3.3.5.3.1: #3, #4, #5
REQ-BO-RF-004	The system shall determine whether refund approval is required and submit the refund request for approval when applicable.	High	3.3.5.3.1: #6
REQ-BO-RF-005	After refund completion, the system shall update the order status and record the refund in relevant operational reports.	High	3.3.5.3.1: #7, #8, #9

Table 3.3.5.3.2: Requirement List of Onsite Refund Process Management

3.3.5.4. Ticket Issuance & Collection Process Management

The ticket issuance and collection process covers the operations of the ticketing system to complete the delivery of tickets to customers, including scenarios such as customers printing paper tickets at the on-site box office when they hold an order, as well as the delivery of electronic tickets.

- **On-site printing:** Customers with an order number or electronic ticket go to the on-site box office, where the operator prints the paper ticket through the system.
- **Electronic Ticket Delivery:** The system supports generating electronic tickets and providing digital delivery options.
- **Courier Ticket Delivery:** The system supports courier delivery configuration including delivery regions and associated fees (Hong Kong local, Mainland China, overseas).

- **Transaction Synchronisation:** After ticket issuance is completed, the system updates ticket fulfilment status and order information to ensure consistent ticket status across system interfaces and sales channels.
- **Operational Support:** The system provides convenient entry points for follow-up actions such as ticket reprint, order lookup, and transaction verification.

3.3.5.4.1. User Process

The following process describes the general user process.

No.	Component	Description
1	Ticket Issuance/Collection Entry	Customers holding the order number or e-ticket go to the on-site box office, and the operator enters the ticket issuance/collection interface.
2	Ticket Verification	The system verifies the order status and ticket validity, including whether the payment has been made and whether the ticket can be retrieved.
3	Verify Reprint Permission	The system verifies whether the operator is authorised to perform ticket reprinting according to system configuration.
4	Initiate Ticket Reprint	After permission verification, the operator initiates the ticket reprint process. If approval is required, the request is submitted for managerial approval before printing.
5	Ticket Issuance/Printing	For paid orders, print paper tickets on-site or generate electronic tickets, and record the status.
6	Update Fulfilment Status	After ticket issuance is completed, the system updates ticket fulfilment status and records printing history.

Table 3.3.5.4.1: Description of the User Process of Ticket Issuance & Collection Process

3.3.5.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-BO-TI-001	The system shall allow staff to retrieve orders using order number, ticket code, or electronic ticket information to initiate the ticket issuance or collection process.	High	3.3.5.4.1: #1
REQ-BO-TI-002	The system shall validate order status and ticket eligibility before ticket issuance or	High	3.3.5.4.1: #2

	collection, including payment confirmation and ticket availability.		
REQ-BO-TI-003	The system shall support on-site printing of paper tickets and record ticket issuance status.	High	3.3.5.4.1: #5
REQ-BO-TI-004	The system shall support electronic ticket generation and delivery.	High	3.3.5.4.1: #5
REQ-BO-TI-005	The system shall support courier ticket delivery according to configured delivery regions and fee rules.	Medium	3.3.5.4.1: #5
REQ-BO-TI-006	The system shall record ticket printing and reprint operations including operator information and print history for audit and operational tracking.	Medium	3.3.5.4.1: #3
REQ-BO-TI-007	After ticket issuance is completed, the system shall update ticket fulfilment status and synchronise order information across system interfaces.	High	3.3.5.4.1: #6

Table 3.3.5.4.2: Requirement List of Ticket Issuance & Collection Process

3.3.5.5. Shift Handover & Settlement Process Management

The shift handover and settlement process is designed to manage box office staff shifts, cash handling, and ticketing transaction records, ensuring transaction security, clear accounting, and consistency across devices and order statuses.

Key features include:

- **Login/Logout Credential Printing:** Allows operators to print credentials upon login or logout to record shift handover information and device alignment details.
- **Transaction Reconciliation and Summary:** The system aggregates sales, refunds, service fees, and ticket stock usage during each shift, with query and review support.
- **Shift Settlement:** At shift handover, each staff member can complete settlement against the ticketing system and connected devices (e.g. Apollo) to ensure financial consistency.
- **Flexible Enquiry:** Supports querying transaction records and discrepancies by staff member or device, providing insights to support ongoing optimisation of shift handover processes.

3.3.5.5.1. User Process

The following process describes the general user process.

No.	Component	Description
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1	Login	Operator logs into the BO system to begin the shift.
2	Login Credential Printing	When logging in, you can choose to print the login credential, which records the time, operator, and device alignment reference text.
3	Transaction Recording	The system begins to record all ticket transactions, refunds, handling fees, and ticket paper usage during the operator's shift.
4	Shift End/Logout	After completing a shift, operators log out of the system and can choose to print a logout credential that includes shift handover settlement information.
5	Settlement Update	The system updates shift settlement records and synchronise transaction summaries with connected devices (e.g. Apollo).
6	Post-Shift Actions	Operators may review shift transaction records and identify discrepancies for reconciliation purposes.

Table 3.3.5.5.1: Description of the User Process of Shift Handover & Settlement Process

3.3.5.5.2. Requirement List

RID	Description	Priority	Related Process
REQ-BO-SH-001	The system shall allow operators to log into the BO system to start a shift and record operator and terminal information.	High	3.3.5.5.1: #1
REQ-BO-SH-002	The system shall support printing login or logout credentials to record operator, timestamp, and terminal information.	High	3.3.5.5.1: #2. #3
REQ-BO-SH-003	The system shall record transaction data during each operator's shift, including ticket sales, refunds, service fees, and ticket stock usage.	High	3.3.5.5.1: #4
REQ-BO-SH-004	At shift end, the system shall generate a shift settlement summary and support reconciliation with connected devices or systems.	High	3.3.5.5.1: #5

REQ-BO-SH-005	The system shall allow querying shift transaction records and discrepancies by operator or device.	Medium	3.3.5.5.1: #6
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Table 3.3.5.5.2: Requirement List of Shift Handover & Settlement Process

3.3.5.6. Box Office User Account Management

The user account management process aims to streamline the login and permission management for box office operators, ensuring that each staff member's operational permissions, transaction records, and sales data are accurate and traceable.

Main functions include:

- **Unified system account:** Each staff member only needs to manage one system user account, eliminating the need to log in repeatedly at different venues.
- **Multi-device login support:** The system may allow the same account to log in on multiple BO devices according to system configuration.
- **Device Dimension Data Statistics:** Supports generating sales statistics and transaction reports by device, facilitating the analysis of sales show across different terminals.
- **Permission Control:** The system can assign different operation permissions based on roles to ensure the security of sensitive operations.

3.3.5.6.1. User Process

The following process describes the general user process.

No.	Component	Description
1	User Account Creation	Administrators create unified system accounts for staff members, assign roles and permissions.
2	Login from BO Device	Staff members log in to the system using a unified account on any BO device to confirm their permissions.
3	Multi-device Login	The system supports simultaneous login of the same account on multiple devices to ensure synchronization of sales data.
4	Credential Verification	The system verifies login credentials. If authentication succeeds, the user enters the system; otherwise, failed login attempts are recorded, and the account may be temporarily locked after repeated failures.
5	Administrator Account Re-enablement	Administrators may re-enable temporarily locked user accounts.

6	Transaction Recording	The system records all transaction operations performed by the user account across devices for traceability and audit purposes.
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Table 3.3.5.6.1: Description of the Box Office User Account Management User Process

3.3.5.6.2. Requirement List

RID	Description	Priority	Related Process
REQ-BO-UM-001	The system shall allow administrators to create unified system accounts for box office staff and assign roles and permissions.	High	3.3.5.6.1: #1
REQ-BO-UM-002	The system shall allow staff to log into the BO system using unified user accounts on BO devices.	High	3.3.5.6.1: #2
REQ-BO-UM-003	The system may allow the same account to log in on multiple BO devices according to system configuration.	Medium	3.3.5.6.1: #3
REQ-BO-UM-004	The system shall verify user credentials during login and temporarily lock the account after repeated failed login attempts.	High	3.3.5.6.1: #4
REQ-BO-UM-005	The system shall allow administrators to re-enable temporarily locked user accounts.	Medium	3.3.5.6.1: #5
REQ-BO-UM-006	The system shall record user account operations and transactions across devices for audit and traceability purposes.	High	3.3.5.6.1: #6

Table 3.3.5.6.2: Requirement List of Box Office User Account Management

3.3.5.7. Seat Reservation and Locking Process Management

The Seat Reservation and Locking Process is designed to provide flexible seat management capabilities, enabling box office staff to temporarily reserve seats for customers or groups for a defined time period. This function supports scenarios such as advanced reservations, group bookings, or special operational arrangements, while reducing the risk of seat conflicts during subsequent ticket sales.

Main functions include:

- **Seat Reservation and Locking:** Staff may reserve selected seats through the reservation mechanism, placing them in a locking status so that they cannot be sold during the reservation period.
- **Reservation Duration Control:** The system shall support configuring reservation duration and notify staff when the reservation is approaching expiration.

- **Flexible Cancellation:** Staff may cancel seat reservations or release held seats based on operational needs.

3.3.5.7.1. User Process

The following process describes the general user process.

No.	Component	Description
1	Ticket Type/Seat Selection	Staff select the seats or ticket types to be reserved for customers or groups in the BO system.
2	Create Seat Reservation	The system places the selected seats into a reservation locking state to prevent them from being sold during the reservation period.
3	Set Reservation Timer	Staff configure the reservation duration or reminder time for the reserved seats.
4	Reservation Expiry Reminder	The system notifies staff when the reservation is approaching its expiration time.
5	Release or Confirm Reservation	Staff decide whether to release the reserved seats or continue locking them, and the system updates seat status accordingly.
6	Update Reporting	The system records seat reservation operations for subsequent enquiry and operational reporting.

Table 3.3.5.7.1: Description of the Reservation & Locking Process User Process

3.3.5.7.2. Requirement List

RID	Description	Priority	Related Process
REQ-BO-RS-001	The system shall allow staff to select and reserve seats or ticket types for customers or groups.	High	3.3.5.7.1: #1
REQ-BO-RS-002	The system shall lock selected seats through the inventory service to prevent duplicate allocation or sales conflicts.	High	3.3.5.7.1: #2
REQ-BO-RS-003	The system shall support configuring reservation duration and notify staff when the reservation is approaching expiration.	High	3.3.5.7.1: #3, #4

REQ-BO-RS-004	Staff shall be able to release or maintain reserved seats based on operational requirements.	Medium	3.3.5.7.1: #5
REQ-BO-RS-005	The system shall record seat reservation and release operations for audit and operational tracking.	Medium	3.3.5.7.1: #6
REQ-BO-RS-006	When reservations are released or converted into ticket orders, the system shall update seat status and synchronise inventory information.	High	3.3.5.7.1: #5

Table 3.3.5.7.2: Requirement List of Reservation & Seat Locking Process

3.3.5.8. Lost Ticket Handling Process Management

The Lost Ticket Handling Process is designed to provide a standardised mechanism for ticket verification and reissuance, enabling box office staff to process cases where customers have lost their tickets or cannot present valid proof. Based on verifiable information, staff may reissue tickets while ensuring ticket usage control and traceability. This function applies to scenarios such as physical ticket reprinting and e-ticket re-delivery, while preventing duplicate usage of original and reissued tickets.

Main functions include:

- Ticket Lookup and Verification: Staff may search and verify ticket validity based on order information, purchaser identity, or supporting credentials.
- Ticket Reissuance: The system supports physical ticket reprinting.
- Original Ticket Status Handling: The system shall invalidate original tickets upon reissuance to prevent duplicate usage.
- Operation Logging: The system records lost ticket handling and reissuance activities for enquiry and audit purposes.

3.3.5.8.1. User Process

The following process describes the general user process.

No.	Component	Description
1	Ticket Lookup	Staff search for the ticket using order number, customer details, or other credentials in the BO system.
2	Ticket Verification	The system verifies ticket status and validity (e.g. unused, in-use).
3	Confirm Reissuance	Staff confirm to proceed with ticket reissuance.
4	Reissue Ticket	The system reissues the ticket via printing.
5	Invalidate Original Ticket	The system marks the original ticket as invalid.

6	Record Operation	The system logs the lost ticket handling and reissuance operation.
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Table 3.3.5.8.1: Description of the Lost Ticket Handling Process User Process

3.3.5.8.2. Requirement List

RID	Description	Priority	Related Process
REQ-BO-LT-001	The system shall allow staff to search tickets based on order or customer information.	High	3.3.5.8.1: #1
REQ-BO-LT-002	The system shall validate ticket status to ensure eligibility for reissuance.	High	3.3.5.8.1: #2
REQ-BO-LT-003	The system shall support physical ticket reprinting.	High	3.3.5.8.1: #3 - #4
REQ-BO-LT-004	The system shall invalidate the original ticket after reissuance.	High	3.3.5.8.1: #5
REQ-BO-LT-005	The system shall log lost ticket handling and reissuance operations.	Medium	3.3.5.8.1: #6

Table 3.3.5.8.2: Requirement List of Lost Ticket Handling Process

3.3.5.9. Membership Sales and Activation

The Box Office governs the end-to-end workflow of onsite membership sales and activation within the Box Office system. It is designed to support efficient frontline processing of membership purchases while ensuring pricing accuracy, correct application of eligible discounts, and proper capture of member information for activation and future membership services.

This module enables box office staff to select the applicable membership product, apply eligible discounts or promotional offers where permitted, complete payment through the standard payment workflow, and perform membership activation after successful payment. By centralising membership product selection, discount handling, payment processing, and activation into a controlled operational flow, the system helps reduce manual errors and ensures that membership records are created and activated in a consistent manner.

The Membership Sales and Activation module serves as the operational foundation for subsequent membership-related services at the box office, such as membership enquiry, benefit validation, and customer support.

3.3.5.9.1. User Process – General Process

The following process describes the user process for membership sales and activation.

No.	Component	Description
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1	Box Office Login	Box office staff logs into the Box Office system using authorised accounts. System validates user roles and related venue access rights.
2	Enter Membership Sales Process	Staff enters the membership sales function from the Box Office system.
3	Membership Product Selection	Staff selects the membership product to be purchased, such as an applicable membership type for M+, HKPM, or PA.
4	Display Membership Product Information	System displays the selected membership product details, including membership name, price, validity, and applicable conditions.
5	Apply Discount/Promotion (Optional)	Staff inputs applicable discount information, promotional code, or other approved pricing adjustment where permitted by business rules.
6	Discount Validation	System validates the entered discount or promotion against applicable rules and scope.
7	Display Membership Order Summary	System displays the payable amount after applying valid discounts or promotional adjustments.
8	Order Confirmation	Staff confirms the membership purchase details with the customer before proceeding to payment.
9	Payment Method Selection	Staff selects the payment method according to customer preference.
10	Payment Processing	System processes payment and handles success, failure, or timeout scenarios.
11	Payment Confirmation	Upon successful payment, system confirms the membership transaction and creates the corresponding order record.
12	Enter Activation Information	Staff inputs the customer's membership activation information, including email address and name.
13	Membership Activation	Staff clicks the activation action. System validates the required information and activates the membership.
14	Activation Confirmation	System confirms successful activation and updates the membership record status accordingly.

15	Transaction Completion	Membership sales and activation process is completed. Staff may proceed to the next customer service action if required.
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Table 3.3.5.9.1: Description of the Membership Sales and Activation User Process

3.3.5.9.2. User Process – Membership Activation

The following process describes the Membership Activation after payment.

No.	Component	Description
1	Enter Activation Process	After successful payment, staff enters the membership activation step.
2	Input Member Email Address	Staff inputs the customer's email address required for membership activation.
3	Input Member Name	Staff inputs the customer's name required for membership activation.
4	Validate Activation Information	System validates whether the required activation information has been provided in the correct format.
5	Submit Activation Request	Staff clicks the activation action to submit the membership activation request.
6	Activate Membership Record	System activates the membership and updates the membership status in the membership record.
7	Display Activation Result	System displays the activation result to staff.
8	Handle Activation Failure (if any)	If activation fails, system notifies staff and allows correction or retry where applicable.

Table 3.3.5.9.2: Description of Membership Activation User Process

3.3.5.9.3. Requirement List

RID	Description	Priority	Related Process
REQ-BO-MS-001	The system shall allow box office staff to access the membership sales function using authorised accounts with appropriate access rights.	High	3.3.5.9.1: #1, #2
REQ-BO-MS-002	The system shall support the sale of applicable membership products for M+, HKPM, and PA.	High	3.3.5.9.1: #3

REQ-BO-MS-003	The system shall display membership product information including product name, price, validity, and applicable conditions.	High	3.3.5.9.1: #4
REQ-BO-MS-004	The system shall allow staff to input applicable discounts, promotional codes, or approved pricing adjustments during membership sales.	Medium	3.3.5.9.1: #5
REQ-BO-MS-005	The system shall validate discounts or promotions against configured business rules before application.	High	3.3.5.9.1: #6/ 3.3.5.9.2: #3
REQ-BO-MS-006	The system shall recalculate and display the payable amount after valid discount application.	High	3.3.5.9.1: #7/ 3.3.5.9.2: #5
REQ-BO-MS-007	The system shall allow staff to confirm the membership order before proceeding to payment.	High	3.3.5.9.1: #8
REQ-BO-MS-008	The system shall support payment processing for membership sales through applicable payment methods configured for the Box Office system.	High	3.3.5.9.1: #9, #10
REQ-BO-MS-009	The system shall create a membership sales order record upon successful payment.	High	3.3.5.9.1: #11
REQ-BO-MS-010	The system shall allow staff to input the customer's email address and name for membership activation after successful payment.	High	3.3.5.9.1: #12/ 3.3.5.9.2: #2, #3
REQ-BO-MS-011	The system shall validate required activation information before membership activation is submitted.	High	3.3.5.9.2: #4
REQ-BO-MS-012	The system shall activate the membership record upon successful submission of valid activation information.	High	3.3.5.9.1: #13/ 3.3.5.9.2: #6
REQ-BO-MS-013	The system shall update the membership status and activation result in the membership record after activation.	High	3.3.5.9.1: #14/ 3.3.5.9.2: #7
REQ-BO-MS-014	The system shall notify staff when membership activation fails and allows correction or retry where applicable.	Medium	3.3.5.9.2: #8

REQ-BO-MS-015	The system shall maintain transaction and activation records for audit and operational tracking purposes.	Medium	3.3.5.9.1: #11-#15
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Table 3.3.5.9.3: Requirement List of Membership Sales and Activation

3.4. Website

This section outlines the end-to-end business processes for the Website integration. It covers the business requirement statement, background and objectives, high level and detailed business requirements, the fulfilment approach, customer journey, and the functional specifications for both frontend and backend workflows.

3.4.1. Business Requirements

From a business perspective, WestK requires ticketing and membership card websites to integrate with AnTank's backend ticketing capabilities to support the ticketing purchase and membership card purchase needs of multi-channel users, while ensuring that operational processes are efficient and data processing is secure and compliant, in line with the operational specifications of the website platform and business development goals.

The following business requirements have been identified.

3.4.1.1. User Registration & Login

WestK requires the ticketing website to support two login modes:

one is registered user login, including official website registration login and co-branded authorization login via third-party platforms, and to achieve single sign-on (SSO) synchronization with the existing WestK Account, while promoting the linkage between the ticketing website and the Customer Relationship Management(CRM) system to achieve synchronization of membership data (identity information, membership level, privilege records, etc.), improve data statistics and tracking capabilities, and ensure the identity consistency of order management and ticket ownership.

the second mode is to support ticket purchase in guest mode, which does not require registering a website account and only requires filling in necessary contact information during the payment process. It must also have a secure session management function to strictly protect users' personal data, and prompt users to read and agree to the privacy protection policy published by WestK before registration and payment.

3.4.1.2. Ticketing Website Ticketing Channel

WestK requires the ticketing website to integrate the event resources of each business unit and centrally display the events of M+, HKPM and PA. It is necessary to design an intuitive homepage navigation, a global search function, as well as clear entry points for detailed pages and mainstream media sharing functions to help users quickly discover target events. The event details page needs to fully present core information such as event title, description, poster, event date, session time,

venue information, ticket type and price, seating availability, etc., providing sufficient basis for user's ticket purchase decisions. Supports users to purchase exhibition tickets and show tickets online and supports online seat selection. Users can select tickets for multiple sessions and multiple ticket types in a single order at once, eliminating the need for repeated orders and improving ticket purchasing efficiency.

3.4.1.3. Electronic Ticketing Wallet

WestK requires users to obtain an electronic ticket (including a unique QR code or verification code) after successfully purchasing through the ticketing website, and they can directly enter without having to exchange for a physical ticket. The website must set up a centralised electronic ticket management entry, where users can view all unused and used electronic tickets in the "My Orders" or dedicated electronic ticket page, clearly displaying the corresponding event information, entry time, seat information, and verification status of the electronic tickets, facilitating users to retrieve and use them at any time.

3.4.1.4. Order Management

WestK requires users to be able to centrally view details of all orders through the "My Orders" module on the ticketing website, including order status (pending payment, paid, cancelled), ticket information, payment amount, payment time, e-tickets, etc. It also allows registered users to apply for online ticket refunds, upgrade ticket types, or change tickets according to the rules within the permitted time frame.

3.4.1.5. Payment

WestK requires ticketing websites to complete online payments through third-party payment platforms. Currency settlement must be uniformly conducted in Hong Kong dollars as the transaction currency.

3.4.1.6. Preferential Policy

WestK requires the ticketing website to provide members with rich preferential benefits and marketing strategy support. Users (including registered website users and visitors) can enjoy package ticket discounts and promotion code discounts when purchasing tickets. Website members can enjoy member discounts, point deductions, and other benefits when purchasing tickets, and at the same time, the website can offer diverse marketing activities, including priority ticket purchase with bank cards, bank card full-reduction discounts, "ticket+coupon" add-on purchase, balloting for high-demand events by randomly selecting ticket buyers, and cross-session package tickets for music seasons. When purchasing tickets, the system automatically identifies package ticket discounts and member discounts and automatically calculates the discount amount, thereby enhancing users' willingness to purchase tickets and member stickiness.

3.4.1.7. Ticket Verification and Entry Control

WestK requires the ticketing website to display the ticketing delivery methods supported by the event (including e-tickets, on-site ticket collection, and paper ticket mailing), and users can choose one as needed when placing an order. Each ticket

corresponds to a unique admission QR code or verification code, the QR code should support static and dynamic code for. During admission verification, the ticket/order status of the website and the AnTank system is synchronized in real-time to prevent repeated entry and ensure traceability of operations. Users can directly verify and enter with the e-ticket code or collect paper tickets at the designated ticket collection machine/ticketing centre with the ticket collection code and then enter or directly enter with the paper tickets delivered by mail.

3.4.1.8. Membership Card Purchase and Activation

WestK requires the ticketing website to integrate AnTank membership card resources, centrally display membership card products of M+ and HKPM, and provide clear entry to the detailed information page. The membership card detailed information page needs to present core information such as name, list of benefits, pricing, validity period, etc., to provide a basis for user decision-making. It supports online purchase, payment, and activation, eliminating the need for offline processing and improving usage efficiency.

3.4.1.9. Accessibility

WestK requires the ticketing website and membership card website to provide an inclusive and accessible user experience for all users, including persons with disabilities and users with different assistive needs. The website shall be designed and implemented to support accessible access to ticketing, membership purchase, order management, and related customer journeys, so that users can complete key online transactions independently and effectively.

The website shall support accessible design features including clear page structure, keyboard-operable navigation, screen reader compatibility, sufficient colour contrast, scalable text display, meaningful form labels, and clear error messages. Interactive elements such as event search, session selection, seat selection where applicable, shopping cart, payment steps, e-ticket display, and membership activation shall be designed to remain usable and understandable for users relying on assistive technologies.

Accessibility requirements shall be considered throughout the design, development, and testing process to support compliance with WCAG 2.1 Level AA and other applicable accessibility standards adopted by WestK.

3.4.1.10. Privacy, Security, Scalability & Inclusive Accessibility

WestK requires the compliant handling of personal data, including minimising data collection, restricting usage to authorised purposes, encrypting sensitive information, and implementing strict access controls. The system must also support high concurrency during peak sales periods to ensure a stable experience for all users. It should achieve multi-device self-adaptation, be compatible with different terminals such as computers and mobile phones, and ensure consistent page layout, operation process, and interactive experience across all devices. It supports multi-language switching among Traditional Chinese, Simplified Chinese, and English.

3.4.2. Business Requirement Analysis

This section further analyses and clarifies the business requirements described in Section 3.4.1, with each subsection focusing on the implementation of core functions and clarifying the key points for the implementation of requirements.

3.4.2.1. User Registration & Login

Map to Business Requirement 3.4.1.1 and 3.4.1.10.

- The ticketing website supports visitors to register and log in natively through the official website.
- The ticketing website supports co-branded authorised login via mainstream third-party platforms (WeChat, TikTok (Douyin), Facebook, X, Google accounts).
- The ticketing website has implemented SSO synchronization with WestK Account. After users log in to any official website, their identity information is automatically associated with the ticketing website.
- The ticketing website is integrated with the CRM system to synchronize member data (including identity information, membership level, privilege records, etc.), thereby improving data statistics and tracking capabilities.
- The ticketing website supports guest mode, eliminating the need to register an account. Simply fill in your contact information (mobile number/email) during the payment process to complete the ticket purchase.
- Maintain user identity consistency for order management and ticket ownership management.
- The system ensures secure session management and user data protection.

3.4.2.2. Website Content

Map to Business Requirement 3.4.1.2.

- The ticketing website designs an intuitive homepage navigation, a global search function, and a clear entry to the event details page.
- The ticketing website integrates resources from various events and centrally showcases exhibition and performances of M+, HKPM and PA.
- The event details page fully presents core information such as title, description, poster, event date, session time, venue information, ticket price, seating availability, etc.
- Provides mainstream third-party media sharing functionality.

3.4.2.3. Ticket Purchase Process

Map to Business Requirement 3.4.1.2, 3.4.1.4, 3.4.1.5, and 3.4.1.7.

- The ticketing website should support two types of tickets:
 - Admission tickets: Selection of Ticket Types and Quantities.

- Seat selection tickets: Provide an interactive online seat map to support users in selecting seats online or use the system's automatic seat selection method for seat selection.
- The ticketing website should support two purchase processes:
 - Standard Settlement: Allows users to settle the selected item immediately after choosing its seats or ticket types.
 - Shopping Cart Settlement: Allows users to select tickets from multiple sessions and ticket types at once and add them to the shopping cart for settlement together.
- The order confirmation page should include the following information and functions:
 - Details of the selected ticket, including ticket type or seat information.
 - For events that require real-name authentication, real-name verification must be supported.
 - Provide available preferential policy options for users to choose and use at their own discretion, and the system will automatically calculate and display the preferential amount based on the selection.
 - Provide available delivery method options, allowing users to choose and fill in relevant information themselves.
 - should clearly display the total amount payable for the order and its details, including ticket price, discounted price, and delivery fee.
 - Supports online payment.
- Inventory lock validity period during the ticket purchase process:
 - Registered website users: After locking inventory, they must complete payment within the specified time; otherwise, the inventory will be unlocked, and shopping cart data will be cleared.
 - Guest Mode: After successfully logging in to Guest Mode, users must complete order payment within the specified time; otherwise, the inventory will be unlocked, the shopping cart and user data will be cleared, and the user will be logged out.
 - Inventory Lock Time Explanation: In the immediate settlement scenario, inventory is locked upon user confirmation; for shopping cart orders, inventory is locked when the ticket is added to the shopping cart.

3.4.2.4. Electronic Ticketing Wallet

Map to Business Requirement 3.4.1.4.

- The ticketing website sets up a centralised e-ticket management, which can be accessed via a dedicated e-ticket page.
- The e-ticket page is categorised and displayed by "unused/used", clearly marking the corresponding item information, admission time, seat information, and verification status.
- Each electronic ticket shall contain a unique QR code for admission.

3.4.2.5. Order Management

Map to Business Requirement 3.4.1.4.

- The orders module of the ticketing website supports viewing historical order data of logged-in users.
- The order details fully display information such as ticket details, payment amount, payment time, and e-tickets.
- Users can reopen the payment processes for unpaid orders.
- Allows performing relevant after-sales operations on ticket orders:
 - Allows users to apply for ticket refunds online.
 - Allows users to upgrade or change ticket types online.

3.4.2.6. Payment

Map to Business Requirement 3.4.1.4 and 3.4.1.5.

- The ticketing website integrates mainstream third-party payment platforms via specific payment gateway, supporting payment methods such as WeChat Pay, Alipay, and bank card payment.
- The payment process uses encrypted transmission technology to ensure transaction security.
- During the payment process, users are allowed to complete the payment within the pending payment time window or cancel the order before it expires.
- The payment result is immediately call backed to the website and displayed to the user, while synchronously updating the order status.
- The transaction currency is uniformly denominated in Hong Kong dollars to ensure consistent settlement standards.

3.4.2.7. Preferential Policy

Map to Business Requirement 3.4.1.6.

- The ticketing website's preferential rights cover two user modes, and both registered users and visitors can enjoy package ticket discounts and promo code discounts.
- Website members enjoy exclusive benefits such as member discounts, promotion code, etc.
- When a user purchases tickets, the system automatically identifies available package ticket discounts and membership discounts and provides optional preferential policies for the user to choose from. The system calculates the discount amount in real-time based on the latest selection.
- Supports diversified marketing campaign configurations, including priority ticket purchase with bank cards, bank card full amount reduction discounts, ticket add-on purchases, balloting ticket purchases, and cross-session package tickets.

3.4.2.8. Ticket Verification and Admission Control

Map to Business Requirement 3.4.1.7.

- The ticketing website labels the supported ticketing delivery methods (including e-tickets, on-site ticket collection, and paper ticket mailing) by event.
- When placing an order, users can only select one of the delivery methods supported by the event.
- Each ticket corresponds to a unique admission QR code or verification code, and the ticket/order status is real-time synchronized with backend ticketing system during admission verification.
- Users can directly verify and enter with the electronic ticket code or obtain a paper ticket at the designated location with the ticket collection code and then enter or directly enter with the mailed paper ticket.

3.4.2.9. Membership Card Purchase and Activation

Map to Business Requirement 3.4.1.8.

- The ticketing website integrates membership card resources, centrally displays M+, HKPM and PA related membership card products, and provides clear entry to the detailed information page.
- The membership card details page clearly presents core information such as name, benefits list, pricing, and validity period, providing support for user decision-making.
- Supports users to complete membership card selection, payment, and activation online, eliminating the need for offline processing steps.

3.4.2.10. Accessibility

Map to Business Requirement 3.4.1.9.

The ticketing website and membership card website shall support an inclusive and accessible user experience for all users, including persons with disabilities and users relying on assistive technologies.

The website shall support accessible access to key user journeys, including user registration and login, event browsing, search, ticket selection, shopping cart review, payment, e-ticket access, membership card purchase, and membership activation.

The website shall support accessible design and interaction features, including:

- Clear and consistent page structure.
- Keyboard-operable navigation and interaction.
- Screen reader compatibility.
- Sufficient colour contrast between text and background.
- Meaningful labels and instructions for form fields and interactive controls.
- Clear and understandable validation messages and error prompts.
- Accessible presentation of important transaction information, including ticket details, pricing, discounts, payment status, and ticket verification information.

- Accessibility requirements shall be considered throughout the design, development, and testing process to support compliance with WCAG 2.1 Level AA.

3.4.2.11. Privacy, Security, Scalability & Inclusive Accessibility

Map to Business Requirement 3.4.1.10.

- Ticketing websites comply with regulations in handling personal data, reducing unnecessary collection, restricting the scope of use, encrypting sensitive information, and implementing strict access controls.
- The system supports high-concurrency access during peak sales periods to ensure stable operation.
- Implement multi-device self-adaptation, compatible with terminals such as computers and mobile phones, to ensure consistent page layout, operation process, and interactive experience.
 - Compatible with iOS at 13.0 or above and Android OS at 12.0 or above.
 - Compatible with browsing on mainstream browsers, including Microsoft Edge at 108 or above, Google Chrome at 108 or above, Firefox at 108 or above, Safari at 16 or above.
- The ticketing website supports language switching between Simplified Chinese, Traditional Chinese, and English, allowing users to choose their preferred language and instantly switch the website content to the selected language for display.

3.4.3. Operating Model

This initiative adopts a collaborative operating model to support the ticketing website and membership card website as digital sales and service channels for WestK.

The operating model defines the respective roles and responsibilities of the key parties involved in website operations, with AnTank providing and supporting the website platform and related ticketing capabilities, and WestK managing the business content, sales policies, and day-to-day channel operations.

AnTank – Ticketing Technology Provider

AnTank is responsible for providing and supporting the ticketing website platform and its related backend capabilities.

Key responsibilities include:

- Providing the website platform and related functional capabilities required to support online ticket sales, membership card purchase, order management, e-ticket access, and related customer journeys.
- Supporting website configuration, including event setup, session setup, ticket pricing, seating, inventory control, delivery methods, promotional rules, and membership card product setup, in accordance with approved business requirements.
- Supporting integration with relevant systems and services, including payment gateways, WestK Account, CRM, and other approved third-party platforms required for website operations.

- Maintaining system availability, operational monitoring, logging, and technical support in accordance with agreed service levels.
- Supporting secure transaction processing, data protection controls, and system operations in accordance with approved technical and compliance requirements.

WestK – Website Business Operations

WestK is responsible for managing the business operations of the ticketing website and membership card website.

Key responsibilities include:

- Managing website business content, including event information, membership card product information, and other customer-facing content required for online sales and service.
- Defining and managing business rules and policies applicable to the website channel, including ticket sales rules, refund and exchange policies, delivery options, membership rules, and promotional arrangements.
- Managing day-to-day website operations, including campaign planning, sales arrangements, benefit configuration requirements, and coordination of business changes across venues and programmes.
- Managing customer service and business handling related to website transactions, including enquiries, order-related follow-up, and coordination of after-sales processes in accordance with business policies.
- Reviewing website performance, sales outcomes, and operational data to support ongoing channel optimisation and business development.

3.4.4. Customer Journey

This section describes what users can do after entering the website.

The user story is used to describe the user's needs, goals, and expected outcomes, providing clarity on what the user wants to achieve and why the functionality is required. When users enter the website, they can complete the entire ticket purchasing and admission process without leaving the platform.

No.	Category	Role	User Story
1	Website Access	As a Visitor/Registered User	I visit the official WestK website or WestK Ticketing Website to browse upcoming exhibitions, shows, or cultural events.
2	Registration/Login	As a Visitor/Registered User	I can choose to continue in visitor mode or log in using a WestK Account or a third-party account (such as WeChat, Google, etc.) to enjoy full ticketing and order management features.

Technology Partnership Package 2
Requirement Document

3	Event Discovery	As a Visitor/Registered User	I search for exhibitions, or events of interest through homepage recommendations, category navigation, or the search function.
4	Event Information Review	As a Visitor/Registered User	I check the event detail page, including the event description, date, session, venue, ticket price, seating chart (if applicable), and related notes to decide whether to purchase tickets.
5	Session & Ticket Selection	As a Visitor/Registered User	I select a specific session and, depending on the type of event, choose seats or ticket types and quantity, with the system displaying real-time stock status.
6	Shopping Cart Review	As a Visitor/Registered User	I review the selected tickets in the shopping cart, confirming the session, seats, quantity, price, and applicable discounts or membership benefits.
7	Order Confirmation	As a Visitor/Registered User	I confirm the order details, choose the ticket delivery method (e.g., e-ticket), and proceed to the payment process.
8	Promotion & Discount Application	As a Visitor/Registered User	Before payment, I can review and apply available promotional offers such as membership discounts, coupons, promotional codes, or other eligible benefits, and the system calculates and displays the updated payable amount accordingly.
9	Payment Processing	As a Visitor/Registered User	I complete the payment through supported online payment methods, with the system locking inventory and processing the transaction result within the specified time.
10	Ticket Issuance	As a Visitor/Registered User	After successful payment, I receive an e-ticket (QR Code) and can view ticket details in "My Orders" or the e-ticket wallet.
11	Order & Ticket Management	As a Registered User	I can check the order status, e-tickets, and payment records in the personal centre, and apply for

			refunds, rescheduling, or upgrades within policy limits.
12	Venue Admission	As a Ticket Holder	Upon arriving at the venue, I use the e-ticket QR code for ticket verification and entry, with the system synchronizing ticket and entry status in real time.
13	Post-Purchase Engagement	As a Registered User	After completing the ticket purchase, I may receive related event recommendations, membership benefit prompts, or marketing information to enhance the overall experience.

Table 3.4.4: Description for the User Story

3.4.5. Business Scenarios

3.4.5.1. Registration and Login

This process defines how users can access the ticketing website securely and in compliance with privacy regulations.

- **Registration process:** Visitors can complete the registration of their ticketing website account by registering natively through the official website, filling in the necessary information, and completing verification.
- **User Login Process:** Users support three login methods: Joint Login (authorised direct login via mainstream third-party platforms WeChat, Douyin, Facebook, X, Google accounts), Registered User Login (login with an account registered on the official website, synchronised with the SSO of WestK's official websites, and the identity is automatically associated with the ticketing website after logging in to any official website), Guest Mode Login (after completing non-robot verification, the system assigns a temporary account to complete the login).
- **Privacy Policy:** Before user registration or payment, users must view and check to agree to WestK Privacy Policy; otherwise, they cannot proceed. The page provides a "Privacy Policy" link that can be clicked to view the full content, ensuring users' Right to be informed.

3.4.5.1.1. User Process

The following table describes the user flow for registration and login.

No.	Component	Description
1	User Registration	Visitors register natively through the official website, fill in the necessary information and complete verification, thus completing the registration for a ticketing website account.

2	Joint Login	Visitors log in through joint authorisation via mainstream third-party platforms (WeChat, TikTok (Douyin), Facebook, X, Google accounts) and directly complete the login operation on the ticketing website after authorization.
3	Registered User Login	Visitors can log in using their registered accounts on the official ticketing website; meanwhile, the ticketing website has implemented SSO single sign-on synchronization with WestK Account. After users log in on either official website, their identity information will be automatically associated with the ticketing website, eliminating the need for repeated logins.
4	Login in Guest Mode	The visitor selects the guest mode to log in. After passing the non-robot verification, the system automatically assigns a temporary account, and the guest mode login is successful.
5	Privacy Protection Policy Checkbox	Before users register or make a payment, they must view and check to agree to the Privacy Protection Policy published by WestK; otherwise, they cannot proceed with subsequent operations.
6	Privacy Policy Link	The page provides a clickable "Privacy Policy" link. After clicking, users can jump to the full privacy policy page to view detailed content such as data processing and rights protection.

Table 3.4.5.1.1: Description of the User Process of Registration and Login

3.4.5.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-WEB-REG-001	Supports visitors to complete the account registration on the ticketing website by registering natively through the official website, filling in necessary information, and completing verification.	High	3.4.5.1.1:#1
REQ-WEB-REG-002	Supports visitors to log in via joint authorisation through mainstream third-party platforms (WeChat, TikTok (Douyin), Facebook, X, Google accounts) and directly complete the login to the ticketing website after authorisation.	High	3.4.5.1.1:#2

REQ-WEB-REG-003	Supports visitors to log in using their official website registered accounts; meanwhile, the ticketing website and WestK Account implement SSO single sign-on synchronisation, so that after users log in to any official website, their identity information is automatically associated with the ticketing website, eliminating the need for repeated logins.	High	3.4.5.1.1:#3
REQ-WEB-REG-004	Supports guest mode login. After visitors complete non-robot verification, the system automatically assigns a temporary account to complete the login.	High	3.4.5.1.1:#4
REQ-WEB-REG-005	Before users register or make a payment, they must view and check to agree to the Privacy Protection Policy published by WestK; otherwise, they cannot proceed with the operation.	High	3.4.5.1.1:#5
REQ-WEB-REG-006	The page provides a clickable "Privacy Policy" link, allowing users to jump to the full privacy policy page to view details.	High	3.4.5.1.1:#6

Table 3.4.5.1.2: Requirement List of Registration and Login

3.4.5.2. Website Content

This business scenario defines the core operational logic for content display and resource integration on ticketing websites, clarifies the operating specifications for event resource presentation, navigation search, information display on detail pages, and sharing functions, achieves standardized and visual display of ticketing event information, helps users efficiently discover target events and obtain core ticketing information, and enhances the user experience in website browsing and ticketing decision-making.

3.4.5.2.1. User Process

The following table describes the website content.

No.	Component	Description
1	Event Resource Integration and Display	The ticketing website integrates event resources from various business units and centrally showcases the exhibition and shows of M+, HKPM and PA.
2	Homepage Navigation and Search Function	The ticketing website designs an intuitive homepage navigation, a global search function, and provides a clear entry to the event details

		page, helping users quickly locate their target events.
3	Core Information Display on Event Details Page	The event details page fully presents core information such as title, description, poster, event date, session time, venue information, ticket price, seating availability, etc., providing a basis for users' ticket purchase decisions.
4	Third-party media sharing function	The ticketing website provides mainstream third-party media sharing functionality, enabling users to share their favourite events on social platforms.

Table 3.4.5.2.1: Description of the User Process of Website Content

3.4.5.2.2. Requirement List

RID	Description	Priority	Related Process
REQ-WEB-CONT-001	The ticketing website needs to integrate event resources from various business units and centrally display exhibition and performing arts (PA) show events of M+ and HKPM.	High	3.4.5.2.1:#1
REQ-WEB-CONT-002	Design intuitive home page navigation and global search functionality, while providing clear entry points to event detail pages, to help users quickly locate target events.	High	3.4.5.2.1:#2
REQ-WEB-CONT-003	The event details page shall fully present core information such as title, description, poster, event date, session time, venue information, ticket price, seating availability, etc.	High	3.4.5.2.1:#3
REQ-WEB-CONT-004	Provides mainstream third-party media sharing functionality, enabling users to share events to social platforms.	High	3.4.5.2.1:#4

Table 3.4.5.2.2: Requirement List of Website Content

3.4.5.3. Ticket Purchase

This business scenario defines the core operational logic of the entire ticketing process on the ticketing website, clarifies the standard processes for ticket type selection, shopping cart management, order confirmation, and inventory locking, achieves a closed loop for the entire ticketing process from ticket type selection to pre-payment, ensures the standardisation and efficiency of ticketing operations, and meets the diverse needs of users for quick ticketing for a single item and combined ticketing for multiple sessions and ticket types.

3.4.5.3.1. User Process

The following table describes the user process for ticket purchase.

No.	Component	Description
1	Ticket Selection	The ticketing website supports ticket selection, allowing users to choose the corresponding ticket type and set the quantity to purchase.
2	Seat Selection Ticket Selection	The ticketing website supports the purchase of seat selection tickets, providing an interactive seating chart for self-selection or system automatic seat selection.
3	Settle Immediately	After the user selects the ticket type/seat, they can directly initiate immediate settlement and enter the order confirmation page.
4	Shopping Ticket Type Operations	Supports adding multi-session, multi-ticket type tickets to the shopping cart, allowing the addition of new items, seats, and ticket types, modification of the price of selected seat tickets, and modification of the quantity of selected ticket types.
5	Delete/Empty Cart Items	Allows users to delete selected items, seats, and ticket types in the shopping cart, or clear all contents of the shopping cart with one click.
6	Unified Settlement of Shopping Cart	After the user finishes editing the shopping cart contents, they initiate unified settlement and enter the order confirmation page.
7	Content displayed on the order confirmation page	The order confirmation page displays details of the selected tickets, clearly indicating the total amount payable and breakdown of costs such as ticket price, discounted price, etc.
8	Order Confirmation Real-Name Entry	For events that require real-name authentication, the order confirmation page supports users to complete the entry of real-name information for ticket purchasers.
9	Order Confirmation - Preferential Policy Selection	The order confirmation page provides available preferential policy options for users to choose and use at their own discretion.
10	Order Confirmation Delivery Method Selection	The order confirmation page provides options for delivery methods supported by the event, allowing users to select and fill in relevant delivery information.

11	Order Confirmation Online Payment	The order confirmation page clearly indicates the total amount payable and supports users in completing online payment through third-party payment platforms.
12	Check the Privacy Policy for Order Confirmation	Users must check and agree to the Privacy Policy on the order confirmation page; otherwise, they will be unable to submit the order and proceed with subsequent payment.
13	Inventory Locking Validity Period	After locking the inventory, if payment is not completed within the specified time, the inventory will be automatically unlocked, the shopping cart will be emptied, and unpaid orders will be cancelled.

Table 3.4.5.3.1: Description of the User Process of Ticket Purchase

3.4.5.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-WEB-BUY-001	Supports ticket selection, allowing users to choose the corresponding ticket type and set the purchase quantity.	High	3.4.5.3.1:#1
REQ-WEB-BUY-002	Supports the purchase of seat selection tickets, providing an interactive seating chart for self-service seat selection or a system automatic seat selection function.	High	3.4.5.3.1:#2
REQ-WEB-BUY-003	Supports the immediate settlement function, allowing users to directly initiate settlement and enter the order confirmation page after selecting the ticket type/seat.	High	3.4.5.3.1:#3
REQ-WEB-BUY-004	Supports adding multi-session, multi-ticket type tickets to the shopping cart, allowing the addition of new items/seats/ticket types, modification of the price of selected seat tickets, and supports modifying the quantity of selected ticket types.	High	3.4.5.3.1:#4
REQ-WEB-BUY-005	Allows users to delete selected items/seats/ticket types in the shopping cart or clear all contents of the shopping cart with one click.	High	3.4.5.3.1:#5
REQ-WEB-BUY-006	Supports the unified settlement function of the shopping cart. After user finish editing the shopping cart content, they can initiate	High	3.4.5.3.1:#6

	settlement and enter the order confirmation page.		
REQ-WEB-BUY-007	The order confirmation page shall display the details of the selected tickets, clearly indicating the total amount payable and the breakdown of expenses such as ticket price, discounted price, etc.	High	3.4.5.3.1:#7
REQ-WEB-BUY-008	For events that require real-name verification, the order confirmation page supports users to complete the entry of real-name information for ticket purchasers.	High	3.4.5.3.1:#8
REQ-WEB-BUY-009	The order confirmation page shall provide available preferential policy options for users to choose and use at their own discretion.	High	3.4.5.3.1:#9
REQ-WEB-BUY-010	The order confirmation page shall provide delivery method options supported by the event, allowing users to select and fill in relevant delivery information.	High	3.4.5.3.1:#10
REQ-WEB-BUY-011	The order confirmation page clearly displays the total amount payable and supports users to complete online payment through the integrated third-party payment platform.	High	3.4.5.3.1:#11
REQ-WEB-BUY-012	Requires users to check and agree to the privacy policy on the order confirmation page; otherwise, they cannot submit the order or proceed with subsequent payment operations.	High	3.4.5.3.1:#12
REQ-WEB-BUY-013	After locking the ticket inventory, if payment is not made within the specified time, the inventory will be automatically unlocked, shopping cart data will be cleared, and the corresponding unpaid order will be cancelled.	High	3.4.5.3.1:#13

Table 3.4.5.3.2: Requirement List of Ticket Purchase

3.4.5.4. Electronic Ticket Wallet

This business scenario defines the core management logic of the e-ticket wallet on the ticketing website, clarifies the operating specifications for e-ticket generation, centralized management, classified display, and voucher validity, realizes the full lifecycle visualisation management of e-tickets from generation to use, facilitates users to quickly retrieve and view e-ticket information, and improves the efficiency of entry verification.

3.4.5.4.1. User Process

The following table describes the user process of the e-ticket wallet.

No.	Component	Description
1	Electronic ticket automatically generated	After the user completes the ticket purchase and the payment is successful, the system immediately generates the corresponding electronic ticket. Each electronic ticket contains a unique entry QR code, which serves as a valid entry voucher.
2	Access to the website's e-ticket management entry	Users can directly access the e-ticket wallet via the website's dedicated e-ticket page or jump to the corresponding e-ticket management page from the "My Orders" module.
3	Access to Other E-Ticket Management Entries	Users can jump to the corresponding e-ticket management page by clicking on the encrypted link provided in the email or SMS.
4	Electronic Ticket Classification Display	The e-ticket wallet automatically classifies and displays all e-tickets by "unused/used", clearly distinguishing the current status of the tickets and enabling users to view them intuitively.
5	View Core Information of Electronic Ticket	In the e-ticket wallet, each e-ticket clearly indicates the corresponding event information, entry time, seat information, and verification status, displaying complete voucher information.

Table 3.4.5.4.1: Description of the User Process of Electronic Ticket Wallet

3.4.5.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-WEB-TIC-001	After the user successfully pays for the ticket purchase, the system immediately generates an electronic ticket, and each electronic ticket contains a unique admission QR code.	High	3.4.5.4.1:#1
REQ-WEB-TIC-002	The ticketing website has set up a dedicated e-ticket management page and supports jumping to access e-tickets from the "My Orders" module.	High	3.4.5.4.1:#2
REQ-WEB-TIC-003	Users can click on the encrypted link provided in the email or SMS to jump to the corresponding e-ticket management page.	High	3.4.5.4.1:#3

REQ- WEB-TIC- 004	The e-ticket wallet shall automatically classify and display e-tickets according to "unused/used" to clearly distinguish the status of tickets.	High	3.4.5.4.1:#4
REQ- WEB-TIC- 005	Each electronic ticket shall clearly indicate core information such as corresponding event details, admission time, seat information, and verification status.	High	3.4.5.4.1:#5

Table 3.4.5.4.2: Requirement List of Electronic Ticket Wallet

3.4.5.5. Order Management

This business scenario defines the core management logic for the entire lifecycle of orders on the ticketing website, clarifies the standard processes for order viewing, handling of unpaid orders, and after-sales operations, enables order information to be retrievable, unpaid orders to be resumed for payment, and compliant orders to be eligible for after-sales service, enhances users' experience in self-service order management, and ensures the standardization and efficiency of order processing.

3.4.5.5.1. User Process

No.	Component	Description
1	View Order List	Logged-in users enter the "My Orders" module of the ticketing website to view the basic information and real-time status of all historical orders under their name.
2	View Order Details	The user selects a specified order from the order list, enters the details page to view the complete order content such as ticket information, payment amount, payment time, e-ticket, etc.
3	Online payment for unpaid orders	The user initiates an online payment operation for an unpaid order on the order list/detail page, jumps to the payment page to complete the payment, and synchronously updates the order status.
4	Online Refund Application	Logged-in users initiate an online refund application for orders that meet the rules within the permitted time frame and wait for system/backend review and processing after submission.
5	Ticket Upgrade/Rescheduling Application	Logged-in users initiate an online ticket type upgrade/change application for eligible orders within the permitted time frame, select the target time and ticket type, complete the payment of

		the price difference, and thus complete the ticket type upgrade/change.
6	Online order inquiry is not supported for guests	When logged in as a guest, the website does not provide functions related to viewing and managing ticket orders.

Table 3.4.5.5.1: Description of the User Process of Order Management

3.4.5.5.2. Requirements List

RID	Description	Priority	Related Process
REQ-WEB-ORD-001	Supports logged-in users to view the basic information and real-time status of all historical orders under their name in the "My Orders" module.	High	3.4.5.5.1:#1
REQ-WEB-ORD-002	Supports users to enter the order details page to fully view the core order content such as ticket information, payment amount, payment time, e-ticket, etc.	High	3.4.5.5.1:#2
REQ-WEB-ORD-003	Supports users to initiate online payment operations for unpaid orders on the order list/detail page and synchronously updates the order status after payment is completed.	High	3.4.5.5.1:#3
REQ-WEB-ORD-004	Supports logged-in users to initiate online refund requests for orders that meet the rules within the permitted time frame.	High	3.4.5.5.1:#4
REQ-WEB-ORD-005	Supports logged-in users to initiate online ticket type upgrade/change requests for orders that meet the rules within the permitted time frame and complete the payment of the price difference.	High	3.4.5.5.1:#5
REQ-WEB-ORD-006	When logged in as a guest, the website does not provide functions related to viewing and managing ticketing orders.	High	3.4.5.5.1:#6

Table 3.4.5.5.2: Requirement List of Order Management

3.4.5.6. Personal Centre

This scenario defines the business logic for registered users to perform self-service management across all dimensions, including account permissions, order ticket folders, real name addresses, membership cards, etc., within the personal centre of the ticketing website. The core objective is to adapt to the full lifecycle service and privacy compliance requirements of users, thereby realising the business value of

enabling users to manage independently and conveniently, improving operational efficiency, and ensuring data security.

3.4.5.6.1. User Process

The following table describes the user flow for individuals.

No.	Component	Description
1	Enter Personal Centre	After a registered user logs in successfully, they can enter through the personal centre entrance on the website. After tourists log in, related functions of the personal centre are not provided.
2	Order Management and Ticket Wallet Entrance	Provides access to ticket order management and ticket wallet and clicking it will lead to the corresponding function page.
3	Real-name Management	Provides real-name information management functionality, supporting registered users to add and edit real-name information, and supporting real-name verification. Sensitive information such as ID numbers in real-name information is desensitised and displayed.
4	Address Management	Provides common address management functionality, supporting registered users to add, edit, and delete address information. Sensitive information such as contact phone numbers and emails in address information is desensitised and displayed.
5	Favourites	After registered users use the collection function on the website, they can view/delete the information of the collected items in the collection folder and directly click to jump to the event details.
6	Account Cancellation	Allows users to voluntarily apply to cancel their ticketing website accounts. After the cancellation is completed, only the ticketing website account information will be cancelled. Meanwhile, cancelled users can re-register for a new account after a specified period.
7	Coupon Viewing	Registered users can view the coupons in their accounts here, including the type of discount, expiration date, usage rules, etc.
8	Membership Card Management	Registered users can view information about their membership cards here, including card details, activation status, validity period, benefit

		descriptions, etc., and can perform operations such as activation and card renewal.
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Table 3.4.5.6.1: *Description of the User Process of Personal Centre*

3.4.5.6.2. Requirement List

RID	Description	Priority	Related Process
REQ-TKT-USER-001	Supports registered users to enter the personal centre page through the personal centre entry on the website after successful login, while no personal centre-related functions are provided for visitors after login.	High	3.4.5.6.1:#1
REQ-TKT-USER-002	Provides access to ticket order management and ticket wallet, allowing users to jump to the corresponding functional page after clicking.	High	3.4.5.6.1:#2
REQ-TKT-USER-003	Supports registered users to add and edit real-name information and complete real-name verification, and the system desensitizes and displays sensitive information such as ID numbers in real-name information.	High	3.4.5.6.1:#3
REQ-TKT-USER-004	Supports registered users to add, edit, and delete frequently used address information, and the system desensitizes and displays sensitive information such as contact phone numbers and emails in the address information.	High	3.4.5.6.1:#4
REQ-TKT-USER-005	Supports registered users to view and delete information of favoured items in the favourites and allows direct click to jump to the event details page.	High	3.4.5.6.1:#5
REQ-TKT-USER-006	Registered users are allowed to voluntarily apply to cancel their ticketing website accounts. After cancellation, only the ticketing website account information will be cleared, and users can re-register for a new account after a specified period.	Medium	3.4.5.6.1:#6
REQ-TKT-USER-007	Supports registered users to view coupon information within their accounts, including details such as discount type, expiration date, usage rules, etc.	High	3.4.5.6.1:#7

REQ-TKT-USER-008	Supports registered users to view information about their membership cards (including card information, activation status, validity period, benefit descriptions, etc.) and perform operations such as activation and card renewal.	High	3.4.5.6.1:#8
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Table 3.4.5.6.2: Requirement List of Personal Centre**3.4.5.7. Payment**

This business scenario defines the core operational logic of the entire online payment process on ticketing websites, clarifies the regulatory requirements for payment channel integration, transaction security assurance, pending payment order processing, payment result synchronisation, and currency valuation, achieves the security, efficiency, and compliance of the payment process, and safeguards users' transaction experience and the security of funds and data.

3.4.5.7.1. User Process

No.	Component	Description
1	Payment Channel Selection and Encrypted Payment	The user selects a payment method from the integrated mainstream third-party payment platforms on the order confirmation page. After initiating the payment, the system performs encrypted transmission processing on the entire payment process and completes the security verification of transaction data.
2	Online payment for pending orders	The user completes the payment operation for the selected payment method and submits the payment instruction within the time window when the order is pending payment.
3	Active cancellation of pending payment orders	Before the order expires, the user can initiate an active cancellation operation for an unpaid order to terminate the current ticketing transaction.
4	Real-time callback for payment results	The payment result is call backed by the payment platform to the ticketing website in real time, and the system automatically displays the transaction result and synchronously updates the order status.
5	Hong Kong Dollar unified pricing and settlement	Throughout the ticketing transaction process, Hong Kong dollars are used as the unit of account, and both the order amount and payment amount are uniformly settled in Hong Kong dollars.

Table 3.4.5.7.1: Description of the User Process of Payment

3.4.5.7.2. Requirement List

RID	Description	Priority	Related Process
REQ-WEB-PAY-001	Integrates mainstream third-party payment platforms, supporting multiple online payment methods such as WeChat Pay, Alipay, and bank card payment (redirecting to third-party platforms for selection); the entire payment process uses encrypted transmission technology to securely protect transaction data and ensure the security of payment data.	High	3.4.5.7.1:#1
REQ-WEB-PAY-002	Supports users to complete online payment operations using the selected payment method within the time window when the order is pending payment.	High	3.4.5.7.1:#2
REQ-WEB-PAY-003	Allows users to initiate an active cancellation operation on unpaid orders before the order expires.	High	3.4.5.7.1:#3
REQ-WEB-PAY-004	The payment result can be call backed by the payment platform to the ticketing website in real time, and the system will automatically display the result and synchronously update the order status.	High	3.4.5.7.1:#4
REQ-WEB-PAY-005	Ticket transactions are uniformly denominated in Hong Kong dollars, ensuring consistency in the settlement basis for orders and payments.	High	3.4.5.7.1:#5

Table 3.4.5.7.2:Requirement List of Payment

3.4.5.8. Diverse Marketing Activities

This business scenario defines the core logic for participating in diversified marketing activities on ticketing websites, clarifies the participation rules and operating procedures for activities such as priority ticket purchase with bank cards, full reduction discounts on bank cards, and balloting ticket purchase for popular events, enriches ticketing marketing forms, and enhances members' willingness to purchase tickets and platform user stickiness.

3.4.5.8.1. User Process

The following table describes the user flow of diversified marketing activities.

No.	Component	Description
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1	Participation in priority ticket purchase with bank card	Members can participate in the priority ticket purchase event for bank cards and snap up tickets for popular events through the platform's exclusive channel within the specified event time.
2	Enjoy Bank Card Full Amount Reduction Promotion	When users choose a bank card discount when purchasing tickets, they can successfully pay at the discounted price after the third-party payment platform verifies and approves the payment.
3	Add to Cart	When purchasing tickets, users can add an item (the item can be a coupon, merchandise, donation, etc.) on the order confirmation page and settle it together with the tickets at a discounted price.
4	Participation in balloting ticket purchasing for popular events	Members can sign up to participate in the balloting ticket purchase event for popular events during the event period and will obtain exclusive ticket purchase eligibility for the event upon winning the balloting.
5	Cross-session package ticket purchase	Users can purchase cross-session festival tickets during the event and enjoy corresponding discounts after reaching the specified quantity.

Table 3.4.5.8.1: Description of the User Process of Diversified Marketing Activities

3.4.5.8.2. Requirement List

RID	Description	Priority	Related Process
REQ-WEB-MAR-001	Supports configuring bank card priority ticketing activities, opening exclusive rush purchase channels for members, and allowing members to participate in ticketing for popular events at specified times.	High	3.4.5.8.1:#1
REQ-WEB-MAR-002	When users choose bank card discounts when purchasing tickets, they can successfully pay at the discounted price after the third-party payment platform verifies and approves the payment.	High	3.4.5.8.1:#2
REQ-WEB-MAR-003	When purchasing tickets, users can add a product (dependent on the optional module - Mall System) to their order on the order confirmation page and settle it together with the tickets at the additional purchase price.	High	3.4.5.8.1:#3

REQ-WEB-MAR-004	For popular events, balloting ticket purchase is supported, allowing members to register and participate during the event period, and upon winning the balloting ticket, members will be granted the ticket purchase qualification for the corresponding event.	High	3.4.5.8.1:#4
REQ-WEB-MAR-005	Users can participate in the purchase of cross-session festival tickets during the event and enjoy corresponding discount after reaching the specified quantity.	High	3.4.5.8.1:#4

Table 3.7.8.2: Requirement List of Diversified Marketing Activities

3.4.5.9. Access Control

This business scenario defines the core operational logic for ticket delivery method selection, entry verification, and cross-system status synchronisation on ticketing websites, clarifies the standard processes for delivery method annotation, verification credential generation, entry verification, and status tracing, achieves the uniqueness and traceability of ticket entry verification, prevents repeated entry, and ensures the standardisation of ticketing operations.

3.4.5.9.1. User Process

The following table describes the user process of admission control.

No.	Component	Description
1	View Event Delivery Method	When purchasing tickets, users can view the clearly marked supported delivery methods (e-ticket, pick-up entry, paper ticket mailing) for the event on the event details page/ticket purchase page.
2	Delivery Method Selection	During the order confirmation process, the user selects one of the available delivery methods for this event to complete the delivery method confirmation.
3	Unique Verification Credential Generation	After the order payment is completed, the system generates a unique admission QR code or verification code for each ticket, which serves as the verification credential for admission.
4	Direct admission verification with e-ticket code	Users complete scanning/verification at the admission verification point using the unique QR code/verification code in the e-ticket, and the system synchronizes the verification results.

5	Electronic Ticket Code Collect Paper Ticket for Admission	Users can also choose to collect paper tickets at the designated ticket vending machines/ticket centres using the electronic ticket code and complete the entry verification at the entry verification points with the paper tickets.
6	Mail paper tickets for direct entry	Users shall complete direct entry verification at the entry verification point with the paper tickets delivered by mail.
7	Admission Status Data Synchronization	After completing the entry verification, the ticketing website synchronizes the entry status of the ticket/order with the AnTank system in real time and records the verification information.

Table 3.4.5.9.1: Description of the User Process of Access Control**3.4.5.9.2. Requirement List**

RID	Description	Priority	Related Process
REQ-WEB-ACC-001	The ticketing website shall mark the supported ticketing delivery methods (electronic tickets, on-site ticket collection, and paper ticket mailing) by event.	High	3.4.5.9.1:#1
REQ-WEB-ACC-002	Supports users to select only one of the delivery methods available for this item when placing an order to complete the delivery method confirmation.	High	3.4.5.9.1:#2
REQ-WEB-ACC-003	After the order payment is completed, a unique admission QR code or verification code will be generated for each ticket as the admission verification credential.	High	3.4.5.9.1:#3
REQ-WEB-ACC-004	Supports users to complete direct verification and enter at the entry verification point using the unique QR code/verification code in the e-ticket.	High	3.4.5.9.1:#4
REQ-WEB-ACC-005	Supports users to collect paper tickets at designated locations using the ticket collection code and complete entry verification with the paper tickets.	High	3.4.5.9.1:#5
REQ-WEB-ACC-006	Supports users to complete direct entry verification at the entry verification point with the paper ticket delivered by mail.	High	3.4.5.9.1:#6

REQ- WEB- ACC-007	After the admission verification is completed, real-time synchronization of ticket/order status between the ticketing website and the AnTank system is achieved, and verification information is recorded.	High	3.4.5.9.1:#7
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Table 3.4.5.9.2: Requirement List of Access Control

3.4.5.10. Membership Card Purchase and Activation

This business scenario defines the core operational logic for the integration, detailed display, and online purchase and activation of membership card resources on ticketing websites, clarifies the standardized processes for membership card resource presentation, core information display, online selection and payment, and automatic activation, achieves a full online closed-loop process for membership card handling without the need for offline operations, and improves user handling efficiency and usage experience.

3.4.5.10.1. User Process

The following table describes the user process for membership card purchase and activation.

No.	Component	Description
1	Membership Card Resource View	The ticketing website integrates AnTank membership card resources, allowing users to view M+ and HKPM-related membership card products centrally displayed on the platform.
2	View Membership Card Details	The user clicks on the entry to the details page of the target membership card product, enters the page to view core information such as name, list of benefits, pricing, and validity period.
3	Online Purchase of Membership Cards	After the user confirms the intention to apply for the membership card on the membership card details page, they initiate the online purchase operation for the membership card product.
4	Online payment for membership card	The user selects a suitable payment method and completes the online payment operation for the selected membership card product fee.
5	Membership Card Activation	After the user completes the membership card payment, the membership card can be automatically activated or the user can complete the activation online, eliminating the need for offline processing steps.

Table 3.4.5.10.1: Description of the User Process for Membership Card Purchase and Activation

3.4.5.10.2. Requirement List

RID	Description	Priority	Related Process
REQ-WEB-MEM-001	The ticketing website needs to integrate AnTank membership card resources and display them centrally M+, HKPM, PA-related membership card products.	High	3.4.5.10.1:#1
REQ-WEB-MEM-002	The membership card details page should clearly present core information such as name, list of benefits, pricing, and validity period.	High	3.4.5.10.1:#2
REQ-WEB-MEM-003	Supports users to initiate online purchase operations for target membership card products on the membership card details page.	High	3.4.5.10.1:#3
REQ-WEB-MEM-004	Supports users in selecting a payment method and completing the online payment operation for the selected membership card product fee.	High	3.4.5.10.1:#4
REQ-WEB-MEM-005	Supports automatic activation of the membership card by the system after the user completes payment and supports users to click to activate it themselves.	High	3.4.5.10.1:#5

Table 3.4.5.10.2: Requirement List of Membership Card Purchase and Activation

3.4.5.11. Privacy, Security, Scalability & Inclusive Accessibility

This business scenario defines the core operational specifications of the ticketing website in personal data processing, system security operation, functional expansion adaptation, multi-terminal compatible access, and accessible user experience design, clarifies the implementation requirements for data compliance management and control, high-concurrency support, multi-device adaptation, multi-language switching, and accessible user experience experience optimisation, achieves full-process compliant operation and stable operation of the website, covers diverse user groups, and reserves expansion capabilities to meet long-term business development needs.

3.4.5.11.1. User Process

The following table describes the user flows for privacy, security, scalability, and inclusive accessibility.

No.	Component	Description
1	Multi-Device Self-Adaptation Access	Users can complete the entire ticketing process on desktop and mobile devices with different

		resolutions, where page information is fully displayed, core function entrances are clear, and no manual zooming is required.
2	Multilingual Self-Switching	Users can independently select any language from Simplified Chinese, Traditional Chinese, or English on the website, and the system will immediately switch the entire site's content to the selected language for display.
3	Compatible access with mainstream browsers	Users access the website via mainstream browsers (Microsoft Edge at 108 or above, Google Chrome at 108 or above, Firefox at 108 or above, Safari at 16 or above), and both page rendering and functional interaction remain consistent without any abnormalities.
4	System support for high-concurrency scenarios	During peak ticket sales periods, the system automatically supports high-concurrency access requirements, ensuring that all users' operations such as browsing and ticket purchasing are stable without lag and that functions operate without anomalies.
5	System Function Expansion Adaptation	The website reserves standardized functional interfaces, enabling subsequent rapid implementation of new feature expansions such as RSVP and workshop reservation based on these interfaces without affecting the operation of the existing system.
6	Accessibility Support	Users, including persons with disabilities and users relying on assistive technologies, can access and operate key website functions such as registration, login, event browsing, ticket selection, shopping cart review, payment, e-ticket access, membership card purchase, and membership activation through accessible design and interaction support. The website shall support accessibility features including keyboard-operable navigation, screen reader compatibility, sufficient colour contrast, meaningful labels and instructions, and clear validation or error messages, in compliance with WCAG 2.1 Level AA.

Table 3.4.5.11.1: *Description of the User Process of Privacy, Security, Scalability & Inclusive Accessibility*

3.4.5.11.2. Privacy, Security, Scalability & Inclusive Accessibility Requirement List

RID	Description	Priority	Related Process
REQ-WEB-PSI-001	Supports multi-device self-adaptation access, ensuring that when users complete the entire ticketing process on desktop and mobile devices with different resolutions, page information is fully displayed, core function entrances are clear, and no manual zooming is required.	High	3.4.5.11.1:#1
REQ-WEB-PSI-002	Supports multi-language self-switching among Simplified Chinese, Traditional Chinese, and English. After the user selects any language on the website, the system will immediately switch the entire site's content to display in the selected language.	High	3.4.5.11.1:#2
REQ-WEB-PSI-003	When users access the website through mainstream browsers (Microsoft Edge at 108 or above, Google Chrome at 108 or above, Firefox at 108 or above, Safari at 16 or above), page rendering and functional interaction remain consistent and error-free.	High	3.4.5.11.1:#3
REQ-WEB-PSI-004	The system supports high-concurrency access requirements during peak ticket sales periods, ensuring that all users' operations such as browsing and ticket purchasing are stable without lag and that functions operate without anomalies.	High	3.4.5.11.1:#4
REQ-WEB-PSI-005	The website reserves standardized functional interfaces, supporting the rapid implementation of new feature expansions such as RSVP and workshop reservation based on the interfaces, and the process of feature expansion does not affect the normal operation of the existing system.	High	3.4.5.11.1:#5
REQ-WEB-PSI-006	The website shall support accessible access to key user journeys, including registration, login, event browsing, ticket selection, shopping cart review, payment, e-ticket access, membership card purchase, and membership activation, so that users with different assistive needs can complete core online transactions independently and effectively.	High	3.4.5.11.1:#6

REQ- WEB-PSI- 007	The website shall support accessibility features including keyboard-operable navigation, screen reader compatibility, sufficient colour contrast, meaningful labels and instructions, and clear validation or error messages.	High	3.4.5.11.1: #6
REQ- WEB-PSI- 008	The website shall be designed, developed, and tested to support compliance with WCAG 2.1 Level AA.	High	3.4.5.11.1: #6

Table 3.4.5.11.2: Requirement List of Privacy, Security, Scalability & Inclusive Accessibility

3.5. Kiosk

3.5.1. Business Requirements

From a business perspective, WestK requires the Kiosk (self-service ticketing machine) system to integrate the existing AnTank backend ticketing capabilities, support core needs such as self-service ticket purchasing and collection for on-site visitors, and at the same time ensure efficient operational processes, accurate data interaction, and service security compliance, in line with the standardised operational specifications of multiple venues and business development goals.

The following are the main business requirements identified for this module.

3.5.1.1. Multilingual Support

WestK requires the Kiosk system to support the switching of three language interfaces: Simplified Chinese, Traditional Chinese, and English, meeting the usage needs of local and international visitors. The language selection entry is prominent, and after switching, the interface is globally synchronised, with a simple and intuitive operation process.

3.5.1.2. Accessibility adaptation

WestK requires the Kiosk system to adapt to the needs of accessible user such as wheelchair users and optimize the design of accessible user experience operations. The interface layout, interactive buttons, and operation processes comply with accessible user experience usage specifications to ensure that accessible user can independently complete core operations such as ticket purchasing and payment process, thereby enhancing service inclusiveness.

3.5.1.3. Self-service ticketing

WestK requires the Kiosk system to support filtering of exhibitions and shows by event type, date, etc. The system is required to support both seat selection and admission ticket to meet the ticketing needs of different event types, and support seat inventory verification and seat locking mechanisms.

3.5.1.4. Payment Service

For Hong Kongers, Mainland Chinese, and global tourists, WestK requires the Kiosk system to support multiple mainstream payment methods such as Visa, Master, Octopus, Alipay, WeChat Pay, etc., to adapt to the payment preferences of different visitors. The payment process is clear and convenient, with real-time display of payment status and operation instructions, and supports retry in case of payment exceptions.

3.5.1.5. Preferential Use

WestK requires the Kiosk system to support self-service redemption of two types of discounts: membership and discount codes. The system can automatically recognise and calculate the discount amount, with only one type of discount allowed per order, and the discount information and the total price after discount are clearly displayed.

3.5.1.6. Self-service ticket service

WestK requires the kiosk system to support ticket collection via code while ensuring code security. The system must allow batch selection and printing, restrict each ticket to a single print, and provide high-quality physical tickets suitable for both entry and souvenir preservation.

3.5.1.7. Contact Information Retention

WestK requires the Kiosk system to provide an option for visitors to leave their contact information after the order is completed. Visitors can choose whether to fill it out voluntarily, which does not affect the normal ticketing process. The system guides in a friendly manner, fully respecting the visitors' right to choose.

3.5.1.8. Display and access to terms and conditions

WestK requires the Kiosk system to set up a compliant access for viewing terms and conditions, safeguarding visitors right to be informed and complying with the compliance operation specifications for ticketing services. The product flow must be easy to find and support visitors to click and view the terms and conditions of the corresponding venue.

3.5.1.9. Maintenance Mode

The system shall provide a Maintenance Interface accessible via a hidden or specific entry point upon entering a valid admin password. Key functional requirements within this mode include a paper quantity calibration tool to sync physical stock with the system record, and a printer test module for verifying connectivity and output. This ensures the kiosk remains operational and reduces onsite troubleshooting time.

3.5.2. Business Requirement Analysis

This chapter further analyses and explains the business requirements described in Section 3.5.1. Each subchapter will provide supplementary explanations for the requirements of the Kiosk system in on-site ticket purchase and collection scenarios

from aspects such as business objectives, operational considerations, and functional implementation expectations.

3.5.2.1. Multilingual and Interface Operations

Map to Business Requirement 3.5.1.1.

The system should support flexible switching of multi-language interfaces and standardized operations, adapt to the language usage habits of different domestic and international visitors, ensure that operations are easy to understand and processes are unified across all scenarios, and enable language switching without manual assistance.

Key capabilities include:

- The Kiosk system supports free switching among Simplified Chinese, Traditional Chinese, and English, and the interface content updates synchronously after switching.
- The language entry is prominently located on the homepage, with simple operations, allowing visitors to complete language selection without guidance.

3.5.2.2. Accessible User Experience

Map to Business Requirement 3.5.1.2.

The system should support accessible user experience operation, fully adapt to the usage needs of accessible user, ensure accessible user such as wheelchair users can smoothly complete self-service operations, and enhance service inclusiveness and humanistic care.

Key capabilities include:

- Elements such as kiosk interfaces and buttons are adapted for use by accessible user such as wheelchair users.
- The accessible user experience switching entry is prominently located on the homepage, with simple operations, allowing visitors to switch accessibility mode without guidance.
- The system adheres to general accessibility design specifications, considering the usage needs of accessible user to ensure that special visitors can use the equipment smoothly and independently.

3.5.2.3. Self-service ticketing

Map to Business Requirement 3.5.1.3.

The system should support on-site visitors to independently complete event filtering and ticket purchasing, covering multiple types of events such as shows, cinemas, and exhibitions, adapting to different ticketing models, while implementing ticketing compliance requirements and safeguarding visitors right to be informed, thereby achieving an efficient and standardized self-service ticketing loop.

Key capabilities include:

- The Kiosk system supports quick filtering of events by events type, date, etc.

- The system clearly displays key information such as event name, time, venue, and available-for-sale status.
- Supports seat selection ticket purchase and admission ticket purchase in two modes.
- Seat selection mode: Select the ticket price, display the seat map, and customers can choose available seats and select the ticket type (student ticket, senior ticket, etc.).
- The system should support seat inventory verification and seat locking mechanisms.
- For seats with special conditions such as obstructed views, once selected by the customer, the system will automatically pop up a prompt window to truthfully inform the customer of the relevant seat conditions, fully safeguarding the customer's Right to be informed.
- Regarding Class III films in theatres, the system clearly informs minors that they are not allowed to watch, and visitors can only make a purchase after confirmation.
- Admission ticket: display ticket types (student tickets, senior tickets, etc.), and visitors select the ticket type and quantity.

In accessibility mode, to ensure operational convenience, only the automatic seat selection function is supported, and manual seat selection operations are prohibited.

3.5.2.4. Payment Service

Map to Business Requirement 3.5.1.4.

The system should support the integration of diverse mainstream payment methods with specific payment gateway, create a convenient, secure, and efficient self-service payment process, synchronize payment status in real-time, ensure smooth transactions, quickly connect to the subsequent ticketing process after payment is completed, and improve the overall efficiency of self-service.

Key capabilities include:

- The Kiosk system supports multiple mainstream payment methods with specific payment gateway such as Visa, Master, Octopus, Alipay, Wechat Pay etc., comprehensively covering the payment preferences of local and international visitors.
- The payment interface provides clear and easy-to-understand guidance, displays transaction progress and status in real-time, and provides instant feedback on results such as payment success, failure, and timeout.
- After successful payment, the system automatically redirects to the ticketing or printing process, with smooth process flow and no additional redundant operations.

3.5.2.5. Discount Usage

Map to Business Requirement 3.5.1.5.

The system should support self-service verification and efficient redemption of preferential rights and interests, simplifying the process of using preferential benefits.

Key capabilities include:

- The system supports the verification and use of two types of discounts, membership discount and discount codes, with instant feedback on verification results.
- After the discount verification is passed, the system automatically calculates the discount amount and displays the discount information and the amount payable.

3.5.2.6. Self-service ticket collection

Map to Business Requirement 3.5.1.6.

The system should support secure and convenient self-service ticket collection, verify ticket passwords, meet the needs of both daily entry ticket and souvenir preservation, standardize ticket printing rules, ensure the ticket collection process is efficient and controllable, and avoid resource waste and ticketing disputes.

Key capabilities include:

- Visitors can quickly query the corresponding order using the code of ticket collection. The ticket collection process is efficient and convenient, supporting verification by entering the ticket collection password to prevent the leakage of the code of ticket collection.
- Supports checkmark printing to meet the needs of visitors for batch ticket collection.
- Support for only one print per ticket is provided to prevent resource waste and ticketing disputes caused by duplicate printing.
- Supports customers in printing souvenir tickets using a QR code after admission.

3.5.2.7. Contact Information Retention

Map to Business Requirement 3.5.1.7.

The system should support non-mandatory contact information retention services and guide visitors to complete their contact information.

Key capabilities include:

- After the ticket purchase is completed, the system will pop up an option to leave contact information, and visitors can choose to fill it out or skip it directly.
- Guide visitors to scan the QR code to complete the form on their mobile phones, ensuring the efficiency of on-site self-service.
- The system adopts a friendly and gentle guiding approach, without forcing intervention in visitors' choices, which aligns with the convenience principle of self-service.

3.5.2.8. Terms and Conditions Inquiry Function

Map to Business Requirement 3.5.1.8.

The system should provide an entry for terms and conditions on the home page and event details page, allowing visitors to review relevant rules, avoid ticketing disputes, safeguard the rights and interests of both parties, and not interfere with the normal operation process.

Key capabilities include:

- Both the home page of the Kiosk device and the details page of the ticket purchase item are provided with dedicated terms and conditions entry points, supporting one-click access.
- The content of the terms is synchronised and adapted to multi-language switching. After the visitor selects the corresponding language, the terms and conditions will automatically switch to the same language version.

3.5.2.9. Maintenance Mode

Map to Business Requirement 3.5.1.9.

The system shall provide a dedicated Maintenance Mode designed for on-site operational staff to perform device-level maintenance and configuration without impacting normal user operations.

- Paper Count Configuration (Initial Ticket Paper Quantity Reset)
The Maintenance Mode interface shall provide functionality for managing the kiosk's ticket paper count:
- Modify Initial Paper Count
Operators shall be able to view and reset the current ticket paper count through the Maintenance Mode interface.
The kiosk's paper count shall be synchronized with the AnTank TAS backend system.
- Backend Synchronization Logic
Upon resetting the paper count, the system shall update the backend with the newly configured current paper count value.
For every ticket printed, the backend shall decrement the current paper count by one (-1).
- Paper Count Threshold & Alerts
The backend system shall allow configuration of a paper-low warning threshold.
When the current paper count becomes less than or equal to the configured threshold, the system shall automatically send an email alert to designate WestK staff, informing them that the ticket printer requires paper replacement.
- Test Print Function
Maintenance Mode shall include a Test Print feature for verification and diagnostics:
- Test Page Printing
The interface shall provide a button for operators to print a test ticket.
- Paper Count Adjustment

Triggering the test print shall reduce the backend's current ticket paper count by one (-1), consistent with normal ticket printing logic.

3.5.3. Operating Model

This event adopts a collaborative operation model, supporting on-site self-service ticketing and ticket collection services at each venue of WestK through the Kiosk system.

This operating model defines the roles and responsibilities of all parties involved in the Kiosk self-service business.

AnTank – Ticket Technology Provider

AnTank is responsible for providing and operating the ticketing technology platform and backend system that support the operation of Kiosk self-service business, ensuring stable system docking and full availability of functions.

- Provides full-process system functions for Kiosk, including self-service ticket purchase, seat selection, self-service ticket collection, discount verification, and ticket printing functions.
- Supports docking with Kiosk hardware devices (including payment terminals, printers, touch screens) to ensure data interaction and hardware linkage.
- Synchronizes ticket data, event information, and order status in real-time with backend systems to ensure data consistency.
- Responsible for on-site technical operation and maintenance of Kiosk devices, including device maintenance, device inspection, and fault handling.
- Ensure system availability, show monitoring, operation logging, and provide technical support within the agreed service level.

WestK On-site Operations Team

WestK box office staff use this system to provide ticketing and customer service on-site at the venue.

- WestK on-site staff are responsible for the daily on-site management, hardware operation and maintenance, and visitor guidance of Kiosk self-service equipment, ensuring the normal self-service experience for visitors.
- Responsible for the daily operational coordination of Kiosk devices, including the replacement and replenishment of consumables such as ticket paper, to ensure continuous serviceability of the devices.
- Promptly report equipment malfunctions, system abnormalities, and hardware damage issues to the technical team, and collaborate with AnTank to conduct troubleshooting and repairs.

3.5.4. Customer Journey

This section describes the entire process of visitors using the Kiosk self-service ticketing machine to complete self-service ticket purchasing, self-service ticket collection, and related supporting operations, clearly explaining the interaction logic between visitors and self-service devices, and covering the core operational steps from device usage to transaction completion.

This process demonstrates how visitors can independently complete operations such as event filtering, ticket purchase, payment, ticket printing, and subsequent information retention through Kiosk devices, without any manual intervention throughout the entire process, thus achieving one-stop self-service ticketing.

The following process is only used to illustrate the core stages of Kiosk self-service operation. For detailed functional rules and implementation requirements, please refer to Section 3.5.2.

No.	Category	Role	Operational Scenario
1	Device Usage and Mode Selection	As a Visitor	I click on the home page of the Kiosk device to wake up the system, select the appropriate language version as needed, or directly switch to the accessibility mode to ensure that the language and interaction form of the subsequent operation interface conform to usage habits, and then enter the main interface of self-service.
2	View Terms and Conditions	As a Visitor	On the device's home page or the ticket purchase item details page, I can click the "Terms and Conditions" entry at any time to view the relevant rules and return to the original interface to continue operating after viewing.
3	Event Screening	As a Visitor	I filter target events by event type, date, and venue, quickly locate the required tickets for shows, cinemas, exhibitions, etc., and check the core information of the event to confirm whether the session meets the requirements.
4	Seat Selection Ticket Purchase	As a Visitor	For seat selection events, I view the visual seat map to select available seats. When encountering special seats or Class III films in theatres, after confirming the system prompt information, I select the corresponding ticket type.
5	Admission Ticket Purchase	As a Visitor	For Admission ticket events, I select the required ticket type and the quantity to purchase, and the system checks the inventory in real time. After confirming that everything is correct, it proceeds to the subsequent process.
6	Preferential Verification and Usage	As a Visitor	I input or scan the membership code and discount code as needed, complete the discount verification, and confirm that the discount amount automatically calculated by the system and the actual payment amount are correct.

7	Order Confirmation	As a Visitor	I checked the ticket details, promotional information, and the amount payable in the order. After confirming the accuracy of the information, I proceeded to the payment process.
8	Payment Processing	As a Visitor	I choose a suitable payment method to complete online payment, wait for the system to provide real-time feedback on the payment result, and after successful payment, jump to the ticket issuance process. In case of an abnormal situation, payment can be re-initiated.
9	Self-Service Ticket Collection and Ticket Printing	As a Visitor	After successful payment, you can directly print the admission ticket, or query historical orders by the ticket collection code to print the ticket, check and print as needed, confirm that the ticket information is complete, and each ticket can only be printed once.
10	Printing of Commemorative Tickets	As a Visitor	After the event, I used the original ticket collection code to query the admitted orders, printed the commemorative tickets for retention, and completed the commemorative ticket printing process.
11	Contact Information Retention	As a Visitor	After all transactions are completed, I can independently choose whether to scan the QR code to fill in my contact information, and I can directly skip this step without affecting the completed ticketing transactions.
12	Maintenance Mode	As a Staff	As a Staff I should be able to reset the ticket quantity and print the testing ticket. Also I should be able to receive an alert email to notify me to change the ticket when ticket quantity is low.

Table 3.5.4: Description for the User Story

3.5.5. Business Scenarios

This chapter focuses on the core service scenarios of Kiosk self-service ticketing machines and is divided into three major modules according to the logic of visitor operations on kiosks: service entry selection, self-service ticket purchase, and self-service ticket collection. It clarifies the operation processes, functional requirements, and demand specifications for each scenario, conforms to the self-service operation habits of on-site visitors, covers the entire self-service closed loop from device interaction to business completion, and all functional rules are consistent with the business requirements in 3.5.1 and 3.5.2 of the previous text.

3.5.5.1. Service Entry and Mode Selection

This business scenario defines the initial operation logic for visitors using Kiosk devices, with its core covering three major core entry operations: ticket purchase/ticket collection service selection, multilingual switching, and accessible user experience mode switching, ensuring that different types of visitors can quickly access the corresponding service modules, taking into account the usage needs of both ordinary visitors and special groups, and enhancing the usability of self-service.

3.5.5.1.1. User Process

The following table describes the user process of Service Entry and Mode Selection.

No.	Component	Description
1	Multilingual Switching	Visitors can access the language switch page on the homepage, select Traditional Chinese, Simplified Chinese, or English as needed, and all interface text will be updated synchronously after the switch.
2	Accessibility Mode Toggle	Special visitors can switch to the accessible user experience operation mode with one click during use, and the interface is adapted for wheelchair use, making it convenient for visitors.
3	View Terms and Conditions	Visitors can click on the fixed entry on the homepage to view the terms and conditions, the content of which is adapted to the current language and operating mode and return to the service selection interface after viewing.
4	Service Type Selection	Visitors can independently choose the self-service ticket purchase or self-service ticket collection service entry, enter the corresponding business operation module, and support returning midway to switch services.

Table 3.5.5.1.1: Description of the User Process of Service Entry and Mode Selection

3.5.5.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-KSK-ENT-001	Supports switching between Traditional Chinese, Simplified Chinese, or English, with global interface synchronously updated.	High	3.5.5.1.1:#1

REQ-KSK-ENT-002	Supports one-click switching to accessibility mode, adapts to the operating habits of special groups, and optimises interface elements.	High	3.5.5.1.1:#2
REQ-KSK-ENT-003	Supports displaying the entry for terms and conditions on the home page and ticket purchase item page. Clicking on it displays the detailed content without interrupting the operation process.	High	3.5.5.1.1:#3
REQ-KSK-ENT-004	Supports the selection of dual service entrances for self-service ticket purchase and self-service ticket collection.	High	3.5.5.1.1:#4

Table 3.5.5.1.2: Requirement List of Service Entry and Mode Selection

3.5.5.2. Self-service ticketing

This business scenario defines the core operational logic of the entire process of ticket purchasing via Kiosk self-service ticketing machines, covering all aspects including seat selection ticket purchase, admission ticket purchase, automatic seat selection, discount verification, order confirmation, payment processing, and ticket printing. It clarifies the specifications for each operational step, achieves a closed-loop for single-session self-service ticket purchasing from event screening to payment and ticket issuance, aligns with on-site self-service scenarios, ensures that ticket purchasing operations are simple, efficient, and compliant, and at the same time implements requirements for special seats, compliance reminders for third-level cinema films, and optional contact information retention for visitors, meeting the core needs of visitors for quick self-service ticket purchasing.

3.5.5.2.1. User Process

The following table describes the user process of Self-Service Ticketing.

No.	Component	Description
1	Event Screening	The Kiosk system supports filtering target events by event type, date, and venue, and displays the core information of the events.
2	Purchase Restrictions for Special Ticketing Items	When a visitor enters the ticket purchase interface for special items (such as Class III films), a pop-up window will prompt the ticket purchase instructions, and the visitor can only proceed with the purchase after confirmation.
3	Admission Ticket Choice	For Admission ticket events, visitors can choose the corresponding ticket type (student ticket, senior ticket, adult ticket, etc.) and set the corresponding purchase quantity, and the system will verify the inventory in real time.

4	Manual Selection of Seat Selection Tickets	For seat selection events, a visual seat map is provided, allowing visitors to independently select available seats. Special seats and Class III films in cinemas will pop up a prompt and require confirmation.
5	Automatic Seat Selection for Seat Selection Tickets	For seat selection events, a convenient automatic seat selection function has been added. visitors can independently choose between manual or automatic seat selection; in accessible mode, the system enforces automatic seat selection as the default and only supports this single method. Once the visitor selects the number of tickets to purchase, the system will automatically allocate available seats according to the seat allocation rules.
6	Order Confirmation	The order confirmation page displays details of the selected tickets, clearly indicating the total amount payable and breakdown of costs such as ticket prices.
7	Order Discount Policy Selection	On the order confirmation page, visitors can independently choose to use a membership code or discount code, complete the discount verification by scanning a code or manual input, and the system will automatically calculate the discount amount.
8	Lock the Tickets after the Order is Confirmed	When initiating settlement payment after the order is confirmed, the system temporarily locks the selected tickets.
9	Online Payment	Visitors choose payment methods such as Visa, Master, Octopus, Alipay, etc. to complete online payment, and the system provides real-time feedback on the payment result.
10	Payment Exception Handling	Supports visitors to initiate payment again in case of payment failure, interruption, or timeout.
11	Select to Print Receipt	During the process of printing admission tickets, it supports selecting whether to print receipts
12	Retain Contact Information	Supports asking visitors whether they are willing to leave their contact information after a successful payment, allowing them to choose to fill in their contact information or skip it, facilitating order notifications and after-sales communication.

13	Ticket Printing	After successful payment, the system will automatically redirect to the printing process to print the admission ticket.
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Table 3.5.5.2.1: Description for the User Process of self-service ticketing

3.5.5.2.2. Requirement List

RID	Description	Priority	Related Process
REQ-KIOSK-BUY-001	The ticket purchase item list displays the core information of items and supports filtering.	High	3.5.5.2.1:#1
REQ-KIOSK-BUY-002	Special ticketing items (age restrictions) set purchase instructions, and the ticketing instructions will pop up upon entering the interface. Visitors can only proceed with subsequent ticketing operations after confirming that they are aware of them.	High	3.5.5.2.1:#5
REQ-KIOSK-BUY-003	Supports the purchase of admission tickets, allowing visitors to independently select ticket types, set the quantity to purchase, and the system will verify inventory in real-time and provide feedback on inventory status.	High	3.5.5.2.1:#2
REQ-KIOSK-BUY-004	Supports manual seat selection for seat selection tickets. In normal mode, visitors can choose seats independently; seats with obstructed views will pop up a prompt and require confirmation, while the manual seat selection function is disabled in accessibility mode.	High	3.5.5.2.1:#3
REQ-KIOSK-BUY-005	Supports automatic seat selection for seat selection tickets. Accessibility mode only supports automatic seat selection. After entering the ticket price and quantity, the system will assign seats and display details.	High	3.5.5.2.1:#4
REQ-KIOSK-BUY-006	The order confirmation page displays details of the selected tickets, clearly indicating the total amount payable and breakdowns of expenses such as ticket price, discounted price, etc.	High	3.5.5.2.1:#6
REQ-KIOSK-BUY-007	The order confirmation page supports the selection of preferential policies, allowing verification to be completed via membership codes or discount codes. The system	High	3.5.5.2.1:#7

	automatically calculates the preferential amount and updates the total actual payment.		
REQ-KIOSK-BUY-008	When initiating settlement, the ticket inventory is automatically and temporarily locked, retained within the lock time limit, and automatically released if payment is not successfully made after the timeout.	High	3.5.5.2.1:#8
REQ-KIOSK-BUY-009	Supports multiple mainstream online payment methods such as Visa, Master, Octopus, Alipay, etc., allowing visitors to choose as needed.	High	3.5.5.2.1:#9
REQ-KIOSK-BUY-010	Payment exception handling, targeting scenarios of payment failure, interruption, and timeout, supports visitors to directly initiate payment again and prompts the reason for the exception.	High	3.5.5.2.1:#10
REQ-KIOSK-BUY-011	After successful payment, visitors are supported to independently choose whether to print the payment receipt.	High	3.5.5.2.1:#11
REQ-KIOSK-BUY-012	After successful payment, it supports visitors to retain their contact information.	High	3.5.5.2.1:#12
REQ-KIOSK-BUY-013	Printing tickets: single admission tickets support only one print, and reprinting is prohibited.	High	3.5.5.2.1:#13

Table 3.5.5.2.2: Requirement List of Self-service Ticket Purchase

3.5.5.3. Self-service ticket collection

This business scenario defines the core operational logic of the entire process for ticket collection at Kiosk self-service ticket machines, covering all aspects including ticket collection information entry, order query, ticket printing, and commemorative ticket printing. It clarifies the ticket collection operation specifications, considers the needs of both on-the-spot ticket purchase and collection, historical order ticket collection, ordinary admission ticket printing, and commemorative ticket printing, ensuring the efficiency and convenience of the ticket collection process and avoiding duplicate ticket printing and resource waste.

3.5.5.3.1. User Process

The following table describes the user process of Self-service Ticket Collection.

No.	Component	Description
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1	Ticketing Information Entry	After visitors enter the self-service ticket collection module, they can submit queries through two methods: manually entering the ticket collection code or quickly scanning the code to input the ticket collection code. It supports additional input of the ticket collection password to complete identity verification, ensuring the security of order information and preventing non-owner ticket collection.
2	Order Inquiry and Information Display	The system, based on the entered ticket collection code information, quickly matches the corresponding valid order in the background, accurately queries and then fully displays the core information of the order, including the event name, session time, seat information (reserved seat tickets), printing status, etc., to facilitate visitors' verification, and immediately pops up a prompt when there is no corresponding order.
3	Batch-Wise Ticking and Printing of Bills	Visitors can independently check one or multiple tickets to be printed according to their own needs, supporting selective printing in batches without having to print all tickets at once, flexibly adapting to the needs of different ticket collection scenarios.
4	Admission Ticket Printing Control	After the visitor confirms printing the selected ticket, the system will link with the device printer to quickly output the admission ticket, which contains complete admission verification information; strictly enforce the rule that each admission ticket can only be printed once, and after printing is completed, the system will automatically mark it as printed, prohibiting repeated printing to avoid ticketing disputes and resource waste.
5	Printing of Souvenir Tickets	For tickets that belong to an event that has ended, have completed on-site entry verification, and have not been printed, the permission to print Souvenir tickets will be granted, and visitors can print commemorative tickets using the original ticket collection code.
6	Print Completion and Process Exit	After the ticket printing is completed, the system immediately pops up a prompt indicating successful printing, clearly informing the visitor that the operation is completed. The device automatically returns to the home page, waiting for the next visitor to use it. The entire process is a closed loop with simple operations.

Table 3.5.5.3.1: Description for the User Process of Self-service Ticket Collection

3.5.5.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-KIOSK-PICK-001	Supports both manual keyboard input of the ticket collection code and quick scanning for entry.	High	3.5.5.3.1:#1
REQ-KIOSK-PICK-002	The system should support ticket collection identity verification by using the last four digits of the reserved mobile phone number or the last four digits of the part before the "@" in the reserved email address to reduce the risk of accidental collection by non-owners or fraudulent claims.	High	3.5.5.3.1:#1
REQ-KIOSK-PICK-003	After verification passes, quickly match valid orders, fully display the core information of the orders, immediately pop up a prompt if there is no corresponding order, ensuring that visitors can quickly verify information and the process is not laggy.	High	3.5.5.3.1:#2
REQ-KIOSK-PICK-004	Supports visitors to independently select and print tickets in batches	High	3.5.5.3.1:#3
REQ-KIOSK-PICK-005	Printed tickets strictly adhere to the single-print rule, automatically marking the printed status after printing, and the system blocks the permission for repeated printing.	High	3.5.5.3.1:#4
REQ-KIOSK-PICK-006	For orders that have completed the event, passed the admission verification, and have not been printed, the printing permission for souvenir tickets will be opened.	High	3.5.5.3.1:#5
REQ-KIOSK-PICK-007	After printing is completed, a success prompt will pop up, and the device will automatically return to the initial home page.	High	3.5.5.3.1:#6

Table 3.5.5.3.2: Requirement List of Self-service Ticket Collection

3.5.5.4. Maintenance Mode

This business scenario defines the core operational logic of the Maintenance Mode available on Kiosk self-service ticket machines, covering essential device-level configuration and diagnostic capabilities required for stable daily operations. It outlines the hidden activation gesture used to enter Maintenance Mode and clarifies the operational specifications for modifying and resetting the ticket paper count, ensuring full synchronization between the kiosk device and the AnTank TAS backend

system. The scenario also incorporates mechanisms for real-time paper consumption tracking and automated low-paper warnings to prevent service interruptions and resource shortages. In addition, the Maintenance Mode provides a test-print function that enables operators to verify printer performance while maintaining accurate paper-count deduction. Together, these capabilities ensure operational reliability, support timely maintenance intervention, and safeguard the efficiency of onsite ticketing services.

3.5.5.4.1. User Process

The following table describes the user process of Maintenance Mode.

No.	Component	Description
1	Paper Count Configuration	The Maintenance Mode interface shall provide functionality for managing the kiosk's ticket paper count:
2	Modify Initial Paper Count	Operators shall be able to view and reset the current ticket paper count through the Maintenance Mode interface. The kiosk's paper count shall be synchronized with the AnTank TAS backend system.
3	Backend Synchronization Logic	Upon resetting the paper count, the system shall update the backend with the newly configured current paper count value. For every ticket printed, the backend shall decrement the current paper count by one (-1) include testing ticket.
4	Paper Count Threshold & Alerts	Paper Count Threshold & Alerts The backend system shall allow configuration of a paper-low warning threshold. When the current paper count becomes less than or equal to the configured threshold, the system shall automatically send an email alert to designate WestK staff, informing them that the ticket printer requires paper replacement.
5	Testing Ticket Function	Maintenance Mode shall include a Test Print feature for verification and diagnostics:

Table 3.5.5.4.1: Description of User Process of Maintenance Mode

3.5.5.4.2. Requirement List

RID	Description	Priority	Related Process
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REQ-KIOSK-MAIN-001	The Maintenance Mode interface shall provide functionality for managing the kiosk's ticket paper count:	High	3.5.5.4.1: #1
REQ-KIOSK-MAIN-002	Operators shall be able to view and reset the current ticket paper count through the Maintenance Mode interface. The kiosk's paper count shall be synchronized with the AnTank TAS backend system.	High	3.5.5.4.1: #2
REQ-KIOSK-MAIN-003	Upon resetting the paper count, the system shall update the backend with the newly configured current paper count value. For every ticket printed, the backend shall decrement the current paper count by one (– 1) include testing ticket.	High	3.5.5.4.1: #3
REQ-KIOSK-MAIN-004	Paper Count Threshold & Alerts The backend system shall allow configuration of a paper-low warning threshold. When the current paper count becomes less than or equal to the configured threshold, the system shall automatically send an email alert to designate WestK staff, informing them that the ticket printer requires paper replacement.	High	3.5.5.4.1: #4
REQ-KIOSK-MAIN-005	Maintenance Mode shall include a Test Print feature for verification and diagnostics:	High	3.5.5.4.1: #5

Table 3.5.5.4.2: Requirement List of Maintenance Mode

3.6. OTA Integration

Following the clarification of the OTA integration event's objectives, requirement classification framework, and technical effort evaluation criteria in the Introduction and Requirement Catalogue sections, this chapter provides a detailed description of the end-to-end business processes involved in integrating Online Travel Agency (OTA) platforms with the Ticketing and Admission System (TAS).

As the central component of this Business Requirements Document (BRD), this section translates the event's overarching objectives—expanding ticketing sales channels and ensuring that ticket sales made via OTA platforms, ticket admissions managed by WestK, and related data reports are automatically transmitted and accurately reflected within WestK's Ticketing and Admission System (TAS) and internal systems, including SIP.

This section outlines the end-to-end business processes for OTA (Online Travel Agency) integration. It covers the business problem statement, background and objectives, high level and detailed business requirements, the fulfilment approach,

customer journey, and the business scenarios for the OTA platform and Ticketing and Admission System (TAS).

3.6.1. Business Requirements

From a business perspective, OTA platforms may selectively integrate the following Application Programming Interfaces (APIs) based on their operational requirements and technical capabilities. This flexibility ensures smooth channel operations and an optimal user experience.

The following business requirements have been identified.

3.6.1.1. Ticketing Event Information Display

Ensure that the OTA platform accurately displays WestK ticketing event information to support customer ticket purchases.

Due to functional differences across OTA platforms, the system must support variations in how event information, ticketing options, and order identifiers are processed. These requirements ensure compatibility with each OTA's operational rules and technical constraints.

3.6.1.2. Ticket Purchase and Order Synchronization

Enable seamless integration of OTA platform orders with the TAS system, ensuring that TAS can efficiently receive, process, and manage the OTA initiated orders.

3.6.1.3. Ticketing Order Management

Ensure that electronic ticket information is generated and the order is accessible from both OTA platform and the TAS system after a successful OTA purchase with payment.

3.6.1.4. Admission Verification

Support onsite admission verification of OTA ticket orders at WestK.

3.6.1.5. Data Push Service

Ensure data consistency between the OTA platform and the TAS system.

3.6.1.6. Ticket Return

The OTA ticket return process must be designed to remain fully compatible with WestK's ticket return policy. This ensures that all refund requests initiated on the OTA platform follow WestK's established return rules and can be validated and processed consistently within TAS.

3.6.2. Business Requirements Analysis

This section provides further analysis and clarification of the business requirements described in Section 3.6.1. Each subsection elaborates on the business intent, operational considerations and functional expectations underpinning the OTA Integration ticketing solution.

3.6.2.1. Ticketing Event Information Display

The OTA platform retrieves WestK ticketing event information through standardized APIs. The data includes event details such as event name, stadium, venue, event period (start and end date & time), ticket price range, event description, posters, session information (e.g., session name, schedule, permitted admissions), and ticket type attributes (e.g., ticket type name, price, entry time window, early/late check-in rules, ticket category, and inventory availability).

Data Update Mechanism:

The TAS system supports real time data push capabilities. Whenever event, session, or ticket information is updated in TAS, the system automatically synchronizes the updated data to OTA platforms to ensure customers always view accurate and up to date information.

Platform-Specific Display Constraints:

To accommodate functional differences across OTA platforms, the TAS system must support configurable display rules. These include handling OTA-specific text-length limits for event fields, enabling or disabling seat-map exposure depending on whether an OTA supports marked-seat selection, and ensuring OTA-provided identifiers (such as frontend order numbers) remain searchable in the TAS backend for operational support. Additional OTA-specific requirements are summarized in Appendix C.

3.6.2.2. Ticket Purchase and Order Synchronization

Enable seamless integration of OTA platform orders with the TAS system, ensuring that TAS can efficiently receive, process, and manage the OTA initiated orders.

The OTA platform supports customers in purchasing both admission tickets and marked-seat tickets. After a ticket purchase is initiated, the OTA platform pushes the order information - including the OTA order number, customer details, event and session information, ticket type, order amount, and payment status - to the TAS system, which returns a unique TAS order number. Once OTA payment is successfully completed, the payment result must be synchronized to TAS, after which TAS generates and returns the e-ticket information to the OTA platform.

For marked-seat tickets, once the customer selects a seat and submits an order on the OTA platform, the OTA order information is synchronized to the TAS system, where TAS performs seat locking. If payment of the order on the OTA platform times out, TAS automatically releases the selected seat.

3.6.2.3. Ticketing Order Management

Order management on OTA platform: the order must include electronic ticket details, admission QR code, event information, admission time. For the marked seat tickets, the seat information must display as well.

Order management on the TAS system: after a customer completes the ticket purchase, the TAS system provides order query capabilities, supporting TAS Users to query the customer order status, ticket information, admission status, and other relevant details. The TAS system must retain complete order records to support subsequent admission processing, and reconciliation workflows, per WestK requirements.

3.6.2.4. Admission Verification

On-site verification staff scan the QR code of the ticket presented on the OTA platform order using the TAS admission devices. The system validates ticket authenticity in real time, including admission status and validity period. Once ticket admitted, the admission process is completed, the order status is updated, and the admission result is pushed to the OTA platform in real time.

3.6.2.5. Data Push Service

The TAS system will push relevant updates to OTA platforms, including:

- Changes of event activation status and session activation status
- Changes of new ticket type creation
- Changes of ticket price
- Changes of order creation
- Changes of order status
- Changes of admission status
- Changes of return status

3.6.2.6. Ticket Return

The OTA platform may initiate ticket return requests, which are transmitted to the TAS system. The TAS Users performs the ticket return review in the TAS system. The TAS system updates the order status and releases the inventory accordingly. Return results are synchronized back to the OTA platform, which is responsible for completing the final customer return process for the customer.

3.6.3. Operating Model

This OTA integration event operates under a Tripartite Collaboration model involving WestK, AnTank, and the OTA partners. The model defines the responsibilities of each party—such as system integration, ticket distribution, and operational support—and outlines how the three parties coordinate throughout the event lifecycle. Detailed responsibilities are provided in the table below.

Party	Role	Responsibility
WestK	Business Owner	1. Define information such as events, sessions, ticket prices, and inventory; 2. Be responsible for admission verification

		and on-site operations; 3. Be responsible for review and approval for ticket return.
OTA Platform	Distribution Channel Operator	1. Provide platform traffic and user reach channels; 2. Obtain and display event information, inventory, and electronic tickets through interfaces; 3. Provide the platform for customer ticket purchase, payment, and ticket return operations.
TAS Supplier (AnTank)	Technical Solution Provider	1. Provide the capability to onboard events for WestK; 2. Provide standard APIs for OTA integration and data synchronization mechanisms; 3. Maintain the order management, electronic ticket generation, admission verification, and ticket return review; 4. Retain customer order data per WestK requirements.

Table 3.6.3: Role and Responsibility of the Parties

3.6.4. Customer Journey

This section describes the end-to-end ticketing process from the perspectives of both OTA customers and WestK staff/TAS users. The user stories outline the users' needs, goals, and expected outcomes, providing clarity on what each user intends to achieve and why the corresponding system capabilities are required.

The following user stories collectively illustrate how OTA customers interact with the ticketing journey—from browsing and purchasing to admission and refund—and how TAS users support and enable these processes through backend operations and system functions.

No.	Role	System	User Story
1	As an OTA Platform Customer	OTA	I can discover WestK events and view event descriptions, images, and key details on the OTA platform.
2	As an OTA Platform Customer	OTA	I can purchase admission tickets on the OTA platform, by selecting events, sessions, and ticket types, submitting orders and completing payment.
3	As an OTA Platform Customer	OTA	I can purchase marked-seat tickets on the OTA platform, by selecting event, session, and ticket type and seat, submitting orders and completing payments.

4	As an OTA Platform Customer	OTA	I can view the purchased order on OTA platform.
4	As an OTA Platform Customer	OTA	I can receive an electronic ticket (QR code) in the OTA app after a successful purchase.
5	As an OTA Platform Customer	OTA	I can apply for ticket return on the OTA platform based on the predefined ticket policy.
6	As an OTA Platform Customer	OTA	I can receive refund approval/rejection results and refund confirmation on the OTA platform.
7	As a WestK TAS Staff	TAS	I can onboard events in the TAS system, after which customers can browse the events and purchase tickets through the OTA platform.
8	As a WestK TAS Staff	TAS	I can update the event information in the TAS system, it will be synchronized to the OTA platform, including changes to information such as events, ticket types, prices, ticket return status, etc.
9	As a WestK TAS Staff	TAS	I can admit the tickets by scanning the QR code of OTA ticket order using admission devices.
10	As a WestK TAS Staff	TAS	I can query and views order information in the TAS system for the OTA orders.
11	As a WestK TAS Staff	TAS	I can review the return request and approval/reject based on WestK's ticket return policy.
11	As a WestK Finance Staff	TAS	I can handle the refund process after receiving the refund request from Operation team.
12	As a WestK Finance Staff	TAS	I can view the sales and operational reports, including sales show, inventory status, and admission analytics, and the TAS system allows me to export them for financial reconciliation and management review.
13	As a WestK Finance Staff	TAS	I can export financial and sales reports for OTA orders for reconciliation purposes.

Table 3.6.4: Description for the User Story

3.6.5. Business Scenarios

This chapter outlines the complete set of business scenarios that define how WestK collaborates with AnTank's Ticketing and Admission System (TAS) and external OTA platforms to deliver a seamless ticketing experience for customers. The scenarios describe both the end-to-end workflow encountered by OTA customers and the supporting operational processes within TAS that enable the overall system to function effectively.

From the moment an event is onboarded into TAS, to its display on OTA platforms, customer ticket purchase, venue admission, refund handling, order management, and reporting, each scenario represents a critical component of the integrated journey. While OTA customers interact directly with browsing, purchasing, e-ticket access, and admission, TAS plays an essential enabling role by synchronizing event information, managing inventory and orders, processing refund validations, and providing sales insights for operational and financial users.

The Diagram 3.6.5 below illustrates how these customer-facing interactions and TAS supporting processes connect across the end-to-end workflow, providing a consolidated view of the overall integration model.

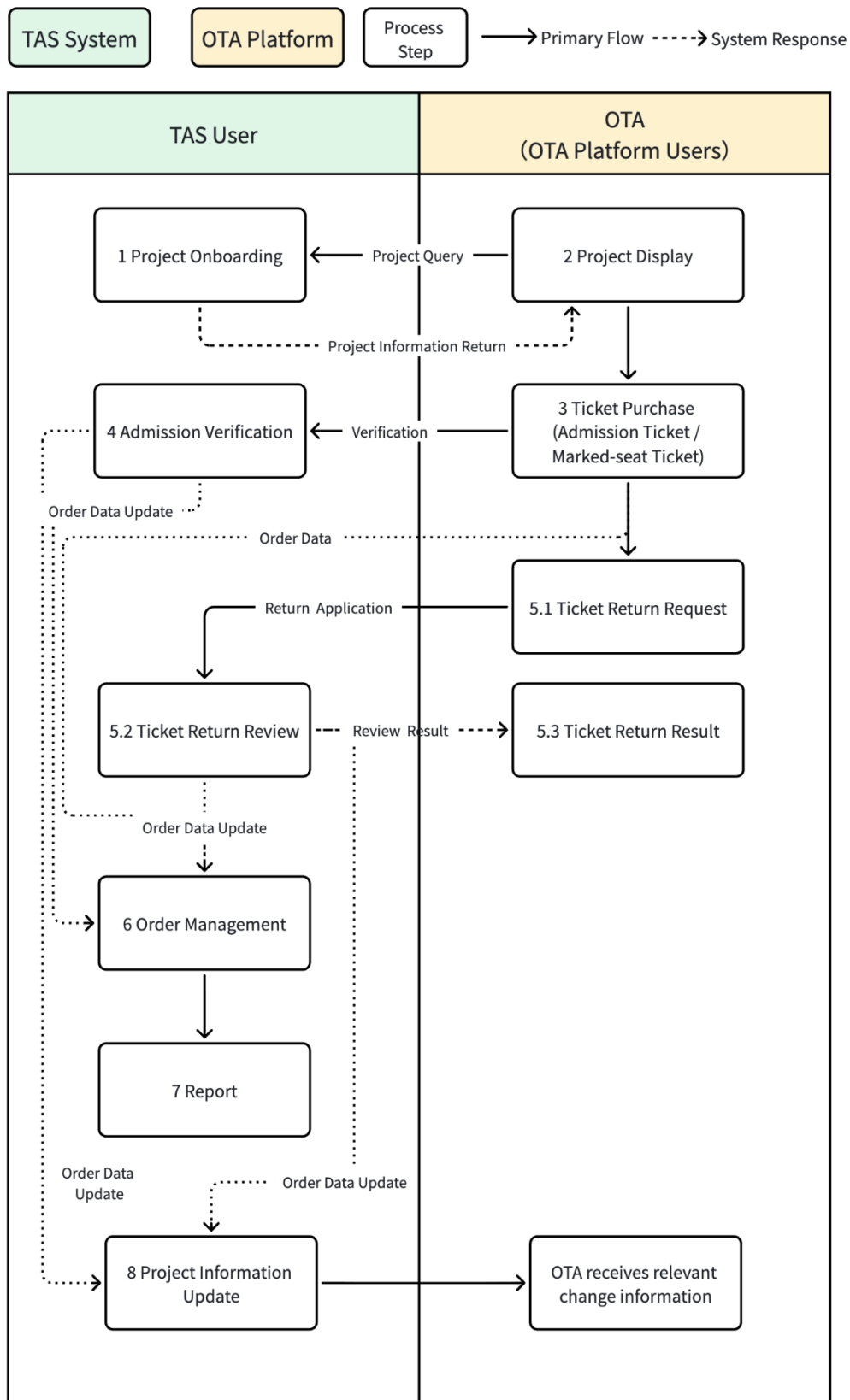


Diagram 3.6.5: Business Process Between TAS System and OTA Platform

3.6.5.1. Event Onboarding

This scenario describes the requirements for creating and configuring event, session, and ticket information within the TAS system. Event onboarding serves as the foundational step of the OTA integration flow, as it enables WestK events to be published and made available on OTA platforms. Only after onboarding is completed can OTA customers discover the events, view available sessions and ticket types, and proceed with ticket purchases.

RID	Component	Description
REQ-OTA-001	Event Onboarding	The TAS User enables the OTA channel, configures and activates the event within the TAS system to facilitate synchronization with the OTA platform. This includes setting up event details including but not limited to: event name, stadium, venue, event period (start date & time, end date & time), ticket price range, event description, poster), session information (including but not limited to: session name, schedule (start date & time, end date & time), number of permitted admissions, and ticket-related attributes, including but not limited to: ticket type name, price, entry time window (start time and end time), permitted early check-in period, permitted late check-in period, ticket classification (standard or concession), ticket inventory.

Table 3.6.5.1: Requirement Description of Event Onboarding

3.6.5.2. Event Display

This scenario describes the requirement about event, session, and ticket information is initially created and configured within the TAS system. Event onboarding is the foundational step that enables WestK events to be published to OTA platforms. Only after onboarding is completed can OTA customers discover and purchase tickets for these events.

After the event is onboarded in the TAS system, OTA will retrieve the information through the relevant APIs and display it to customers:

RID	Component	Description
REQ-OTA-002	Event Information	The customers can view the event information including but not limited to event name, stadium, venue, event period (start date & time, end date & time), ticket price range, event description, poster on the OTA platform.
REQ-OTA-003	Session Information	The customers can view the session information including but not limited to session name, schedule (start date & time, end date & time), and number of permitted admissions in the event on the OTA platform.
REQ-OTA-004	Ticket Type Information	The customers can view ticket-related attributes, including but not limited to ticket type name, price, entry time window (start time and end time), permitted early check-in period, permitted late check-in period, ticket

		classification (standard or concession), ticket inventory in the event on the OTA platform.
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Table 3.6.5.2: Requirement Description of Event Display**3.6.5.3. Ticket Purchase and Order View**

This scenario describes the end-to-end process enabling OTA customers to purchase the admission ticket and marked-seat ticket.

RID	Component	Description
REQ-OTA-005	Admission Ticket Order Placement	The OTA customers can place admission ticket orders with successful payment on the OTA platform. The OTA customers can place admission ticket order with payment on the OTA platform.
REQ-OTA-006	View Admission Ticket Order Information	After ticket purchase, the customers can view admission ticket order information on the OTA platform, including the QR code.
REQ-OTA-007	Marked-Seat Ticket Order Placement	The OTA customers can place marked-seat ticket order with successful payment on the OTA platform.
REQ-OTA-008	View Marked-Seat Ticket Order Information	After ticket purchase, the customers can view marked-seat ticket order information on the OTA platform, including the QR code.

Table 3.6.5.3: Requirement Description of Ticket Purchase and Order View**3.6.5.4. Admission Verification**

This scenario explains how OTA-issued electronic tickets are validated at WestK venues through TAS admission devices. It ensures smooth customer entry and accurate admission status updates. It corresponds to requirements governing QR code validation, ticket authentication, and admission record synchronization.

After the ticket is admitted, the TAS system will update the verification status to the OTA platform.

RID	Component	Description
REQ-OTA-009	Verification Operation (Admission Ticket/Marked-Seat Ticket)	When the customer arrives at WestK, he/she shall present the QR code in OTA ticket order from the OTA platform, and WestK onsite staff will scan the QR code through the TAS admission device for verification.

Table 3.6.5.4: Requirement Description of Admission Verification

3.6.5.5. Ticket Return

This scenario describes how OTA customers initiate refund requests and how TAS validates refund eligibility based on WestK policies. It aligns with requirements covering refund rules, order status updates, and OTA-TAS refund synchronization.

RID	Component	Description
REQ-OTA-010	Customer Application	When the customer submits a ticket return request on the OTA platform, the request will be synchronized to the TAS system.
REQ-OTA-011	WestK Review & Approval	WestK staff review and approve the return request in the TAS system in accordance with the business operational rules, and the results pushed to the OTA platform. Upon approval, the TAS system releases the inventory.
REQ-OTA-012	OTA Order Status Change	OTA users can see the order status updated to returned on the OTA platform.

Table 3.6.5.5: Requirement Description of Ticket Return

3.6.5.6. Order Management in TAS

This scenario details how WestK operational staff view and manage OTA orders through the TAS system. It supports requirements related to order querying, operational oversight, exception handling, and channel-level monitoring across OTA integrations.

RID	Component	Description
REQ-OTA-013	Query and View Admission Ticket Orders	The TAS User can query and view the details of the admission ticket orders placed from OTA in the TAS system with OTA order information including but not limited to: Ticket Type, Stadium, Venue, Session, Order Number, Ticket Price, Ticket Quantity, Payment/Refund Time, Ticket Status, Channel.
REQ-OTA-014	View Marked-Seat Ticket Orders	The TAS User can query and view the details of the marked-seat ticket orders placed from OTA in the TAS system with OTA order information including but not limited to: Ticket Type, Stadium, Venue, Session, Order Number, Ticket Price, Ticket Quantity, Payment/Refund Time, Ticket Status, Channel.

Table 3.6.5.6: Requirement Description of Order Management in TAS

3.6.5.7. Reporting

This scenario outlines the reporting capabilities required for finance and operational users, including OTA sales, inventory usage, admission data, and revenue reconciliation. It directly corresponds to reporting-related requirements, ensuring

WestK staff can generate accurate sales and financial reports for management and settlement.

RID	Component	Description
REQ-OTA-015	Report	The system must provide WestK staff with access to centralized sales and operational reports, covering OTA order information including but not limited to: Ticket Type, Stadium, Venue, Session, Order Number, Ticket Price, Ticket Quantity, Payment/Refund Time, Ticket Status, Channel.

Table 3.6.5.7: Requirement Description of Report in TAS

3.6.5.8. Event Information Change

This scenario describes how modifications to event, session, ticket type, pricing, or refund rules made within TAS are synchronized to OTA platforms in real time. It links to requirements governing data update mechanisms, accuracy of customer-facing event information, and real-time OTA synchronization.

RID	Component	Description
REQ-OTA-016	Event Information Change	When event information changes in the TAS system, the updated information is pushed to the OTA platform in real time, including changes to event activation status, session activation status, new ticket type creation, ticket price, order creation, order status, admission status, return status.

Table 3.6.5.8: Requirement Description of Event Information Change

3.7. Ticketing Admission System

This section sets out the business processes of the ticketing system. It covers the business requirements, event background and objectives, high-level and detailed requirements, implementation approach, customer journey, and the functional specifications for both front-end and back-end workflows.

3.7.1. Business Requirements

From a business perspective, WestK requires the ticketing system to cover the entire process of ticketing for all types of events such as exhibitions, shows, workshops, etc., adapt to diverse ticketing forms, integrate various core ticketing capabilities, ensure system compliance, efficiency, and traceability, and meet the goals of refined operation.

The following business requirements have been identified.

3.7.1.1. Ticket Configuration Management

WestK requires the ticketing system to implement refined configuration and classified management of all types of tickets, support multi-dimensional customisation and multi-scenario adaptation, as follows:

- Basic configuration: Supports the configuration of regular ticket types such as standard tickets, student tickets, and senior tickets, and allows customizing the display order of ticket types.
- Exclusive ticket type configuration: Supports independent configuration of special ticket types such as sponsorship tickets and complimentary tickets, allowing for separate association of validity periods or configuration of channel sales permissions.
- Ticket type classification: divided into flexible tickets (controlled by total inventory and validity period) and calendar tickets (controlled by daily independent inventory and specific dates), suitable for different types of activities.
- Shared quota configuration: Supports shared quotas for preferential ticket types, with a configurable total admission limit; supports shared inventory for different preferential ticket prices.

3.7.1.2. Ticket Pricing and Service Fees

WestK requires the ticketing system to achieve refined ticketing pricing, flexible configuration of service fees, adapt to the differentiated needs of different seats, audiences, and channels, and ensure that pricing is compliant and tailored to operational scenarios, as follows:

- Regional Pricing: Seated tickets support differentiated pricing by seat zones, enabling refined management and control.
- Differential Pricing for Different Customer Groups: The same seat can be settled at different discounted fares (e.g., student tickets, senior tickets).
- Service fee: A fixed fee is charged per ticket (independent of ticket price), and some items may have a single-transaction cap; it supports differentiated settings by channel, item, and ticket type, and can also be waived.

3.7.1.3. Management of Admission Time Slots and Ticket Validity

WestK requires the ticketing system to implement flexible configuration of admission time slots, multi-mode management and control of ticket validity periods, and adapt to the management and control requirements of different events, as follows:

- General Configuration: All tickets must have entry time slots, support all-day access, time-slot entry, and "entry at any time on the day of purchase", and allow customization of early/late entry rules.
- Flexible Ticket Management: Flexible tickets allow the bearer to enter directly or require prior reservation, and the system manages passenger flow according to the quota.
- Multiple entries: Supports the same ticket being used to enter through different entrances and supports multiple entries with the same ticket.

- **Special Scenario Adjustment:** When the venue closes due to inclement weather, the validity period of tickets will automatically be extended by 2 months regardless of whether the bearer has entered or not.

3.7.1.4. Workshop Activity Registration Management

WestK requires the ticketing system to implement integrated management of Workshop registration and ticketing, as follows:

- **Business Adaptation:** Workshop is classified as a fixed-session event, compatible with paid/free and seat selection/Admission ticket.
- **Registration Form:** The registration form can be customized, and users must fill it out and submit it before entering the ticketing/reservation process.
- **Registration Review:** It can be configured whether review is required. For items that require review, users can only make payment (for paid items) after passing the review.
- **Purchase Limit Control:** Single-user purchase limits can be set by event, session, and ticket type, and a maximum purchase limit per order can also be set.

3.7.1.5. Inventory and Quota Management

WestK requires the ticketing system to implement refined control of inventory and quota, ensuring data synchronization across all channels, as follows:

- **Quota Coordination:** Unified management of all channel quotas, supporting flexible adjustment and sharing, with separate rules for special promotional activities.
- **Ticket Pool Management (Exclusive to Seat Selection Tickets):** Supports customizing ticket pools and sales permissions for each channel and supports multiple OTAs to select seats simultaneously (first come, first served).
- **Data Synchronization:** Adjustments to core ticketing data (such as seat data, ticket pool allocation, inventory, admission time slots, etc.) take effect immediately, and OTA channels complete data synchronization once every 5 - 10 minutes.

3.7.1.6. Seat Management (Exclusive to Seat Selection Tickets)

WestK requires the ticketing system to implement refined management of the entire process for seat selection tickets, meeting the requirements of show events, as follows:

- **Seating layout:** Supports customising theatre seating layouts, allowing uploading of base maps, setting of partitions and seat arrangements, and supporting the arrangement of odd-even numbers and sequential numbers.
- **Mixed Seating:** The same seating chart can simultaneously configure seated areas (define layout) and standing areas (set a maximum number of people), supporting simultaneous settlement for standing and seat selection.
- **Special Seats:** Support special seat configurations such as wheelchair seats, allow setting sales rules according to business characteristics, and can be converted to regular seats for sale.

- Sales Function: Supports independent sales of individual seats and bulk purchase configurations (such as bundled sales of couple seats or a table with four seats).
- Reserved Seats: Exclusive seats can be locked, and only Box Office can unlock and issue tickets after locking.

3.7.1.7. Sales Channel and Sales Rule Management

WestK requires the ticketing system to uniformly manage and control all sales channels, support flexible configuration rules, and the details are as follows:

- Channel Coverage: Covers all channels including Kiosks, official website, mini-program, OTA, and Box Office.
- Permission Control: Customize sales permissions for each channel, support OTA exclusive ticket pools and priority for self-owned channels and allow temporary suspension of sales or adjustment of permissions.
- Sales Period: Configure the available sales scope by channel, for example, OTA only sells tickets within 1 month, and Kiosks only sell same-day tickets.
- Data Synchronization: Adjustments to the system's core data take effect immediately, and the OTA channel synchronizes every 5-10 minutes.

3.7.1.8. Combined Sales Function

WestK requires the ticketing system to support diversified combined sales models, enhance marketing conversion efficiency through functions such as package tickets and add-on purchases, and adapt to the sales scenarios of different events, as follows:

- Same-event combo tickets: Support the combined sales of multiple tickets for the same item and ticket type, with combo ticket discounts covering all channels.
- Cross-session combo tickets: Can bundle and sell specified seat selection items and sessions. Users who meet the session and ticket price thresholds can enjoy discounts.
- Add-to-cart feature: Users can add items to their purchase (to be implemented in conjunction with the mall system) and enjoy corresponding discounts when checking out.

3.7.1.9. System Linkage and Supporting Functions

WestK requires the ticketing system to improve its supporting functions, enhance user experience and operational efficiency, as follows:

- Ticket Style: E-tickets support portrait orientation and clause redirection, with customizable layout; physical tickets support trilingual display, custom design, and can add sponsor logos and adapt to printing.
- Message push: Scenarios such as successful ticket purchase, event start reminder, and event change are pushed to users, covering email, WeChat, and SMS channels; scenarios such as successful ticketing for special seats like wheelchair seats and the expiration of reservation/seat lock orders are pushed to operators, covering email, SMS (to be implemented in conjunction with the Message Push Management System and third-party Message Platform) channels;

- **Queue Management:** For popular seatless events, queue numbers are generated based on the ticket purchase time, supporting internal reserved numbers (not publicly displayed).

3.7.2. Business Requirement Analysis

This section conducts in-depth decomposition and refinement of the business requirements in Section 3.7.1, with each sub-module corresponding one by one, clarifying the business intent, implementation rules, operational considerations, etc. of each requirement, providing guidance for the development, acceptance, and omnichannel operation of the AnTank ticketing system, adapting to the omnichannel collaborative operation requirements of WestK Douyin Mini Program, independent ticketing official website, etc., and aligning with the core objectives of the digital ticketing strategy of WestK.

3.7.2.1. Ticket Type Configuration Management

Map to Business Requirement 3.7.1.1.

The core is to build a standardised and customisable ticket type configuration system, adapt to the differentiated operational requirements of multiple business units and multiple event types of WestK, achieve unified management and control of ticket type rules across all channels, and meet the hierarchical service needs of different customer groups.

1) Basic Ticket Type Configuration: Supports full lifecycle management of regular ticket types, allows adding new custom ticket types and adjusting front-end display order.

2) Exclusive Ticket Type Configuration: Special ticket types such as sponsorship tickets and complimentary tickets have independent configuration entries, with separate settings for validity period, sales channels, etc., isolated from publicly available ticket type data, and can restrict specified sales channels.

3) Ticket Type Classification: Clarify the underlying configuration logic for flexible tickets and calendar tickets. The former adapts to long-term exhibitions through total inventory management and control, while the latter adapts to fixed-session events through single-session/single-day inventory management and control, with the underlying rules isolated.

4) Shared Quota Configuration: Discount ticket types support the shared quota model, where a total admission cap is set, allowing multiple discount ticket types to share the underlying inventory and avoid the risk of overbooking.

3.7.2.2. Ticket Pricing and Service Fees

Map to Business Requirement 3.7.1.2.

Through a refined pricing and service fee configuration system, it adapts to the operational needs of multiple venues and channels, achieving segmented operation of customer groups.

1) Regional Differential Pricing: Seat selection tickets support setting standard ticket prices by seat zones, and different ticket types within the same zone can have different discounted prices, achieving a two-tier system of "zone-based pricing + group-specific discounts".

2) Differential Pricing for Different Customer Segments: For the same seat and session, different settlement prices can be set for different ticket types, and identity verification rules for discounted ticket types can be configured to prevent abuse of discounts.

3) Flexible configuration of service fees: Supports charging a fixed service fee per ticket, with the ability to set a cap on the service fee for a single order; Differentiated service fee rules can be set based on channels, events, and ticket types, and rules for exempting service fees can also be set for specific scenarios.

3.7.2.3. Ticket Validity and Entry Time Slot Management

Map to Business Requirement 3.7.1.3.

Through multi-modal validity period and admission time slot management, refined passenger flow management is achieved, addressing uneven passenger flow and ensuring on-site operational efficiency.

1) Universal Time Slot Configuration Rules: All tickets are associated with admission time slots, supporting three major modes: all-day admission, time-slot-based admission, and admission at any time within the validity period of the time slot, all of which allow for customizing early/late admission rules.

2) Exclusive Time Slot Management for Flexible Tickets: Supports two modes: direct entry without reservation and entry by reservation. The latter manages passenger flow according to reserved quotas, meeting the needs of large-scale passenger flow exhibitions.

3) Multiple-entry ticket configuration: Supports multi-user group verification mode, allowing entry from different entrances, and supports setting the number of entries based on ticket types, with different entrances able to set different numbers of entries.

4) Special Scenario Validity Adjustment: When the venue is closed due to bad weather, the validity period of tickets for unadmitted spectators can be extended; for those who have entered but are forced to leave, the system will mark and record this, allowing them to re-enter within the extended period and reset the entry count.

3.7.2.4. Event Registration Management

Map to Business Requirement 3.7.1.4.

For fixed-session events with strong registration attributes and high information collection requirements, it implements integrated management of registration and ticketing, addresses pain points such as fragmented processes and untraceable information, and adapts to diverse operational models and segmented event needs.

1) Activity Attribute Adaptation: Classify Workshop as an independent fixed-session event type, isolate it from regular event configurations, and be compatible with free/paid and seat selection/Admission ticket selection.

2) Custom Registration Form Configuration: Provides a visual form configuration function, allowing customization of fields, required/optional settings, and validation rules. Users can only proceed to the ticketing/reservation stage after filling out and submitting the form.

3) Registration Review Process Configuration: Supports two modes: "Direct Purchase without Review" and "Purchase after Review Approval". If the review fails,

the reason must be marked and the user notified, and the registration process will be terminated.

4) Purchase Limit Rule Management: Based on events, sessions, and ticket types, set a maximum purchase quantity limit per user, and also limit the number of tickets per order, compatible with the general purchase limit system.

3.7.2.5. Inventory and Quota Management

Map to Business Requirement 3.7.1.5.

Establish a unified inventory and quota management and control system across all channels, with the AnTank system as the sole data source for inventory, to address pain points such as out-of-sync inventory across multiple channels and adapt to the requirements of omnichannel ticketing and differentiated operations.

1) Omnichannel Quota Overall Management: Achieves unified management of quotas for self-owned, OTA, and other channels, supports flexible adjustment and cross-channel sharing, and allows separate quota rules to be set for special promotional activities.

2) Exclusive Ticket Pool Management for Seat Selection Tickets: Divide independent ticket pools by seating area, customize channel sales permissions, and support two allocation modes: first-come, first-served seat selection across multiple OTAs and exclusive ticket pool division by OTA.

3) Omnichannel data synchronization mechanism: After core ticketing data is adjusted, it takes effect immediately on self-owned channels, while OTA channels synchronize every 5-10 minutes to ensure consistent inventory data and avoid the risk of overselling.

3.7.2.6. Seat Management (Exclusive to Seat Selection Tickets)

Map to Business Requirement 3.7.1.6.

Implement refined management of seat layout, special seats, and sales rules for reserved-seat tickets, adapt to the operational needs of different venues, ensure the service needs of special groups, and enhance user experience.

1) Custom Configuration of Seat Layout: Supports visual design of seat layout, allows uploading of base maps, custom partitioning and arrangement, supports single/double numbering/sequential numbering, and is accompanied by front-end display of seat numbering.

2) Mixed Seating Configuration: The same seating chart can simultaneously configure seated and unseated areas, with the unseated area having a maximum number of people, supporting simultaneous settlement of two types of tickets, and adapting to the needs of mixed operations.

3) Special Seat Management: Independently configure wheelchair seats, companion seats, etc., set exclusive sales rules, and automatically notify operating personnel upon sale to ensure follow-up of accessible user experience services.

4) Seat Sales Function Configuration: Supports single seat sales and multiple bulk purchases, allows configuration of bundled products such as couple seats and multi-seat tables, which cannot be purchased separately.

5) Reserved Seat Management: Reserved seats can be locked for camera positions, VIPs, etc. Only the offline ticketing centre can unlock and issue tickets, and they are not available for sale through public channels.

3.7.2.7. Sales Channel and Sales Rule Management

Map to Business Requirement 3.7.1.7.

Implement unified management and control across all sales channels, along with differentiated rule configuration, to adapt to multi-channel operational requirements, implement the operational strategy of prioritizing self-owned channels, and gain the initiative in ticket operations.

1) Omnichannel Coverage Adaptation: Fully cover channels such as self-service terminals, ticketing websites, various mini-programs, OTAs, offline ticketing centres, etc., with all ticketing data uniformly connected to the AnTank system for management.

2) Refined control of channel permissions: Customize the sales permissions of each channel from the dimensions of event, session, seat, and ticket type, and support adjusting the ticket pool, configuring the priority sales rights of self-owned channels, etc.

3) Differential Configuration of Sales Period: Configure the available ticket sales time by channel, for example, OTA only sells tickets within 1 month, and self-service terminals only sell same-day tickets, adapting to the operational positioning of each channel.

3.7.2.8. Combined Sales Function

Map to Business Requirement 3.7.1.8.

Enrich marketing strategies through diversified combined sales models, enhance the value per customer and conversion efficiency, meet the needs of multi-venue and multi-event coordinated operation, and drive the sales of niche events and related products.

1) Same-session ticket package sales: Supports combined pricing of multiple tickets for the same item, same session, and same ticket type. The discount rules cover all channels, and the system automatically calculates the discount amount.

2) Cross-show package ticket sales: For seat selection ticket shows, cross-show package tickets can be configured. Users who purchase package tickets that meet the conditions can enjoy discounts, suitable for joint marketing such as series shows.

3) Add-on Purchase Function Configuration: During the order settlement process, it supports adding on products such as peripherals, catering, and membership benefits, allowing users to enjoy add-on purchase discounts and enabling the linked sales of tickets and peripherals.

3.7.2.9. Ticketing supporting functions

Map to Business Requirement 3.7.1.9.

Improve ticketing supporting functions to enhance the user experience throughout the process and the work efficiency of operational staff.

1) Refined Management of Ticket Styles: Electronic tickets can customize layout information by ticket type and event, while physical tickets support traditional Chinese and English bilingual, custom design, and are compatible with printing; electronic tickets are sent via email and SMS in the form of a link containing a QR code.

2) Full-scenario message push function: Supports push notifications from both the user and operator sides. The user side covers scenarios such as successful ticket purchase and shows start reminder, while the operator side covers scenarios such as special seat ticket issuance, reaching users through multiple channels.

3) Popular Event Queue Management: For popular seatless events, queue numbers are generated based on the ticket purchase time, supporting internal number reservation, controlling ticket purchase order, and avoiding system congestion and ticket-grabbing chaos.

3.7.3. Customer Journey

The Ticketing System functions as the central backend platform, supporting unified ticket lifecycle management, inventory control, promotion rules, and admission validation for both internal operators and end users.

The journey illustrates how administrator interact with the ticketing system through multiple channels while the Ticketing System ensures consistency, accuracy, and real-time synchronisation throughout the process.

No.	Category	Role	User Story
1	System Access	As an Administrator/Operator	I log into the Ticketing System using an authorised account so that I can access configuration and operational functions based on my role and permissions.
2	Event & Session Configuration	As an Administrator	I create and maintain events, sessions, ticket types, pricing rules, quotas, and sales policies so that ticketing information is correctly defined in the system.
3	Channel Enablement	As an Administrator	I configure ticket availability, quotas, and channel-specific rules (e.g. Website, Box Office, Kiosk, OTA) so that tickets can be distributed consistently across all sales channels.
4	Inventory Control	As an Administrator	I manage events inventory, seat map, and reservation rules to ensure accurate ticket availability and prevent overselling.
5	Order Monitoring	As an Operator	I monitor ticket orders created from different channels so that I

			can track sales progress and identify exceptions or issues.
6	Admission Validation	As an Operator	I use the Ticketing System or connected admission tools to verify tickets at the venue so that only valid tickets are admitted and entry data is recorded.
7	Exception Handling	As an Operator	I handle operational exceptions such as ticket exchanges, refunds, invalid tickets, or special admission scenarios according to configured business rules.
8	Reporting & Analysis	As an Administrator	I review consolidated sales, admission, and operational reports generated by the system so that management decisions can be supported by accurate data.

Table 3.7.3: Description for the Ticketing User Story

3.7.4. Business Scenarios

3.7.4.1. Ticket Management

This scenario defines the full-process business logic for operators to set up basic information for ticket events, configure ticket types and package ticket rules, manage time slots and inventory, and enable sales channels and queuing numbers. The core objective is to address the pain points of scattered multi-dimensional ticket configuration and inconsistent rules, adapt to the refined operation requirements of multi-channel and multi-form tickets, achieve standardised and efficient management of ticket configuration, and improve operational efficiency and the accuracy of rule execution.

3.7.4.1.1. User Process

The following table describes the user processes for ticket item management, ticket type management, time slot and inventory management, and sales permission configuration.

No.	Component	Description
1	Basic Information Management of Ticket Items	In the ticket ticketing system, you can create and modify the basic information of ticket items, including name, affiliated venue, validity period, item classification, etc.
2	Basic Ticket Type Management	Supports the configuration of regular ticket types such as standard tickets, student tickets, senior tickets, etc.

Technology Partnership Package 2
Requirement Document

		Define basic information such as the name, description, selling price, admission time, number of admissible people, display order, etc. of the basic ticket types.
3	Same-Event Ticket Package Configuration	Define the discount rules for multi-ticket combination packages, set the total inventory limit for the packages, and support single-ticket type combinations and multi-ticket type combinations.
4	Basic Configuration of Time Period Management	Define basic information such as the start and end times, name, ticket sale starts time, real-name rules, entry and ticket collection methods of the admission period.
5	Sales Channel Permission Management	Sales channels can be specified at the event, ticket type, same-event package ticket, and time period dimensions, covering self-service terminals, ticketing websites, various mini-programs, OTA, offline ticketing centres, and other channels.
6	Shared Quota Configuration	The maximum number of people allowed to enter during a set time period can be set, meaning that all ticket types within the time period share this inventory. Multiple ticket types can be specified to share a single inventory.
7	Differential Configuration of Sales Periods	Different sales cycles can be defined for different channels during the same time period.
8	Multiple Entry Configuration for Ticket Types	The event dimension can set up a multi-user group (multi-entry) verification mode, where the same ticket code can be used for multiple entries through different entrances; the ticket type can set the number of entries, and different user groups (entrances) can set different numbers of entries.
9	General Admission Time Slot Configuration	Through the configuration of the start and end times of time periods and the admissible time for ticket types, full-day admission, time-period-based admission, and admission at any time within the validity period of a time period are achieved.
10	Configuration of Flexible Ticket Admission Method	Flexible ticket time slots support two modes: direct entry without reservation and entry by reservation.

11	Inventory of Time Period Ticket Types	Different ticket types of sales inventories can be set for different time periods and different channels.
12	Purchase Limit Rule Management	The time period dimension supports the maximum number of tickets purchasable per order and the maximum cumulative number of tickets purchasable per user. The event dimension supports setting purchase limit rules by event, time period, and ticket type.
13	Time Period Rescheduling or Postponement	Supports modifying the validity period of time slots. After the modification is successful, the system will automatically refresh historical order data and synchronously refresh the validity period of sold tickets.
14	Queue Number	Supports enabling the queuing number function for a specified time period. During the period when queuing numbers are enabled, the system automatically generates numbers in chronological order of orders.

Table 3.7.4.1.1: *Explanation of Ticket Management Process***3.7.4.1.2. Requirements List**

RID	Description	Priority	Related Process
REQ-TKT-ADT-001	Supports operators in creating and modifying basic information of ticket items in the ticket ticketing system, including core content such as name, affiliated venue, validity period, and item classification.	High	3.7.4.1.1:#1
REQ-TKT-ADT-002	Supports operators to configure standard tickets, student tickets, senior tickets and other regular ticket types, and define basic information such as the name, description, selling price, admission time, number of admissible people, display order, etc. for each ticket type.	High	3.7.4.1.1:#2
REQ-TKT-ADT-003	Supports operators in defining multi-ticket combination package ticket discount rules, setting total package ticket inventory limits, and being compatible with package ticket operation scenarios for single-ticket type combinations and multi-ticket type combinations.	High	3.7.4.1.1:#3

REQ-TKT-ADT-004	Supports operators in defining basic configuration information such as the start and end times, name, sales start time, real-name rules, entry and ticket collection methods of admission time periods.	High	3.7.4.1.1:#4
REQ-TKT-ADT-005	It supports full-channel coverage including self-service terminals, ticketing websites, various mini-programs, OTAs, offline ticketing centres, etc., and can specify available sales channels separately at the event, ticket type, same-event package ticket, and time period dimensions, enabling refined channel permission management and control.	High	3.7.4.1.1:#5
REQ-TKT-ADT-006	Supports operators to set the maximum number of people allowed to enter during a time period (all ticket types within the time period share this inventory).	High	3.7.4.1.1:#6
REQ-TKT-ADT-007	Supports operators to specify that several ticket types can share a single inventory.	High	3.7.4.1.1:#6
REQ-TKT-ADT-008	Supports defining differentiated sales cycles for different channels during the same period to adapt to the differential needs of multi-channel operation strategies.	High	3.7.4.1.1:#7
REQ-TKT-ADT-009	Supports operators to define events that support multi-user group verification (multi-entry admission).	High	3.7.4.1.1:#8
REQ-TKT-ADT-010	Supports operators to define the number of admissible entries for different user groups (entrances) of ticket types.	High	3.7.4.1.1:#8
REQ-TKT-ADT-011	Supports three common admission modes, namely all-day admission, time-slot admission, and admission at any time within the validity period of the time slot, by configuring the start and end times of the time slot and the admissible time for ticket types.	High	3.7.4.1.1:#9
REQ-TKT-ADT-012	Supports configuring two modes for flexible ticket time slots: direct entry without reservation and entry by reservation.	High	3.7.4.1.1:#10
REQ-TKT-ADT-013	Supports operators to separately set the sales inventory for various ticket types according to different time periods and	High	3.7.4.1.1:#11

	different channels, ensuring precise control of inventory across all channels.		
REQ-TKT-ADT-014	Supports setting the upper limit for the number of tickets purchased per order and the cumulative number of tickets purchased per user at the time period dimension; supports setting ticket purchase limit rules by event, time period, and ticket type at the event dimension to prevent over-purchase.	High	3.7.4.1.1:#12
REQ-TKT-ADT-015	Supports operators to modify the validity period of time slots. After the modification is successful, the system automatically refreshes historical order data, synchronously updates the validity period of sold tickets, and ensures data consistency.	High	3.7.4.1.1:#13
REQ-TKT-ADT-016	Supports enabling the queuing number function for a specified time period. After enabling, the system automatically generates numbers in chronological order of orders, standardizing the ticketing order during peak hours.	High	3.7.4.1.1:#14

Table 3.7.4.1.2: List of Ticket Management Requirements

3.7.4.2. Seating Plan Management

This scenario defines the full-process business logic for operators to complete the basic setup of seat plans, multi-mode seat editing, special seat definition, seat bundling and mixed configuration, custom attribute settings, and multi-format data export for seat selection ticketing events.

3.7.4.2.1. User Process

The following table describes the user process for seat plan management.

No.	Component	Description
1	Basic Configuration of Seating Plan	Define the name of the seating plan, associate it with the venue and location, upload the base map of the venue, and complete the construction of the basic framework of the seating plan.
2	Method 1: Edit the Seats in the Online Editing Area	Online editing area, seat information, and setting up seating arrangement information.
3	Method 2: Excel Seat Map Upload Validation	Upload a seating chart in Excel format, and the system will automatically verify the integrity of the data. If the verification fails, it supports re-uploading for correction. Once the verification is

		successful, the basic seating chart is successfully set up.
4	Special Seat Definition	Define special seats such as wheelchair seats, companion seats, aisle seats, and visually obstructed seats, and set up associated relationships and rules.
5	Multiple Seats Bundled	Multiple seats can be assigned to the same group, and multiple groups can be set for the same seating plan (e.g., multi-seat tables, couple seats).
6	Hybrid Seating	In the seating plan, both seated and standing areas can be configured simultaneously, with a maximum number of people set for the standing area.
7	Custom Configuration	Supports customizing seat labels, attributes, and aliases, setting automatic seat selection priority, ticket printing order, and hiding specific seats, etc.
8	Multi-Format Data Download	You can download the seat list, statistical data, Excel seat table, PNG seat map, etc., for easy offline verification and archiving.

Table 3.7.4.2.1: Seating Plan Management Process Description**3.7.4.2.2. Requirements List**

RID	Description	Priority	Related Process
REQ-TKT-STM-001	Supports operators in defining the name of the seating plan, associating it with the venue, uploading the venue base map, and completing the construction of the basic framework of the seating plan.	High	3.7.4.2.1:#1
REQ-TKT-STM-002	Supports operators to edit area and seat information online and set seating arrangement information.	Low	3.7.4.2.1:#2
REQ-TKT-STM-003	Supports operators to upload seat maps in Excel format. The system automatically verifies data integrity, supports re-uploading and correction if verification fails, and completes the basic seat map setup after successful verification.	High	3.7.4.2.1:#3

REQ-TKT-STM-004	Supports operators in defining wheelchair seats, companion seats, aisle seats, and special seats with visual obstruction, and allows setting of associated relationships and rules.	Medium	3.7.4.2.1:#4
REQ-TKT-STM-005	Supports operators to assign multiple seats to the same group, and multiple groups can be set for the same seating plan.	Medium	3.7.4.2.1:#5
REQ-TKT-STM-006	Supports operators to configure both seated and standing areas simultaneously in the same seating plan, and the standing area can set a maximum number of people.	Medium	3.7.4.2.1:#6
REQ-TKT-STM-007	Supports operators to customize seat labels, attributes, and aliases, set the priority of automatic seat selection, ticketing order, and hide specific seats, etc.	High	3.7.4.2.1:#7
REQ-TKT-STM-008	Supports operators to download seat lists, statistical data, Excel seat tables, PNG seat maps, etc.	Medium	3.7.4.2.1:#8

Table 3.7.4.2.2: Seating Plan Management Requirements List

3.7.4.3. Seat Selection Ticketing Management

This scenario defines the full-process business logic for operators to set up basic information, configure ticket price and package ticket rules, bind show sessions with seats, manage special seats and reserved seats, divide ticket pools, and configure multi-channel sales cycles for seat selection-based ticketing events. The core objective is to address the pain points of multiple configuration dimensions for seat selection tickets and inconsistent multi-channel inventory and sales rules, and to adapt to the refined operation requirements of theatre shows and seat selection-based events.

3.7.4.3.1. User Process

The following table describes the user process of seat selection ticketing management.

No.	Component	Description
1	Basic Information Management of Seat Selection Event	In the seat selection ticketing system, you can create and modify the basic information of seat selection events, including name, affiliated venue, validity period, event classification, etc.
2	Basic Fare Management	Define basic information such as the colour, description, and selling price of the base fare.

3	Special Ticket Type Management	Based on the base fare, special ticket types can be defined, and special ticket inventory limits can be set to implement sales scenarios for special fares such as concession tickets, senior tickets, and child tickets.
4	Special Ticket Types Share Inventory	Several special ticket types can be specified to share a single inventory in the designated sessions.
5	Same-Event Ticket Package Configuration	Based on the base fare, you can define the preferential rules for single-ticket type package combinations and set the total inventory limit for the package tickets.
6	Basic Configuration for Session Management	Define basic information such as session time, name, start time of ticket sales, real-name rules, admission and ticket collection methods, etc.
7	Sales Channel Permission Management	Comprehensively covers channels such as self-service terminals, ticketing websites, various mini-programs, OTAs, and offline ticketing centres. Sales channels can be specified at the event, same-event package ticket, special ticket type, and event session dimensions.
8	Differential Configuration of Sales Periods	Different sales cycles can be defined for different channels during the same time period. Different sales cycles can be defined for the same fare through different channels.
9	Seat Base Fare Assignment	Assign corresponding base fares to seats, allowing different fares to be assigned to different areas and seats, thus completing the binding and association between seats and fares.
10	Configuration of Special Attribute Seat Sales Rules	Special seats (such as wheelchair seats, couple seats) have dedicated sales rules.
11	Reserved Seat Management	For special business scenario requirements, the following operations can be performed: setting reserved seats (reserving specific seats for specific purposes, such as camera positions), and seat locking operations (temporarily locking seats for VIP customers). Only the offline ticketing centre can unlock and issue tickets, and public channels are not available for sale.

12	Exclusive Ticket Pool Management for Seat Selection Tickets	Supports dividing ticket pools for specified seats, and only the sales channels designated by the ticket pool are allowed to sell the seats in the ticket pool.
13	Hybrid Seating Mode	For ticket templates that match seats well, corresponding base fares can be assigned to seats in the seated area, and available fares and corresponding inventory can be set for the standing area.
14	Purchase Limit Rule Management	The time period dimension supports the maximum number of tickets purchasable per order and the maximum cumulative number of tickets purchasable per user. The event dimension supports setting purchase limit rules by event, time period, and ticket type.
15	Game Rescheduled or Postponed	Supports modifying the validity period of sessions. After the modification is successful, the system will automatically refresh historical order data and synchronously update the validity period of sold tickets.

Table 3.7.4.3.1: Explanation of Seat Selection Ticketing Management Process**3.7.4.3.2. Requirements List**

RID	Description	Priority	Related Process
REQ-TKT-SEAT-001	Supports operators in creating and modifying basic information of seat selection events in the seat selection ticketing system, including core content such as name, affiliated venue, validity period, event classification, etc.	High	3.7.4.3.1:#1
REQ-TKT-SEAT-002	Supports operators in defining basic information such as the colour, description, and selling price of base fares, laying the foundation for subsequent seat pricing and ticket type configuration.	High	3.7.4.3.1:#2
REQ-TKT-SEAT-003	Supports operators to define special ticket types (such as concession tickets, senior tickets, and child tickets) based on the base fare and set inventory limits for special ticket types to adapt to differentiated sales scenarios.	High	3.7.4.3.1:#3

REQ-TKT-SEAT-004	Supports operators to designate several special ticket types, enabling them to share a single inventory in designated sessions, thus avoiding operational inefficiencies caused by scattered inventory.	High	3.7.4.3.1:#4
REQ-TKT-SEAT-005	Supports operators to define combined discount rules for single-ticket type package tickets based on the base fare and set the total inventory limit for package tickets.	High	3.7.4.3.1:#5
REQ-TKT-SEAT-006	Supports operators in defining basic information such as session time, name, start time of sales, real-name rules, admission and ticket collection methods, etc.	High	3.7.4.3.1:#6
REQ-TKT-SEAT-007	It supports full-channel coverage including self-service terminals, ticketing websites, various mini-programs, OTAs, offline ticketing centres, etc., and can specify available sales channels separately at the event, same-event package ticket, special ticket type, and session dimensions.	High	3.7.4.3.1:#7
REQ-TKT-SEAT-008	Supports defining differentiated sales cycles for different channels within the same time period and different channels with the same fare, respectively.	High	3.7.4.3.1:#8
REQ-TKT-SEAT-009	Supports operators to assign corresponding base fares to seats, with different fares configurable for different areas and seats.	High	3.7.4.3.1:#9
REQ-TKT-SEAT-010	Supports operators in setting exclusive sales rules for seats with special attributes (such as wheelchair seats, couple seats) and specifying the sales channels to which the rules apply.	High	3.7.4.3.1:#10
REQ-TKT-SEAT-011	Supports operators to set reserved seats and perform seat locking operations. After locking, only the offline ticketing center can unlock and issue tickets, and public domain channels are not available for sale.	High	3.7.4.3.1:#11
REQ-TKT-SEAT-012	Supports operators to divide ticket pools for designated seats, and only the sales channels associated with the ticket pool can sell the corresponding seats.	High	3.7.4.3.1:#12

REQ-TKT-SEAT-013	Supports operators to assign corresponding base fares to seats in the seated area and set available fares and corresponding inventory for the standing area for ticket templates in the mixed seating mode, adapting to the "seated + standing" mixed operation scenario.	Medium	3.7.4.3.1:#13
REQ-TKT-SEAT-014	Supports setting the upper limit for the number of tickets purchased per order and the cumulative number of tickets purchased per user at the time period dimension; supports setting ticket purchase limit rules by event, time period, and ticket type at the event dimension to prevent over-purchase.	High	3.7.4.3.1:#14
REQ-TKT-SEAT-015	Supports operators to modify the validity period of sessions. After the modification is successful, the system automatically refreshes historical order data and synchronously updates the validity period of sold tickets.	High	3.7.4.3.1:#15

Table 3.7.4.3.2: *Seat Selection Ticketing Management Requirements List*

3.7.4.4. Seat Selection Multi-Session Package Ticket Management

This scenario defines the full-process business logic for operators to create activities, screen participating event sessions, configure tiered discount rules, and query data for seat selection-based multi-session package tickets, adapting to the operational needs of multi-session linked activities such as series shows and art festivals.

3.7.4.4.1. User Process

The following table describes the user process for managing the seat selection multi-session package ticket event.

No.	Component	Description
1	Basic Attribute Configuration for Seat Selection Multi-Session Package Tickets	The operator creates a new seat selection cross-session package ticket event, defining basic information such as name, sales time, rules, and description. The operator sets the upper limit on the number of single-user purchases of the ticket package.
2	Set the Ticket Price for the Event Sessions Participating in the Package Ticket Event	Select the events, sessions, and ticket price ranges participating in the current package ticket event.

3	Set the Discount Rules for Package Tickets	Supports setting corresponding tiered preferential policies based on the number of events or the number of sessions.
4	Query of Package Ticket Purchase Data	Operators can view the order data of cross-session ticket package activities in the backend.

Table 3.7.4.4.1: Description of the User Process for Seat Selection and Multi-Session Package Tickets

3.7.4.4.2. Requirements List

RID	Description	Priority	Related Process
REQ-TKT-SST-001	Supports operators in creating new seat selection cross-session ticket package activities, defining basic information such as event name, sales time, rule description, event description, etc., and setting the maximum number of ticket packages that a single user can purchase.	High	3.7.4.4.1:#1
REQ-TKT-SST-002	Supports operators in selecting the target events, corresponding sessions, and ticket price ranges participating in the current package ticket event.	High	3.7.4.4.1:#2
REQ-TKT-SST-003	Supports operators to set corresponding tiered preferential policies based on the number of events or the number of sessions participated in.	High	3.7.4.4.1:#3
REQ-TKT-SST-004	Supports operators to view in real-time the participation order data of cross-session package ticket activities in the backend.	High	3.7.4.4.1:#4

Table 3.7.4.4.2: List of Management Requirements for Seat Selection Multi-Session Package Tickets

3.7.4.5. Event Management

This scenario defines the full-process business logic for operators to configure basic information, customise registration forms, set purchase limit rules, conduct registration review and notification, and manage user registration information for activities that require registration (such as workshops and trial courses), achieving integrated control of registration - review - ticketing - data management, and enhancing operational efficiency and user experience.

3.7.4.5.1. User Process

The following table describes the user process for event management.

No.	Component	Description
1	Event Basic Attribute Configuration	Operators create a new event and define basic information such as event name, venue, paid/free, description, number of sessions, number of registrants, etc. Operator configuration for registration moderation mode: direct purchase/reservation without moderation, or payment and successful registration can only be completed after moderation approval.
2	Custom Configuration for Registration Form	Operators customize the fields of the registration form, set the required/optional attributes and data validation rules of the fields, and complete the configuration of the registration form through the visual configuration function.
3	Quota Limit Rule Configuration	Operators configure the purchase limit rules by setting the upper limit on the number of registrations per user and the total number of tickets per order, both on an event and session basis.
4	Registration Review	For activities that require moderation, the operations staff will complete the registration moderation in the backend. After the moderation is approved or rejected, the system will automatically change the status of the event order, and the rejected order will be marked with the reason for rejection.
5	User Ticket Purchase/Reservation Completed	Users approved in moderation mode enter the ticketing/reservation process and complete payment/reservation confirmation within the specified time; users in the no-moderation mode directly enter the ticketing/reservation process after submitting the form, and the system simultaneously verifies the purchase limit rules.
6	Registration Information Management and Export	Operators can view the information and ticket purchase status of all registered users for a single session in the backend, and support batch export of registration information for on-site check-in and user operation.

Table 3.7.4.5.1: Event Management User Process

3.7.4.5.2. Requirements List

RID	Description	Priority	Related Process
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REQ-TKT-WKS-001	Supports operators in creating new activities, defining basic information such as event name, venue, paid/free attribute, event description, number of sessions, maximum number of participants, etc.; and allows configuration of the moderation mode for registration.	High	3.7.4.5.1:#1
REQ-TKT-WKS-002	Supports operators to customize registration form fields, set the required/optional attributes of fields, and configure data validation rules through the visual configuration feature.	High	3.7.4.5.1:#2
REQ-TKT-WKS-003	Supports operators to set the upper limit of the number of registrations per user and the limit on the total number of tickets purchased per order according to the dimensions of events and sessions.	High	3.7.4.5.1:#3
REQ-TKT-WKS-004	For activities that require moderation, it supports operators to complete the registration moderation process in the backend. After the moderation is approved or rejected, the system automatically updates the status of the event order, and the rejected order is marked with the reason for rejection.	High	3.7.4.5.1:#4
REQ-TKT-WKS-005	Users approved by moderation need to complete payment/booking confirmation within the specified time; users in the moderation-free mode directly enter the payment/booking process after submitting the form, and the system simultaneously verifies the purchase limit rules.	High	3.7.4.5.1:#5
REQ-TKT-WKS-006	Supports operators to view registration user information and registration status in the backend and supports batch export of registration information.	High	3.7.4.5.1:#6

Table 3.7.4.5.2: *Event Management Requirements List***3.7.4.6. Ticket Management**

This scenario defines the full-process business logic for operators to manage basic information and perform visual configuration on the ticket surfaces of paper tickets and electronic tickets.

3.7.4.6.1. User process

The following table describes the user process for ticket management.

No.	Component	Description
1	Paper Ticket Surface Management	Operators can add and manage paper ticket vouchers, including basic information such as names. The printing content of the ticket, sponsor logo, ticket layout, etc. can be set through the visual configuration interface.
2	Electronic Ticket Face Management	Operators can add and manage e-ticket vouchers, including basic information such as names. The display content, layout, etc. of the ticket can be set through the visual configuration interface

Table 3.7.4.6.1: Description of Ticket Management Process

3.7.4.6.2. Requirements List

RID	Description	Priority	Related Process
REQ-TKT-TKL-001	Supports operators to add and manage the basic information of paper tickets; through the visual configuration interface, users can customize the printing content, sponsor logo, and layout of the ticket surface to meet the ticket design requirements of different events.	High	3.7.4.6.1:#1
REQ-TKT-TKL-002	Supports operators to add and manage the basic information of e-ticket vouchers; the display content and layout of vouchers can be customized through a visual configuration interface.	Low	3.7.4.6.1:#2

Table 3.7.4.6.2: List of Ticket Management Requirements

3.7.4.7. Ticket Scenario Message Push Management

This scenario defines the full-process business logic for operators to perform scenario definition, template configuration, and push channel selection for message push in the ticketing scenario, adapting to the message requirements of multiple scenarios such as ticket purchase confirmation, show start reminder, and operational early warning.

3.7.4.7.1. User Process

The following table describes the user process for message push management in the ticketing scenario.

No.	Component	Description
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1	User-Oriented Message Push	Can define the scenarios and templates for message push, including successful ticket purchase and show start reminder. Supports push methods for SMS and email channels.
2	Message Push for Operators	Can define the scenarios and templates for message push, such as wheelchair seat sales and reminder for reserved order expiration Supports push methods for SMS and email channels.

Table 3.7.4.7.1: Description of the Message Push Management Process for Ticketing Scenarios

3.7.4.7.2. Requirements List

RID	Description	Priority	Related Process
REQ-TKT-MSM-001	Supports operators to define user-facing message push scenarios and define message template content; supports two push channels: SMS and email.	High	3.7.4.7.1:#1
REQ-TKT-MSM-002	Supports operators to define message push scenarios for operators and define the content of message templates; supports two push channels: SMS and email.	High	3.7.4.7.1:#2

Table 3.7.4.7.2: List of Requirements for Ticket Scenario Message Push Management

3.8. Reporting

This chapter outlines the end-to-end business process of WestK Reporting System, covering business pain points and objectives, high-level and detailed business requirements, implementation methods, user journeys, and functional specifications of front-end and back-end workflows.

3.8.1. Business Requirements

From a business perspective, WestK requires this system to support basic operational reports and customized reports for Ticketing, BU Operation, Membership, and Finance, while meeting the requirements for multi-BU permission management, Data Transmission Service, system interaction, etc.

WestK requires the ability to view AnTank system transaction data and be able to support downloads. Report requirements:

- **Seat Selection Statistics:** The system can clearly count the detailed sales of seat selection for each venue, intuitively present the sales status of each area

and seat, and facilitate the understanding of seat sales distribution and demand.

- **Ticket Statistics:** The system can provide comprehensive ticket statistics reports, covering sales details with and without refunds, daily channel sales, ticket transaction summaries, and comparison data between sales and admission verification, facilitating a comprehensive understanding of the overall ticketing situation.
- **Membership Report:** Focuses on membership lifecycle management, covering growth, retention, consumption, and benefit analysis. Statistics are conducted by dimensions such as membership card type and channel, including indicators such as membership retention and growth, consumption contribution, and promotion usage.
- **Merchandise:** The system can provide detailed reports on the sales of merchandise, which are used to accurately track product sales data and intuitively analyse the actual sales situation of various types of products.
- **Marketing Promotions:** The system can provide a detailed coupon report to track the issuance and verification of coupons, to evaluate the marketing effectiveness of various ticketing promotion activities.
- **BU Operation Report:** Focusing on multi-dimensional aspects of ticketing business, conducting refined statistical analysis, and performing statistics by dimensions such as channels and events, including indicators such as total revenue, number of tickets sold, inventory sell-out rate, discount rate, and number of tickets use.
- **Financial Statements:** Focus on financial data statistics, covering business scenarios such as fund settlement, order reconciliation, revenue allocation, etc., to meet the needs of financial analysis.

3.8.2. Business Requirements Analysis

This section further analyses and elaborates on the business requirements described in Section 3.8.1, clarifying the business intent and functional expectations.

Map to Business Requirement 3.8.1.

- **Seat Selection Statistics**

The system can accurately count and clearly present the detailed seat selection and sales information for each event, display the sold status and sales data of each area and seat, intuitively reflect the distribution of seat sales and user ticketing demand, facilitate operators to quickly grasp sales dynamics, reasonably analyse seat popularity, and provide reliable data support for pricing adjustment, session planning, and on-site management.

- **Ticket Statistics**

The system can automatically generate comprehensive and complete ticket statistical reports, support multi-dimensional data summarization and analysis, including sales details in both refund-included and refund-excluded calibres, and can also count daily sales show by each channel and provide an overview of ticket payment transactions. Meanwhile, it offers comparative analysis between sales data and on-site admission verification data, facilitating operators to comprehensively grasp the overall picture of ticket operations and promptly monitor the progress of sales and verification.

- **Membership Report**

The system can generate detailed and comprehensive sub-card sales statements, accurately recording key data such as customer information, purchase quantity, card opening time, usage status, and consumption details for each sub-card sale. The data is updated in a timely manner and presented clearly, providing reliable data support for operators to carry out refined membership management, customer loyalty maintenance, and subsequent marketing activities.

- **Merchandise**

The system can generate comprehensive online mall product sales detail reports, accurately recording core data such as product information, sales quantity, transaction amount, order time, and purchasing users for each order, tracking product sales dynamics, clearly presenting the sales volume and revenue of various products, facilitating intuitive analysis of sales show, and providing data basis for product operation and inventory management.

- **Marketing Promotion**

The system can generate detailed coupon statement reports, comprehensively recording the number of coupons issued, users who received them, usage rules, verification time, and corresponding order information. Through multi-dimensional statistics, it clearly reflects the verification rate and conversion status, providing reliable data support for accurately evaluating the marketing effectiveness of various ticketing promotion activities.

- **BU Operation Analysis**

The system supports viewing the overall sales volume, inventory status, discount usage, and admission verification status of each business unit (BU). Through multi-dimensional data aggregation, it intuitively presents the operational situation, providing decision-making basis for ticket inventory allocation and ticket price strategy optimization. The report supports flexible setting of custom query conditions and allows one-click data export to meet diverse statistical and archiving needs.

- **Special Financial Report**

Supports daily generation of the previous day's fund settlement report, facilitating timely reconciliation of accounts by financial staff; monthly, it can automatically aggregate and generate detailed reports on member revenue, ticket advance receipts, coupon and gift card liability details, profit and loss statements for each department, and detailed revenue reports for product lines, achieving systematic and standardized management of financial data and providing comprehensive and accurate data support for business analysis and cost accounting.

3.8.3. Operating Model

This system builds long-term stable fixed reports based on business requirements, mainly supporting the viewing and analysis of daily operational data at various dimensions and levels:

- **AnTank Team:** Responsible for data interface development, data warehouse construction, Extra-Transform-Load (ETL) process maintenance, and system underlying permission synchronization, achieving report delivery, ensuring the accuracy and timeliness of data sources.

- WestK Business Users: As data users, use the reporting system provided by the system for report viewing and data insights, without relying on the technical team for temporary data retrieval.

3.8.4. Customer Journey

This section describes the typical journeys of users in different roles when using WestK Reporting System. User Stories are used to clarify user requirements, goals, and expected outcomes.

No.	Category	Role	User Story
1	Data Access	As a BU operator	I hope to log in to the system through a unified account to quickly access the exclusive Data Dashboard of my business unit.
2	Ticketing Analysis	As a ticketing manager	Wishes to view sales reports broken down by channel and ticket type to analyse the sales show of each channel and optimize strategies.
3	Ticketing Analysis	As a ticketing manager	Hope to see the situation of popular items on the report and adjust inventory or marketing strategies accordingly.
4	Member Analysis	As a member operation	Wishes to view the analysis report on member growth and retention to evaluate the effectiveness of member activities.
5	Member Analysis	As a member operation	When drilling down to view member consumption contributions, we hope to be able to filter by card type to identify high-value member groups.
6	Financial Analysis	As a financial analyst	We hope to reduce the cost of data extraction across systems for the finance team and avoid inconsistencies in the data calibre among different Lines of Business.
7	Data Export	As a financial analyst	Hope to export the detailed report in Excel format for offline secondary accounting and archiving.
8	Permission Management	As a system administrator	It is hoped that the system can automatically synchronize AnTank's permission changes to reduce the workload of manual maintenance and ensure data security.
9	Decision Support	As the Group's management	Hope to see the summary of core operating indicators across BUs on the home page

			dashboard to keep track of the overall operational status.
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Table 3.8.4: Description of Customer Journey

3.8.5. Business Scenarios

This section defines the reports that users can access.

- User Process: After logging in to the AnTank backend management page, users can click on the main menu "Data Reporting System" to access the menus of the following report categories and export data.

3.8.5.1. User Process

The following table describes the report categories that users can access.

No.	Report Classification	Description
1	Seat Selection Report	Centring on the full-process data statistics of seat selection ticketing, it covers overall operations, daily sales reports, channel sales, ticket issuance statistics, seat sales details, and segmented scenarios such as refund/reprint/reservation/seat locking, and includes special sales statistics for Douyin, major clients, user groups, etc.
2	Ticket Report	Full-dimensional data statistics for admission tickets, including core reports such as overall ticket operations, daily sales reports, channel sales, ticketing statistics, sales details (including/excluding ticket refunds), etc., while also covering both Chinese and English bilingual versions of the reports.
3	Membership Report	Integrate the full lifecycle data reports of three types of membership cards, namely sub-cards, privilege cards, and stored-value cards, covering core dimensions such as sales, gifting, usage details/summaries, stored-value transactions, and member stored-value inquiries.
4	Merchandise Order Report	Covers the full-process statistics of merchandise orders including sales statistics, added-to-cart products, distribution statistics, order details, special refund details, etc.
5	Coupon Usage Statistics Report	Focuses on statistical analysis of marketing promotion data, with the core being the coupon verification details table, which records key data such as coupon usage and verification, and serves as the core report for analysing the effectiveness of marketing activities.

6	BU Operation Report	Focuses on event ticket sales, inventory, and admission verification statistics, with indicators including event sales revenue, sell-out rate, and number of admissions.
7	Special Financial Statements	Focus on fund settlement and order payment status, member income details, ticket advance receipts, coupon and gift card liabilities, profit and loss statistics for each department, and income details reports for product lines.

Table 3.8.5.1: Report Classification Requirements

3.8.5.2. Requirements List

After users log in to the AnTank backstage management page, they can click on the main menu of "Data Report System" to access the menus of the following report categories and support data export.

RID	Name/Description	Priority	Related Process
REQ-RPT-SET-001	Event Channel Seat Ticket Sales Report Provide detailed sales data for seat tickets across different sales channels.	High	3.8.5.1: #1
REQ-RPT-SET-002	Seat Ticket Sales Details Report Details the seat sales data, which may be counted by individual ticket type or session.	High	3.8.5.1: #1
REQ-RPT-SET-003	Event Sales Details Report records the ticket sales of reserved seats for each event or event.	High	3.8.5.1: #1
REQ-RPT-SET-004	Channel Sales Report Aggregate seat ticket sales data from different channels.	High	3.8.5.1: #1
REQ-RPT-SET-005	Ticket Transaction Summary Aggregate all payment transaction data related to seat ticket sales.	High	3.8.5.1: #1
REQ-RPT-SET-006	Real-time Report for Seat Selection Ticket Sales Provide daily order quantity, conversion rate, and inventory level for channels and events.	High	3.8.5.1: #1
REQ-RPT-SET-007	Seat Selection Ticket Channel Sales Analysis Report	High	3.8.5.1: #1

	Provide daily revenue and its contribution by dimensions such as BU, channel, event, etc.		
REQ-RPT-SET-008	Seat Selection Ticket Usage Status Table Provides daily attendance rates for admission in dimensions such as events and ticket types.	Medium	3.8.5.1: #1
REQ-RPT-TKT-001	Detailed Admission Ticket Sales (excluding Refunds) Provide detailed sales data for tickets without refund.	High	3.8.5.1: #2
REQ-RPT-TKT-002	Detailed Admission Ticket Sales (including Refunds) Provide detailed ticket sales data including refunded tickets.	High	3.8.5.1: #2
REQ-RPT-TKT-003	Ticket Channel Sales Report Provide detailed sales data for tickets from different sales channels.	High	3.8.5.1: #2
REQ-RPT-TKT-004	Summary of Ticket Transaction Aggregate ticket sales data from different channels for summary.	High	3.8.5.1: #2
REQ-RPT-TKT-005	Summary of Ticket Sales and Verification Aggregate ticket sales and verification data summaries from different channels.	High	3.8.5.1: #2
REQ-RPT-TKT-006	Ticket Usage Status Report Provides daily attendance rates for admission in dimensions such as events and ticket types.	Medium	3.8.5.1: #2
REQ-RPT-MEM-001	Secondary card Sales Details Report Provide a detailed sales statement for secondary cards by querying order type, card type, and name.	High	3.8.5.1: #3
REQ-RPT-MEM-002	Member Growth and Retention Report Provides statistics on the total number of members, the number of new members, the number of renewed members, etc., in dimensions such as monthly BU and sales channels.	High	3.8.5.1: #3
REQ-RPT-MEM-003	Member Value Analysis Report	Medium	3.8.5.1: #3

	Provides statistics on the total number of members, the number of new members, the number of renewed members, etc., in dimensions such as monthly BU and sales channels.		
REQ-RPT-MTD-001	Online Mall Product Sales Details Provide detailed sales information of online mall products by different channels and order status.	High	3.8.5.1: #4
REQ-RPT-CPN-001	Coupon Details Report Provide detailed sales information of online mall products for different coupon types, activities, and statuses.	High	3.8.5.1: #5
REQ-RPT-BU-001	BU Sales Daily Report Provides daily sales, monthly sales, per customer transaction, and other indicators for dimensions such as daily BU, events, sales channels, etc.	High	3.8.5.1: #6
REQ-RPT-BU-002	Inventory and Pricing Optimization Report Provides sales volume, inventory, start of sessions and other dimensions of sales, inventory, sell-out rate and other indicators.	High	3.8.5.1: #6
REQ-RPT-BU-003	Ticket Redemption Statistics Provide statistics on the daily number of ticket verifications.	Low	3.8.5.1: #6
REQ-RPT-FIN-001	Daily Report of Fund Settlement Provides statistics such as total transaction amount and number of transactions on a daily basis, including dimensions like BU and payment channels.	High	3.8.5.1: #7
REQ-RPT-FIN-002	Order and Payment Reconciliation Statement Provides statistics on order amounts and other dimensions such as daily order numbers, transaction times, product lines, and payment statuses.	High	3.8.5.1: #7
REQ-RPT-FIN-003	Member Revenue Allocation Report Provide member information such as membership type, expiration date, and sales amount for the membership number.	High	3.8.5.1: #7
REQ-RPT-FIN-004	Ticket Contract Liability Statement	High	3.8.5.1: #7

	Provide statistics such as the number of tickets sold, the number of tickets sold, and the number of tickets verified for dimensions such as exhibition/show name, BU, and ticket type.		
REQ-RPT-FIN-005	Coupons/Gift Cards Balance Report Provide statistics on the number of issued, used, and the amount used for coupon types (general coupons/BU exclusive coupons/co-branded coupons).	High	3.8.5.1: #7
REQ-RPT-FIN-006	BU Profit and Loss Summary Report Provides revenue and cost statistics for dimensions such as BU, revenue accounts, and cost accounts.	High	3.8.5.1: #7
REQ-RPT-FIN-007	Detailed Income Statement of Product Lines Provide statistics such as sales volume and sales amount by dimensions such as product line.	High	3.8.5.1: #7
REQ-RPT-FIN-008	Ticket Contract Liability Report Provide statistics on abnormal ticketing.	Medium	3.8.5.1: #7
REQ-RPT-FIN-009	Event Preferential Sales Statistics Report Provide all amounts generating discounts and summarize them by categories such as promotions, memberships, coupons, points, etc.	High	3.8.5.1: #7
REQ-RPT-FIN-010	Event Settlement Statistics Report Provide the settlement statement for a single event.	High	3.8.5.1: #7
REQ-RPT-FIN-011	Tour/School/Group Booking Business Report Provides statistics for Tour/School/Group reservation orders.	Medium	3.8.5.1: #7
REQ-RPT-FIN-012	Key Account Prepayment Balance Detail Provides statistics on the balance of large customer prepayments function.	High	3.8.5.1: #7
REQ-RPT-FIN-013	Key Account Receivables & Collection Tracking Provide statistics on the amount and details of unpaid ticketing orders for large customers.	High	3.8.5.1: #7

Table 3.8.5.2: Basic Report Detail Requirements

3.9. Admission

3.9.1. Business Requirements

From a business perspective, WestK requires the Admission System to support a wide range of onsite admission validation scenarios, while meeting operational requirements for robust admission permission management, effective special scenario handling, comprehensive audit logging, and high execution efficiency.

Based on these needs, the following key business requirements have been identified.

3.9.1.1. Admission Account and Permission Management

WestK requires the system to support the configuration and management of admission accounts and associated permissions, ensuring that users can perform admission operations and access admission data strictly within their authorised scope.

3.9.1.2. Admission Account Switching-over and Authorised Scope Reviewing

WestK requires the system to enable onsite staff to view the venues and events available for admission under the currently logged in account directly from the admission terminals.

The system must also support rapid switching-over between different admission accounts and, upon account switch, allow users to clearly review the authorised admission scope of the active account. This capability is essential to support high frequency account switching scenarios and to enhance onsite operational efficiency and accuracy.

3.9.1.3. Admission Validation for Multiple Scenario

WestK requires the system to support onsite admission validation by frontline staff based on various types of credentials presented by visitors. These include QR codes for admission tickets and marked-seat tickets, membership cards, and coupons.

The system should return clear and unambiguous admission results in accordance with predefined business rules, enabling orderly admission release while effectively intercepting exceptions during onsite operations.

3.9.1.4. Special Scenarios Requiring Manual Secondary Confirmation

WestK requires the system to support a manual secondary confirmation mechanism for designated special admission scenarios, enabling controlled and compliant handling of admission for the special scenarios, including, but are not limited to:

- Invalid tickets
- Admission using concession tickets
- Membership based or VIP admission
- Coupon redemption

This mechanism is intended to allow frontline staff to address exceptional cases with appropriate flexibility while ensuring operational control, consistent decision making, and full auditability of admission actions.

3.9.1.5. Group Admission Handling

WestK requires the system to support group admission scenarios in which multiple attendees share a single admission credential and may enter the venue in separate batches, in order to meet operational requirements for flexible and controlled onsite admission management.

3.9.1.6. Admission Terminal Environment Support

WestK requires the system to support multiple types of admission terminal environments to accommodate different onsite operating models, while ensuring consistent admission validation logic, reliable admission outcomes, and clear, standardised operational feedback across all supported terminals.

3.9.1.7. Admission Records, Statistics and Reporting

WestK requires the system to maintain complete and comprehensive admission logs throughout the entire admission validation process. The system must also provide robust admission statistics and reporting capabilities, including enabling frontline staff to view admission reports and same day admission summaries directly on admission terminals.

These capabilities are essential to support post event review, day to day operational management, show monitoring, and audit or compliance enquiries.

3.9.2. Business Requirement Analysis

This section provides further analysis and clarification of the business requirements outlined in Section 3.9.1. Each subsection elaborates on the underlying business intent, key operational considerations, and corresponding functional expectations of the Admission Control System in support of onsite admission operations.

3.9.2.1. Admission Account and Permission Management

Map to Business Requirement 3.9.1.1.

The system shall leverage a unified backend user account framework to configure and manage admission permissions for accounts assigned with Admission System responsibilities. Through centralised backend administration, the system shall define and enforce the admission scope and admission types authorised for each account, ensuring that onsite staff may only perform admission operations within their approved authority.

Key Capabilities

- Support configuration of admission permissions at the individual account level.
- Enable permission configuration by venue, space, and admission type.

- For seated events, support granular permission configuration by seating plan and designated seating areas.
- Control the visibility scope of admission data, reports, and operational functions accessible to each account on admission terminals.

This design ensures clear accountability, controlled access, and consistent execution of onsite admission validation operations.

3.9.2.2. Admission Account Switching-over and Authorised Scope Reviewing

Map to Business Requirement 3.9.1.2.

The system shall enable onsite staff to view, via the admission terminal, the venues and events that the currently logged in account is authorised to validate. The system shall also support rapid switching between multiple admission accounts and, upon account switching, automatically display and enforce the authorised admission scope associated with the active account.

This capability is essential to support high frequency account switching scenarios and to improve onsite execution efficiency while maintaining permission accuracy and operational control.

Key Capabilities

- Display, on the admission terminal, the venues and events that the active account is authorised to admit.
- Support fast and seamless switching between different admission accounts on the admission terminal.
- Automatically refresh and apply the corresponding admission permission scope—including authorised venues and events—immediately after account switching.

3.9.2.3. Admission Validation for Multiple Scenarios

Map to Business Requirement 3.9.1.3.

The system shall support onsite admission validation by frontline staff using multiple types of admission scenarios including tickets, membership cards, and coupons. Based on predefined and standardised admission rules, the system shall return clear, accurate, and actionable admission results, enabling consistent and efficient admission validation across different business scenarios.

Key Capabilities

- Support admission validation through both QR code scanning and manual input of admission codes.
- Support admission of ticketing credentials for both seated events and general admission events.
- Support admission of membership cards and coupons.
- Automatically apply the appropriate admission rules based on credential type.
- Perform admission checks including, but not limited to:

- QR code validity
- Ticket status
- Event or session date and time
- Membership validity
- Coupon validity
- Return clear success confirmations for valid credentials, and proactively intercept invalid credentials with explicit prompts and error messaging to guide onsite handling

3.9.2.4. Special Scenarios Requiring Manual Secondary Confirmation

Map to Business Requirement 3.9.1.4.

The system shall provide clear exception prompts and a controlled manual secondary confirmation mechanism for special admission scenarios in which a final admission decision cannot be determined solely through automated rules. This capability shall enable consistent, auditable, and well governed handling of exceptional cases during onsite admission operations.

Key Capabilities

- Intercept invalid tickets and display clear prompts with relevant admission details.
- Provide prompts and require manual secondary confirmation for discounted ticket admission.
- Provide prompts and require manual secondary confirmation for membership-based or VIP admission.
- Provide prompts and require manual secondary confirmation for cross area admission in seated show events.
- Provide prompts and require manual secondary confirmation when an electronic admission code is presented after a paper ticket has already been collected.
- Provide prompts and require manual secondary confirmation for coupon redemption.

3.9.2.5. Group Admission Handling

Map to Business Requirement 3.9.1.5.

The system shall support group admission validation scenarios and provide dedicated handling mechanisms to accommodate cases where multiple attendees share a single admission credential and may enter the venue in separate batches. This capability is required to ensure accurate admission control, orderly entry, and reliable attendance tracking for group-based admissions.

Key Capabilities

- Identify group admission validation scenarios based on admission credential attributes.

- Display the total booked admission quantity and the quantity already admitted.
- Allow frontline staff to input the actual number of attendees entering during the current batch.
- Update admission and validation records in real time upon each admission release.
- Support batch-based admission processing for group entries.
- Enforce admission rules to ensure that the entered admission quantity does not exceed the remaining allowable admission count.

This functionality ensures accurate attendance recording, prevents over admission, and provides a reliable data foundation for downstream operational reporting and analysis.

3.9.2.6. Admission Terminal Environment Support

Map to Business Requirement 3.9.1.6.

The system shall support different types of admission terminal environments to accommodate diverse onsite operating models, including WeChat Mini Program and admission handheld device. While supporting different terminal platforms, the system shall ensure consistent application of admission logic, uniform admission outcomes, and clear, standardised operational feedback across all terminal environments.

3.9.2.6.1. Admission WeChat Mini Program

The system shall support deployment of admission terminals on the WeChat Mini Program platform, enabling frontline staff to access the Admission System through the WeChat client for onsite admission validation operations.

The Admission Mini Program shall support multiple language options—including Simplified Chinese, Traditional Chinese, and English—to meet operational needs in different environments. In addition, configurable admission sound prompts shall be provided to assist frontline staff in efficiently identifying admission outcomes during high throughput admission scenarios.

Key Capabilities

- Support deployment on the WeChat Mini Program platform.
- Enable access to the Admission System via the WeChat client.
- Allow staff to log in through the Mini Program to perform admission validation operations.
- Support Traditional Chinese, Simplified Chinese and English user interface languages.
- Provide configurable audio prompts for admission results to enhance onsite operational efficiency.
- Maintain consistent admission logic, rule application, and result determination with other admission terminal environments.

3.9.2.6.2. Admission Handler Device (Handheld Terminal)

The system shall support deployment on the admission handheld devices via an Android application package (APK), allowing staff to access the admission application installed on the device for ticket admission. The terminal shall support Traditional Chinese, Simplified Chinese and English, and comply with specified Android version requirements. Configurable admission sound prompts shall be supported.

Key Capabilities

- Deployment on admission handheld devices via Android APK.
- Terminal deployment to be handled by WestK.
- Staff login through the installed admission application to perform ticket admission.
- Support for Traditional Chinese, Simplified Chinese and English.
- Support for Android version from Android 8 (API 26) to Android 14 (API 34).
- Maintain consistent admission logic and result determination with other terminal environments.

3.9.2.6.3. Languages on Terminals

In accordance with WestK system usage standards, the primary system language is English for admission.

During admission on terminals, event-related information including event name and session details, and business data displayed on admission statistics pages shall be presented using the system's primary language version.

3.9.2.7. Admission Records, Statistics and Reporting

Map to Business Requirement 3.9.1.7.

The system shall maintain complete admission logs throughout the admission process and provide admission statistics and reporting capabilities to support post event admission, operational management, audit enquiries, and frontline needs for terminal based statistical viewing.

Key Capabilities

- Record and allow backend viewing of ticket admission statistics, for both seated and Admission ticket events.
- Record and allow backend viewing of membership card and coupon admission statistics.
- Record and allow backend viewing of group admission validation statistics.
- Allow frontline staff to view ticket admission statistics on admission terminals.
- Allow frontline staff to view today's admission summary on admission terminals.
- Support real time viewing of admission data in the backend.

3.9.3. Operating Model

This initiative adopts a collaborative operating model to support Admission Control operations across WestK venues and event sites. Different parties assume responsibility for system platform provision and onsite admission execution, jointly ensuring the smooth delivery of admission validation operations.

AnTank: Admission Control Technology Provider

AnTank is responsible for providing and maintaining the ticketing system and admission technology platform that supports Admission Control operations.

Key Responsibilities

- Provide Admission Control system capabilities, including:
 - Admission account and permission configuration
 - Admission terminal integration
 - Admission validation
 - Exception prompts and secondary confirmation
 - Group admission handling
 - Admission records and statistical reporting
- Provide data support related to the ticketing system, including:
 - Events and sessions
 - Ticketing credentials
 - Membership cards
 - Coupons
- Support multiple admission terminal environments, including:
 - WeChat Mini Program
 - Admission handheld terminals
 - Ensure consistent admission logic and results across all terminals
- Provide admission logs, real time statistics, and reporting to support operational review, onsite management, and data enquiries.
- Ensure system availability, show monitoring, operation logging, and technical support in accordance with agreed Service Level Agreements (SLA).

WestK: Admission Control Operations

WestK operations staff utilise the Admission Control system to perform onsite admission validation and related operations during events.

Key Responsibilities

- Perform onsite admission validation operations, including:
 - Credential scanning
 - Admission result confirmation
 - Exception prompt handling
 - Admission release or interception
- Handle special scenarios in accordance with business rules, including:

- Concession tickets
- Membership cards
- Cross-area admission
- Use of electronic codes after paper ticket collection
- Perform group admission validation, including:
 - Reviewing booked quantities
 - Entering actual admission quantities per batch
 - Managing batch-based entry release
- View admission statistics and Today's admission summary via admission terminals to support onsite operational management.
- Perform daily onsite operational activities, including:
 - Switching admission entrances
 - Switching admission accounts
 - Reviewing related admission records

3.9.4. Customer Journey

This section describes the customer journey of admission staff by using the admission system to perform onsite admission validation, special scenarios handling, and admission statistics enquiries.

No.	Category	Role	User Story
1	System Access	As an Admission Staff	I log in to the admission system on site using an authorised admission terminal in order to perform daily admission validation tasks.
2	Language Support	As an Admission Staff	When using the admission terminal, I can select Traditional Chinese, Simplified Chinese, or English as needed to operate effectively in different language environments.
3	Admission Scope Review	As an Admission Staff	After logging in, I review the venues and events that my account is authorised to validate, in order to confirm my permitted scope of operation.
4	Account Switching	As an Admission Staff	I can quickly switch between different admission accounts on the terminal to perform the appropriate admission tasks in different scenarios.
5	Admission Mode Selection	As an Admission Staff	I select either the standard admission entry or the group admission entry, based on the actual on site admission scenario.

Technology Partnership Package 2
Requirement Document

6	Admission Method Selection	As an Admission Staff	After entering the admission flow, I choose either QR code scanning or manual code input to complete admission, depending onsite conditions.
7	Credential Admission	As an Admission Staff	I scan the QR code presented by the visitor or manually input the admission code to allow the system to verify the ticket or credential.
8	Admission Result Review	As an Admission Staff	I review the admission result returned by the system to determine whether the visitor passes admission and may be admitted.
9	Invalid Ticket Interception	As an Admission Staff	When the system identifies an invalid, expired, or ineligible ticket, I receive an exception prompt to prevent unauthorised entry.
10	Multiple Entry Admission	As an Admission Staff	When a multi entry ticket is presented again, the system confirms success if remaining entries exist or displays failure if usage has been exhausted.
11	Concession Ticket Admission Handling	As an Admission Staff	When a Concession ticket requires further confirmation, I receive a prompt and perform secondary confirmation according to onsite rules.
12	Membership/ VIP Admission Handling	As an Admission Staff	When a membership or VIP credential requires further confirmation, I receive a prompt and perform secondary confirmation based on authorisation rules.
13	Cross Area Admission Handling	As an Admission Staff	When a visitor for a show event attempts entry at an incorrect area, I receive a prompt and perform secondary confirmation to handle the exception in a controlled manner.
14	Paper vs Electronic Ticket Conflict	As an Admission Staff	When a paper ticket has already been collected but an electronic code is presented, I receive a prompt and perform secondary confirmation to prevent duplicate use.
15	Coupon Admission	As an Admission Staff	When admission involves coupons, I follow system prompts to verify whether the coupon is valid or has already been redeemed.

16	Sound Prompt Configuration	As an Admission Staff	I can select different admission sound prompt schemes to distinguish between success, failure, and exception states.
17	Today's Admission Statistics	As an Admission Staff	I view today's admission statistics on the terminal to understand current admission progress under my account.
18	Admission Reports	As an Admission Staff	I view admission reports on the terminal to understand event admission status and support on site operations.
19	Admission Audit Log	As an Admission Staff	After admission, the system automatically records admission time, operator, terminal, and outcome for future reference.

Table 3.9.4: Description for the Customer Journey

3.9.5. Business Scenarios

3.9.5.1. Admission Account and Permission Management

This section defines the permission configuration and maintenance process for admission accounts. Admission accounts are managed within a unified backend user system, and fine-grained permission controls ensure that only authorised personnel can perform admission operations within specified scopes.

Overview

- **Permission Initiation:** An administrator logs into the backend system and selects the target account requiring permission configuration.
- **Access to Admission Management:** Navigate to the Admission Management System module, which serves as the exclusive configuration entry point for admission related permissions.
- **Admission Permission Configuration:** Within Admission Management, the administrator assigns the permitted admission business types (e.g. ticket, membership card, coupon) and specifies authorised venues and spaces.
- **Terminal Report Visibility Configuration:** Configure which admission reports are visible and accessible to the account on admission terminals.
- **Seated Event Area Configuration (Optional):** For seated events, configure permitted admission areas by seating plan through Seated Ticketing – Area Admission Management.
- **Permission Update and Activation:** Permissions can be amended as needed, with changes taking effect after saving.

3.9.5.1.1. User Process

The following table describes the user process of Admission Account and Permission Management.

No.	Component	Description
1	Log in to the Admission Management System	Administrators can use the back-end management system to configure admission-related permissions for admission accounts.
2	Admission Permission Configuration	The system supports the configuration of admission accounts across multiple business types, including ticket admission, membership card admission, and coupon admission, while defining the scope of authorized venues and facilities and specifying the types of admission reports accessible on the terminal.
3	Seat Selection Ticket - Area Admission Management	For seat selection-based shows/activities, the system supports assigning specific verifiable areas (such as Area A, Area B, VIP Area) to admission accounts, clearly defining the boundaries of area permissions.

Table 3.9.5.1.1: Description for the User Process of Admission Account and Permission Management

3.9.5.1.2. Requirement List of Admission Account and Permission Management

RID	Description	Priority	Related Process
REQ-AC-AM-001	The system backend must provide an admission permission management interface for configuring the business types (tickets/membership cards/coupons) that can be verified and the stadium and venue permissions for accounts.	High	3.9.5.1.1: #1, #2, #3
REQ-AC-AM-002	The system backend must provide the ability to set the types of admission reports that a specified account can view on the terminal.	Medium	3.9.5.1.1: #1, #2
REQ-AC-AM-003	The system backend must provide a seat selection ticketing - area admission management interface, and for seat selection events, configure verifiable area permissions for accounts according to seat plans.	High	3.9.5.1.1: #1, #3
REQ-AC-AM-004	The system must support setting whether to allow cross-regional entry. When cross-regional entry is allowed, cross-regional tickets enter the manual judgment process; when cross-regional entry is prohibited, cross-regional tickets are directly invalidated with failure.	Medium	3.9.5.1.1: #1, #3

Table 3.9.5.1.2: *Requirement List of Admission Account and Permission Management*

3.9.5.2. Admission Account Switching-over and Authorised Scope Reviewing

This scenario enables admission staff to quickly review authorised admission scope and efficiently switch between previously logged in accounts in multivenue or multievent environments, improving onsite efficiency and reducing operational errors.

Overview

- Terminal Login: Staff log in via WeChat Mini Program or admission handheld terminal using authorised credentials.
- Authorised Scope Display: Upon login, the terminal displays authorised venues, events, and admission business types.
- Quick Account Switching: Multiple accounts may use the same terminal; staff can quickly switch to a previously logged in account, with permissions reloaded automatically.

3.9.5.2.1. User Process

The following table describes the user process of Admission Account and Permission Management.

No.	Component	Description
1	Terminal Login	Staff log into the admission system via WeChat Mini Program or admission device with authorised credentials.
2	View Admission Scope	Staff review authorised venues and events on the terminal.
3	Quick Account Switching	Staff switch between previously logged in accounts; permissions and rules reload automatically.

Table 3.9.5.2.1: *Description for the User Process of Admission Account Switching-over and Authorised Scope Reviewing*

3.9.5.2.2. Requirement List of Admission Account Switching-over and Authorised Scope Reviewing

RID	Description	Priority	Related Process
REQ-AC-TS-001	Terminal shall display authorised venues, events, and business types after login.	High	3.9.5.2.1: #1, #2

REQ-AC-TS-002	Terminal shall support quick switching among previously logged in accounts.	High	3.9.5.2.1: #1, #3
REQ-AC-TS-003	After account switching, terminal shall reload all permissions, rules, and menus for the new account.	High	3.9.5.2.1: #1, #3

Table 3.9.5.2.2: Requirement List of Admission Account Switching-over and Authorised Scope Reviewing

3.9.5.3. Admission Validation for Multiple Scenarios

This scenario covers the onsite admission process for multiple types of admission credentials. It supports admission through QR code scanning or manual code entry, with the system automatically verifying credential validity based on predefined rules and returning clear admission results to ensure orderly admission.

Overview

Admission Initiation:

Visitors present their admission credentials (ticket QR code, membership credential, or coupon credential). Frontline staff initiate admission by selecting either scanning or manual input.

- **Credential Data Capture:**
The admission code is captured through scanning or manual entry, and the terminal uploads the credential information to the backend system.
- **Credential Admission:**
The system identifies the credential type and performs rule-based admission accordingly:
 - Ticket credentials: validate QR code validity, ticket status, and event/session time.
 - Membership credentials: validate membership status, applicable events, and eligibility.
 - Coupon credentials: validate coupon code validity and expiry date.
- **Result Feedback and Recording:**
 - If admission is successful, the system prompts release of entry.
 - If admission fails, the system displays the reason and blocks entry.
 - If secondary confirmation is required, the process proceeds to the corresponding confirmation flow.

After admission is completed, the system updates the credential status and records key admission data.

3.9.5.3.1. User Process

The following table describes the user process of Admission Account and Permission Management.

No.	Component	Description
1	Credential Presentation	Visitors present the QR codes of tickets, membership cards, or coupons; staff initiate admission.
2	Credential Capture	QR scanning or manual code input uploads credential data to the backend.
3	Credential Admission	System validates based on credential type and rules (ticket, membership, coupon).
4	Result Feedback	Terminal displays success, failure, or secondary confirmation prompt with sound/vibration cues.
5	Record Synchronization	System updates credential status and records admission data in real time.

Table3.9.5.3.1: Description for the User Process of Admission Validation for Multiple Scenarios

3.9.5.3.2. Requirement List of Admission Validation for Multiple Scenarios

RID	Description	Priority	Related Process
REQ-AC-ET-001	The admission terminal supports two admission methods: scanning (identifying QR codes) and manual entry of admission codes, adapting to different onsite scenarios.	High	3.9.5.3.1: #1, #2
REQ-AC-ET-002	The system supports the admission of three types of vouchers: ticketing vouchers (seat selection/ admission ticket selection), membership vouchers, and coupon vouchers, and completes automatic admission according to the rules of each type of voucher.	High	3.9.5.3.1: #1, #3
REQ-AC-ET-003	Ticket admission requires checking the validity of the QR code, the status of the ticket, and the session time window to ensure that only eligible tickets pass the admission.	High	3.9.5.3.1: #1, #3
REQ-AC-ET-004	Admission of membership card requires checking the member status, applicable items/benefit scope, and other necessary admission conditions, and only authorized members are allowed entry.	High	3.9.5.3.1: #1, #3

REQ-AC-ET-005	Coupon admission requires checking the validity of the coupon code and expiration date to ensure the compliant use of coupons.	High	3.9.5.3.1: #1, #3
REQ-AC-ET-006	When an invalid ticket (such as a forged ticket) is verified, the system must trigger a clear audible alarm and text prompt on the admission device to support staff interception.	High	3.9.5.3.1: #1, #4
REQ-AC-ET-007	The admission result feedback is clear, distinguishing "Pass/Fail/Require Secondary Confirmation" through text prompts, dedicated prompt tones, and vibrations, thereby enhancing the onsite result recognition.	High	3.9.5.3.1: #1, #4
REQ-AC-ET-008	The system records admission data in real time and supports queries.	High	3.9.5.3.1: #1, #5

Table 3.9.5.3.2: Requirement List of Admission Validation for Multiple Scenarios

3.9.5.4. Special Scenarios Requiring Manual Secondary Confirmation

This scenario addresses special cases that cannot be conclusively determined through automated admission rules alone. It provides exception prompts and a manual secondary confirmation mechanism to ensure controlled handling of such scenarios, balancing operational flexibility with compliance and consistency.

Overview

- **Basic Admission:**
Frontline staff scan or manually input credential information. The system performs basic checks, including QR code format admission and admission permission verification for the current account.
- **Special Rule Evaluation:**
The system evaluates whether the credential attributes match any predefined rules that require secondary confirmation (e.g. concession tickets, membership/VIP admission, cross area admission).
- **Rule Prompt Display:**
When a special rule is triggered, the admission terminal prominently displays the corresponding prompt to alert the staff.
- **Manual Confirmation and Logging:**
Based on onsite conditions, staff select "Confirm" or "Cancel":
 - Selecting "Confirm" executes the admission and records the operation for audit purposes.
 - Selecting "Cancel" does not complete admission, but the action and decision are still recorded for traceability.

3.9.5.4.1. User Process

The following table describes the user process of Admission Account and Permission Management.

No.	Component	Description
1	Basic Admission Initiation	Frontline staff scan or manually input credential information. The system performs basic admission checks, such as QR code format validity and confirmation that the account has the appropriate admission permissions.
2	Special Rule Trigger Evaluation	The system checks whether the credential attributes match any predefined rules that require secondary confirmation, including but not limited to: - Concession ticket types (e.g. elderly tickets, student tickets); - Abnormal ticket status (e.g. paper ticket already collected but an electronic code is presented); - Membership/VIP admission (identity verification required); - Cross area admission for seated events (ticket assigned area does not match the current admission entrance); - Coupon redemption (confirmation of applicable discount entitlement).
3	Rule Type Identification and Prompt Display	If a special rule is triggered, the system clearly identifies the rule type and displays a highlighted prompt on the admission terminal (e.g. "Please confirm: Elderly ticket – age verification required", "Please confirm: Paper ticket already collected – allow electronic code entry?"). For cross area admission in seated events, system behaviour depends on configuration: - If cross area admission is allowed, the system enters the secondary confirmation flow; - If cross area admission is not allowed, admission fails directly and entry is denied with a clear prompt.
4	Manual Secondary Confirmation	Based on onsite circumstances (e.g. verifying the visitor's age or membership identification), staff select "Confirm" or "Cancel" on the terminal: - Confirm: The system completes admission, updates the credential status, displays "Admission Successful", and records the operation for audit purposes; - Cancel: The system does not complete admission, displays "Admission Cancelled", and

		retains the relevant operation or exception handling record for traceability.
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Table 3.9.5.4.1: *Description for the User Process of Special Scenarios Requiring Manual Secondary Confirmation*

3.9.5.4.2. Requirement List of Special Scenarios Requiring Manual Secondary Confirmation

RID	Description	Priority	Related Process
REQ-AC-SC-001	The system shall be able to identify concession ticket types (e.g. child tickets, elderly tickets, student tickets), trigger prompts on the admission terminal, and require secondary confirmation to ensure compliant use of concession tickets.	High	3.9.5.4.1: #1, #2, #3
REQ-AC-SC-002	The system shall be able to detect scenarios where a paper ticket has already been collected but an electronic code is presented for entry, trigger terminal prompts, and require secondary confirmation to prevent duplicate admission.	High	3.9.5.4.1: #1, #2, #3
REQ-AC-SC-003	The system shall support secondary confirmation for membership/VIP admission, prompting staff to verify identity to ensure accurate fulfilment of membership benefits.	High	3.9.5.4.1: #1, #2, #3
REQ-AC-SC-004	For cross area admission in seated events, the system shall support configurable behaviour: if cross area admission is allowed, secondary confirmation is triggered; if not allowed, admission fails directly and the terminal clearly displays the reason.	Medium	3.9.5.4.1: #1, #2, #3
REQ-AC-SC-005	The system shall support triggering secondary confirmation for coupon redemption scenarios.	High	3.9.5.4.1: #1, #2, #3
REQ-AC-SC-006	The secondary confirmation interface shall restrict user actions to core options only (displaying only “Confirm” and “Cancel” buttons), with prominent, highlighted prompts to minimise the risk of incorrect operation.	High	3.9.5.4.1: #1, #4

Table 3.9.5.4.2: *Requirement List of Special Scenarios Requiring Manual Secondary Confirmation*

3.9.5.5. Group Admission Handling

This scenario addresses group admission cases where multiple attendees share the same admission credential and may enter the venue in multiple batches. A dedicated admission workflow is provided to support visualised management of booked quantity and admitted quantity, ensuring that batch-based admission is orderly and controllable.

Overview

- **Process Initiation:**
Staff initiate the group admission process via a dedicated “Group Admission” entry on the admission terminal.
- **Information Review:**
The terminal displays key information associated with the group, including event name, booking time, total booked quantity, and number of attendees already admitted.
- **Attendance Input and Admission:**
Staff enter the actual number of attendees entering in the current batch. The value must be a positive integer greater than zero and must not exceed the remaining admissible quantity (i.e. total booked quantity minus already admitted quantity). The system validates the input in real time.
- **Operation Confirmation:**
After verifying the entered number, staff choose either “Cancel” or “Confirm Entry”. Upon confirmation, the system prompts a final confirmation dialog.
- **Admission Execution and Accumulation:**
After final confirmation, the system updates the admission records and displays a success message. Batch admission is supported until the total admitted quantity reaches the booked limit.

3.9.5.5.1. User Process

The following table describes the user process of Admission Account and Permission Management.

No.	Component	Description
1	Enter the Group Admission Entry	Staff initiate the group admission process through the dedicated "Group Admission" entry on the terminal.
2	Group Information Display	The interface displays core information: event information such as the event name corresponding to the group, reservation time, total number of reservations, and number of people who have entered.
3	Actual Attendance Entry	Staff members input the number of people entering this time (should be a positive integer greater than 0 and not exceeding the difference between the total number of reserved people and the number of people who have already entered) into the terminal based on the actual number of

		people present on site, and the system verifies the rationality of the entered value in real time.
4	Operation Confirmation	After the staff verifies the number of people entered, they select an operation: - Cancel: Abandon the current admission, the terminal will prompt "Admission Cancelled", and the entry record will not be updated; - Confirm entry: Submit the number of people to be recorded, and the system will pop up a final confirmation dialog box.
5	Final Confirmation and Admission Execution	When the staff clicks "Final Confirmation", the system performs the following operations: - Update the group entry record (number of people already admitted = original number of people already admitted + number of people entered this time); - The terminal prompts "Admission successful", accompanied by a prompt tone and vibration.
6	Batch Entry Support	If the group needs to enter in batches, subsequent batches shall repeat the above process, and the system will continuously accumulate the number of people who have entered until the number of people who have entered reaches the total number of reserved people.

Table 3.9.5.5.1: Description for the User Process of Group Admission Handling

3.9.5.5.2. Requirement List of Group Admission Handling

RID	Description	Priority	Related Process
REQ-AC-GA-001	The system provides a dedicated admission entry for group admission, which is distinguished from the regular admission process.	High	3.9.5.5.1: #1
REQ-AC-GA-002	The group admission interface should clearly display information such as event name, reservation time, total number of booked people, number of people who have entered, etc.	High	3.9.5.5.1: #2
REQ-AC-GA-003	The admission terminal shall support the input of the actual number of individuals admitted for each entry instance, validate that the number does not exceed the remaining allowable capacity, and update the	High	3.9.5.5.1: #3, # 4, #5

	admission records upon successful admission verification.		
REQ-AC-GA-004	The system should support group entry in batches and accumulate the number of people who have entered until the upper limit of the total number of reserved people is reached.	High	3.9.5.5.1: #6
REQ-AC-GA-005	Group admission requires recording admission results and supporting query of admission data.	High	3.9.5.5.1: #5

Table 3.9.5.5.2: Requirement List of Group Admission Handling

3.9.5.6. Admission Terminal Environment Support

This scenario supports two core admission terminal environments—the WeChat Mini Program and the admission handheld terminal—to ensure consistent admission logic and results across different onsite operating models, while also meeting usability requirements such as language switching and configurable prompt sounds.

Overview

- **WeChat Mini Program Terminal:**
Staff access the Admission WeChat Mini Program through the WeChat client. After logging in, they can switch languages, select admission prompt sounds, perform admission following standard workflows, and synchronise data with the backend system.
- **Admission Handheld Terminal:**
WestK is responsible for installing the admission application on admission handheld devices. After logging in, staff can switch languages, select prompt sounds, and perform admission operations. The admission logic remains consistent with that of the WeChat Mini Program.
- **Terminal Data Synchronisation:**
Both terminal types share the same backend database. Admission records, credential status, and related data are synchronised in real time to ensure consistency and avoid data conflicts across terminals.

3.9.5.6.1. User Process

The following table describes the user process of Admission Terminal Environment Support.

No.	Component	Description
1	Admission WeChat Mini Program	Staff open the WeChat client and access the admission Mini Program, then log in using authorised account credentials (with support for saving previously logged in accounts). The Mini Program supports Traditional Chinese, Simplified Chinese, and English, allowing users to switch languages. Staff can also select different

		admission prompt sounds, perform admission operations according to defined workflows, and synchronise data with the backend system in real time.
2	Admission Handheld Device Environment (Handheld Terminal)	WestK installs the admission application (Android APK) on admission handheld devices (compatible with Android 8 (API 26) or higher, with Android 11 (API 30) as the target version). Staff open the application and log in to the system. The terminal supports Traditional Chinese, Simplified Chinese, and English, allows language switching and prompt sound selection, and enables admission operations using logic consistent with the WeChat Mini Program. Data is synchronised in real time.
3	Common Terminal Rules	Both terminal types share the same backend database. Admission records, credential status, and statistical data are synchronised in real time to ensure consistency and prevent data conflicts across terminals.

Table 3.9.5.6.1: Description for the User Process of Admission Terminal Environment Support

3.9.5.6.2. Requirement List of Admission Terminal Environment Support

RID	Description	Priority	Related Process
REQ-AC-TE-001	Supports WeChat Mini Program admission terminals, deployed on the WeChat platform, accessed via the WeChat Client, and supports admission by scanning codes and manual input.	High	3.9.5.6.1: #1
REQ-AC-TE-002	Supports admission 50 handheld terminals, deployed via Android APK, with system version Android 11.	High	3.9.5.6.1: #2
REQ-AC-TE-003	Supports admission T1Pro handheld terminals, deployed via Android APK, with system version Android 14.	High	3.9.5.6.1: #2
REQ-AC-TE-004	The WeChat Mini Program Client supports three languages: Traditional Chinese, Simplified Chinese, and English. Users can switch between them, and after switching, the interface elements will be updated synchronously.	High	3.9.5.6.1: #1, #2

REQ-AC-TE-005	The admission handheld terminal supports three languages: Traditional Chinese, Simplified Chinese, and English. Users can switch between them, and after switching, the interface elements will be updated synchronously.	High	3.9.5.6.1: #1, #2
REQ-AC-TE-006	Both types of terminals support selecting different admission prompt tones to distinguish between success/failure/secondary confirmation states.	Medium	3.9.5.6.1: #1, #2
REQ-AC-TE-007	The two types of terminals shall follow unified admission business rules, result judgment logic, and data synchronization mechanisms to ensure operational consistency.	High	3.9.5.6.1: #3

Table 3.9.5.6.2: Requirement List of Admission Terminal Environment Support

3.9.5.7. Admission Records, Statistics and Reporting

This scenario requires the system to fully record core admission data and provide real time statistical viewing on admission terminals as well as reporting and analysis capabilities on the backend. These capabilities support onsite operations management, post event review, and data enquiry.

Overview

- **Terminal based Statistical Viewing:**
Frontline staff access the “Statistics” entry on the admission terminal to view today’s admission statistics (total admission count and counts by credential type), or access ticket admission reports to view data at the event and session levels.
- **Backend Report Enquiry:**
Administrators and operations staff log in to the system backend and access the “Admission Statistics and Reports” module to query reports using multiple filtering dimensions.
- **Data Refresh and Consistency:**
Report data is updated in a timely manner to ensure alignment with the latest admission data, supporting various management and operational needs.

3.9.5.7.1. User Process

The following table describes the user process of Admission Records, Statistics and Reporting.

No.	Component	Description
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1	Frontline Staff View Admission Statistics on the Terminal	From the admission terminal home interface, staff access the statistics page to view “Today’s Admission Statistics”, including total admission count and admission counts by credential type. Data is updated in real time.
2	Frontline Staff View Ticket Admission Statistical Reports on the Terminal	Staff access the ticket admission statistics report page on the terminal (limited to ticket admission data). The event list by default displays events with sessions in the current month (supporting search by event name and filtering by month). After selecting an event, the system displays the total tickets sold/total tickets validated for the event, as well as tickets sold/validated per session (time filtering supported, default starting from the current day).
3	Backend Administrators /Operations Staff	Administrators or operations staff log in to the system backend and access the “Data Reports” module to view admission statistics reports for tickets, membership cards, and coupons. A function is provided to retrieve the latest report data in real time, ensuring consistency with the most recent admission records.

Table 3.9.5.7.1: Description for the User Process of Admission Terminal Environment Support

3.9.5.7.2. Requirement List of Admission Records, Statistics and Reporting

RID	Description	Priority	Related Process
REQ-AC-RS-001	The system records the admission data of various admission businesses (tickets, membership cards, coupons).	High	3.9.5.7.1: #1, #2, #3
REQ-AC-RS-002	The backend supports viewing admission statistics reports for types such as tickets, membership cards, and coupons.	High	3.9.5.7.1: #3
REQ-AC-RS-003	The admission terminal supports onsite staff to view "Today's Admission Statistics", including the total number of admissions and the number of admissions for each type (ticket/membership card/coupon), with data updated in real time.	High	3.9.5.7.1: #1
REQ-AC-RS-004	The admission terminal supports viewing terminal reports - Ticket Admission Statistical Report (only for ticket admission data): The event list defaults to displaying events with sessions in the current month	High	3.9.5.7.1: #2

	(supports searching by event name and filtering by month). After selecting a specific event, it displays the total number of tickets sold/total number of admitted tickets for the event and the number of tickets sold/admitted for each session (supports filtering by time, defaulting to starting from today).		
REQ-AC-RS-005	The admission report data should support real-time query and be consistent with the latest admission data.	High	3.9.5.7.1: #1, #2, #3

Table 3.9.5.7.2: Requirement List of Admission Records, Statistics and Reporting

3.10. Membership and Membership Card

3.10.1. Business Requirements

3.10.1.1. Membership Card Purchase and Order Management

WestK requires the system to provide a centralised membership card purchase channel, enabling customers to conveniently purchase membership cards of different Business Units (BUs) through WestK platform without separately accessing the independent sales channels of each BU.

Key requirements include:

- Customers can purchase membership cards for various venues (such as M+, HKPM, PA, etc.) in one-stop on WestK Cultural District platform, without the need to separately access the independent channels of each BU.
- The system shall support functions such as querying, refunding, and related operational management of membership card orders.

3.10.1.2. Membership Card Types

WestK requires the system to support multiple membership card levels and types in order to meet the differentiated pricing, eligibility conditions, and service arrangements for different customer groups.

Key requirements include:

- The system shall support at least two membership levels: individual membership and patron membership.
- The system shall support differentiated membership card types for different target groups, such as youth cards, senior cards, children's cards, and personal cards.
- The system shall support flexible membership card schemes for family customers and paired customers, including family cards and other multi-person membership products where applicable.
- Family membership cards shall support a base configuration such as 2 adults + 1 child, with the ability to add members subject to the membership product rules.

- Each membership card type shall have its own pricing strategy and applicable business rules.
- The system shall support validation of customer eligibility for the selected membership card type, including checks based on age or other required criteria.

3.10.1.3. Membership Card Gifting and Transfer

WestK requires the system to support membership card gifting and transfer scenarios, allowing customers to purchase membership cards as gifts and transfer them to designated recipients.

Key requirements include:

- Customers may purchase eligible membership cards as gifts for relatives or friends.
- The system shall support the transfer of an inactivated gifted membership card to a designated recipient.
- The recipient shall be required to accept and activate the gifted membership card within a defined validity period before use.
- Unactivated gifted membership cards shall automatically expire after a configured number of days from the date of purchase.

3.10.1.4. Membership Card Activation and Supplementary Card Binding

WestK requires the system to support membership card activation and related member information capture, while also supporting supplementary card binding for applicable multi-person membership products. This is intended to ensure proper member identification, enable benefit entitlement control, and support the delivery of high-quality membership services.

Key requirements include:

- When activating a membership card, customers are required to fill in necessary personal information, and the system will automatically verify the uniqueness of user information to ensure that each customer can only hold one activated membership card at the same venue and time
- When activating multi-person card members, each member needs to separately enter personal information and undergo corresponding verification, including mobile phone number/email uniqueness and age verification.
- After the family card is activated, members of the family card can bind the supplementary membership card to the accounts of family members. Supplementary card members can enjoy the same promotional discounts as the primary card of the family card and can also receive coupons specific to the supplementary card.

3.10.1.5. Membership Benefits to Drive Repeat Purchase and Customer Engagement

WestK requires the system to encourage repeat purchases by providing members with differentiated benefits and incentives throughout their customer lifecycle.

Key requirements include:

- After the membership card is activated, customers will automatically receive corresponding coupons and enjoy ticket purchase discounts.
- Member customers can enjoy exclusive discount when purchasing tickets, and the system will automatically calculate and display the discounted price.
- Member customers enjoy priority ticketing rights and can have priority ticketing opportunities at designated venues.

3.10.1.6. Convenient Membership Card Usage Experience

WestK requires the system to enable members to easily access and use their membership cards for ticket purchase, admission, and benefit redemption without unnecessary friction. Consistent interaction design and streamlined operational flows shall be adopted to ensure that membership card usage remains intuitive, efficient, and accessible across different channels.

Key requirements include:

- Customers can view the dynamic QR code of their membership card via their personal terminals and quickly verify it when entering or making a purchase.
- The multi-person card supports one person, one code, allowing each family member to use it independently without affecting each other, and also supports separate statistics on usage by member.
- The child card in the family card can be displayed in any adult's account, facilitating parents to bring their children into the venue together.

3.10.1.7. Membership Validity, Renewal, and Rights Protection

WestK requires the system to support automated tracking of membership status, configurable renewal rules, and accurate enforcement of member entitlements. Membership rights shall be consistently applied throughout the membership period to ensure fairness, transparency, and continuity of benefits for members. Before the membership card expires, the customer can choose to renew the card to continue their membership, and the renewal fee can be set separately.

Key requirements include:

- Customers who successfully renew their cards will receive additional coupons as a reward.
- Under special circumstances, to safeguard the security of customer rights and interests, membership cards may be frozen. During the freezing period, the use of relevant rights and interests will be suspended and will resume normal operation after the freezing is lifted. Family cards can be frozen as a whole, or the designated primary card and secondary card can be frozen separately according to.

3.10.1.8. Comprehensive Membership Data Management

WestK requires the system to maintain a single, authoritative source of membership data, including profile information, membership status, benefit eligibility, and transaction history. Proper data governance, integrity controls, and auditability shall

be supported to enable reliable operations, reporting, and future data-driven initiatives.

Key requirements include:

- Operators can view the basic information of members and the status of their membership cards.
- Supports editing and maintaining member profiles to ensure information accuracy.
- Can query members' historical orders (tickets, merchandise, membership cards, etc.) and operation records (registration, transfer, coupon acquisition, etc.).
- Supports multi-dimensional screening of members by card type, registration time, etc., and can export data for further analysis. The exported data will be desensitized to protect customer privacy.

3.10.1.9. Unified Member Data Integration

Member data will be integrated with the enterprise's existing IT systems (such as CRM) to ensure that customers enjoy a consistent service experience across all systems, and operators can globally view and manage customer assets in each system.

3.10.2. Business Requirement Analysis

This section further analyses and explains the business requirements described in Section 3.10.1. Each subsection elaborates on the business intent, operational considerations, and functional expectations behind the marketing promotion system solution.

3.10.2.1. Membership Card Purchase

Map to Business Requirement 3.10.1.1.

1) Membership cards of each BU can be purchased uniformly on WestK platform

Membership cards created by BU (such as M+, HKPM, PA) in their systems can be synchronized to WestK system for unified sales. WestK platform can perform secondary configuration on membership cards to better meet the operational needs of unified sales channels, such as displaying attractive membership card images, associating exclusive coupons, linking promotional activities, and specifying the scope of available venues, etc.

2) Provide customers with convenient card refund services

For membership cards purchased on WestK Platform, customers can easily find the corresponding order in the order management and apply for a refund. After the refund is successfully processed, the funds will be returned to the customer's payment account via the original payment channel, ensuring the security of the funds.

3.10.2.2. Membership cards Configuration

Map to Business Requirement 3.10.1.2.

1) Supports configuration of different card types

Provide customers with a variety of card types to choose from (Youth Card/Senior Card/Children's Card/Personal Card/Family Card), meeting the needs of different age groups and life scenarios

- Each card type has independent inventory management and validity period settings to ensure service quality.
- Each card type adopts an independent pricing strategy, allowing customers to choose the plan that best suits their budget. However, pricing must comply with the payment limit range of WeChat Mini Programs; otherwise, customers will not be able to make purchases.
- Each card type is bound to different preferential rights and interests, providing customers with differentiated value experiences.
- Configure corresponding age limits based on the characteristics of card types to ensure that customers choose the appropriate membership card type.

2) Supports flexible configuration of family cards

- Family Card adopts the basic pricing model (2 adults + 1 child), and customers can flexibly add members according to the actual needs of their family, with fees accumulated based on individual member types.
- Family cards can set an upper limit for additional members, ensuring the rational allocation of service resources.
- Each member of the Family Card needs to fill in their birthday information, and the system will automatically verify whether their age meets the requirements of this card type.

3.10.2.3. Membership Cards Support the Gifting Function

Map to Business Requirement 3.10.1.3.

1) The membership card can be configured to enable or disable the "Support Transfer Function"

- Some membership cards can choose to enable the gifting function, allowing customers to share cultural experiences as gifts with their relatives and friends.
- Membership cards without the transfer function enabled are only available for use by the purchaser themselves.

2) Convenient transfer operation process

- Customers can select the membership card to be transferred in the membership card list of the WeChat Mini Program and transfer it via WeChat friends.
- The recipient must accept within 24 hours; otherwise, the transfer will automatically expire.
- After the recipient successfully receives it, they can activate the membership card and enjoy the corresponding benefits.

3.10.2.4. Membership Activation

Map to Business Requirement 3.10.1.4.

1) Activation of membership card after purchase

- To ensure service quality, each customer may only hold one activated membership card at the same venue and time
- When a customer activates, they need to fill in their mobile phone number and email address. The system will automatically verify their uniqueness. If there is already an active membership card, it cannot be reactivated.
- Depending on the characteristics of the card type, customers need to fill in the corresponding information fields (adults need to fill in mobile phone number/email/birthday/name, while children only need to fill in name and birthday)
- If the card type has age requirements, the system will automatically verify eligibility based on the birthday information provided by the customer

2) Binding of Supplementary Cards for Family Cards

- The first adult in the Family Card serves as the primary cardholder, and the remaining members are secondary cardholders.
- Non-child type secondary card members can be invited by the primary cardholder via WeChat Mini Program, and the secondary cardholder can complete the binding after accepting the invitation
- The mini-program account of the secondary cardholder must match the mobile phone number of the secondary cardholder filled in by the primary cardholder during activation to receive messages normally.
- After the secondary card is bound to its own account, the primary cardholder will no longer display the dynamic code of the secondary card, and the secondary cardholder can enjoy the same preferential discounts as the primary cardholder.

3) Coupon Model of Family Card

- Family Card offers two coupon issuance modes to meet the usage scenarios of different families:
 - One Card, One Set Mode: One membership card is issued with one set of coupons.
 - One Set per Person Mode: According to the number of family members, each person will be issued a set of coupons, supporting the configuration of exclusive coupon combinations for child members.
- Under the "one person, one set" model, two sharing models are further supported:
 - Family Sharing Mode: All family members can view and use all coupons within the family.
 - Child-only coupon sharing mode: Adult coupons can only be viewed and used in individual accounts, while child coupons can be shared, viewed, and used across all adult accounts.

4) Regarding the Child Supplementary Card in the Family Card

- Child sub-cards cannot be bound to independent personal accounts and will be displayed in the accounts of all primary and sub-card members within the family card.
- When children enter, adults must use their secondary cards to present the QR code for verification and cancellation on their behalf.

3.10.2.5. Membership Benefits

Map to Business Requirement 3.10.1.5.

1) Coupon Benefits

- Each card type can be individually configured with a combination of coupons to be given away, which customers can receive upon successful activation.
- Coupons can be individually set with an expiration date, flexibly paired with the expiration date of the membership card.

2) Preferential Promotion Rights

- Each card type can be individually configured with one or more promotional discounts (immediate reduction or discount method).
- Customers can enjoy this promotional discount after activating their membership cards.
- Promotional discount activities can be individually set with an expiration date, which can be flexibly paired with the expiration date of the membership card.

3) Preferential Purchase Rights

- Each card type can be individually configured with priority purchase rights, including the number of available uses and the scope of applicable venues.
- Customers can enjoy the privilege of priority ticket purchase after activating their membership cards.

3.10.2.6. Usage and Verification of Membership Cards

Map to Business Requirement 3.10.1.6.

1) Admission via dynamic QR code verification

- Each card type can be configured with applicable venue scope and entry limit.
- After the customer activates the membership card, they can view the dynamic QR code (refreshed regularly for security) on their personal terminal.
- Customers can complete entry verification by presenting a QR code.

2) Family Card Entry

- Both the primary and secondary card members of the family card have independent dynamic QR codes for admission.
- Children's QR codes can be displayed and verified for admission in any adult's primary or secondary card account.

3) Entry Record Query

- The system records all entry information, including entry time, entry personnel, and verification group information.
- Operators can query statistics at any time, providing data support for subsequent operational analysis.

3.10.2.7. Membership Card Renewal and Freezing

Map to Business Requirement 3.10.1.7.

1) Membership Card Renewal

- Customers can apply for card renewal within the specified time before the membership card expires to extend their membership status.
- Renewal fees can be set separately, distinct from the initial purchase price
- Customers who successfully renew their membership cards will receive additional coupons as a reward to encourage continued membership.

2) Membership Card Freeze

- When an abnormal situation is detected, the operator can freeze the membership card to ensure the safety of customer rights and interests.
- After freezing, this card will suspend enjoying promotional benefits, priority purchase rights, and will not be able to enter normally.
- The frozen state can be lifted, and after that, the membership card will resume normal use.
- Family cards support overall freezing or individual freezing for single primary/secondary cards.

3.10.2.8. Member Basic Information Management

Map to Business Requirement 3.10.1.8.

1) User List Overview

- Operators can view the complete list of all registered customers.
- Supports multi-dimensional customer filtering by mobile phone number, email, registration time, etc.
- Supports manually adding a single customer or batch importing customer information.
- Supports exporting customer data, automatically performing desensitization during export to protect customer privacy.

2) User Information

- The system presents customer information from multiple dimensions:
 - Basic Information: Name, Mobile Phone Number, Email, Gender, Date of Birth, etc.
 - Membership Card Information: Membership card assets and status already owned by the customer.

- Order Information: All orders (tickets, merchandise, membership cards, etc.) of the customer in the system, including detailed information such as order time, amount, etc.
- Behaviour Information: Registration Time, Transfer Operation Records, etc.
- Operators can edit and maintain customer information, such as changing mobile phone numbers, email addresses, etc.

3.10.2.9. Connecting Members with Third-Party Systems

Map to Business Requirement 3.10.1.9.

- The system provides standard interfaces to support the integration of member information with the enterprise's existing IT systems (such as CRM).
- The specific docking plan will be adjusted according to the system actually used by WestK to ensure that customers enjoy a consistent service experience across all systems.

3.10.3. Customer Journey

No.	Category	Role	User Story
1	Create Membership Card	As an operator	I want to configure the membership card's benefit rules and pricing strategy to attract users to purchase and enhance the sense of membership value.
2	Multi-Person Card Type of Membership Card	As an operator	I want to create different types of membership cards (single-person cards/multi-person cards) to design differentiated membership products that meet the needs of different users.
3	Configuration of Membership Card Activation Information	As an operator	I want to configure the information that must be filled in for activation and the verification rules for this information to ensure that users can normally use their membership benefits after purchasing or receiving a card as a gift.
4	Management of the Relationship Between the Primary and Secondary Membership Cards	As an operator	I want to handle the relationship between the primary card and the secondary card to support multi-person sharing scenarios such as family cards and corporate cards.
5	Membership Card Cancellation,	As an operator	I want to manage operations such as card renewal, card cancellation, card freezing, and card extension of membership cards to

	Freezing, and Extension		flexibly respond to various business needs of users.
6	Membership Card Renewal	As an operator	I would like to configure the renewal card opening time, renewal card discount amount, and renewal card coupon giveaway to enhance member continuity and loyalty.
7	Membership Card Binding Related Benefits	As an operator	I would like to configure the benefit combination bound to the membership card so that discounts, coupons and other benefits can be automatically distributed when the user activates it.
8	Membership Card Verification and Cancellation	As an operator	I want to perform membership card verification and cancellation to verify the member's identity and record usage when the user makes a purchase.
9	Membership Card Mapping	As an operator	I would like to sell membership cards of M+ and HKPM on WestK client.
10	Registered User Data List	As an operator	I would like to view the list of registered user data to gain a comprehensive understanding of the user scale and composition of the platform.
11	Manually Add User	As an operator	I want to manually add users and complete their basic information to handle user entry for offline channels or special scenarios.
12	Manual Adjustment of User Basic Information	As an operator	I would like to manually adjust the user's basic information to update data such as the user's name, mobile phone number, email, gender, remarks, etc., in case the client's information changes.
13	Conduct Data Integration with the Enterprise's Existing IT Systems	As an operator	I want to achieve data integration with the enterprise's existing IT systems to ensure real-time synchronization of user data and guarantee the consistency of enterprise customer resources.

Table 3.10.3: User Journey Enumeration

3.10.4. Business Scenarios

This scenario defines the business logic for operators to manage membership cards and the membership system throughout their lifecycle. The core objective is to enhance user loyalty and re-purchase rate through membership system operations, achieving precise user management and differentiated services.

3.10.4.1. Membership Card Management

This scenario defines the business logic for operators to manage membership cards throughout their lifecycle. Operators create membership card templates and configure card types (single-person card/multi-person card), benefit rules, validity periods, and pricing strategies; manage the membership card activation process, supporting both purchase activation and gift activation; handle the relationship between primary and secondary cards, allowing the primary card to send secondary cards to designated users and configure secondary card benefits; manage status change operations such as card renewal, card cancellation, card freezing, and card extension; and configure membership card binding benefits, including discount benefits, coupon packages, and point rewards. The core objective is to enhance user loyalty and Re-purchase Rate through the operation of the membership card system, thereby achieving differentiated membership services.

3.10.4.1.1. User Process

No.	Component	Description
1	Create Membership Card	Operators can create membership cards in the system and configure coupons in multiple aspects such as basic information, inventory, usage restrictions, etc.
2	Multi-Person Card Type of Membership Card	Membership cards support single-person and multi-person card types. Multi-person cards include family cards and support adding additional family members on top of the basic structure.
3	Activate Membership Card	After purchasing, users need to fill in activation information. Only after the system verifies it can users successfully use the membership card.
4	Membership Card Transfer	Users can purchase membership cards and transfer them as gifts to my friends.
5	The Primary Card Sends the Secondary Card to the Secondary Card User	After successfully activating the card, the primary cardholder (purchaser) of the multi-person card can send the secondary card to the designated member.
6	Membership Card Binding Related Benefits	Operators can bind corresponding benefits, such as promotional activities, coupons, etc., to the membership card, and users can obtain such benefits after purchase.
7	Membership Card Renewal	Operators can configure membership card renewal in the backend.
8	Membership Card Cancellation	Operators can process card cancellation for membership card orders in the backend.

9	Membership Card Frozen	Operators can freeze membership cards in the backend.
10	Membership Card Extension	Operators can process the extension of membership cards in the backend.
11	Membership Card Verification and Cancellation	Operators can configure membership card verification and cancellation in the backend.
12	Membership Card Mapping	The Group can map the membership cards created by BU to the Group's system and list them for sale.

Table 3.10.4.1.1: Business Requirements Specification for Membership Card Management

3.10.4.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-MC-001	Create Membership Card	High	3.10.4.1.1: #1, #2
REQ-MC-002	Multi-Person Card Type of Membership Card	High	3.10.4.1.1: #2
REQ-MC-003	Activate Membership Card	High	3.10.4.1.1: #3
REQ-MC-004	Membership Card Transfer	High	3.10.4.1.1: #4
REQ-MC-005	The Primary Card Sends the Secondary Card to the Secondary Card User	High	3.10.4.1.1: #5
REQ-MC-006	Membership Card Binding-Related Benefits	High	3.10.4.1.1: #6
REQ-MC-007	Membership Card Renewal	High	3.10.4.1.1: #7
REQ-MC-008	Membership Card Cancellation	High	3.10.4.1.1: #8
REQ-MC-009	Membership Card Frozen	High	3.10.4.1.1: #9
REQ-MC-010	Membership Card Extension	High	3.10.4.1.1: #10

REQ-MC-011	Membership Card Verification and Cancellation	High	3.10.4.1.1: #11
REQ-MC-012	Membership Card Mapping	High	3.10.4.1.1: #12

Table 3.10.4.1.2: Membership Card Management Business Requirements List

3.10.4.2. Member Management

This scenario defines the business logic for operators to centrally manage member user data. Operators maintain a list of registered user data, support conditional filtering and querying; manually add users and complete their basic information; manually adjust user basic information; implement data integration with the enterprise's existing IT systems to ensure real-time synchronization of user data; manage the binding relationship between users and membership cards, supporting unbinding and rebinding operations. The core objective is to achieve precise user management and data connectivity, providing data support for personalized marketing and services.

3.10.4.2.1. User Process

No.	Component	Description
1	Registered User Data List	Operators can view the data of all registered users in the system.
2	Manually Add User	Operators can manually add single or multiple users in the system.
3	Manual Adjustment of User Basic Information	Operators can manually modify the basic information of a single user in the system.
4	Conduct Data Integration with the Enterprise's Existing IT Systems	The system supports data integration with the enterprise's existing IT systems.

Table 3.10.4.2.1: Member Management Business Requirements Specification

3.10.4.2.2. Requirement List

RID	Requirement Description	Priority	Related Process
REQ-MC-013	Registered User Data List	High	3.10.4.2.1: #1
REQ-MC-014	Manually Add User	High	3.10.4.2.1: #2

REQ-MC-015	Manual Adjustment of User Basic Information	High	3.10.4.2.1: #3
REQ-MC-016	Conduct Data Integration with the Enterprise's Existing IT Systems	High	3.10.4.2.1: #4

Table 3.10.4.2.2: List of Member Management Business Requirements

3.11. Marketing, Promotion and RSVP

This system constructs a complete marketing and promotion system for the ticketing platform, covering modules such as coupons, promotional activities, and marketing treasure chests. Coupons support two types: full reduction and discount and can be obtained through multiple channels such as system distribution, membership benefits, and card purchase gifts; promotional activities include forms such as immediate reduction and discount, and support multiple participation methods such as public, channel exclusive, and promotional code; the marketing treasure chest supports bank card cooperation activities and invitation activities. The bank cooperation module is linked with designated bank cards to provide exclusive discounts such as immediate payment reduction; the invitation function supports the complete process of targeted invitation, registration review, and waitlist replenishment. The system aims to improve user conversion rate and re-purchase rate, while reducing customer acquisition cost, providing flexible marketing tools for operators, and achieving precise customer acquisition and user retention.

3.11.1. Business Requirements

WestK hopes to build a complete ticketing marketing system through the marketing promotion system, centring on user needs, providing flexible and diverse forms of promotions (cash vouchers, discount vouchers, exchange vouchers) and acquisition channels (registration gifts, membership benefits, ticketing rewards, event participation), lowering the threshold for users to purchase tickets, and motivating order conversion. It also supports precise marketing strategies, achieving precise customer acquisition, increasing the re-purchase rate, and raising the per customer transaction through differentiated promotion forms (public promotions, channel-exclusive promotions, member-exclusive promotions, promotional codes, bank partnerships) and refined rule configurations (usage thresholds, validity periods, inventory, stacking rules), and providing exclusive event experiences such as invitations for VIP users, ultimately achieving the dual goals of optimizing customer acquisition costs and enhancing user value.

3.11.1.1. Promotional Event

WestK hopes that the system can provide flexible discount combinations, such as coupon giveaway upon order placement, coupon giveaway upon membership card purchase, multi-buy multi-discount, new user registration rewards, etc.

The goal is to motivate users to place orders, increase conversion rates, boost new user registrations, and simultaneously drive repeat purchases and membership card purchases.

3.11.1.2. Coupon Configuration

Coupons can support different types of deductions (such as full reduction, discount, etc.) and can be applied to specified ticketing products or online merchandise.

Multiple usage methods: can be used alone or combined with other promotional activities to ensure that users receive the best experience and maximum benefits during the payment process.

3.11.1.3. Promotional Activities

WestK requires the system to support various promotional activities:

- **Public Promotion:** Visible and available to all users, such as roadshows, major promotional events, and holiday activities, maximizing event exposure.
- **Channel Exclusive:** Exclusive to users of designated channels (e.g., those entering through a specific event page), enabling differentiated channel operations.
- **Member Exclusive:** Exclusive to users of designated membership cards, enhancing members' sense of privilege and willingness to renew.
- **Channel cooperation:** For example, joint activities between the Palace Museum exhibition and UnionPay cards, and designated discounts with banks as sponsors, to expand cooperation channels.
- The goal is to reach a broader user base through differentiated promotional strategies while enhancing partner value.

3.11.1.4. Member Loyalty

When C-end users purchase and activate membership cards, they can enjoy personalized discount packages such as discounts + coupon gifts + point rewards.

Members enjoy exclusive promotional activities when purchasing tickets or merchandise, enhancing their sense of identity value and willingness to make repeat purchases.

3.11.1.5. Coupon Management

Users can quickly locate coupons in the card package, clearly view the validity period and usage conditions, and conveniently select, quickly display, and use them.

Provide intuitive and convenient operation experience, reducing the cognitive cost and operational resistance of using discounts.

When C-end users purchase tickets, the system automatically selects the optimal coupon or promotional discount, streamlining the entire order process and enhancing the user experience.

3.11.1.6. Unified Backstage Management

WestK can uniformly view and manage the status of bound coupons, including unused, used, expired, etc.

Supports manual verification or adjustment based on actual circumstances to ensure the controllability and flexibility of the event execution process.

3.11.1.7. RSVP Event

WestK hopes that the system will support precision marketing activities in the form of RSVP to enhance the exclusive experience of VIP users and specific groups:

- Member activities support targeted invitations: In scenarios such as offline salons, workshops, and member activities, WestK hopes to invite users from specific user groups via email to participate in exclusive activities. After receiving the email, users can click on the event link to perform operations such as free acquisition or ticket purchase.
- Offline Event Invitation: In scenarios such as meet-and-greet sessions and media tickets, WestK will send event invitations to designated users. Users who receive the invitations can fill in information such as the guests' personal information and the number of participants through a form. For different events, the information to be filled in can be customized. Meanwhile, WestK staff can collect and review the information. After the review is approved, admission tickets will be automatically issued. The tickets can be quickly verified on-site.

3.11.2. Business Requirement Analysis

This section further analyses and explains the business requirements described in Section 3.11.1. Each subsection elaborates on the business intent, operational considerations, and functional expectations behind the marketing promotion system solution.

3.11.2.1. Promotional Events

Map to Business Requirement 3.11.1.1.

1) Support coverage of multiple promotional event scenarios

Coupons need to support configuration of applicable scenarios: online ticket purchase, offline ticket purchase, membership card purchase, and online product purchase

The system shall support unified management of multiple business scenarios. Operators can select the applicable business type when creating coupons, and the system will automatically control the display and usage scope of coupons based on the scenario type.

2) Purchase a membership card and receive coupons

After the membership card configuration is activated, coupon issuance is enabled, and the coupon will be automatically bound to the user's account upon successful activation of the membership card.

The membership card type can be configured to activate and receive specified coupons, including multiple coupon combinations. After the user purchases and activates the membership card, the system automatically binds the coupons in the gift package to the user's account, and the user can view and use the coupons on the C-end.

3) Offline/Online Scenario Implementation

Coupon configuration usage channel attributes: online only, offline only, online and offline universal.

Each coupon can be configured with usage channel restrictions, and the system will control the display and usage of coupons accordingly. Offline scenarios need to support QR code/barcode verification, while online scenarios will automatically deduct at checkout.

3.11.2.2. Coupon Configuration

Map to Business Requirement 3.11.1.2.

1) Coupons with different types of deductions

The system supports three types of coupons: full reduction coupons (set the full reduction threshold and face value), discount coupons (set the discount rate), and exchange coupons (set the specified goods/tickets that can be exchanged).

- Full reduction coupons: Set the minimum consumption amount and the reduction amount, such as "200 minus 50".
- Discount coupons: Set the discount rate, such as an 80% discount coupon (the maximum discount amount can be set).
- Exchange coupons: Specify the specific goods or tickets that can be exchanged, with no additional payment required.

2) Coupons can be applied to all or specified products

Coupon configuration scope of application: applicable to the entire venue, specified show events, specified sessions, specified venues, and specified online products.

Support multi-level range configuration, which can be precise to specific sessions or products. The system will automatically verify the applicable range of coupons when users place orders, and coupons that do not meet the conditions will be automatically filtered and not displayed.

3) Coupons can be flexibly configured with full reduction conditions

Supports setting a minimum consumption amount threshold and supports setting dimensions such as total order amount or number of coupons.

4) Limited distribution creates scarcity

Supports configuring the total issuance inventory of coupons and the limit on the number of coupons each user can claim.

You can set the total inventory of coupons (e.g., issue 10,000 coupons), the limit per person (e.g., limit 1 or 3 coupons per person). When the inventory is insufficient, the system will automatically stop issuing.

Release and display the "fully claimed" status on the front end.

5) Supports the combined use of coupons

Supports configuring stacking rules: cannot be stacked with other discounts, can be stacked with promotional activities, can be stacked with specified types of coupons.

Each coupon can be configured with stacking rules. The system will automatically calculate the optimal discount combination at checkout to ensure that users receive the maximum discount while guaranteeing the platform's revenue.

3.11.2.3. Promotional Activities

Map to Business Requirement 3.11.1.3 and 3.11.1.4.

1) Public Promotion

Promotional activities are visible to all logged-in users by default, and users can enjoy the benefits without additional conditions.

Public promotions are the most basic form of promotion, and all logged-in users can see and enjoy the discounts on the ticket purchase page. They are suitable for large-scale marketing activities, such as holiday promotions, anniversary celebrations, etc.

2) Channel Exclusive

Promotional activities can be bound to channel identification parameters and are only visible to users who enter through that channel.

Supports configuring channel-exclusive activities, where only users from the specified channel can see the offer. The system tracks user sources through URL parameters to achieve channel-differentiated operations.

3) Member Exclusive

Promotional activities can be configured with membership card restrictions and are only visible and available to users who hold membership cards of specified levels.

Supports setting up member-exclusive activities, such as only gold card members being eligible for a 20% discount. The system will automatically verify the user's membership level, and users who do not meet the criteria will not see the event or be able to enjoy the discount.

4) Channel Cooperation

Supports configuring bank card promotional activities: specified bank list, instant payment reduction amount/ratio, card number verification rules (such as verifying the last 4 digits entered).

Supports cooperation with designated banks to carry out payment instant discount activities. When users choose to pay with the corresponding bank card, the system automatically verifies the card number and applies the discount. Supports setting an instant discount amount (e.g., 50 yuan off) or an instant discount percentage (e.g., 10% off).

5) Promo Code

Supports generating unique promotional codes (random or custom codes), validates and applies corresponding discounts after user input, and supports batch generation, export, and distribution.

Supports generating promotional codes (e.g., XJ2024VIP), allowing users to enter the promotional code on the checkout page to enjoy corresponding discounts. Supports batch generation, Excel export, and distribution to partner channels. The system records the usage count and associated orders for each promotional code.

3.11.2.4. Coupon Management

Map to Business Requirement 3.11.1.2.

1) Coupon Display

The user personal centre provides a coupon list, categorized by status: unused (sorted in descending order by expiration date), used, and expired.

Users can view all coupons in the "My Coupons" section of the Personal Centre, where the system displays them categorized by status. Unused coupons are sorted in descending order of validity period (with those about to expire listed first), making it convenient for users to use them in a timely manner. The coupon details page clearly shows all key information, including: coupon name (e.g., "New User Exclusive Full Reduction Coupon"), denomination (e.g., "Full 200 Minus 50"), validity period (e.g., "2024.01.01 - 2024.06.30"), usage conditions (e.g., "Only available for online ticket purchases"), and scope of application (e.g., "Valid for the entire venue" or "Designated event: XX Show").

2) Automatic Optimal Selection

The system automatically calculates the deduction amount of each available coupon based on the order amount, and by default selects the option with the highest deduction.

The system intelligently recommends the optimal coupon, defaulting to selecting the coupon that can save the user the most amount. Users can manually switch to select other coupons, and the system will update the discount amount in real time.

3.11.2.5. Unified Backend Management

Map to Business Requirement 3.11.1.6.

1) View coupon status uniformly

The background management page provides a coupon query function, with filtering conditions including user account, coupon type, status (unused/used/expired/voided), and issuance time range.

Operators can query the issuance and usage of all coupons in the backend. It supports multi-dimensional filtering, such as querying all coupons obtained by a specific user within a specific time period or querying the usage rate of a certain batch of coupons. The backend list displays the complete lifecycle information of coupons, including unique coupon code, bound user, coupon type, face value/discount, validity period, current status (unused/used/expired/voided), usage time, associated order number, etc.

2) Adjustment and Cancellation

Supports the cancellation of unused coupons, including single-item cancellation and batch cancellation. After cancellation, they will no longer be displayed on the user side.

When there is an error in the event configuration or an urgent stop is required, unused coupons can be voided. It supports single-item voiding and batch voiding. After voiding, the coupon will no longer be displayed on the user side, and voided coupons cannot be restored for use.

3.11.2.6. RSVP Event

Map to Business Requirement 3.11.1.7.

1) Targeted Invitation:

Operators can select the target user group in the backend to send invitations, supporting uploading Excel files to batch import specified user email addresses or mobile phone numbers.

2) Email Template Configuration

The system provides an email template editor that supports rich text editing. It allows configuration of email subject, event name, time, location, registration link, event description, and other content. It supports inserting variables (such as username) to enable personalized email sending.

3) Registration Information Collection and Summarization

The invitation event supports custom registration forms. In addition to basic fields (name, phone number), custom questions (such as "From which channel did you learn about this event?") can be added. The registration information collected by the system can be used for subsequent user profile analysis and event optimization.

The background displays the registration list, with fields including registrant, contact information, submission time, status (pending review/approved/rejected/cancelled), and supports filtering by status.

4) Audit Management

Operators can review registrations one by one or in batches. Users who pass the review will receive tickets. Reasons for rejection (such as "quota is full") need to be filled in. The upper limit of the number of participants can be set to ensure the controllability of the event.

5) Automatic Ticket Issuance and On-site Verification

After the review is approved, an electronic ticket (including a QR code) will be automatically generated, and an email will be sent to notify the user. The user can view the ticket in the personal centre. The ticket can be verified for entry.

The electronic ticket displays the QR code, which supports on-site staff to use scanning devices or mobile phones for verification. After verification, the ticket status is updated to use to prevent reuse.

3.11.3. Customer Journey

No.	Category	Role	User Story
1	Create Coupon	As an operator	I want to create different types of coupons (full reduction/discount/exchange) to match different marketing campaign strategies and attract users to place orders.
2	Issue Coupons	As an operator	I want to distribute coupons in batches to a specified user group to achieve targeted marketing and improve conversion rates.
3	Coupon Binding Record	As an operator	I would like to view the user's coupon binding records and usage status to monitor the effectiveness of the campaign and conduct Data Analysis.

4	Coupon Exception Management	As an operator	I would like to reissue abnormal coupons to address customer complaints or operational errors.
5	Coupon Combination Management	As an operator	I want to configure coupon combination rules to implement a differentiated marketing strategy and enhance user engagement.
6	Create a Promotional Campaign	As an operator	I want to create diverse promotional activities (instant discounts/percentage discounts/threshold discounts) to increase event exposure and order conversion rates.
7	Create a Promotional Campaign	As an operator	I would like to configure channel-exclusive promotional activities to track the conversion effectiveness of different channels and optimize the placement strategy.
8	Create a Promotional Campaign	As an operator	I would like to configure bank cooperation preferential activities to expand cooperation channels and reduce customer acquisition costs.
9	View Orders Associated with Promotions	As an operator	I would like to view the order data associated with the promotional event to evaluate the event's ROI and optimize subsequent event plans.
10	Create Invitation	As an operator	I want to create an invitation event and customize the registration form to collect personalized requirement information from VIP users.
11	Invitation Push	As an operator	I want to perform targeted ops push of invitations to specific user groups to enhance the exclusive experience of high-end users.
12	Invitation Registration Management and Review	As an operator	I want to manage the registration review process to efficiently control event quotas and automatically handle supplementary enrolment.
13	Automated Email Push	As an operator	I want to batch automate pushing invitations or other event links to users at a specified time.

Table 3.11.3: User Story Description

3.11.4. Business Scenarios

This scenario defines the business logic for operators to centrally manage the marketing promotion system, covering three major modules: coupon management, promotional event management, and invitation event management. Operators create

coupon templates and configure their types, denominations, and usage rules, distribute them to user accounts through multiple channels, view binding records, and handle exceptional situations; create promotional activities and configure the form of discounts and participation methods, set payment discount rules for bank partnerships, view event-related orders for effectiveness analysis; Create invitation activities, configure form fields, perform targeted ops push to Target Users, and manage the registration review process and the waitlist and supplementary enrolment mechanism. Through diverse forms of promotions and precise operational measures, combined with channel cooperation and targeted invitation strategies, the system promotes user order conversion, enhances the exclusive experience of VIP users, and achieves the dual goals of optimizing customer acquisition costs and enhancing user value.

3.11.4.1. Coupon Management

This scenario defines the business logic for operators to create, configure, and distribute coupons, as well as to view binding records after users claim them, handle exceptions, and perform combined configuration. When creating a coupon template, operators configure the type (full reduction/discount/exchange), face value, usage threshold, validity period, scope of application, and stacking rules; bind coupons to user accounts through channels such as system batch distribution, user-initiated claiming, automatic member distribution, and card purchase gifts; and view user coupon binding records centrally in the background, supporting filtering by user, type, status, and time range. Manually verify, void, or adjust the status of abnormal coupons; configure coupon combination rules to implement differentiated marketing strategies. The core objective is to use the means of issuing coupons to support marketing activities and encourage users to place orders.

3.11.4.1.1. User Process

No.	Component	Description
1	Create Coupon	Operators can create coupons in the system and configure them in multiple aspects such as basic information, inventory, usage restrictions, etc.
2	Issue Coupons	Operators can issue coupons to specified users in the system or generate online promotional event pages for users to claim coupons voluntarily.
3	Coupon Binding Record	Operators can view and manage the data of coupons already claimed by users in a unified manner.
4	Coupon Exception Management	Operators can view the reasons for and reissue coupons that failed to be issued.
5	Coupon Combination Management	Operators can combine multiple coupons and distribute them together.

Table 3.11.4.1.1: Business Requirements Specification for Coupon Management

3.11.4.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-MP-001	Create Coupon	High	3.11.4.1.1: #1
REQ-MP-002	Issue Coupons	High	3.11.4.1.1: #2
REQ-MP-003	Coupon Binding Record	High	3.11.4.1.1: #3
REQ-MP-004	Coupon Exception Management	High	3.11.4.1.1: #4
REQ-MP-005	Coupon Combination Management	High	3.11.4.1.1: #5

Table 3.11.4.1.2: *Coupon Management Business Requirements List*

3.11.4.2. Promotion Event Management

This scenario defines the business logic for operators to create, configure, and view associated orders for promotional activities. When creating a promotional event, operators configure the type (immediate discount/discount/conditional discount), discount strength, event duration, participation method (public/channel/member/promotional code), scope of application, and stacking rules; configure payment immediate discount or discount rules for bank partnerships, set the card type range and limit control; view associated orders of promotional activities through the backend, supporting filtering by time, status, user type, and data export. The core objective is to utilize diverse promotional forms in conjunction with channel partnerships to enhance event conversion rates and order volume.

3.11.4.2.1. User Process

No.	Component	Description
1	Create a Promotional Campaign	When creating a promotional event, the operation staff configures the type (immediate reduction/discount/full reduction), preferential strength, event time, participation method (public/channel/member/promotional code), scope of application, and stacking rules; for bank cooperation, configures payment immediate reduction or discount rules, and sets the card type range and limit control.
2	View Orders Associated with Promotions	View promotion-related orders through the backend, supporting multi-dimensional filtering.

Table 3.11.4.2.1: *Business Requirements Description for Promotional Activities*

3.11.4.2.2. Requirement List

RID	Description	Priority	Related Process
REQ-MP-006	Create a Promotional Campaign	High	3.11.4.2.1: #1
REQ-MP-007	View Orders Associated with Promotions	High	3.11.4.2.1: #2

Table 3.11.4.2.2: List of Business Requirements for Promotional Activities

3.11.4.3. RSVP Event

This scenario defines the business logic for operators to create, push, manage registrations, and review invitation activities. When creating an invitation event, operators configure event information, quota limits, registration deadline, and custom form fields; select target user groups for targeted ops push, support email template configuration and automated email sending; view registration information in the background, support status filtering for pending review/approved/rejected and single/batch review; automatically issue admission tickets after approval and support on-site QR code verification. The core objective is to enhance the exclusive experience of VIP users and specific groups through targeted invitation methods.

3.11.4.3.1. User Process

No.	Component	Description
1	Create Invitation	When creating an invitation event, the operator configures event information, quota limits, registration deadline, and custom form fields.
2	Invitation Push	Operators can perform targeted ops push to users within a specified range to restrict access to the invitation to those users.
3	Invitation Registration Management and Review	Operators can view registration information in the backend, supporting status filtering for pending review/approved/rejected and single/batch review.
4	Automated Email Push	Operators can configure email templates in the backend and complete scheduled email sending to specified users.

Table 3.11.4.3.1: RSVP Event Business Requirements Description for You

3.11.4.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-MP-008	Create Invitation	High	3.11.4.3.1: #1
REQ-MP-009	Invitation Push	High	3.11.4.3.1: #2
REQ-MP-010	Invitation Registration Management and Review	High	3.11.4.3.1: #3
REQ-MP-011	Automated Email Push	High	3.11.4.3.1: #4

Table 3.11.4.3.2: RSVP Active Business Requirements List

3.12. Customer Data Platform

3.12.1. Business Requirements

From a business perspective, WestK requires the Mini Program to have unified Customer Data Platform (CDP) event tracking capabilities to support data recording, statistics, and subsequent analysis of key business scenarios. This document only defines the scope of event tracking capabilities and business requirements; the specific event tracking list, event names, and attribute fields will be further elaborated in the corresponding detailed business requirement documents.

The event tracking data we push is based on the actual scenario the user is currently using, without translation or secondary processing.

If the user is browsing the content of the English version, then the event tracking data is in English.

If the user is browsing content on the Chinese version, then the event tracking data will be in Chinese.

3.12.1.1. Member Card Tracking Capability

The system shall support recording key events and user behaviours in membership card-related scenarios, including but not limited to membership card display, detail browsing, benefit viewing, claiming, activation, usage, and related conversion behaviours, to meet the needs of membership operation analysis. All relevant events shall be synchronized to WestK's CDP system.

3.12.1.2. Ticketing Tracking Capability

The system shall support recording key events and user behaviours in ticketing-related scenarios, including but not limited to ticket browsing, session viewing, ticket type viewing, adding to cart, placing an order, payment, and order-related behaviours, to meet the needs of ticketing operation analysis. All relevant events shall be synchronized to WestK's CDP system.

3.12.1.3. Campaign Tracking Capability

The system shall support recording key events and user behaviours in scenarios related to special events, including but not limited to special page exposure, content browsing, interactive clicks, event participation, and event conversion behaviours, to meet the operational analysis needs of special events. All relevant events shall be synchronized to WestK's CDP system.

3.12.1.4. Merchandise Information Tracking Capability

The system shall support recording key events and user behaviours in scenarios related to product information, including but not limited to product exposure, product detail browsing, attribute selection, adding to cart, collection, and related conversion behaviours, to meet the needs of product operation analysis. All relevant events shall be synchronized to WestK's CDP system.

3.12.1.5. User Information Capability

The system shall support compliant recording of basic information, identification information, and information required for group analysis of mini-program users to meet the operational analysis needs of user information. All relevant events shall be synchronized to WestK's CDP system.

3.12.1.6. User Behaviour Tracking Capability

The system shall support the recording of behaviours such as page access, click, stay, jump, search, share, favourite, and authorization of mini-program users to meet the operational analysis requirements of user behaviour. All relevant events shall be synchronized to WestK's CDP system.

3.12.2. Business Requirement Analysis

This section further analyses and clarifies the business requirements described in 3.12.1, explaining how WestK expects the system to address various event tracking requirements at the capability level. This document remains at the business requirement analysis level and does not delve into functional fields, event tracking parameters, or Technology Implementation details.

3.12.2.1. Business Tracking Capability

Membership cards, tickets, special events, and product information are the core business objects through which WestK provides services to users and reaches out to them. They are widely present in front-end systems such as mini-programs, Mobile Website, and websites, and serve as the main carriers for users to browse, understand, participate in, and initiate business operations.

From a business perspective, the browsing, viewing, clicking, and subsequent operational behaviours generated by users around the above Business Objects constitute important business touchpoints of WestK in digital channels. If the data of these touchpoints fails to be effectively recorded in each front-end system and

uniformly pushed to WestK's CDP system, WestK will find it difficult to fully precipitate data related to Business Objects from different channels in the Customer Data Platform, and it will also be difficult to ensure the consistent collection of front-end touchpoint data.

Therefore, WestK hopes that front-end systems such as mini-programs, Mobile Website, and websites can establish unified data recording and pushing capabilities around core Business Objects, enabling key business records in scenarios related to membership cards, tickets, special events, and product information to enter WestK's CDP system as part of the unified precipitation of customer data.

The focus of this requirement is to ensure that the business touchpoints formed by the front-end system around core Business Objects can be continuously recorded and uniformly pushed, rather than requiring the front-end system to directly assume the responsibility of Data Analysis. As for the specific event details, field definitions, naming rules, and Technology Implementation methods that need to be recorded under each Business Object, they will be further defined in subsequent special business requirement documents and technical requirement documents, and this document will not elaborate on them.

3.12.2.2. User Identification and User Information

- The system should be able to identify users under the premise of compliance and establish user dimensions that match behavioural records.
- User information records should serve statistical analysis and operational insights and should not be collected beyond the scope of authorization.
- Behaviour records of the same user under different business scenarios should support unified identification and aggregation.

3.12.3. Operating Model

The business model of this event is as follows: WestK, as the initiator of business requirements and data user, clarifies the business scope, management requirements, and analysis objectives of the event tracking capabilities on the mini-program side; AnTank, as the builder and deliverer, undertakes the design and implementation of the event tracking capabilities on the mini-program side; relevant business teams, operation teams, and data teams conduct analysis and operation work based on unified event tracking data.

In this mode, the event tracking capabilities on the mini-program side should serve as a unified data foundation, connecting business domains such as membership, ticketing, activities, products, and user behaviour, to support subsequent data statistics, operational analysis, and cross-business insights. Specific event definitions, attribute specifications, and data consumption methods will be refined in each dedicated document.

3.12.4. Customer Journey

This section outlines, in the form of a user journey, the main behaviours of Mini Program users across different business domains, as well as the key touchpoints that WestK hopes to record.

No.	Category	Role	User Story
1	Member Card Tracking	As WestK Data Analysis Team	I hope that the system backend can automatically collect key data across the whole-link, including membership card exposure, detail browsing, benefit viewing, activation upon receipt, and benefit usage, and provide this data to WestK's CDP system.
2	Ticketing Tracking	As WestK Data Analysis Team	I hope the system backend can automatically collect whole-link key data such as ticket list browsing, session and ticket type viewing, adding to cart, placing orders, payment, and order status, and provide the data to WestK's CDP system.
3	Campaign Tracking	As WestK Data Analysis Team	I hope that the system backend can automatically collect whole-link key data such as exposure of special event pages, content browsing, interactive clicks, event participation, and conversion, and provide the data to WestK's CDP system.
4	Merchandise Tracking	As WestK Data Analysis Team	I hope that the system backend can automatically collect key whole-link data such as product exposure, detail browsing, attribute selection, collection, adding to cart, and conversion, and provide the data to WestK's CDP system.
5	User Identification	As WestK Data Analysis Team	I hope that the system, on the premise of compliance, automatically collects and associates the basic information and unique identification of mini-program users in the background, and provides the data to WestK's CDP system.
6	General User Behaviour Tracking	As WestK Data Analysis Team	I hope that the system backend can automatically collect full-dimensional behavioural data of mini-program users, including page visits, sources, stays, jumps, searches, shares, collections, authorizations, etc., and provide the data to WestK's CDP system.

Table 3.12.4: Description for the User Story

3.12.5. Business Scenarios

The following scenarios illustrate the main requirements of WestK for event tracking capabilities in different business domains. Each scenario only describes the scope of capabilities that need to be recorded and supplements relevant business objectives,

such as understanding user interests, evaluating content reach effectiveness, identifying conversion paths, and supporting subsequent operational segmentation; specific event lists, fields, and attributes are not elaborated. Relevant details will be specified in the specific business requirements document.

3.12.5.1. Membership Card

The system shall support recording key events and user behaviours in membership card-related scenarios, including but not limited to membership card display, detail browsing, benefit viewing, claiming, activation, usage, and related conversion behaviours, to meet the needs of membership operation analysis. All relevant events shall be synchronized to WestK's CDP system.

3.12.5.1.1. User Process

No.	Component	Description
1	Membership Card Exposure	Record the exposure of membership card entry, list, and related display positions.
2	Membership Card Details Viewing	Record the user's behaviour of entering the membership card details page and viewing relevant benefit content.
3	Membership Card Collection/Activation	Record user actions such as claiming, activating, or related confirmation.
4	Behaviours Related to Membership Card Usage	Record the key touchpoints of users when using membership card benefits.

Table 3.12.5.1.1: Description for the User Process

3.12.5.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-CDP-MEM-001	The system should support recording the exposure of membership card-related pages and entry points.	High	3.12.5.1.1: #1
REQ-CDP-MEM-002	The system shall support recording the behaviour of browsing membership card details and viewing membership card benefits.	High	3.12.5.1.1: #2
REQ-CDP-MEM-003	The system should support recording membership card collection, activation, and related conversion behaviours.	High	3.12.5.1.1: #3

REQ-CDP-MEM-004	The system should support recording key behaviours related to membership card usage.	Medium	3.12.5.1.1: #4
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Table 3.12.5.1.2: Requirement List of Membership Card CDP**3.12.5.2. Ticketing**

The system shall support recording key events and user behaviours in ticketing-related scenarios, including but not limited to ticket browsing, session viewing, ticket type viewing, adding to cart, placing an order, payment, and order-related behaviours, to meet the needs of ticketing operation analysis. All relevant events shall be synchronized to WestK's CDP system.

3.12.5.2.1. User Process

No.	Component	Description
1	Ticket List and Details Browsing	Record the exposure of the ticket list, browsing of details, and viewing of show sessions.
2	Ticket Types and Purchase Actions	Record key behaviours such as ticket type viewing, adding to cart, placing an order, and payment.
3	Order-Related Actions	Record relevant actions such as order viewing, cancellation, and payment results.

Table 3.12.5.2.1: Description for the User Process**3.12.5.2.2. Requirement List**

RID	Description	Priority	Related Process
REQ-CDP-TIC-001	The system should support recording ticket list, ticket details, and session browsing behaviour.	High	3.12.5.2.1: #1
REQ-CDP-TIC-002	The system should support recording behaviours related to ticket type viewing, adding to cart, placing orders, and payment.	High	3.12.5.2.1: #2
REQ-CDP-TIC-003	The system should support recording behaviours related to order viewing and order status.	High	3.12.5.2.1: #3

Table 3.12.5.2.2: Requirement List of Ticketing CDP

3.12.5.3. Campaign

The system shall support recording key events and user behaviours in scenarios related to special events, including but not limited to special page exposure, content browsing, interactive clicks, event participation, and event conversion behaviours, to meet the operational analysis needs of special events. All relevant events shall be synchronized to WestK's CDP system.

3.12.5.3.1. User Process

No.	Component	Description
1	Exposure and Browsing of Special Topic Page	Record the display of the special topic page and content browsing behaviour.
2	Special Interactive Behaviour	Record button clicks, event entry clicks, and related interactive behaviours.
3	Event Participation and Conversion	Record behaviours related to event participation, redirection, and conversion results.

Table 3.12.5.3.1: Description for the User Process

3.12.5.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-CDP-CAM-001	The system should support recording the exposure and browsing behaviour of the special event page.	High	3.12.5.3.1: #1
REQ-CDP-CAM-002	The system should support recording key interactive behaviours in special events.	High	3.12.5.3.1: #2
REQ-CDP-CAM-003	The system should support recording event participation and related conversion behaviours.	High	3.12.5.3.1: #3

Table 3.12.5.3.2: Requirement List of Campaign CDP

3.12.5.4. Merchandise Information

The system shall support recording key events and user behaviours in scenarios related to product information, including but not limited to product exposure, product detail browsing, attribute selection, adding to cart, collection, and related conversion behaviours, to meet the needs of product operation analysis. All relevant events shall be synchronized to WestK's CDP system.

3.12.5.4.1. User Process

No.	Component	Description
1	Product Exposure and Detail Viewing	Record the exposure of the product list and the browsing behaviour of product details.
2	Product Interaction Behaviour	Record key interactive behaviours such as attribute selection, collection, and adding to cart.
3	Product Conversion Behaviour	Record key behaviours related to the transaction or conversion of goods.

Table 3.12.5.4.1: Description for the User Process

3.12.5.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-CDP-PRO-001	The system should support recording product exposure and product detail browsing behaviour.	High	3.12.5.4.1: #1
REQ-CDP-PRO-002	The system should support recording behaviours such as product attribute selection, collection, and adding to cart.	High	3.12.5.4.1: #2
REQ-CDP-PRO-003	The system should support recording commodity-related conversion behaviours.	High	3.12.5.4.1: #3

Table 3.12.5.4.2: Requirement List of Merchandise Information

3.12.5.5. User Information

The system shall support compliant recording of basic information, identification information, and information required for group analysis of mini-program users to meet the operational analysis needs of user information. All relevant events shall be synchronized to WestK's CDP system.

3.12.5.5.1. User Process

No.	Component	Description
1	User Basic Information	Record the basic information of authorized mini-program users required for business purposes.
2	User Identification and Aggregation	Supports associating and aggregating user information with their behaviour records.

3	Foundation of User Segmentation Analysis	Supports providing basic data for subsequent user tagging, stratification, and analysis.
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Table 3.12.5.5.1: Description for the User Process

3.12.5.5.2. Requirement List

RID	Description	Priority	Related Process
REQ-CDP-USE-001	The system shall support compliance records of relevant basic information of Mini Program users.	High	3.12.5.5.1: #1
REQ-CDP-USE-002	The system should support the association and aggregation of user information with user behaviour.	High	3.12.5.5.1: #2
REQ-CDP-USE-003	The system should support providing basic data for user stratification and analysis.	Medium	3.12.5.5.1: #3

Table 3.12.5.5.2: Requirement List of User Information

3.12.5.6. User Behaviour

The system shall support the recording of behaviours such as page access, click, stay, jump, search, share, favourite, and authorization of mini-program users to meet the operational analysis requirements of user behaviour. All relevant events shall be synchronized to WestK's CDP system.

3.12.5.6.1. User Process

No.	Component	Description
1	Page Access and Stay	Records page visits, entry sources, duration of stay, and exit status.
2	Interactive Click and Jump	Record button clicks, module clicks, and page navigation actions.
3	Search, Share, Collect, Authorize	Record high-value behaviours and related outcomes.

Table 3.12.5.6.1: Description for the User Process

3.12.5.6.2. Requirement List

RID	Description	Priority	Related Process
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REQ- CDP- BEH-001	The system should support recording behaviours related to page access, source, stay, and exit.	High	3.12.5.6.1: #1
REQ- CDP- BEH-002	The system should support recording interactive clicks and page jump behaviours.	High	3.12.5.6.1: #2
REQ- CDP- BEH-003	The system should support high-value actions such as record search, sharing, collection, and authorization.	High	3.12.5.6.1: #3

Table 3.12.5.6.2: Requirement List of User Behaviour

3.13. CRM and ERP

3.13.1. Business Requirements

From a business perspective, WestK requires the system to have the capabilities of report generation, scheduled transmission, format control, field control, and scope control for CRM/ERP, so that the internal system can stably receive and use relevant data.

3.13.1.1. ERP Report Generation

WestK requires the system to generate report files for ERP use, which are used to carry data related to ticket sales and settlement, in order to support financial reconciliation, revenue verification, and subsequent ERP accounting processing.

The report should output transaction-related information according to the agreed data scope, and follow the file format, field structure, and statistical calibre agreed upon on the ERP side.

3.13.1.2. CRM/Product Master Report Generation

WestK requires the system to generate CRM/Product Master reports to provide event Master Data and product Master Data to CRM, in order to support the synchronization of event, session, and related product information.

The report should cover all Master Data of products within the agreed scope and support subsequent CRM consumption and Master Data management.

3.13.1.3. CRM/Transaction Report Generation

WestK requires the system to generate CRM/Transaction Report to provide transaction data to CRM, supporting customer business analysis, transaction tracking, and subsequent business analysis.

The report should output transaction details on a scheduled cycle and provide information related to ticketing transactions according to established field rules.

3.13.1.4. CRM/Admission Report Generation

WestK requires the system to generate CRM/Admission Report statements to provide verification and admission-related data to CRM, supporting on-site analysis, user behaviour analysis, and subsequent business use.

The report should output verification data on a scheduled cycle and reflect the recorded entry and verification records.

3.13.1.5. CRM/Inventory Report Generation

WestK requires the system to generate CRM/Inventory Report to provide CRM with data related to inventory and available sales balance, in order to support inventory monitoring, sales analysis, and business decision-making.

The report should output inventory and available ticket quantity data according to the agreed scope and reflect business scenarios such as tickets and reserved inventory.

3.13.1.6. Daily scheduled SFTP transfer

WestK requires the system to transfer the above reports to the designated SFTP location through daily scheduled tasks, as the standard method for CRM/ERP data delivery at the current stage.

This transmission method should be automatically executed at the agreed fixed time, and support ERP and CRM to distinguish and receive according to directory or file rules.

3.13.1.7. File Format and Encryption Requirements

WestK requires the system to output reports in accordance with the agreed file naming, delimiter, encoding, header and footer rules, and encryption method to ensure that the target system can stably parse and receive them.

Among them, CRM reports need to be encrypted and transmitted as agreed; ERP reports should be output in the file structure agreed upon by the ERP side.

3.13.1.8. Field and Mapping Requirements

WestK requires the system to provide agreed-upon fields in the report and output values in accordance with the field dictionary, code mapping table, and business calibre requirements to ensure that CRM/ERP can correctly consume relevant data.

The naming, value range, mapping source, language, encoding, and statistical calibre of the field shall remain consistent, and the detailed field list will be further defined in subsequent documents.

3.13.1.9. API Interface Data Transmission

WestK requires the system to further provide the ability to transmit ticketing sales data, inventory data, and verification data via API in the future to support a more automated and integrated data consumption model.

This capability has not been implemented yet.

During the implementation cycle of this event, if the interface between the CRM and ERP system parties reaches a connectable state, AnTank will schedule the interface connection for CRM/ERP. The interface fields shall be consistent with the CRM/ERP report fields.

3.13.1.10. OTA Data Does Not Fall within the Scope

WestK requires that OTA data does not enter the scope of ERP report data to avoid data that does not meet the ERP integration criteria being passed into ERP.

For CRM-related reports, the data scope shall be implemented in accordance with the business criteria agreed upon on the CRM side; the data scope of ERP and CRM shall not be considered completely consistent.

3.13.2. Business Requirements Analysis

This section further analyses and clarifies the aforementioned business requirements, with a focus on explaining the business intent, scope of application, and current status.

3.13.2.1. Report Scope Analysis

Map to Business Requirement 3.13.1.1 to 3.13.1.5.

The business scenarios covered in this document have expanded from general requirements to five specific reports but are still described at the business level without delving into the levels of field dictionaries, exception cases, interface messages, or program implementation.

3.13.2.2. Analysis of Report Generation Mechanism

Map to Business Requirement 3.13.1.1 to 3.13.1.5.

The five reports serve two types of target systems, ERP and CRM, respectively. Among them, ERP focuses more on financial and settlement perspectives, while CRM focuses more on the continuous consumption of business data such as product master data, transactions, write-offs, and inventory.

3.13.2.3. Requirements Analysis for Scheduled SFTP Transfer

Map to Business Requirement 3.13.1.6.

SFTP transfer capabilities are already in place and can serve as the standard delivery method for providing data to CRM/ERP during the current phase. Compared to ad-hoc exports or manual uploads, daily scheduled tasks are more conducive to enhancing delivery stability, traceability, and consistency of data reception.

3.13.2.4. Analysis of File Format and Encryption Requirements

Map to Business Requirement 3.13.1.7.

Different target systems have differences in file structure, delimiters, header and footer rules, encoding, and encryption methods. Therefore, report output cannot simply meet the requirement of "having data"; it must also meet the requirements of "being parsable, verifiable, and storable in the database".

3.13.2.5. Field and Mapping Requirement Analysis

Map to Business Requirement 3.13.1.8.

Field requirements include not only whether a field is output, but also where the field value comes from, how it is mapped, whether it needs to be converted according to a dictionary, whether it needs to meet language or encoding specifications, and whether it needs to comply with established statistical criteria.

3.13.2.6. Analysis of Data Scope Differences between ERP and CRM

Map to Business Requirement 3.13.1.1 to 3.13.1.5.

ERP and CRM have different business objectives, so their data scopes are not entirely consistent. Especially in the OTA scenario, WestK has clearly stated that OTA data will not enter the ERP system, while the scope of data entry for CRM-related reports should be separately confirmed according to the CRM business criteria.

3.13.2.7. API Transmission Requirement Analysis

Map to Business Requirement 3.13.1.9.

Compared with the file-based transfer method, the interface method is more suitable for supporting automated integration between systems, on-demand data retrieval, near real-time synchronization, and subsequent expansion. WestK has therefore proposed API data transfer requirements, but the current stage still mainly relies on the SFTP report method.

During the implementation cycle of this event, if the interface between CRM and ERP system parties reaches a connectable state, AnTank will schedule the interface connection for CRM/ERP. The interface fields shall be consistent with the CRM/ERP report fields.

3.13.2.8. Business Scenario Boundary Explanation

Map to Business Requirement 3.13.1.10.

Although this document has supplemented specific business scenarios, it still focuses on the business requirements themselves. For the detailed field definitions, test cases, boundary conditions, exception handling, scheduling details, and interface protocols of each report, further explanations are required in subsequent specialized business documents or interface specification documents.

3.13.3. Operating Model

This event can be understood as the following operating model at the data delivery level:

- As the source system, the ticketing platform is responsible for generating and maintaining reports related to sales, inventory, verification, and product Master Data for ERP/CRM.
- As the business owner, WestK defines data usage objectives, business criteria, transmission methods, field rules, and scope control principles.
- As target systems, ERP and CRM respectively receive and use relevant data according to their own data structures and business requirements.
- At the current stage, daily scheduled SFTP report transfer has been implemented as the delivery method; in subsequent stages, the API interface transfer capability will be evaluated and developed.
- In terms of data scope control, ERP and CRM can adopt different business scopes, among which the rule that OTA data does not enter ERP should be clearly enforced.

3.13.4. Customer Journey

The following user story is used to summarize the current business requirements from the perspective of WestK's internal users.

No.	Category	Role	User Story
1	ERP Data Reception	As WestK finance or ERP user	I want to receive the ERP report through the scheduled SFTP delivery every day, so that my team can use the data for reconciliation and ERP processing.
2	CRM Data Receipt	As WestK CRM user	I want to receive Product Master, Transaction Report, Admission Report and Inventory Report on a scheduled basis, so that my team can consume the agreed data in CRM.
3	File and Field Management and Control	As WestK business or IT user	I want the report files to follow the agreed file format, encryption method and field mapping rules, so that the downstream systems can parse and use the data correctly.
4	Data Range Control	As WestK finance user	I want OTA data to be excluded from ERP data scope, so that ERP only receives the data that matches the agreed business rules.
5	Future Integration and Evolution	As WestK CRM/ERP user	I want the system to provide API-based data delivery in the future, so that CRM/ERP can consume ticketing data in a more automated way.

Table 3.13.4: Description of User Story

3.13.5. Business Scenario

This document supplements the following specific business scenarios as the scope of application for this CRM/ERP data delivery requirement. Each scenario is only for business-level summary and explanation, without delving into field implementation details.

3.13.5.1. User Process

No.	Business Scenario	Scene Description	Remarks
1	ERP Report Generation	The system generates ERP reports according to the agreed scope to support ERP financial reconciliation and related subsequent processing.	Focus on ERP data consumption scenarios.
2	CRM/Product Master Report Generation	The system generates a Product Master Data Report to provide event, session, and product master data to CRM.	belongs to the full Master Data scenario.
3	CRM/Transaction Report Generation	The system generates transaction reports to provide transaction data within the agreed scope to CRM.	Focus on sales and trading analysis.
4	CRM/Admission Report Generation	The system generates verification reports to provide admission and verification data to CRM.	Focus on verification and on-site analysis.
5	CRM/Inventory Report Generation	The system generates inventory reports to provide CRM with data related to inventory and remaining quantity.	Focus on inventory monitoring and sales analysis.
6	Daily Scheduled SFTP transfer	The system automatically transfers ERP and CRM reports to the specified SFTP directory according to the fixed scheduling time.	Standard delivery method at the current stage.
7	File Format, Encryption, and Field Control	The system outputs report in accordance with the agreed file format, encoding, encryption method, and field mapping rules.	Detailed field rules will be separately explained later.

8	API Data Transmission	Provide relevant data to CRM/ERP through an interface.	During the implementation cycle of this event, if the interface between CRM and ERP system parties reaches a connectable state, AnTank will schedule the interface connection for CRM/ERP. The interface fields shall be consistent with the CRM/ERP report fields.
9	OTA Data Does Not Enter ERP	The system shall ensure that OTA data does not fall within the scope of ERP reports.	The scope of ERP and CRM can be different.

Table 3.13.5.1: User Process of ERP Report

The detailed fields, processing rules, update frequency, responsibility division, and exception handling methods involved in the above business scenarios will be elaborated in the specific business requirement documents, field dictionaries, or interface specification documents, and will not be expanded upon in this document.

3.13.5.2. Requirement List

RID	Description	Related Process
REQ-CRM-ERP-001	The system shall support generating report data for ERP use to meet the requirements of ERP financial reconciliation and related processing.	3.13.5.1: #1
REQ-CRM-ERP-002	The system shall support generating CRM/Product Master reports to provide event and product Master Data to CRM.	3.13.5.1: #2
REQ-CRM-ERP-003	The system shall support generating CRM/Transaction Report to provide transaction data to CRM.	3.13.5.1: #3
REQ-CRM-ERP-004	The system shall support generating CRM/Admission Report to provide verification data to CRM.	3.13.5.1: #4

REQ-CRM-ERP-005	The system shall support generating CRM/Inventory Report to provide inventory data to CRM.	3.13.5.1: #5
REQ-CRM-ERP-006	The system shall support providing ERP and CRM related report data to WestK via SFTP through daily scheduled tasks.	3.13.5.1: #6
REQ-CRM-ERP-007	The system shall support outputting and transmitting relevant reports in accordance with the agreed file naming, delimiter, encoding, header and footer rules, and encryption method.	3.13.5.1: #7
REQ-CRM-ERP-008	The system shall support outputting report data according to the agreed fields, field mapping tables, and business criteria.	3.13.5.1: #7
REQ-CRM-ERP-009	The system shall support controlling the entry scope of ERP data according to WestK rules to ensure that OTA data does not enter ERP reports.	3.13.5.1: #9
REQ-CRM-ERP-010	The system shall support subsequent transfer of ticket sales data, inventory data, and verification data for CRM/ERP use to WestK via an interface.	3.13.5.1: #8

Table 3.13.5.2: Requirement List

3.14. Payment

This chapter focuses on the construction of WestK's payment business capabilities, covering core business processes such as payment method expansion, channel differentiated access, transaction status synchronization, refund processing, authentication activation, daily settlement, and financial reconciliation.

3.14.1. Business Requirements

From a business perspective, WestK's core requirement for the payment system is no single-channel access, but rather the ability to provide appropriate payment methods across different business terminals, while ensuring that transaction, refund, settlement, and authentication management all adhere to unified business rules and traceability.

3.14.1.1. Online Bank Card Payment

I hope that the online ticketing scenario can continue to use Cybersource to complete bank card payments, and support payment result confirmation, payment inquiry, refund, and exception compensation processing.

3.14.1.2. BO Apollo POS Payment

I hope that the Apollo POS handheld device can be used to complete payments in scenarios such as back-office ticketing or manual services, so that WestK staff can carry the device to flexibly complete payments at different locations.

3.14.1.3. Kiosk Apollo POS Payment

I hope that the Kiosk self-service machine scenario can be connected to Apollo POS payment so that visitors can complete payments at fixed terminals.

3.14.1.4. Octopus Payment

I hope the system can support Octopus payment to meet the common payment habits of local WestK users in offline scenarios.

3.14.1.5. Payment Status Query and Synchronization

I hope the system can record and synchronize transaction status under each payment method and still be able to query the final result when there is a delay in third-party callback, interface timeout, or unclear status.

3.14.1.6. Timeout Auto Close Order

I hope that orders with incomplete payment can be automatically closed after the specified time limit to avoid long-term occupation of orders or inventory.

3.14.1.7. Refund Processing

I hope that back-office staff can initiate refunds for eligible orders and view the refund process, results, and abnormal status.

3.14.1.8. Certification and Activation Preparation

I hope that each payment method will complete the corresponding certification before going live, among which Apollo and Cybersource will be certified by GP, and Octopus will be certified by Octopus Company.

3.14.1.9. Settlement and Reconciliation

I hope that all payment methods can complete daily settlement, with Apollo and Octopus supporting manual settlement and Cybersource supporting automatic settlement, to meet the requirements of financial operations.

3.14.2. Business Requirement Analysis

3.14.2.1. Payment Channel and Terminal Adaptation

Map to Business Requirement 3.14.1.1.

WestK's payment capabilities need to cover online, BO, Kiosk, and local payment scenarios. The online front-end mainly corresponds to Cybersource; the back-end BO can correspond to Apollo POS; Kiosk can correspond to Apollo POS; Octopus can be enabled for on-site or designated scenarios. Due to the different operating roles and business environments of each scenario, actual integration requires design in combination with the scenario.

3.14.2.2. Apollo Differences between BO and Kiosk

Map to Business Requirement 3.14.1.1 and 3.14.1.2.

Apollo payments for BO scenarios and Kiosk scenarios are designed for different user roles and operating environments. BO is more inclined towards manual services and on-site collection, while Kiosk is more focused on self-service ticketing for visitors. The system must ensure that payments can be smoothly completed in both types of scenarios and are seamlessly integrated with the corresponding ticketing processes.

3.14.2.3. Transaction Status Management

Map to Business Requirement 3.14.1.5.

All payment methods shall maintain a unified mapping relationship among payment receipts, business orders, and third-party transaction records. Regardless of whether the transaction result is obtained through asynchronous notification, POS return result, or active query, the system shall ensure that the background status is ultimately consistent and provide queryable records for customer service, operations, and finance.

3.14.2.4. Refund and Exception Compensation

Map to Business Requirement 3.14.1.7.

Refund capabilities should support executing corresponding processes according to payment methods. Cybersource still needs to support the processing strategy of Void first and then Refund; offline payment methods should also support retaining refund records in the backend and providing a manual intervention mechanism for abnormal transactions.

3.14.2.5. Authentication Dependency Management

Map to Business Requirement 3.14.1.8.

Before the official launch of each payment method, WestK must complete the corresponding third-party certification. Apollo and Cybersource must pass GP certification, and Octopus must pass the certification of the Octopus Company. The progress of certification will directly affect the launch schedule of payment functions, so it needs to be managed as a clear precondition dependency in the event.

3.14.2.6. Settlement and Financial Reconciliation

Map to Business Requirement 3.14.1.9.

All payment systems need to be included in daily settlement management. Apollo and Octopus adopt the manual settlement mode, requiring WestK or operations personnel to perform daily settlement; Cybersource adopts the automatic settlement mode. The system supports outputting reconciliation files.

3.14.3. Operating Model

This project adopts a multi-party collaborative payment operation model to achieve payment access, authentication, settlement, and operation management under different business scenarios of WestK.

WestK, as the business owner, is responsible for defining the scope of payment services, applicable scenarios, daily operational rules, and financial settlement requirements, and for promoting the activation of corresponding payment methods and business acceptance.

AnTank is responsible for providing payment access capabilities, back-end management capabilities, transaction status management, refund processing, payment inquiry, log retention, and report support, and cooperating with WestK to complete joint debug, testing, and pre-launch preparations with third-party institutions.

GP is responsible for the certification process related to Apollo and Cybersource, ensuring that payment access meets their certification requirements. Octopus Company is responsible for the certification process related to Octopus payments.

At the daily operation level, although online payment, back-end POS payment, and Kiosk payment are targeted at different roles and terminals, they should be uniformly incorporated into WestK's transaction management, daily settlement, and financial reconciliation systems.

3.14.4. Customer Journey

This section describes payment-related User Stories from the perspective of roles, reflecting WestK's business requirements in different payment channels.

No.	Category	Role	User Story
1	Online Payment	As an online ticketing user	I hope to securely complete bank card payment via Cybersource and promptly see accurate results after payment to confirm whether the order is effective.
2	BO Apollo Payment	As a frontline staff member of WestK	I hope that in the background or on-site service scenarios, the Apollo handheld POS machine can be used to complete payment collection, so as to flexibly handle the ticketing or supplementary payment requirements at different locations.
3	Kiosk Apollo Payment	As a self-service ticketing visitor	I hope that payment can be completed via Apollo POS on the Kiosk, so that the ticket purchase process can be independently completed at the fixed terminal.

4	Octopus Payment	As an on-site visitor	I hope to use Octopus to complete the payment so that I can use a familiar local payment method.
5	Payment Status Management	As a WestK operator	I hope to be able to query the final payment result even when the transaction status is unclear, so as to handle customer inquiries and abnormal orders.
6	Refund Processing	As a WestK backstage staff member	I hope to initiate a refund for eligible orders and view the refund progress and results to handle after-sales requirements.
7	Certification and Activation	As a project manager for WestK project	I hope to clearly understand the certifying parties and preconditions for going live corresponding to each payment method, so as to reasonably arrange the project plan.
8	Settlement and Reconciliation	As a financial staff member of WestK	I hope that all payment methods will be included in the scope of daily settlement and reconciliation, and that manual settlement and automatic settlement modes will be distinguished to facilitate financial reconciliation and auditing.

Table 3.14.4: Description for the User Story

3.14.5. Business Scenarios

3.14.5.1. Online Cybersource Payment Scenario

This scenario describes the business requirements for users to initiate order payment on the online front end and complete business operations such as bank card payment, payment confirmation, payment inquiry, and refund processing through Cybersource.

3.14.5.1.1. Requirement List

RID	Description	Priority
REQ-PAY-001	The system must support online front-end to initiate and complete bank card payments through Cybersource.	High
REQ-PAY-002	The system must support recording the mapping relationships between payment orders, business orders, and third-party transaction flows.	High
REQ-PAY-003	The system must support confirming Cybersource payment results through asynchronous notifications and proactive queries.	High

REQ-PAY-004	The system must support automatically closing unpaid orders after they time out.	High
REQ-PAY-005	The system must support executing refunds for Cybersource transactions in the background and adopt a processing strategy of Void first and then Refund.	High

Table 3.14.5.1.1: Requirement List of Online Cybersource Payment Scenarios

3.14.5.2. BO Apollo POS Payment Scenario

This scenario describes the business requirements for WestK staff to use the Apollo handheld POS machine to complete payment collection in the background or on-site service scenarios.

3.14.5.2.1. Requirement List

RID	Description	Priority
REQ-PAY-006	The system must support the integration of Apollo POS payment in the BO scenario.	High
REQ-PAY-007	The system must support the BO scenario using Apollo handheld POS machines for mobile payment collection.	High
REQ-PAY-008	The system must support background viewing of Apollo BO transaction results and status records.	High
REQ-PAY-009	The system must support the inclusion of Apollo BO transactions in the daily manual settlement process.	High

Table 3.14.5.2.1: Requirement List of BO Apollo POS Payment Scenario

3.14.5.3. Kiosk Apollo POS Payment Scenario

This scenario describes the business requirements for visitors to complete payments via Apollo POS on the Kiosk self-service machine.

3.14.5.3.1. Requirement List

RID	Description	Priority
REQ-PAY-010	The system must support accessing Apollo POS payment in Kiosk scenarios.	High
REQ-PAY-011	The system must support viewing Kiosk Apollo transaction results and status records in the background.	High

Table 3.14.5.3.1: Requirement List of Kiosk Apollo POS Payment Scenario

3.14.5.4. Octopus Payment Scenario

This scenario describes the business requirements for WestK to connect to Octopus payment in a specified on-site business scenario.

3.14.5.4.1. Requirement List

RID	Description	Priority
REQ-PAY-012	The system must support integration with Octopus payment.	High
REQ-PAY-013	The system must support viewing Octopus transaction status and related records in the background.	High
REQ-PAY-014	The system must support the inclusion of Octopus transactions in the daily manual settlement process.	High

Table 3.14.5.4: Requirement List of Octopus Payment Scenarios

3.14.5.5. Certification, Settlement, and Reconciliation Scenarios

This scenario describes the business requirements for pre-launch certification, daily settlement, and financial reconciliation of each payment method.

3.14.5.5.1. Requirement List

RID	Description	Priority
REQ-PAY-015	GP certification must be completed before the launch of Apollo's payment capabilities.	High
REQ-PAY-016	GP certification must be completed before the launch of Cybersource payment capabilities.	High
REQ-PAY-017	Octopus payment capability must complete Octopus Company certification before going live.	High
REQ-PAY-018	The system must support Cybersource in adopting the automatic settlement mode.	High
REQ-PAY-019	The system must support Apollo and Octopus in using the manual settlement mode.	High
REQ-PAY-020	The system must support the generation of payment and refund documents to meet the needs of operational and financial audits.	High

Table 3.14.5.5.1: Requirement List of Certification, Settlement, and Reconciliation Scenarios

3.15. Ticketing Operation Centre

3.15.1. Business requirements

From a business perspective, WestK requires this system to implement a visual dashboard through BI tools, while meeting the requirements for multi-BU permission management, data integration, system interaction, etc.

WestK requires the Ticketing Operation Centre system to achieve visualization capabilities through BI tools, to display data in charts, view sales distribution, trends, member growth and retention, as well as on-site operational status. The BI dashboard includes:

- **Ticketing Dashboard:** Through the board display omni-channel ticket sales trends, distribution, channel proportion and ticket conversion, through multi-dimensional data analysis quickly review sales weaknesses, timely support marketing decision-making, achieve dynamic iteration of operational strategies, effectively improve ticket sales and overall revenue.
- **User and Membership Dashboard:** The dashboard centrally displays core indicators such as overall users, new users, new members, retention, renewal, churn, and paid card members, intuitively presenting the health of the member lifecycle. By analysing the member revenue structure, relying on data insights to optimize benefit configuration, precise outreach, and renewal incentive strategies, it improves member operational efficiency, continuously enhances user stickiness, and increases the overall member lifecycle value and retention rate.
- **Daily Operations Dashboard:** Real-time summary of daily entry passenger flow, time flow, support operation personnel to intuitively grasp the on-site congestion and service load. Dynamically adjust ticket inventory and release rhythm based on real-time sales progress, optimize on-site resource allocation such as security check, guide, merchants, and security based on passenger flow distribution, improve on-site order and visitor experience, and ensure safe and efficient operation.
- **Event Details Dashboard:** Through the Dashboard with the event as the core dimension to centrally display the sales progress, enabling intuitive understanding of the operational status of each event, helping managers timely monitor event progress, sales dynamics, and resource input-output, and improving the efficiency of overall decision-making and refined management.
- **Equipment monitoring dashboard (Monitoring Dashboard):** Through the dashboard overview of the operation of the equipment, the online status of the equipment can be monitored in real time, and the abnormal disconnection of the equipment can be found in time.

3.15.2. Business Requirements Analysis

This section further analyses and elaborates on the business requirements described in Section 3.15.1, clarifying business intent, operational considerations, and functional expectations.

Maps to Business Requirement 3.15.1.

- **Ticketing Dashboard:** By consolidating scattered data from sales tables, inventory tables, and verification tables onto a single BI Dashboard , using statistical methods such as calculating absolute values, relative values, and cumulative values, it centrally presents the trends, distributions, and interrelationships of sales, inventory, and verification in the form of charts, revealing the effectiveness of ticketing operations to improve decision-making efficiency and accuracy in ticketing operations.
- **User and Membership Dashboard:** By consolidating scattered data from sales tables and member tables onto a single BI dashboard, using statistical methods such as calculating absolute values, relative values, and cumulative values, it presents in a graphical manner overall users, new users, paid card-purchasing members, member retention growth, member consumption, trends and distribution of member discount usage, reveals the effectiveness of member operations, and continuously improves member operation services.
- **Daily Operations Dashboard:** By loading in real-time the scattered data dispersed across sales tables, inventory tables, and verification tables, and unifying them on a single BI Dashboard , through statistical methods of calculating absolute values, relative values, and cumulative values, it centrally displays the trends of sales, inventory, and verification over time, as well as the distribution of business dimensions in the form of charts, to reflect on-site operations in real-time and flexibly improve on-site operational efficiency.
- **Event Details Dashboard:** To achieve refined management of event sales, it is necessary to support display and analysis by event dimension through the Dashboard, intuitively presenting the sales order volume, ticket booking quantity, and sales amount of each event. The system should provide clear data visualization views to facilitate managers in grasping the overall event situation in real time, promptly detecting anomalies, tracking sales progress, and providing data support for event decision-making, resource allocation, and sales strategy optimization, thereby enhancing the timeliness and effectiveness of event management.
- **Equipment Monitoring Dashboard (Monitoring Dashboard):** The system shall provide an equipment monitoring dashboard to support centralized and unified overview of on-site equipment. Managers can use this dashboard to monitor the online status of equipment in real time, quickly identify issues such as equipment disconnection and abnormal offline status, and ensure timely handling. Meanwhile, the dashboard can intuitively display the corresponding remaining ticket quantity of each device, monitor in real time whether the remaining tickets are sufficient, avoid situations where insufficient remaining tickets affect business processing, achieve integrated monitoring of equipment status and ticket inventory, and ensure the continuous, stable, and efficient operation of ticketing services.

3.15.3. Operation Model

This project adopts the operation mode of "fixed Dashboard analysis as the main + self-service analysis as a supplement":

- **AnTank Team:** Responsible for data interface development, data warehouse construction, Extract-Transform-Load (ETL) process maintenance, dashboard development, and system underlying permission synchronization to ensure the accuracy and timeliness of data sources.

- WestK Business Team: As core data users, relying on the BI tools provided by the system to conduct various types of dashboards viewing, data insights, support daily operational decision-making, and significantly improve data utilization efficiency and business response speed.

3.15.4. Customer Journey

This section describes the typical journeys of users in different roles when using WestK BI Dashboard System. User Stories are used to clarify user requirements, goals, and expected outcomes.

No.	Category	Role	User Story
1	Data Access	As a BU operator	I hope to log in to the system through a unified account to quickly access the exclusive Data Dashboard of my business unit.
2	Ticketing Analysis	As a ticketing manager	Desire to view sales data broken down by channel and ticket type to analyse the sales show of each channel and optimize strategies.
3	Ticketing Analysis	As a ticketing manager	We hope to see real-time trends on the sell-out rate dashboard to promptly identify popular events and adjust inventory or marketing strategies.
4	Member Analysis	As a member operation	Wish to view the analysis of member growth and retention dashboard to evaluate the effectiveness of member activities.
5	Member Analysis	As a member operation	When drilling down to view member consumption contributions, we hope to be able to filter by card type to identify high-value member groups.
6	Permission Management	As a system administrator	It is hoped that the system can automatically synchronize the permission changes of AnTank to reduce the workload of manual maintenance and ensure data security.
7	Decision Support	As the Group's management	Hope to see the summary of core operating indicators across BUs on the home page dashboard to keep track of the overall operational status.

Table 3.15.4: Description of User Story

3.15.5. Business Scenarios

3.15.5.1. Data Dashboard Classification

The following table describes the Dashboard board categories that users can access.

No.	Dashboard Classification	Description
1	Ticketing (Ticketing Dashboard) Board	Visualization of ticket sales and usage statistics, with indicators including sales revenue, sales conversion rate, and admission verification status.
2	User and Member (Membership Dashboard) Dashboard	Visualization of ticket overall users, new users, member growth and retention, and member consumption statistics, with indicators including overall users, new users, new members, paid card-purchasing members, renewed members, member consumption amount, and member discount usage.
3	On-Site Operations (Daily Operations Dashboard) Dashboard	Statistics on ticket sales, inventory, and admission verification, with indicators including event sales revenue, inventory level, and number of admissions.
4	Project Details (Event Dashboard) Dashboard	Analyse event sales data and order data from the event dimension to analyse the event's operating status.
5	Device Monitoring (Monitoring Dashboard) Board	Monitor the operating status of the device and promptly observe whether any device has gone offline.

Table 3.15.5.1: Dashboard Classification Requirements

3.15.5.2. Requirement List

The following table describes the user BI Dashboard detailed requirements.

RID	Description	Priority	Related Process
REQ-DSB-TKT-001	Ticketing (Ticketing Dashboard) Board 1. Display verification indicators for ticketing operations such as ticket sales revenue, sales volume, attendance rate, etc., and show their year-on-year and month-on-month comparisons. 2. Display the sales ranking of events. 3. Display the verification ranking of entry points.	High	3.15.5.1: #1

	4. Sales trends and distribution of each BU, sales channel, and ticket type. 5. Multi-dimensional cross-analysis with BU, sales channels, and ticket types as the core dimensions.		
REQ-DSB-MEN-001	User and Member (Membership Dashboard) Dashboard 1. Displays core indicators of membership operations, such as total number of users, new users, paid card-purchasing members, active members, new members, renewed members, etc., as well as their year-on-year and month-on-month comparisons. 2. Display the distribution of each membership status for each BU. 3. Comparison of consumption situations among BU members. 4. Comparison of consumption situations of various membership cards. 5. Usage of each member's discounts.	High	3.15.5.1: #2
REQ-DSB-OST-001	On-site Operations (Daily Operations Dashboard) Dashboard 1. Display the core indicators of daily operations, including real-time ticket sales, planned number of attendees, actual number of attendees, and overall inventory levels. 2. Trends of ticket sales volume, conversion rate, and inventory level by time period. 3. Inventory levels by dimensions such as sales channels, events, ticket types, etc. 4. Verification trends for admission during each time period. 5. Ranking of real-time event ticket sales volume.	High	3.15.5.1: #3
REQ-DSB-EDL-001	Event Details (Event Dashboard) Dashboard 1. Core event indicators such as the number of sales orders and the number of sales tickets with the event as the core dimension. 2. Trends in the number of visitors during each time period. 3. Sales proportion of each channel. 4. Session sales situation.	High	3.15.5.1: #4

REQ- DSB- MNT-001	Device monitoring (Monitoring Dashboard) board 1. Monitoring of the online and offline status of each hardware device. 2. Operational monitoring of key components such as CPU and memory of each hardware device. 3. Error log statistics for each hardware device. 4. Response capabilities of each hardware device.	High	3.15.5.1: #5
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Table 3.15.5.2: Dashboard Detailed Requirements

3.16. B2B Portal (Tour, School Tour, and Academy)

3.16.1. Business Requirements

3.16.1.1. User Registration and Organization Authentication

WestK requires the establishment of a dedicated B2B Portal to provide registration and identity authentication services for different types of institutional users, including schools, travel agencies, NGOs, and other B2B partners. The system shall support account creation based on institutional information and provide appropriate verification and approval processes to confirm institutional identity.

The Portal shall assign operational permissions according to institution types to ensure users can only access functions relevant to their business roles, and account activation shall follow WestK's established onboarding and approval procedures.

3.16.1.2. School Booking

WestK requires the system to provide dedicated visit booking services for schools in Hong Kong, China. Teachers or authorized school representatives shall be able to schedule student visits, educational programmes, and related activities through the Portal. The system shall support school directory management, including the entry and maintenance of school information. The system shall also allow configuration of annual visit quotas and daily booking capacities for each school, enabling users to view the current remaining availability during the booking process.

Teachers or school representatives shall be able to select visit dates, time slots, and participant numbers, and complete the required information forms with supporting documents where applicable. The system shall support submission of both preferred and alternative booking time slots and provide a waiting list mechanism when capacity limits are reached.

Booking requests shall become effective after review and approval by authorized staff. Once approved, users shall be able to complete the required payment to confirm the reservation. The system shall support the issuance of admission credentials and record actual attendance during admission.

The system shall support recording actual attendance to track the utilization of schools' annual visit quotas. Reasonable deviations between booked and actual

attendance may be permitted, while quota usage may be considered for future quota allocation. If booking adjustments are required, schools may submit updated booking requests for further review, and backend staff may also handle such adjustments when necessary. NGO organizations that have completed identity verification and account activation may also submit booking requests through the Portal.

3.16.1.3. Tour Booking

WestK requires the system to provide group visit booking services for travel agencies and other organized groups. After completing identity authentication, institutional users shall be able to browse available group programmes through the B2B Portal, select visit dates and participant numbers, and optionally add services such as guided tours, workshops, parking, or tour bus arrangements. The system shall support combined booking of tickets and related services.

For programmes with capacity limits, the system shall prompt users to adjust their booking arrangements when the number of participants exceeds the allowed capacity.

After a group booking request is submitted, backend staff may communicate with the institution to confirm or adjust the booking details. Once the application is approved, users shall be able to complete online payment to confirm the reservation. The system shall support the generation of admission credentials and record actual attendance during admission. If the actual number of participants exceeds the booked quantity, the additional charges may be handled according to operational policies.

For certain B2B partners such as small travel agencies or hotel concierge services, the system shall support institutional account management and different settlement arrangements to accommodate partner operations. Institutions may complete ticket issuance or booking confirmation through the group booking portal or other operational channels. If a group fails to attend as scheduled, the situation may be handled by staff in accordance with operational policies. Unconfirmed or released quotas shall become available to users on the waiting list.

3.16.1.4. Academy Programme Registration

WestK requires the system to support online application workflows for Academy short-term educational programmes. These programmes are typically organized on a weekly basis and primarily target professionals in the cultural and creative industries who possess a bachelor's degree or equivalent professional qualifications. Applicants shall be able to browse programme information, schedules, and venues before submitting applications through the system.

The system shall support multilingual application forms. Applicants shall complete personal information and submit relevant supporting documents for eligibility verification. Proxy applications shall not be allowed, and each applicant may submit only their own application.

Applications shall undergo a review process conducted by backend staff, including document verification and interview evaluation where applicable. Review outcomes may include admitted, waiting list, or not approved. The system shall support sending notifications to applicants regarding review outcomes and allow staff to send

additional communications when required. Applicants who are admitted shall complete online payment to confirm their place in the programme.

Each programme shall have a defined enrolment capacity. When the capacity limit is reached, the system shall support a waiting list mechanism and notify applicants when places become available. Backend staff shall be able to maintain application records and add review notes. The system shall also support recording special participant requirements, such as accessibility needs. Online refund is generally not supported, and exceptional cases may be handled by staff.

3.16.1.5. Booking Approval and Order Management

WestK requires the system to support comprehensive booking approval and order management workflows. After institutional users submit booking or registration requests, backend staff shall be able to review the relevant order information and perform approval or rejection actions. If an application is rejected, institutional users may revise the information and resubmit the request.

The system shall support order status management to reflect the progress of orders across different processing stages, such as pending review, rejected, confirmed, pending payment, paid, and completed. The system shall also provide notification mechanisms to inform relevant users at key stages of the workflow, with configurations aligned to the responsibilities of different operational teams.

Institutional users shall be able to view their order status through the Portal and download related documents such as e-tickets or payment records. If order modifications are required, institutional users may submit an updated booking request and re-enter the approval workflow. After approval, backend staff may send order confirmation notifications to the institution. The system shall also support multiple staff accounts to accommodate approval and management activities across different operational teams.

3.16.1.6. Inventory and Quota Management

WestK requires the system to provide centralized inventory and quota management for different booking programmes. The system shall allow administrators to configure booking capacity limits for different programmes and time slots. When capacity limits are reached, the system shall stop accepting new booking requests or provide a waiting list mechanism.

The system shall support reserving quotas during the order review process to prevent booking conflicts. If a reservation is not ultimately confirmed, the system shall allow the corresponding quota to become available to users on the waiting list. The system shall also support temporarily suspending booking for specific time slots when required for operational purposes.

For school bookings, the system shall support configuration of annual visit quotas and daily booking capacities, with the option to set certain quotas as unlimited where appropriate. For group bookings, the system shall support separate capacity limits for different programme types. When the number of tickets selected by individual visitors meets the group booking threshold, the system shall prompt users to proceed through the group booking channel.

After a booking request is submitted, the system shall update the corresponding quota status so that users can view the current remaining availability.

Administrators shall also be able to adjust inventory in the backend to accommodate cross-channel booking conflicts or other operational requirements.

3.16.1.7. Payment and Financial Handling

WestK requires the system to support online payment capabilities. After booking or registration requests are approved, institutional users shall be able to complete payments directly through the system. The system shall support mainstream electronic payment methods such as credit cards, Alipay, and WeChat Pay, leveraging the existing payment channels used for individual ticketing.

The system shall be capable of processing orders of varying amounts, including high-value group bookings. Appropriate payment handling mechanisms shall be supported to ensure that high-value transactions can be completed successfully.

The system shall support order payment status management and allow backend staff to view orders with incomplete payments for follow-up actions. The system shall also support sending payment notifications to institutions after booking applications are approved.

The system shall support multiple payment and settlement methods to accommodate the business needs of different types of institutional users. The system shall also support offline payment scenarios and allow order payment status to be updated after financial confirmation.

3.16.1.8. Invoice, Receipt and Financial Records

WestK requires the system to support invoice and receipt management functions. After institutional users complete online payments, the system shall automatically generate electronic receipts for download. The system shall also allow institutional users to request standardized electronic invoices to support financial reimbursement requirements for schools, enterprises, and other organizations.

The system shall maintain complete financial records related to orders, including payment records, invoice issuance records, and refund records, ensuring traceability and auditability. For large-value group orders, review records and payment notification logs shall also be retained to support financial reconciliation and auditing processes.

The system shall support periodic reconciliation with B2B partners and generate corresponding financial records or invoices based on actual ticket issuance to ensure consistency between financial data of both parties.

3.16.1.9. Reporting and Operational Data

WestK requires the system to provide operational reporting and analytics capabilities to support business analysis and operational management. The system shall be able to generate reports across multiple dimensions, including booking type (such as school bookings, group bookings, and Academy programmes), time, venue, and programme categories. Key operational metrics may include booking volumes, actual attendance, sales revenue, and quota utilization.

The system shall support the generation of standardized reports to meet the needs of operational monitoring and financial analysis. Different business teams shall be able to access relevant reports according to their operational requirements.

The system shall also support data statistics and list exports for different business scenarios, such as school bookings and Academy programme registrations, to facilitate operational analysis and management.

3.16.2. Business Requirement Analysis

This section provides further analysis and clarification of the business requirements described in Section 3.16.1. Each subsection elaborates on the business intent, operational considerations and functional expectations supporting the B2B Portal solution for Tour Booking, School Booking and Academy Programme Registration.

3.16.2.1. User Registration & Organisation Authentication

Map to Business Requirement 3.16.1.1.

The B2B Portal shall support registration and authentication mechanisms for institutional users, including team accounts associated with organizations such as travel agencies, schools, NGOs, or other B2B partners.

- **Team User Registration:** Users representing institutional teams shall be able to register accounts through the Portal using their email addresses. The system shall support verification of user identity during the registration process.
- **Organisation Creation:** If a corresponding organisation does not yet exist in the system, users shall be able to submit organisation information (such as organisation name, organisation type, and contact information) together with supporting documents. The submission shall initiate an organisation registration request.
- **Organisation Approval Workflow:** Organisation registration requests shall undergo manual review by backend staff. Upon approval, the system shall create the organisation record and associate the initial registered email account with the organisation. If the request is rejected, the applicant shall receive a notification indicating the reason for rejection.
- **Account Creation under Existing Organisation:** For organisations that already exist in the system, administrators may create additional user accounts under the organisation through the B2B Portal.
- **User Account Approval:** Newly created user accounts shall be subject to a review process before activation. Backend staff shall be able to approve or reject account requests and notify applicants of the review outcome.
- **User Login:** After account approval, users shall be able to log in to the B2B Portal using their registered email address and proceed with relevant booking or application activities.

3.16.2.2. Booking & Enrolment Application Management

Map to Business Requirement 3.16.1.2, 3.16.1.3 and 3.16.1.4.

The B2B Portal shall support institutional users in browsing available programmes and submitting booking or enrolment applications for different service scenarios, including school visits, group bookings, and Academy programme registrations.

- **Programme Discovery:** Users shall be able to browse available programmes, activities, and courses through the Portal, including relevant information such as schedules, venues, and programme descriptions.
- **Booking and Enrolment Application:** Users shall be able to select visit dates, sessions, participant numbers, and other relevant options where applicable, and submit booking or enrolment requests through the Portal.
- **Application Form Submission:** The system shall provide structured application forms to capture required information, such as contact details, organisational information, and programme-specific requirements.
- **Supporting Document Submission:** The system shall support submission of supporting documents where required for verification or eligibility purposes.
- **Waiting List Mechanism:** When programme capacity is reached, the system shall support a waiting list mechanism that allows users to register interest for available slots.
- **Application Status Tracking:** Users shall be able to view the status of submitted booking or enrolment requests through the Portal.
- **Application Update and Resubmission:** Where adjustments are required, users shall be able to submit updated applications and re-enter the approval workflow.

3.16.2.3. Inventory & Quota Management

Map to Business Requirement 3.16.1.6.

The system shall provide unified management of booking capacity and quotas to support different programme types and booking scenarios.

- **Capacity Configuration:** The system shall support configuration of booking capacity limits for different programmes and sessions.
- **Availability Validation:** During the booking process, the system shall validate inventory availability and ensure that booking requests do not exceed the configured capacity.
- **Quota Reservation Mechanism:** The system shall support temporary reservation of quotas during the booking and approval process, with automatic release of reserved quotas if the booking is not confirmed.
- **Waiting List Management:** When available capacity is reached, the system shall support a waiting list mechanism that allows users to register for potential available slots.
- **School Quota Management:** For school bookings, the system shall support management of annual free visit quotas and daily booking quotas.
- **Administrative Quota Adjustment:** Backend administrators shall be able to adjust inventory and quota configurations based on operational requirements.

3.16.2.4. Booking Approval

Map to Business Requirement 3.16.1.5.

Booking or enrolment applications submitted through the Portal shall undergo backend review.

- The system shall support backend manual review of booking applications.
- Approve/Return/Reject Review results shall include Approve/Return/Reject options.
- Returned applications shall allow modification and resubmission.
- Review outcomes shall be communicated through system notifications or email.
- All review actions shall be recorded for operational traceability.
- Upon approval, a formal order shall be generated.

3.16.2.5. Order & Payment Handling

Map to Business Requirement 3.16.1.7.

The system shall support order creation and payment handling for approved bookings.

- Formal orders shall be created upon approval of booking applications.
- Users shall be able to complete online payments through the system.
- Multiple payment methods including credit cards and electronic payments shall be supported.
- Certain B2B partners may settle payments through prepaid accounts or credit limits.
- Upon successful payment, the order status shall be updated, and inventory capacity shall be locked.
- If payment expires, the reserved capacity shall be automatically released.

3.16.2.6. Invoice, Receipt & Financial Record Handling

Map to Business Requirement 3.16.1.8.

The B2B Portal shall support invoice issuance, receipt generation, and financial record management to ensure that institutional users can obtain proper financial documentation while enabling WestK to maintain accurate financial traceability and reconciliation.

- The system shall automatically generate electronic receipts after successful payment and allow institutional users to download the receipts through the Portal.
- The system shall support requests for standard electronic invoices to meet the financial reimbursement requirements of schools, enterprises, and other institutions.
- The system shall maintain complete financial records related to orders, including payment records, invoice issuance records, and refund records, ensuring full traceability.

- For high-value group orders, the system shall retain approval records and payment notification logs to support financial reconciliation and auditing processes.
- The system may support periodic financial reconciliation with B2B partners based on actual ticket issuance or completed orders.

3.16.2.7. Admission & Attendance Management

Map to Business Requirement 3.16.1.2, 3.16.1.3 and 3.16.1.5.

The system shall support admission credential generation and attendance tracking.

- The system shall generate admission credentials such as QR codes or e-tickets.
- Group bookings may generate either group-level or individual admission codes.
- On-site admission validation shall be supported via QR code scanning.
- Actual attendance numbers shall be recorded during admission.
- Admission data shall be synchronised to backend systems for operational reporting.

3.16.2.8. Reporting & Operational Analytics

Map to Business Requirement 3.16.1.9.

The system shall provide reporting and operational analytics capabilities.

- The system shall provide statistical reporting for bookings, applications and orders.
- Reports may be generated by programme, time period, or organisation type.
- Reports shall be exportable in Excel or other standard formats.
- Academy programme enrolment data and application forms shall be exportable.
- Operational data shall be updated in near real time.

3.16.3. Operating Model

This initiative adopts a collaborative operating model to support institutional booking and programme registration through the B2B Portal. The model involves coordinated responsibilities across WestK, the system technology provider, and operational teams responsible for managing bookings and programmes.

System Technology Provider

The system technology provider is responsible for delivering and operating the technology platform that supports the B2B Portal.

- Provide the core booking and enrolment system capabilities, including programme setup, session configuration, inventory and quota management, order lifecycle management, and ticket or admission credential generation.

- Develop and maintain the B2B Portal interface, including user account management, booking workflows, and user interaction interfaces.
- Integrate with external systems where required, such as payment gateways, ticketing systems, or admission validation systems.
- Ensure system availability, security, and show monitoring, and provide technical support in accordance with agreed service levels.

WestK Business and Operations Teams

WestK acts as the business owner and is responsible for defining the operational policies and managing the programmes available through the B2B Portal.

- Define programme offerings, booking rules, pricing strategies, and quota policies for different types of institutional bookings.
- Manage programme information including tour programmes, school visit activities, and Academy courses displayed on the Portal.
- Review booking and enrolment applications submitted by users and manage order approval workflows.
- Manage operational processes such as attendance verification, quota adjustments, and operational reporting.

Finance and Settlement Management

WestK's finance team is responsible for financial settlement, reconciliation, and accounting processes associated with bookings and programme registrations.

- Receive payment records generated through the system and perform financial reconciliation and accounting recognition.
- Manage invoice issuance, receipt management, and financial documentation related to institutional bookings.
- Coordinate with operational teams to resolve payment discrepancies or financial exceptions where required.

3.16.4. Customer Journey

This section describes what users can do after entering the B2B Portal.

User stories are used to describe the needs, goals, and expected outcomes of different user roles interacting with the Portal. These stories illustrate how institutional users, individual applicants, and WestK staff interact with the system to complete bookings, programme registrations, and operational management tasks.

Through the Portal, users can complete the full booking or registration lifecycle, including programme discovery, application submission, approval, payment, and admission management.

3.16.4.1. Tour Booking Customer Journey

No.	Category	Role	User Story
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1	Registration	As a travel agency user	I want to register my organization and verify my identity, so that I can access the B2B Portal and create group bookings.
2	Programme Discovery	As a travel agency user	I want to browse available tour programmes, including visit schedules and event information, so that I can select suitable programmes for my group.
3	Booking Application	As a travel agency user	I want to select visit dates, participant numbers, and optional services such as guided tours or workshops, so that I can submit a group booking request.
4	Booking Application	As a travel agency user	I want to submit my booking request and receive confirmation after review, so that my group visit can be scheduled properly.
5	Payment	As a travel agency user	I want to complete payment after my booking request is approved, so that my reservation can be confirmed.
6	Order Management	As a travel agency user	I want to access my booking details and download admission credentials, so that I can prepare for the group visit.
7	Admission	As a group organiser	I want to receive a group QR code or individual QR codes for my participants, so that my group can enter the venue efficiently.
8	Attendance	As a WestK staff member	I want to scan the QR code during admission and record the actual attendance, so that visit data can be tracked for operational purposes.

Table 3.16.4.1: Description for the Tour Booking Customer Journey

3.16.4.2. School Booking Customer Journey

No.	Category	Role	User Story
1	Registration	As a school representative	I want to log in using my institutional email address and verification code, so that I can access the school booking portal.
2	Programme Discovery	As a school representative	I want to browse available educational programmes and visit sessions, so that I can plan student visits.
3	Quota Checking	As a school representative	I want to view my school's available quotas and remaining capacity, so that I can ensure my booking complies with quota limits.

4	Booking Application	As a school representative	I want to select visit dates, student numbers, and supporting information, so that I can submit a school visit booking.
5	Waitlist	As a school representative	I want to join a waiting list if the quota is full, so that my school may still obtain a visit slot if availability becomes available.
6	Payment	As a school representative	I want to complete payment for any applicable programme fees, so that the visit can be confirmed.
7	Admission	As a school representative	I want to receive group admission credentials, so that students and teachers can enter the venue smoothly.
8	Attendance Tracking	As WestK staff member	I want to record the actual attendance of students and teachers during admission, so that school quota usage can be tracked.

Table 3.16.4.2: Description for the School Booking Customer Journey**3.16.4.3. Academy Programme Registration Customer Journey**

No.	Category	Role	User Story
1	Programme Discovery	As an Academy applicant	I want to browse available Academy courses, including course descriptions and schedules, so that I can determine whether the course is suitable for me.
2	Application	As an Academy applicant	I want to complete an application form and submit personal information, so that I can apply for the programme.
3	Document Submission	As an Academy applicant	I want to upload supporting documents such as academic certificates or employment proof, so that my eligibility can be verified.
4	Application Review	As a WestK staff member	I want to review applications and determine whether applicants should be admitted, waitlisted, or rejected, so that suitable participants can be selected.
5	Admission Confirmation	As an admitted applicant	I want to receive a notification confirming my admission, so that I can proceed with payment.
6	Payment	As an admitted applicant	I want to complete online payment within the required timeframe, so that I can secure my place in the programme.

7	Admission Credential	As a course participant	I want to receive a QR code or admission credential, so that I can access the programme sessions.
8	Attendance	As WestK staff member	I want to verify participant attendance during the course period, so that programme participation can be tracked.

Table 3.16.4.3: *Description for the Academy Programme Registration Customer Journey*

3.16.5. Business Scenarios

This section describes the key business scenarios supported by the B2B Portal.

The scenarios illustrate how different types of users interact with the system across the Tour Booking, School Tour Booking, and Academy Programme Registration services.

Each scenario provides an overview of the operational workflow and key functional components involved in the booking or enrolment process. Detailed process diagrams and requirement mappings are provided in the following subsections to illustrate the end-to-end operational flow supported by the system.

3.16.5.1. Tour Booking

Tour Booking is a group visit and event booking function provided through the B2B Portal.

It is primarily designed for institutional customers such as travel agencies, corporate groups, and other organised visitors.

Through this function, users can browse available guided tour programmes, select visit dates and sessions, specify group size, and submit booking applications through the Portal. After submission, the application is reviewed by backend staff. Once approved, the system generates a formal order and proceeds to subsequent processes including payment, admission credential generation, and on-site admission validation.

The system applies a unified inventory and approval mechanism to manage programme capacity, session availability, and related services. It also supports group-level admission credentials, admission validation at the venue, and recording of actual attendance figures.

All booking applications, orders, and admission records are centrally managed within the backend system, enabling WestK to maintain an end-to-end operational workflow covering booking submission, approval, fulfilment, and admission management.

3.16.5.1.1. User Process - General Process

The following process describes the general user process of tour booking.

No.	Component	Description
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1	Access the Portal and Enter Tour Booking	The user accesses the designated venue Portal website via a web browser, selects the “Tour Booking” entry point, and enters the group visit or event booking flow.
2	View Programmes Available for Tour Booking	The system displays tour programmes available for group booking, including programme name, overview, bookable date range, session time, location, images, and booking fees. Programme information may be browsed without login; login is required to proceed with booking.
3	Group User Login/Organisation Registration	The user enters the login page and logs in using a registered email address associated with an organisation. If the email is not associated with an organisation, the user may initiate an organisation registration request and enter the corresponding registration workflow.
4	Email Verification	The system sends a verification code to the user's email address. The user enters the verification code to complete authentication and access the Portal.
5	Select Programme, Date, Session and Ticket Type	After login, the user enters the booking flow, selects a bookable date from the calendar, views the corresponding sessions, selects a session, and enters the required booking quantity.
6	Check Session and Ticket Type Inventory	The system validates whether inventory for the selected session and ticket types is sufficient. If availability is insufficient, booking is blocked and the user may choose to join the waitlist; if sufficient, the user proceeds to the next step.
7	Complete Booking Details and Reserve Inventory	The system reserves inventory for the selected session and ticket types. The user completes booking details including participant numbers, contact information, additional service requirements (e.g. guided tours or catering), coach information, and supporting remarks or documents, and then submits the booking application.
8	Manual Review of Booking Application	Staff manually review the booking application. Rejected applications are cancelled and inventory is released; returned applications are sent back to the user for amendment; approved applications proceed to order creation.
9	Create Order	The system creates a formal order in pending-payment status based on the approved booking application.

10	Send Review Result and Payment Notification	After the order is created, the system sends an approval notification email to the group contact.
11	Log in to the Portal to Complete Order Payment	The group contact logs in to the Portal and completes payment using supported electronic payment methods. After successful payment the order status is updated to confirmed; if payment is not completed within the allowed time, the order is cancelled and inventory is released.
12	Payment Completed and Admission Credential Issued	After successful payment, the order is confirmed. The system may generate a group admission credential with QR code for collective entry, or support sending individual admission codes to participants via email.
13	On-Site Admission Validation	Upon arrival at the venue, staff scan the admission credential using validation devices and record the actual number of attendees.
14	Print Individual Ticket Code	If the programme involves multiple distributed points within the venue, the group leader may proceed to the box office to print individual ticket codes based on the group order and distribute them to participants.
15	Admission Validation within the Venue Using Individual Ticket Code	Participants use individual ticket codes to complete admission validation at designated locations within the venue and participate in the related activities.

Table 3.16.5.1.1: Description of the User Process of Tour Booking

3.16.5.1.2. User Process - Group Management

The following process describes Group Management.

No.	Component	Description
1	Access the System Backend and Enter the “Group Management” Menu	Backend staff log in to the system backend and access the “Group Management” menu. The system supports searching, editing, and deleting existing group records.
2	Add a Group	Backend staff may add new groups by manually entering group information or importing group data using a predefined spreadsheet template. Basic information must be provided, including group type, qualification information, and contact details.
3	Group Added Successfully	After a group is successfully created, the designated group contact may log in to the B2B

		Portal using the registered email address and perform related operations, such as placing booking orders or managing group member accounts.
4	Select a Group to Perform Related Operations	The system allows staff to select a group and manage related operational data, including viewing group user account applications, waitlist applications, and booking applications associated with the group. Staff may also access group order records and manage user accounts under the group, including adding, editing, or deleting user records.

Table 3.16.5.1.2: Description of the User Process of Group Management

3.16.5.1.3. User Process - Group Registration

The following process describes the Group Registration.

No.	Component	Description
1	Access Portal and Initiate Team Registration	When accessing the B2B Portal, users may select the “Team Registration” entry to create a new team or organisation account for subsequent group booking and related operations.
2	Submit Team/Organisation Information	The user completes the team or organisation information form (such as organisation name, organisation type, and contact details) and uploads supporting qualification documents before submitting the team registration request.
3	Backend Manual Review of Team Qualifications	Backend staff manually review the team registration request, including the submitted organisation information and supporting qualification documents.
4	Team Registration Result Notification	If the application is approved, the system creates the organisation record and sends an approval notification to the applicant. If rejected, a rejection notification stating the reason is sent.
5	Team Contact Login	After the team registration is approved, the team contact may log in to the B2B Portal using the registered email address and proceed with relevant booking or operational activities.
6	Team Administrator Creates Additional User Accounts	After the organisation is created, the team administrator may create additional user accounts for team members through the Portal backend and manage their access permissions.

Table 3.16.5.1.3: Description of User Process of Group Registration

3.16.5.1.4. User Process - Waitlist

The following process describes the Waitlist User Process.

No.	Component	Description
1	Submit Waitlist Registration	When inventory for the selected session or ticket types is insufficient, the group user may submit a waitlist registration to be considered when capacity becomes available.
2	Backend Review of Waitlist Registration	Backend staff review the waitlist registration and determine whether capacity may be allocated to the application based on inventory availability and operational considerations.
3	Waitlist Approval Notification	If the waitlist application is approved, the system reserves the corresponding quota and automatically creates a booking order, and a notification email is sent to the group contact. If rejected, a notification stating the reason is sent.
4	View Order and Proceed with Subsequent Steps	The group user logs in to the Portal and views the order generated from the waitlist application in the order management page and proceeds with subsequent steps such as payment or admission preparation.

Table 3.16.5.1.4: Description of User Process of Waitlist

3.16.5.1.5. User Process - Booking Application Amendment

The following process describes the Booking Application Amendment User Process.

No.	Component	Description
1	Return Booking Application	If the booking application contains incomplete or non-compliant information, backend staff may return the application for amendment. The system sends a notification email to the group contact indicating the reason for the return. During the amendment period, the reserved inventory remains locked.
2	Group Contact Initiates Booking Amendment	Before the order payment is completed, the group contact may also initiate an amendment to the booking application. The user may access the booking management page in the Portal, adjust the booking information, and resubmit the application.

3	Resubmit Booking Application	The group user logs in to the Portal and views the returned or amendable booking application, updates the relevant booking information, and resubmits the application for review.
4	Backend Re-Review of Booking Application	Backend staff review the amended booking application again to verify participant numbers, qualifications, and event requirements. If the application is amended after approval, it must re-enter the approval workflow. If approved, the system creates or updates the order accordingly. If the application still does not meet the requirements, it may be returned for further amendment or rejected.

Table 3.16.5.1.5: Description of User Process of Booking Application Amendment

3.16.5.1.6. Tour Booking Requirement List

RID	Description	Priority	Related Process
REQ-BP-TB-001	The system shall allow users to access the Tour Booking entry point through the B2B Portal and enter the group booking workflow.	High	3.16.5.1.1: #1
REQ-BP-TB-002	The system shall display available tour programmes including programme name, description, bookable date range, session time, location, and booking fees.	High	3.16.5.1.1: #2
REQ-BP-TB-003	The system shall allow programme browsing without login while requiring authentication before submitting a booking application.	High	3.16.5.1.1: #2
REQ-BP-TB-004	The system shall support login to the Portal using an email address associated with an organisation.	High	3.16.5.1.1: #3
REQ-BP-TB-005	The system shall support email-based verification using a one-time verification code.	High	3.16.5.1.1: #4
REQ-BP-TB-006	The system shall allow users to select programme, booking date, session, and booking quantity during the booking process.	High	3.16.5.1.1: #5
REQ-BP-TB-007	The system shall validate availability of session capacity and ticket inventory during the booking process.	High	3.16.5.1.1: #6

REQ-BP-TB-008	The system shall allow users to register for a waitlist when booking capacity is unavailable.	High	3.16.5.1.1/ 3.16.5.1.4
REQ-BP-TB-009	The system shall allow users to submit booking applications including participant information, contact details, and additional service requirements.	High	3.16.5.1.1: #7
REQ-BP-TB-010	The system shall reserve booking capacity when a booking application is submitted.	High	3.16.5.1.1: #7
REQ-BP-TB-011	The system shall support manual review of booking applications by backend staff.	High	3.16.5.1.1: #8
REQ-BP-TB-012	The system shall allow booking applications to be approved, rejected, or returned for amendment during the review process.	High	3.16.5.1.1/ 3.16.5.1.5
REQ-BP-TB-013	The system shall generate a booking order after a booking application is approved.	High	3.16.5.1.1: #9
REQ-BP-TB-014	The system shall notify the group contact of the booking review result and order creation.	High	3.16.5.1.1: #10
REQ-BP-TB-015	The system shall support online payment for approved booking orders through the Portal.	High	3.16.5.1.1: #11
REQ-BP-TB-016	The system shall update the order status and confirm the booking after successful payment.	High	3.16.5.1.1: #11
REQ-BP-TB-017	The system shall generate a group admission credential with a QR code after successful payment.	High	3.16.5.1.1: #12
REQ-BP-TB-018	The system shall support admission validation through scanning QR code credentials at the venue.	High	3.16.5.1.1: #13
REQ-BP-TB-019	The system shall support generating individual admission codes for participants when required.	Medium	3.16.5.1.1: #14
REQ-BP-TB-020	The system shall allow booking applications to be amended before payment and re-enter the approval workflow when required.	High	3.16.5.1.5

Table 3.16.5.1.6: Requirement List of Tour Booking**3.16.5.2. School Booking**

The School Tour Booking function of the B2B Portal provides visit booking services specifically designed for schools in Hong Kong.

The end-to-end workflow includes teacher login, programme and session selection, submission of participant details, booking application submission, backend review, payment processing, admission validation, and operational reporting. This workflow forms a closed-loop management process for school visits.

The system supports key operational capabilities such as identity verification, capacity management, waitlist handling, value-added service selection, multiple payment methods, and flexible admission credential issuance.

Backend operational functions also support report generation, notification management, and traceability of booking and attendance data, ensuring transparency and operational efficiency for school-based bookings.

3.16.5.2.1. User Process – General Process

The following process describes the general workflow of School Booking.

No.	Component	Description
1	Access the Portal and Enter School Booking	A school staff accesses WestK B2B Portal website via a web browser, navigates to the designated venue page, and selects “School Booking” to enter the booking flow.
2	View Programmes Available for School Booking	The system displays tickets, activities, workshops, and other items available for school booking, including event overview, descriptions, time, location, images, and related ancillary information. Teachers may browse basic information without login but must log in to proceed with booking.
3	Teacher Login/Registration	The teacher enters the login page and logs in or registers using an email address. The system determines whether to initiate a login flow or a first-time registration flow based on whether the email address is already registered.
4	Email Verification	The system sends a verification code to the teacher’s email address. The teacher enters the verification code on the login page to complete authentication. If the code is valid, login is successful; otherwise, an error message is displayed, and the user may re-enter or request a new code.
5	Select Programme, Date and Session	The teacher enters the booking flow, selects a bookable date from the calendar, views available sessions, selects a session, and enters the required quantity based on the available ticket types.
6	Check School Free Quota	The system checks whether the school’s remaining free quota is sufficient. If the requested

		quantity exceeds the available free quota, the system indicates insufficient quota and booking cannot proceed.
7	Check Session Inventory	The system checks whether inventory for the selected session and ticket types is sufficient. If inventory is insufficient, booking cannot proceed and the teacher may choose to register for the waitlist.
8	Complete Booking Information and Submit Application	If both the school free quota and session inventory meet the requirements, the teacher completes the booking information and submits the booking application. Required information may include the number of students, number of accompanying teachers, requested services (e.g. guided tours), teacher contact details, coach information if applicable, and supporting documents. The system reserves the required capacity during submission.
9	Backend Review of Booking Application	Backend staff manually review the booking application to verify school identity, participant numbers, qualifications, and event requirements. Applications may be rejected (inventory released), returned for amendment, or approved to proceed to order creation.
10	Create Order	After approval, the system generates a formal order in the backend and deducts the corresponding school free quota and session inventory.
11	Send Review Result Notification	The system automatically sends the review result notification to the teacher's email address.
12	View Order and Obtain Group Admission Pass	The teacher logs in to the Portal, accesses Booking Application Management, views the approved application, and downloads or prints the group admission pass with a QR code.
13	On-Site Admission Validation	On site, admission is validated using the group admission pass. Staff record the actual number of attendees using the validation terminal.
14	Print Individual Ticket Codes	To support scenarios where students participate in dispersed activities, the teacher may print individual ticket codes at the box office using the group order QR code and distribute them to attending students and teachers.

15	Admission Validation within the Venue Using Individual Ticket Codes	Attendees use individual ticket codes for admission validation within the venue and participate in the related activities.
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Table 3.16.5.2.1: Description for the User Process of School Booking

3.16.5.2.2. User Process - School Directory Management

The following process describes the School Directory Management.

No.	Component	Description
1	Access the System Backend and Enter the “School Directory Management” Menu	Backend staff log in to the system backend and access the “School Directory Management” menu. The system supports searching, editing, and deleting existing school records.
2	Add a School	Schools may be added through several methods, including manual entry of school information, or batch import using a predefined spreadsheet template.
3	School Added Successfully	After a school record is successfully created, the designated school contact may log in to the B2B Portal using the registered email address and perform related operations, such as submitting school booking applications or managing teacher account information.
4	Select a School to Perform Related Operations	The system supports configuring available free visit quotas for each school and allows backend staff to manually adjust quotas when required, with change logs retained. The system also supports quick access to school-related operational data, including teacher account applications, waitlist applications, and booking applications, with filtering by teacher email address. Staff may also view booking orders associated with the school and manage teacher account directories under each school, including adding, editing, and deleting teacher accounts.

Table 3.16.5.2.2: Description for the User Process of School Directory Management

3.16.5.2.3. User Process - Teacher Account Review

The following process describes the Teacher Account Review.

No.	Component	Description
1	Teacher Enters Login Page and Submits Email	The teacher enters the B2B Portal login page, selects the corresponding school, enters an email address, and requests a verification code. The

		system checks whether the email address already exists in the school's teacher email directory.
2	System Validates Email Eligibility	If the email address exists in the school's teacher email directory, the system sends a verification code, and the teacher may proceed with login. If the email address is not registered, the teacher is prompted to submit qualification documents for manual review.
3	Submit Qualification Supporting Documents	The teacher uploads qualification supporting documents (such as teacher credentials, school certificates, or other supporting documents) and submits an account review request. The system creates a teacher account review application record.
4	Backend Manual Review of Teacher Qualifications	Backend staff review the application in the system and manually assess the teacher's qualifications. If approved, the email address is added to the school's teacher email directory, and an approval notification is sent. If rejected, a rejection notification stating the reason is sent.
5	Teacher Logs in Again	After the email address is successfully registered in the school's teacher email directory, the teacher may return to the login page, request a verification code, and complete the login process to access subsequent school booking functions.

Table 3.16.5.2.3: Description of Teacher Account Review User Process**3.16.5.2.4. User Process - Waitlist**

The following process describes the Waitlist User Process.

No.	Component	Description
1	Submit Waitlist Registration	When inventory for the selected session is insufficient, the teacher may submit a waitlist registration to be considered when capacity becomes available.
2	Backend Review of Waitlist Registration	Backend staff review the waitlist registration and determine whether capacity may be allocated based on inventory availability or operational arrangements. Inventory may be allocated through released inventory or additional quota configuration.
3	Waitlist Approval Notification	If the waitlist application is approved, the system reserves the corresponding inventory and school free quota and automatically creates a booking order. An approval notification email is sent to

		the teacher. If rejected, a notification email stating the reason is sent.
4	Teacher Logs in to View Order	The teacher logs in to the Portal and views the order generated from the waitlist application in the order management page and proceeds with subsequent steps such as admission preparation.

Table 3.16.5.2.4: Description of Waitlist User Process

3.16.5.2.5. User Process - Booking Application Amendment

The following process describes the Booking Application Amendment User Process.

No.	Component	Description
1	Return Booking Application	If the booking application contains incomplete or non-compliant information, backend staff may return the application for amendment. A notification email indicating the reason for return will be sent to the teacher. The reserved quota associated with the application remains retained during the amendment period.
2	Teacher Amends and Resubmits Application	The teacher logs in to the Portal and accesses the booking application management page to view the returned application and its rejection reason. The teacher may update the booking information and resubmit the application.
3	Backend Re-Review of Booking Application	Backend staff review the amended booking application again to confirm school identity, participant numbers, and event requirements. If approved, the system creates the corresponding order. If the application still does not meet the requirements, it may be returned again for amendment or rejected.

Table 3.16.5.2.5: Description for Booking Application Amendment User Process

3.16.5.2.6. School Booking Requirement List

RID	Description	Priority	Related Process
REQ-BP-SB-001	The system shall allow teachers to access the School Booking entry via the B2B Portal and enter the school booking workflow.	High	3.16.5.2.1: #1
REQ-BP-SB-002	The system shall display programmes available for school booking, including overview, schedule, venue, images, and fee information.	High	3.16.5.2.1: #2

REQ-BP-SB-003	The system shall allow programme information for school booking to be browsed without login.	Medium	3.16.5.2.1: #2
REQ-BP-SB-004	The system shall support teacher login or registration via email address and association with the corresponding school.	High	3.16.5.2.1: #3
REQ-BP-SB-005	The system shall support email verification codes for teacher authentication during login.	High	3.16.5.2.1: #4
REQ-BP-SB-006	The system shall allow teachers to select programme, date, session, ticket type, and booking quantity.	High	3.16.5.2.1: #5
REQ-BP-SB-007	The system shall validate whether the school's free visit quota is sufficient for the requested booking quantity.	High	3.16.5.2.1: #6
REQ-BP-SB-008	The system shall validate whether session inventory and ticket availability are sufficient for booking.	High	3.16.5.2.1: #7
REQ-BP-SB-009	The system shall allow teachers to submit a waitlist registration when inventory is insufficient.	High	3.16.5.2.1: #1
REQ-BP-SB-010	The system shall allow teachers to complete required booking information such as number of students, accompanying teachers, service requirements, and coach information.	High	3.16.5.2.1: #8
REQ-BP-SB-011	The system shall support uploading supporting documents required for school booking applications.	Medium	3.16.5.2.1: #8
REQ-BP-SB-012	The system shall support manual review of school booking applications by backend staff.	High	3.16.5.2.1: #9
REQ-BP-SB-013	The system shall automatically create an order after a booking application is approved.	High	3.16.5.2.1: #10
REQ-BP-SB-014	The system shall send review result notifications to teachers after booking application review.	High	3.16.5.2.1: #11
REQ-BP-SB-015	The system shall allow teachers to view booking applications and orders through the Portal.	High	3.16.5.2.1: #12
REQ-BP-SB-016	The system shall generate a group admission pass with QR code for school group entry.	High	3.16.5.2.1: #12

REQ-BP-SB-017	The system shall support on-site admission validation and recording of actual attendance numbers.	High	3.16.5.2.1: #13
REQ-BP-SB-018	The system shall support printing individual ticket codes when required for admission validation within the venue.	Medium	3.16.5.2.1: #14
REQ-BP-SB-019	The system shall support school directory management, including school information maintenance and configuration of official email domains.	High	3.16.5.2.2: #1
REQ-BP-SB-020	The system shall support teacher account qualification review for teachers whose email addresses are not registered in the school directory.	High	3.16.5.2.3: #1
REQ-BP-SB-021	The system shall support review of waitlist registrations and automatically create booking orders when applications are approved.	High	3.16.5.2.4: #2
REQ-BP-SB-022	The system shall support amendment workflows for booking applications, including return for modification and re-review.	High	3.16.5.2.5: #1

Table 3.16.5.2.6: Requirement List of School Booking

3.16.5.3. Academy Programme Registration

Academy Programme Registration supports enrolment management for Academy programmes operated by WestK.

These programmes are typically short-term courses with a fixed duration, structured learning activities, and predefined schedules. Participants enrol in the programme as a complete course rather than selecting individual modules.

Unlike group bookings or ticketing scenarios, Academy enrolment targets applicants who may need to meet specific eligibility criteria. The enrolment workflow therefore includes application submission, qualification verification, manual review, and capacity control.

After submitting an application, backend staff review the application and determine the outcome, which may include admission, placement on the waitlist, or rejection. Only admitted applicants may proceed to complete payment within the specified timeframe to confirm their enrolment.

3.16.5.3.1. User Process – General Process

The following process describes the general user process.

No.	Component	Description
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1	Access the Academy Programme Registration Entry	The user accesses the Academy Programme Registration entry via the official website or Portal. After browsing course information, the user selects “Apply” to enter the registration workflow.
2	View Available Courses for Enrolment	The system displays the list of Academy courses open for enrolment, including course name, course duration, schedule, venue, partner organisations (if any), course overview, and related images.
3	View Course Details	The user selects a course to view detailed information, including course structure, daily schedule overview, target participants, enrolment requirements, fee information, and important notes.
4	User Login or Registration	Before entering the enrolment flow, the user must complete login or registration. Individual users may register or log in directly using an email address, while institutional users (e.g. schools or NGOs) must associate their account with an organisation according to the B2B Portal logic before logging in.
5	Select Course and Enter Enrolment Form	After confirming the selected Academy course, the user enters the enrolment flow. Courses are enrolled as a complete programme and do not support partial enrolment or module selection.
6	Complete Enrolment Information Form	The user completes the enrolment form by providing required personal information such as name, contact details, education background, work experience, and course-related questions. The form supports multi-language switching.
7	Upload Qualification Supporting Documents	The user uploads required qualification supporting documents, such as academic certificates or employment proof, in supported formats such as PDF.
8	Submit Enrolment Application	After confirming all information, the user submits the enrolment application. The system creates an enrolment record and routes it to the backend review process.
9	Backend Manual Review of Enrolment Application	Backend staff manually review and assess enrolment applications and may record interview results where applicable. Review outcomes include Approved, Waitlisted, or Rejected.
10	Send Review Result Notification	The system automatically sends notification emails based on the review outcome. Approved

		applicants are instructed to complete payment within the specified period; waitlisted applicants receive waitlist notifications; rejected applicants receive rejection notifications, optionally including reasons.
11	User Completes Online Payment	Approved applicants log in to the Portal and complete online payment within the specified time limit.
12	System Confirms Enrolment and Reserves Seat	After successful payment, the system confirms enrolment and reserves the course seat. If payment is not completed within the allowed time, the seat is released and may be allocated to waitlisted applicants according to defined rules.
13	Send Enrolment Confirmation and Admission Pass	The system sends enrolment confirmation notifications and generates admission passes (e.g. QR codes) for course attendance.
14	On-Site Admission Validation	During the course period, participants use their admission passes for on-site check-in or admission validation, such as QR code scanning or device-based validation.

Table 3.16.5.3.1: *Description for the User Process of Academy Programme Registration*

3.16.5.3.2. User Process - User Registration

The following process describes the User Registration User Process.

No.	Component	Description
1	Enter Login/ Registration Page	When a user selects “Apply” in the Academy Programme Registration flow without authentication, the system redirects the user to the login/registration page and indicates that identity verification is required before continuing the enrolment process.
2	Select User Type	The user selects a registration type: Individual Applicant or Institution/NGO Applicant. The system routes the user to the corresponding registration flow.
3A	Individual User Enters Email Address	In the Individual Applicant flow, the user enters a personal email address for registration or login.
3B	Organisation User Selects Organisation and Enters Email	In the Institution/NGO Applicant flow, the user selects the associated organisation (such as a school or NGO) and enters a personal email address for registration or login.

4	Receive and Enter Email Verification Code	The system sends a verification code to the user's email address, and the user enters the code for identity verification.
5	System Validates Verification Code	The system validates the verification code. If verification succeeds, the user proceeds to the next step; if it fails, an error message is displayed with options to re-enter the code or request a new one.
6	Registration Completed and User Logged in	After successful registration or login, the user enters the Academy Programme Registration enrolment workflow and proceeds with the application.

Table 3.16.5.3.2: Description of User Registration User Process

3.16.5.3.3. User Process - Waitlist

The following process describes the Waitlist User Process.

No.	Component	Description
1	Add Enrolment Application to Waitlist	During enrolment review, if course capacity is full or the applicant is not immediately admitted, backend staff may move the enrolment application to the waitlist. The submitted enrolment information is retained for potential allocation when seats become available.
2	Allocate Course Seat from Waitlist	When course seats become available (e.g. due to cancellations or payment timeouts), backend staff may allocate seats to eligible applicants from the waitlist.
3	Send Waitlist Approval Notification	The system sends notification emails to selected waitlisted applicants informing them that enrolment eligibility has been granted and payment must be completed within the specified period.
4	User Logs in to View Enrolment Information	The user logs in to the Portal and accesses the Academy Enrolment Management page to view the updated enrolment status, including course information, fees, and payment deadline.
5	Complete Payment within Specified Deadline	The user must complete payment through the Portal within the specified deadline. If payment is completed on time, enrolment is confirmed. If payment is not completed within the deadline, the waitlist eligibility expires and the course seat is released.

6	Payment Completed and Enrolment Confirmed	After successful payment, the system updates the enrolment status to confirmed and sends enrolment confirmation notifications together with course participation or admission information.
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Table 3.16.5.3.3: Description of Waitlist User Process**3.16.5.3.4. Academy Programme Registration Requirement List**

RID	Description	Priority	Related Process
REQ-BP-AP-001	The system shall allow users to access the Academy Programme Registration entry through the official website or Portal.	High	3.16.5.3.1: #1
REQ-BP-AP-002	The system shall display the list of Academy programmes available for enrolment, including course name, schedule, venue, and overview information.	High	3.16.5.3.1: #2
REQ-BP-AP-003	The system shall allow users to view detailed course information including course structure, target audience, enrolment requirements, and fee information.	High	3.16.5.3.1: #3
REQ-BP-AP-004	The system shall support user login or registration via email before entering the enrolment workflow.	High	3.16.5.3.2: #1
REQ-BP-AP-005	The system shall support user registration for both Individual Applicants and Institution/NGO Applicants.	High	3.16.5.3.2: #2
REQ-BP-AP-006	The system shall support user authentication via email verification code.	High	3.16.5.3.2: #4
REQ-BP-AP-007	The system shall allow users to select a course and enter the enrolment form workflow.	High	3.16.5.3.1: #5
REQ-BP-AP-008	The system shall support enrolment forms capturing applicant information such as personal details, educational background, and course-related questions.	High	3.16.5.3.1: #6
REQ-BP-AP-009	The system shall support multi-language display for enrolment forms.	Medium	3.16.5.3.1: #6

REQ-BP-AP-010	The system shall support uploading supporting documents required for enrolment applications.	High	3.16.5.3.1: #7
REQ-BP-AP-011	The system shall support submission of Academy enrolment applications.	High	3.16.5.3.1: #8
REQ-BP-AP-012	The system shall support manual review and selection of enrolment applications by backend staff.	High	3.16.5.3.1: #9
REQ-BP-AP-013	The system shall support review outcomes including Approved, Waitlisted, and Rejected statuses.	High	3.16.5.3.1: #9
REQ-BP-AP-014	The system shall send notification emails to applicants informing them of enrolment review results.	High	3.16.5.3.1: #10
REQ-BP-AP-015	The system shall allow approved applicants to complete online payment within the specified deadline.	High	3.16.5.3.1: #11
REQ-BP-AP-016	The system shall confirm enrolment and reserve the course seat after successful payment.	High	3.16.5.3.1: #12
REQ-BP-AP-017	The system shall release course seats if payment is not completed within the allowed period and allocate them to eligible waitlisted applicants.	High	3.16.5.3.3: #2
REQ-BP-AP-018	The system shall support waitlist management and allocation of seats when capacity becomes available.	High	3.16.5.3.3: #1
REQ-BP-AP-019	The system shall send enrolment confirmation notifications after successful enrolment confirmation.	High	3.16.5.3.1: #13
REQ-BP-AP-020	The system shall generate admission passes for course attendance or check-in validation.	High	3.16.5.3.1: #13
REQ-BP-AP-021	The system shall support on-site attendance validation using QR codes or scanning devices.	Medium	3.16.5.3.1: #14

Table 3.16.5.3.4: Requirement List of Academy Programme Registration

3.17. WeChat Mini Program

This section describes the overall business processes of the WeChat Mini Program integration, including the customer journey, detailed business requirements, and comprehensive feature specifications covering both frontend and backend workflows.

3.17.1. Business Requirements

From a business perspective, WestK requires the WeChat Mini Program integration to support ticket sales, merchandise sales, membership card sales, and coupon usage, while ensuring operational efficiency, secure data handling, and compliance with platform policies.

The solution shall support users with registered WeChat accounts, including users in Hong Kong, Macao, and Mainland China.

The following business requirements have been identified.

3.17.1.1. Registration & Login

WestK requires the WeChat Mini Program to support user authentication via WeChat native login, with automatic account binding to enable consistent user identity management across orders, memberships, tickets, coupons, and merchandise purchases.

3.17.1.2. WeChat Ticket Sales Channel

WestK requires the WeChat Mini Program to support the listing and sale of exhibition and show tickets for M+ Museum (M+), Hong Kong Palace Museum (HKPM), and Performing Arts (PA).

The Mini Program shall support multiple ticket purchasing models, including general admission tickets and seat-selection tickets, and provide a complete ticket ordering experience within the WeChat Mini Program.

Eligible promotions, including membership benefits and coupons, shall be applicable during the ticket purchasing process in accordance with business rules defined by WestK.

3.17.1.3. Membership Card Sales Channel

WestK requires the WeChat Mini Program to support the sale and management of multiple types of membership cards, including individual memberships, multi-user or family-based memberships, and patron-based memberships.

Membership benefits shall be applicable to eligible ticket purchases, merchandise purchases, and promotional activities, subject to defined business rules.

BU	Card Name	Card Type	Validity	Age Limit	Purchase Limit	Member Benefits
	M+ Member	Individual				

Technology Partnership Package 2
Requirement Document

M+ Member	M+ Young Member		One year	Age 12-17 or full-time student	One Card in one BU	Each member can enjoy unlimited admission to all galleries and exhibitions, M+ Lounge, access to M+ private viewing, priority ticket purchase and much more.
	M+ Senior Member			Age 60+		
	M+ Dual membership			Multi-Member		
	M+ Family Membership	Multi-Member For 2 adults and 1 child or youth (add-ons available for additional family members)		For Child or Youth member: Age 12-17 or full-time student		
		M+ Member and HKPM Friends Combo Membership		13 months		One Card in both BU
M+ Patron	M+ Single Patron	Individual	One Year		One Card in one BU	Each patron member can enjoy unparalleled access to the museum and contribute to building the cultural capital for future generations, unlimited year-round admission to all galleries and exhibitions with guest privileges,
	M+ Young Patron					
	M+ Dual Patron	Multi-Member		Age 18-35		
	M+ and HKPM Combo Patron Membership	Individual Member Cross BU			One Card in both BU	

Technology Partnership Package 2
Requirement Document

						invitations to patron-only events, opening receptions, and much more.
HKPM Friends	HKPM Friends – Individual	Individual	One Year		One Card in each BU	Each Member can enjoy unlimited entry to the Museum’s Thematic exhibitions, receive complimentary admission codes to visit the Special exhibitions, member-exclusive viewing hour and many more privileges.
	HKPM Friends - Youth			Age 7-11 or full-time students		
	HKPM Friends - Seniors			Age 60 or above		
	HKPM Friends - Dual	Multi-Member				
	HKPM Friends - Family	Multi-Member For two adults and one child member (age 7-11) or full-time student. Add-ons are available for additional family members.		For Child or Youth member: Age 7-11 or full-time student		
	HKPM Friends & M+ Member Combo Membership	Individual	13 months		One Card in both BU	Enjoy both M+ and HKPM member benefits.

HKPM Patrons	HKPM Single Patron	Individual	One Year		One Card in each BU	Each patron member can enjoy the Patron-exclusive events, including private previews of new Special exhibitions and curator-led tours and privilege to bring your guests to visit the Museum and galleries for free.
	HKPM Dual Membership Patron	Multi-Member				
	HKPM Friends & M+ member Patron membership	Individual Member Cross BUs			One Card in both BU	

Table 3.17.1.3: Membership Specification cross different BUs

3.17.1.4. Online Merchandise Store

WestK requires the WeChat Mini Program to support online merchandise sales, enabling users to browse products, purchase merchandise, and complete orders within the Mini Program.

Merchandise purchases may be conducted independently or, where applicable, in conjunction with ticket purchases, subject to defined business rules.

3.17.1.5. Coupon Management & Redemption

WestK requires the WeChat Mini Program to support coupon visibility and coupon usage for eligible transactions, including ticket purchases and merchandise purchases.

Coupon applicability and redemption outcomes shall be handled in accordance with business rules defined by WestK.

3.17.1.6. E-Ticket Wallet

WestK requires the WeChat Mini Program to provide electronic ticket management capabilities, enabling users to access their electronic tickets and present valid tickets for venue admission using QR codes.

3.17.1.7. Ticket Verification & Admission Control

WestK requires the WeChat Mini Program to support ticket verification and admission control through QR-code-based validation, with admission status synchronized across relevant backend systems to prevent duplicate entry and ensure operational traceability.

3.17.1.8. Integration with Ticketing System

WestK requires the WeChat Mini Program to integrate with a centralized ticketing system that serves as the authoritative source for event listings, ticket inventory, pricing, order processing, and ticket lifecycle management.

WestK requires that all transactions created through the WeChat Mini Program—including ticket purchases, membership purchases, merchandise orders, and coupon usage—are traceable across both the WeChat platform and backend systems.

3.17.1.9. Privacy, Security, Scalability & Inclusive Accessibility

WestK requires the WeChat Mini Program to comply with applicable privacy, security, and accessibility requirements, and to support stable show during peak access periods.

3.17.2. Business Requirements Analysis

This section provides further analysis and clarification of the business requirements outlined in Section 3.17.1. It elaborates on the functional expectations, operational considerations, and system behaviour required to support the WeChat Mini Program business objectives.

3.17.2.1. Registration and Login

Map to Business Requirement 3.17.1.1.

The WeChat Mini Program shall leverage WeChat native authentication to enable seamless user access without requiring separate account registration.

Upon successful login:

- A unique user identity shall be established and persistently bound to the user's WeChat account.
- User identity shall serve as the primary reference for order ownership, membership entitlement application, coupon usage, and ticket verification.

The system shall ensure secure session handling and consistent user identification across all Mini Program functions.

3.17.2.2. Event Discovery & Event Information

Map to Business Requirement 3.17.1.2.

The WeChat Mini Program provides intuitive event discovery capabilities to enable users to browse, explore, and access exhibitions and shows across M+, HKPM, and PA.

The Home Page shall serve as the primary entry point and provide venue-based main access points for M+, HKPM, and PA, allowing users to initiate exploration from a specific venue context.

The Home Page shall support:

- Event listings and highlighted exhibitions and shows.

- Core transactional functions (e.g. ticket, membership, merchandise).
- Search functionality.
- Relevant venue and visitor information, including site navigation support and on-site dining information.

Users shall be able to navigate from the Home Page or venue-specific listings to individual event detail pages.

Event Detail Pages shall present sufficient information to support informed decision-making, including:

- Event overview and descriptive content.
- Event schedules and venue information.
- Ticket types, pricing, and seat availability (where applicable).
- Ticket availability status.

Where applicable, special notices or eligibility requirements associated with specific events or ticket types, such as age-restricted events (e.g. 18-only events) or special admission conditions, shall be clearly presented to users prior to purchase, subject to event configuration.

3.17.2.3. Ticket Purchasing Process

Map to Business Requirement 3.17.1.2.

The Mini Program shall support multiple ticket purchase modes:

- General Admission Ticket. allowing selection of ticket types and quantities
- Seat-selection purchase, allowing seat selection through a visual seating interface

During the ticket purchase process:

- Eligible membership benefits and coupons shall be identifiable and applicable.
- An order confirmation step shall present the final payable amount prior to payment.

Special event notices (such as audience eligibility requirements or special admission conditions) shall be displayed to users before order submission where applicable.

Ticket payment shall be completed within the WeChat Mini Program using native WeChat Pay.

Where payment is processed in a non-HKD currency due to platform requirements, the Mini Program shall ensure that ticket pricing, settlement, and financial reconciliation are recorded and managed in Hong Kong Dollars (HKD), in accordance with WestK's financial requirements.

3.17.2.4. Membership Card Purchasing & Management

Map to Business Requirement 3.17.1.3.

The WeChat Mini Program shall support membership card purchasing and management functions, providing clear entry points for users to browse and purchase available membership programmes where applicable.

The Mini Program shall support different membership types, including individual memberships and multi-user or family-based memberships, with clearly defined terms and conditions associated with each membership type.

Although patron-based membership card types may be offered by WestK, such membership cards shall not be available for purchase within the WeChat Mini Program due to platform payment limitations on large-value transactions (over 10K). Relevant terms and conditions should clearly inform users that patron-based memberships are to be purchased through designated patrons or sponsorship channels outside the Mini Program.

After successful purchase, users shall be able to manage their memberships within the Mini Program, including viewing membership status, validity period, and applicable benefits.

Each membership card shall be associated with a unique membership identifier. For multi-user or family-based memberships, the shared nature of the membership shall be clearly indicated.

Where applicable, purchased memberships shall be linked to eligible membership discounts or promotional benefits during qualifying transactions, subject to defined business rules.

The Mini Program should provide a dedicated member-exclusive section to present membership-related content and provide access to member-only events or activities, where applicable.

The Mini Program shall support essential membership lifecycle capabilities, including membership activation, renewal, and management, in accordance with WestK membership policies.

3.17.2.5. Online Merchandise Purchasing Process

Map to Business Requirement 3.17.1.4.

The WeChat Mini Program shall support online merchandise purchasing functionality, enabling users to browse and purchase selected merchandise made available by WestK.

The Mini Program should provide clear entry points for users to access the online merchandise store and view product information, including product details, pricing, and availability.

Users shall be able to purchase merchandise independently or, where applicable, in conjunction with ticket purchases, subject to defined business rules.

Merchandise payment shall be completed within the WeChat Mini Program using supported payment methods.

Where applicable, fulfilment options such as in-person pickup or delivery shall be supported in accordance with fulfilment policies defined by WestK.

3.17.2.6. Coupon Management

Map to Business Requirement 3.17.1.5.

The WeChat Mini Program shall support coupon management functions, enabling users to view and use eligible coupons within the Mini Program.

Users shall be able to access their available coupons, including basic information such as coupon validity period, applicable scope, and usage conditions.

During eligible transactions, including ticket purchases and merchandise purchases, the Mini Program shall support the application of applicable coupons in accordance with defined business rules. The discounted payable amount shall be calculated and presented prior to payment confirmation.

Coupon eligibility, application, and redemption status shall be managed consistently to ensure that coupons are applied correctly and used only once, in accordance with coupon policies defined by WestK.

3.17.2.7. Order Management & E-Ticket Wallet

Map to Business Requirement 3.17.1.6.

The WeChat Mini Program shall provide a centralized order management capability, enabling users to view and manage transactions generated within the Mini Program.

Users shall be able to access their orders, including ticket purchases, merchandise orders, and membership purchases, and review relevant order information and status (e.g. Pending Payment, Paid, Cancelled, Refunded).

The Mini Program shall provide an electronic ticket wallet to display electronic tickets associated with ticket purchases. The ticket wallet shall present both unused and used tickets, with individual verification codes (e.g. QR codes) available for admission purposes.

Where applicable, QR code visibility within the ticket wallet may be conditionally controlled based on event type or membership rules, such as member exclusive programmes.

3.17.2.8. Ticket Verification & Admission Control

Map to Business Requirement 3.17.1.7.

The WeChat Mini Program shall support ticket verification and admission control for electronic tickets presented at venue entry points.

Electronic tickets shall be verified using unique verification codes. Upon successful verification, admission status shall be updated in real time to prevent duplicate entry.

Ticket usage and admission status shall be synchronized between the WeChat Mini Program and backend systems, including the AnTank Ticketing and Admission System, to ensure operational traceability and consistency.

Admission control shall support event-specific rules, including eligibility restrictions or special access conditions for selected membership-based events.

3.17.2.9. Integration with Ticketing System & Data Governance

Map to Business Requirement 3.17.1.8 & 3.17.1.9.

The WeChat Mini Program shall integrate with the centralized AnTank Ticketing and Admission System, which serves as the authoritative source of truth for event listings, ticket inventories, pricing, and order lifecycle management.

The Mini Program shall consume master data from AnTank and synchronize transactional and status updates back to backend systems to support centralized reporting, operational oversight, and financial reconciliation.

Data integrity and cross-channel consistency shall be maintained across all integrated systems.

3.17.3. Operating Model

The WeChat Mini Program operates under a multi-party operating model with clearly defined roles and responsibilities across business ownership, technology delivery, and platform operations.

WestK (Business Owner)

WestK is the business owner of the WeChat Mini Program and holds overall accountability for business operations, customer policies, and financial governance.

- Owns all business content and offerings, including events, exhibitions, merchandise, and membership.
- Defines ticket pricing, inventory policies, membership benefits, promotional rules, and customer-facing policies.
- Owns customer relationships and post-purchase policies within the agreed scope.
- Acts as the legal and financial merchant entity operating under the official WestK WeChat account.
- Receives settlement proceeds from WeChat Pay into WestK's official WeChat Pay merchant account.
- Performs accounting recognition, financial reconciliation, and audit activities based on settlement and transaction reports.
- Acts as the primary finance counterparty for payment settlement and reconciliation.

AnTank (Ticketing and Admission System Provider)

AnTank delivers and operates the centralized ticketing technology layer and provides technical development and maintenance for the WeChat Mini Program.

- Provides and operates the ticketing system, including:
 - Event and session configuration.
 - Ticket inventory management.
 - Order lifecycle management.
 - Electronic ticket issuance (QR code/e-ticket).
 - Ticket verification records and admission status.
 - Sales and operational reporting exports.
- Develops and maintains the WeChat Mini Program, including:
 - End-to-end user journeys for ticketing, membership, merchandise, and coupons.
 - WeChat Open Authorisation (OAuth) login integration.

- UI/UX implementation.
- Synchronization of transaction data with backend systems.
- Basic operational and reporting dashboards where applicable.
- Integrates with payment endpoints as required by the solution design, typically via WeChat Pay APIs.
- Ensures system availability, functional correctness, logging, monitoring, and technical support in accordance with agreed service levels.

AnTank acts as the system of record for ticketing and transactional data, while operating under business rules defined by WestK.

WeChat Platform (Channel & Payment Provider)

The WeChat platform provides the consumer-facing Mini Program channel and integrated payment capabilities.

The WeChat Platform:

- Provides the Mini Program runtime environment and user access within the WeChat ecosystem.
- Provides WeChat native authentication (OAuth) and payment services via WeChat Pay.
- Processes payments under WestK's official WeChat Pay merchant account in accordance with platform policies.
- Supports platform-level settlement and transaction reporting, subject to WeChat Pay arrangements.

The WeChat Mini Program functions as a front-end transactional channel only and does not act as a fund-holding or settlement entity.

3.17.4. Customer Journey

This section describes the high-level user journeys supported by the WeChat Mini Program. It outlines how users typically interact with the Mini Program to discover content, complete transactions, and manage post-purchase activities. These user journeys are illustrative and serve to contextualize the business requirements defined in Sections 3.17.1 and 3.17.2.

No.	Category	Role	User Story
1	Entry	As a WeChat user (Use "User" in following Role)	I want to find WestK Mini Program via WeChat search using keywords such as "WestK", "M+", or "HKPM", so that I can quickly access official services.
2	Entry	As a user	I want to access the Mini Program through the official WestK or venue WeChat account menu, so that I know I am entering the official platform.
3	Entry	As a user	I want to scan a Mini Program QR code from online or on-site promotional materials, so

Technology Partnership Package 2
Requirement Document

			that I can directly open the Mini Program without searching.
4	Entry	As a user	I want to open the Mini Program from a link or Mini Program card shared by friends or groups, so that I can easily follow recommendations.
5	Entry	As a returning user	I want to reopen the Mini Program from my recently used Mini Programs list, so that I can quickly continue my previous activities.
6	Registration and Login	As a user	I want to log in using my phone number bound to my WeChat account, so that I can access the Mini Program quickly and conveniently.
7	Home Page	As a user	I want to see clearly categorised entry points on the Home Page, so that I can easily start exploring services.
8	Event Discovery	As a user	I want to browse exhibitions and shows by venue (M+, HKPM, PA), so that I can focus on events at a specific venue.
9	Event Discovery	As a user	I want to view highlighted or recommended exhibitions and shows, so that I can discover popular or featured events.
10	Event Discovery	As a user	I want to search for exhibitions or shows using keywords, so that I can quickly find events I am interested in.
11	Visit Planning	As a user	I want to view site maps, route guidance, and dining information, so that I can plan my visit in advance.
12	Event Information	As a user	I want to view detailed event information, including schedules, venue details, pricing, and availability, so that I can decide whether to attend.
13	Event Information	As a user	I want to see special notices such as age restrictions or eligibility requirements before purchasing tickets, so that I understand whether I am eligible to attend.
14	Ticket Purchasing	As a user	I want to purchase general admission tickets, so that I can attend an exhibition or show without selecting seats.

Technology Partnership Package 2
Requirement Document

15	Ticket Purchasing	As a user	I want to purchase seat-selection tickets, so that I can choose specific seats for a show where applicable.
16	Ticket Purchasing	As a member	I want to apply eligible membership discounts or coupons, so that I can enjoy applicable benefits.
17	Ticket Purchasing	As a user	I want to review my selected tickets, applied benefits, and final payable amount before payment, so that I can confirm my purchase details.
18	Payment	As a user	I want to complete payment using WeChat Pay, so that I can safely and conveniently finalize my purchase.
19	Membership	As a user	I want to browse available membership events and understand their benefits and terms, so that I can decide whether to become a member.
20	Membership	As a member	I want to view my membership status, validity period, and benefits, so that I can manage my membership effectively.
21	Merchandise	As a user	I want to browse and purchase merchandise independently or together with ticket purchases, so that I can complete related purchases conveniently.
22	Coupon Management	As a user	I want to view all my available coupons in one place, so that I know which coupons I can use for eligible purchases.
23	Coupon Management	As a user	I want to see the status of my coupons (e.g. unused, used, expired), so that I understand whether a coupon is still valid.
24	Coupon Management	As a user	I want to see coupon details such as validity period, applicable items, and usage conditions, so that I can decide when and how to use them.
25	Order Management	As a user	I want to view all my orders, including tickets, membership, and merchandise purchases, so that I can track my transactions.
26	E-Ticket Access	As a user	I want to access my electronic tickets within the Mini Program, so that I can enter venues without collecting physical tickets.

27	Venue Admission	As a user	I want to present my electronic ticket QR code at the venue entrance, so that I can enter the venue smoothly
28	Admission	As a WestK operating staff	I want to verify and redeem tickets purchased via the WeChat Mini Program, so that visitors can be admitted smoothly and securely.
29	Admission	As a WestK operating staff	I want to verify membership cards presented via the WeChat Mini Program, so that member eligibility and benefits are correctly honoured.
30	Admission	As a WestK operating staff	I want to verify and redeem merchandise or coupon entitlements purchased via the WeChat Mini Program, so that on-site fulfilment can be properly managed.
31	Financial Reporting	As a WestK finance staff	I want to view sales records for tickets, membership, merchandise sold via the WeChat Mini Program, so that I can perform accounting recognition and reconciliation.
32	Financial Reporting	As a WestK finance staff	I want to access consolidated transaction and settlement reports for the WeChat Mini Program, so that I can complete monthly financial reconciliation and audit preparation.

Table 3.17.4: Description for the User Story

3.17.5. Business Scenarios

This section provides a functional breakdown of the requirements defined in Section 3.4. Each scenario decomposes high-level requirements into implementable components with uniquely identifiable requirement IDs for design, development, and testing purposes.

3.17.5.1. Registration and Login

This scenario defines how users securely access the WeChat Mini Program in compliance with applicable privacy and data protection requirements.

Overview

When an authentication request is initiated, the system verifies whether the user already has an active session.

- If a valid session exists, the user is redirected directly to the Home Page.
- If no valid session is detected, the system prompts the user to complete authentication using the **One-Click Phone Login**, a native login capability supported by the WeChat Platform, before accessing the Mini Program.

Users register or log in using the mobile phone number registered with their WeChat account. To proceed, users must review and agree to the required privacy terms prior to login or registration. Users may also access the full Privacy Policy document via a dedicated link.

3.17.5.1.1. User Process

The following table describes the user process of Registration and Login.

No.	Component	Description
1	Register and Login	Existing users can log in with their WeChat-registered phone number via the One-Click Phone Login button. New users can register using their WeChat-registered phone number.
2	Service Agreement Checkbox	New users must tick the first four privacy clauses to complete registration. Returning users who have logged in before are also required to tick these clauses to log back in after logging out.
3	Optional Consent	The fifth optional checkbox is for users to consent to receiving direct marketing communications, which can be ticked at their discretion during login or registration.
4	Privacy Policy Link	By clicking the Privacy Policy link, users will be redirected to the full Privacy Policy page published by WestK Enterprise Limited (WKEL) for review.

Table 3.17.5.1.1: Description for the User Process of Registration and Login

3.17.5.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-WC-REG-001	Support one-click registration and login using the WeChat registered mobile phone number.	High	3.17.5.1.1: #1
REQ-WC-REG-002	Enforce mandatory checking of four core privacy and sensitive information terms before login.	High	3.17.5.1.1: #2
REQ-WC-REG-003	Provide an optional checkbox for direct marketing communication consent.	Medium	3.17.5.1.1: #3
REQ-WC-REG-004	Provide a clickable link to view the complete WKEL privacy policy.	High	3.17.5.1.1: #4

Table 3.17.5.1.2: Requirement List of Registration and Login

3.17.5.2. Event Discovery & Event Information

This scenario defines how users discover exhibitions and shows, navigate by venue, and review detailed event information within the WeChat Mini Program before making purchase decisions.

Overview

After successful login, users access the Home Page, which serves as the primary entry point for event discovery. The WeChat Mini Program supports venue-based navigation for **M+**, **HKPM**, and **PA**, together with content discovery, search capability, and visitor guidance information.

From the event discovery and event information journey, users may proceed to appropriate transactional flows supported by the Mini Program, including **ticket purchasing**, **membership card purchasing**, and **online merchandise purchasing**, where applicable.

Users may browse exhibitions and shows by venue, search for specific programmes, and access detailed event information pages to review schedules, venue details, ticket pricing, availability, and any applicable eligibility requirements before proceeding.

3.17.5.2.1. User Process

The following table describes the user process of event discovery & event information.

No.	Component	Description
1	Home Page & Venue Navigation	The Home Page provides venue-based main entry points for M+, HKPM, and PA, enabling users to browse exhibitions and show by venue context.
2	Venue-Specific Listings	Selecting a venue displays a list of exhibitions and shows associated only with the selected venue.
3	Event Discovery	The system supports discovery of featured or recommended exhibitions and shows to assist users in identifying highlighted programmes.
4	Search Capability	Users may search for exhibitions and shows using keyword-based search functionality.
5	Visitor Guidance Information	The Mini Program provides visitor information, including site maps, venue location guidance, and dining information, to support visit planning.
6	Event Detail Page	Users access detailed event pages to review programme descriptions, schedules, venue information, ticket pricing, and availability.

7	Eligibility Information Display	Where applicable, eligibility requirements such as age restrictions or special admission conditions are displayed prior to ticket purchase.
8	Membership Purchase Entry	The system provides access points for users to navigate to membership card information and supported membership purchasing flows.
9	Merchandise Purchasing	The system provides access points for users to navigate to the online merchandise catalogue and supported merchandise purchasing flows.

Table 3.17.5.2.1: Description for Event Discovery and Event Information**3.17.5.2.2. Requirement List**

RID	Description	Priority	Related Process
REQ-WC-HOM-001	Provide venue-based navigation entry points for M+, HKPM, and PA.	High	3.17.5.2.1: #1
REQ-WC-HOM-002	Display venue-specific listings of exhibitions and shows based on user selection.	High	3.17.5.2.1: #2
REQ-WC-HOM-003	Support discovery of featured or recommended exhibitions and shows.	Medium	3.17.5.2.1: #3
REQ-WC-HOM-004	Provide keyword-based search capability for exhibitions and shows.	High	3.17.5.2.1: #4
REQ-WC-HOM-005	Display visitor guidance information including site maps and dining information.	Medium	3.17.5.2.1: #5
REQ-WC-HOM-006	Provide event detail pages displaying programme descriptions, schedules, venue information, ticket pricing, and availability.	High	3.17.5.2.1: #6
REQ-WC-HOM-007	Display eligibility restrictions and special admission conditions prior to ticket purchase where applicable.	High	3.17.5.2.1: #7
REQ-WC-HOM-008	Provide user access points to navigate to supported membership card information and purchasing flows.	High	3.17.5.2.1: #8
REQ-WC-HOM-009	Provide user access points to navigate to the online merchandise catalogue and supported purchasing flows.	High	3.17.5.2.1: #9

Table 3.17.5.2.2: Requirement List of Event Discovery & Event Information

3.17.5.3. Ticket Purchasing

This scenario defines how users purchase tickets for exhibitions and shows within the WeChat Mini Program after completing event discovery and reviewing event information.

Overview

From an event detail page, users may proceed to purchase tickets for the selected exhibition or show. The Mini Program supports both **general admission tickets** and **seat-selection tickets**, subject to event configuration.

During ticket purchasing, eligible membership benefits and coupons may be applied in accordance with defined business rules. Users are presented with an order confirmation summary showing selected tickets, applied benefits, and the final payable amount before completing payment through supported payment methods.

3.17.5.3.1. User Process for General Admission Ticket Purchasing

Applicable to:

Exhibitions, date-based admission tickets, flex entry tickets, or events where seat selection is not required.

Users select applicable general admission ticket types and quantities based on the selected event or date, review pricing and availability, and proceed to confirmation and payment.

3.17.5.3.2. General Admission Ticket Purchasing Requirement Lists

RID	Description	Priority	Related Process
REQ-WC-TKT-GA-001	Support selection of general admission ticket types and quantities.	High	3.17.5.3.1
REQ-WC-TKT-GA-002	Support date-based ticket selection or validity period display where applicable.	High	3.17.5.3.1
REQ-WC-TKT-GA-003	Validate ticket availability based on quantity rules prior to order confirmation.	High	3.17.5.3.1

Table 3.17.5.3.2: Requirement List of General Admission Ticket Purchasing

3.17.5.3.3. User Process for Seat-Selection Ticket Purchasing

Applicable to:

Performing arts events, theatrical shows, cinema shows or shows requiring seat allocation.

Users select a specific session, choose available seats, review seat allocation and pricing, and proceed to confirmation and payment.

3.17.5.3.4. Seat-Selection Ticket Purchasing Requirement Lists

RID	Description	Priority	Related Process
REQ-WC-TKT-SS-001	Provide a seat-selection capability for seat-based events.	High	3.17.5.3.3
REQ-WC-TKT-SS-002	Allow users to select specific seats for a chosen session.	High	3.17.5.3.3
REQ-WC-TKT-SS-003	Prevent double booking of seats during the seat-selection process.	High	3.17.5.3.3
REQ-WC-TKT-SS-004	Release unconfirmed seat holds if payment is not completed within a defined time window.	High	3.17.5.3.3

Table 3.17.5.3.4: Requirement List of General Admission Ticket Purchasing

3.17.5.3.5. User Process for Discounts & Payment

Applicable to both General Admission and Seat-Selection tickets.

The following table describes the user process of Discounts & Payment.

No.	Component	Description
1	Discount Identification	The system identifies eligible membership benefits and coupons applicable to the selected tickets.
2	Discount Application	Applicable membership discounts and coupons are applied to the order in accordance with defined business rules.
3	Order Confirmation	Users review selected tickets or seats, applied benefits, and the final payable amount before proceeding to payment.
4	Payment Completion	Users complete ticket payment using supported payment methods within the WeChat Mini Program.

Table 3.17.5.3.5: User Process for Discount & Payment

3.17.5.3.6. Discounts & Payment Requirement List

RID	Description	Priority	Related Process
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REQ-WC-TKT-COM-001	Identify eligible membership discounts applicable to ticket purchases.	High	3.17.5.3.5: #1
REQ-WC-TKT-COM-002	Identify and apply eligible coupons during ticket purchases.	High	3.17.5.3.5: #2
REQ-WC-TKT-COM-003	Identify and apply eligible coupons during ticket purchases.	High	3.17.5.3.5: #3
REQ-WC-TKT-COM-004	Support ticket payment via WeChat Pay and other platform-supported payment methods.	High	3.17.5.3.5: #4
REQ-WC-TKT-COM-005	Where payment is processed in a non-HKD currency, ensure ticket pricing and settlement are recorded and managed in HKD.	High	3.17.5.3.5: #5

Table 3.17.5.3.6: Requirement List of General Admission Ticket Purchasing

3.17.5.4. Membership Card Purchasing & Management

This scenario defines how users purchase, activate, manage, renew, and use membership cards within the WeChat Mini Program, as well as how membership entitlements are applied during eligible transactions.

This section focuses on **user-facing membership purchasing and usage scenarios only**. Membership card creation, configuration, and discount rule management are governed by dedicated backend configuration processes and are referenced separately.

Overview

From the WeChat Mini Program, users may access membership-related services through designated membership entry points. The Mini Program supports the purchase of eligible membership card types, membership activation and renewal, and ongoing membership management, in accordance with WestK's membership policies.

For multi-user membership cards, the primary cardholder may assign or transfer secondary memberships to other users. A dedicated **Member Centre** is provided for members to manage memberships, access member-exclusive content, and participate in member-only activities or promotions.

Certain membership card types, such as patron-based memberships, may be displayed for reference but are not available for purchase within the WeChat Mini Program. In such cases, the Mini Program provides clear guidance directing users to external patron or sponsorship channels.

Membership card definitions (e.g. card types, validity, entitlements) and membership discount rules are created and managed through backend configuration processes and are not defined within this scenario.

3.17.5.4.1. User Process

No.	Component	Description
1	Membership Entry	Users access membership services via dedicated entry points within the Mini Program.
2	Membership Browsing	Users browse available membership card types and review benefits, validity period, and applicable terms and conditions.
3	Membership Purchase	Users purchase supported membership card types (e.g. individual or family-based memberships) within the Mini Program where applicable.
4	Membership Activation	Purchased membership cards need to be activated within a certain time period, and need go through a user-initiated activation process, depending on membership rules.
5	Membership Renewal	Users with expiring or expired memberships may renew eligible membership cards within the Mini Program.
6	Membership Transfer	User can transfer the purchased cards to other WeChat Friends.
7	Multi-User Membership Management	For multi-user or family-based memberships, the primary cardholder manages secondary membership assignments, including transferring or granting access to other users.
8	Membership Status & Identification	Each membership card is associated with a unique membership identifier, with clear indication of primary and secondary memberships where applicable.
9	Member Centre Access	Members access a dedicated Member Centre page to manage memberships, view benefits, and access member-exclusive content.
10	Member-Exclusive Activities & Promotions	Members may register for member-exclusive events or enjoy member-only promotions and discounts for eligible programmes or merchandise.
11	Membership Benefit Usage	Membership benefits are applied automatically during eligible ticket and merchandise purchasing flows in accordance with defined business rules.

12	Patron Membership Notice	Where patron-based membership cards are displayed, the Mini Program provides clear guidance indicating that purchase is not supported and directs users to external channels.
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Table 3.17.5.4.1: User Process for Membership Card

3.17.5.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-WC-MEM-001	Provide entry points to access membership card services within the Mini Program.	High	3.17.5.4.1: #1
REQ-WC-MEM-002	Display available membership card types with benefits, validity period, and applicable terms and conditions.	High	3.17.5.4.1: #2
REQ-WC-MEM-003	Support the purchase of eligible membership card types within the Mini Program.	High	3.17.5.4.1: #3
REQ-WC-MEM-004	Support activation of purchased membership cards in accordance with defined membership rules.	High	3.17.5.4.1: #4
REQ-WC-MEM-005	Support renewal of eligible membership cards within the Mini Program.	Medium	3.17.5.4.1: #5
REQ-WC-MEM-006	Support transfer of eligible membership cards within the Mini Program.	Medium	3.17.5.4.1: #6
REQ-WC-MEM-007	Support management of multi-user memberships, including assignment or transfer of secondary memberships by the primary cardholder.	High	3.17.5.4.1: #7
REQ-WC-MEM-008	Assign a unique membership QR code to each membership card and clearly distinguish primary and secondary memberships.	High	3.17.5.4.1: #8
REQ-WC-MEM-009	Provide a dedicated Member Centre for membership management and member-exclusive content access.	Medium	3.17.5.4.1: #9
REQ-WC-MEM-010	Support registration for member-exclusive events and access to member-only programmes or promotions.	Medium	3.17.5.4.1: #10
REQ-WC-MEM-011	Apply eligible membership benefits automatically during ticket and merchandise purchasing flows.	High	3.17.5.4.1: #11

REQ-WC-MEM-012	Display patron-based membership cards with clear indication that purchase is not supported within the Mini Program.	High	3.17.5.4.1: #12
REQ-WC-MEM-013	Membership card creation, configuration, and lifecycle setup shall be governed by backend membership configuration processes defined in Section 3.10.	Low	3.10
REQ-WC-MEM-014	Membership discount definitions and eligibility rules shall be governed by backend discount configuration processes defined in Section 3.10.	Low	3.10

Table 3.17.5.4.2: Requirement List of Membership Card

3.17.5.5. Online Merchandise Purchasing

This scenario defines how users browse, purchase, and fulfil merchandise orders within the WeChat Mini Program, either independently or in conjunction with other transactions.

This section focuses on **user-facing merchandise purchasing and fulfilment scenarios only**.

Merchandise creation, pricing configuration, inventory setup, and promotion rules are governed by backend configuration processes and are referenced separately.

Overview

From the WeChat Mini Program, users may access online merchandise through dedicated merchandise entry points. The Mini Program supports merchandise purchasing as a standalone flow or bundled with ticket purchasing, where applicable.

Applicable membership prices or promotional discounts are presented to users during eligible merchandise purchase flows. The eligibility logic and pricing rules are configured and managed in backend systems and are not defined within this scenario.

The Mini Program supports merchandise fulfilment via in-person pickup or delivery. Where a shopping cart contains items from multiple merchants, the Mini Program supports a multi-merchant shopping cart, allowing users to review, confirm, and complete purchases in a single checkout flow, subject to merchant and fulfilment rules. Shopping cart support same delivery mode items only.

3.17.5.5.1. User Process

No.	Component	Description
1	Merchandise Entry	Users access the online merchandise catalogue via dedicated entry points within the Mini Program.
2	Merchandise Browsing	Users browse merchandise listings and review product details, pricing, and availability.

3	Merchandise Selection	Users select merchandise items for purchase, either as a standalone transaction or as part of a ticket purchasing flow where applicable.
4	Multi-Merchant Cart	The system supports a shopping cart that may contain merchandise items from one or multiple merchants.
5	Membership Price & Discount Display	Where applicable, eligible membership prices or promotional discounts are displayed and applied during merchandise purchase flows.
6	Order Confirmation	Users review selected merchandise items, merchant information, applicable discounts, fulfilment options, and the final payable amount before payment.
7	Payment Completion	Users complete merchandise payment using supported payment methods within the Mini Program.
8	Fulfilment Method Selection	Users select applicable fulfilment methods, such as in-person pickup or delivery, based on fulfilment policies and merchant rules.
9	Order Tracking	Users view merchandise order status and fulfilment information within the order management section.

Table 3.17.5.5.1: *User Process of Online Merchandise*

3.17.5.5.2. Requirement List

RID	Description	Priority	Related Process
REQ-WC-MER-001	Provide entry points to access the online merchandise catalogue within the Mini Program.	High	3.17.5.5.1: #1
REQ-WC-MER-002	Display merchandise listings with product details, pricing, and availability.	High	3.17.5.5.1: #2
REQ-WC-MER-003	Support standalone merchandise purchase independent of ticket purchasing.	High	3.17.5.5.1: #3
REQ-WC-MER-004	Support merchandise purchase in conjunction with ticket purchasing flows where applicable.	Medium	3.17.5.5.1: #3
REQ-WC-MER-005	Support a shopping cart containing merchandise items from one or multiple merchants.	High	3.17.5.5.1: #4

REQ-WC-MER-006	Display and apply eligible membership prices or promotional discounts during merchandise purchase flows.	High	3.17.5.5.1: #5
REQ-WC-MER-007	Display an order confirmation summary including merchandise items, merchant details, fulfilment options, applied discounts, and final payable amount.	High	3.17.5.5.1: #6
REQ-WC-MER-008	Support merchandise payment via WeChat Pay and other platform-supported payment methods.	High	3.17.5.5.1: #7
REQ-WC-MER-009	Support fulfilment options such as in-person pickup or delivery in accordance with merchant and fulfilment policies.	High	3.17.5.5.1: #8
REQ-WC-MER-010	Allow users to view merchandise order status and fulfilment information within order management.	High	3.17.5.5.1: #9
REQ-WC-MER-012	Membership pricing and promotional discount eligibility applied to merchandise shall be governed by backend discount configuration processes defined in Section 3.10.	Low	3.10

Table 3.17.5.5.2: Requirement List of Online Merchandise

3.17.5.6. Coupon Management

This scenario defines how users view, apply, and redeem coupons within the WeChat Mini Program, and how coupon status is managed throughout the transaction lifecycle.

This section focuses on **user-facing coupon usage and redemption scenarios only**. Coupon creation, eligibility rules, discount definitions, issuance logic, and lifecycle configuration are governed by backend configuration processes and are referenced separately.

Overview

From the WeChat Mini Program, users may access their available coupons through a centralized coupon interface. Coupons may be applicable to eligible ticket purchases, merchandise purchases, or other supported transactions, subject to defined business rules.

During eligible transactions, applicable coupons are identified and applied automatically or through user selection. Coupon usage status is updated upon successful transaction completion to prevent reuse.

Coupon rules, validity, and redemption conditions are configured and managed through backend systems and are not defined within this scenario.

3.17.5.6.1. User Process

No.	Component	Description
1	Coupon Access	Users access their coupons via designated entry points within the Mini Program (e.g. My Coupons/Customer Hub).
2	Coupon Listing	Users view a list of coupons, including key details such as applicability, validity period, and usage conditions.
3	Coupon Status Display	Each coupon displays under the correct status (e.g. unused, used, expired).
4	Coupon Selection	During eligible transactions, users may select applicable coupons where user selection is supported.
5	Automatic Coupon Application	Where applicable, eligible coupons are applied automatically based on configured rules.
6	Coupon Redemption	Selected or automatically applied coupons are redeemed upon successful order completion.
7	Status Update	Coupon status is updated after redemption to prevent reuse and reflect actual usage.

Table 3.17.5.6.1: Description for the User Process of Coupon Management

3.17.5.6.2. Requirement List

RID	Description	Priority	Related Process
REQ-WC-CPN-001	Provide entry points for users to access their available coupons within the Mini Program.	High	3.17.5.6.1: #1
REQ-WC-CPN-002	Display a list of coupons with applicability, validity period, and usage conditions.	High	3.17.5.6.1: #2
REQ-WC-CPN-003	Display coupon status, including unused, used, and expired states.	High	3.17.5.6.1: #3
REQ-WC-CPN-004	Identify eligible coupons during ticket and merchandise purchasing flows.	High	3.17.5.6.1: #4, #5
REQ-WC-CPN-005	Allow user selection of applicable coupons where manual selection is supported.	Medium	3.17.5.6.1: #4
REQ-WC-CPN-006	Support automatic application of eligible coupons based on backend rules.	High	3.17.5.6.1: #5

REQ-WC-CPN-007	Redeem applied coupons upon successful transaction completion.	High	3.17.5.6.1: #6
REQ-WC-CPN-008	Update coupon status after redemption to prevent reuse.	High	3.17.5.6.1: #7
REQ-WC-CPN-009	Coupon creation, issuance, eligibility logic, discount rules, and lifecycle management shall be governed by backend coupon configuration processes defined in Section 3.11.	Low	3.11

Table 3.17.5.6.2: Requirement List of Coupon Management

3.17.5.7. Order Management & E-Ticket Wallet

This scenario defines how users review and manage their orders and access electronic tickets within the WeChat Mini Program, including completed orders and interrupted or pending payment orders.

Overview

After completing a purchase, or where a purchase process is interrupted (e.g. payment not completed), users may access a centralized order management interface within the WeChat Mini Program to view transaction history, order status, and associated details for **tickets**, **memberships**, **merchandise**, and **coupon usage**.

For orders with **pending payment status**, users may resume and complete payment within a configured time window (e.g. 15 minutes from order creation), subject to system and payment platform rules.

For ticket orders with successful payment, users may access an e-Ticket Wallet, where electronic tickets are displayed with unique QR codes for admission. Ticket usage status (e.g. unused, used, expired) is clearly presented to support visit preparation prior to on-site admission.

3.17.5.7.1. User Process

No.	Component	Description
1	Order Management Access	Users access a centralized order management section to view all orders created via the Mini Program.
2	Order Overview & Details	Users view order summaries and detailed information, including order items, transaction amount, order type (ticket, membership, merchandise), and current order status.
3	Pending Payment Handling	For orders in pending payment status, users may resume and complete payment within the allowed payment window (15 minutes) before order expiry.

4	E-Ticket Wallet Access	For completed ticket orders, users access an e-Ticket Wallet displaying associated electronic tickets.
5	Ticket Display for Admission	Users view electronic tickets with usage status and QR codes for admission reference, in accordance with defined admission procedures.

Table 3.17.5.7.1: Description for the User Process of Order Management & E-Ticket Wallet

3.17.5.7.2. Requirement List

RID	Description	Priority	Related Process
REQ-WC-ORD-001	Provide a centralized order management interface for users to access all orders created via the Mini Program.	High	3.17.5.7.1: #1
REQ-WC-ORD-002	Display order summaries and detailed order information, including order items, transaction amount, and current order status.	High	3.17.5.7.1: #2
REQ-WC-ORD-003	Support display and handling of pending payment orders, allowing users to resume payment within the defined payment validity window.	High	3.17.5.7.1: #3
REQ-WC-ORD-004	Provide a dedicated e-Ticket Wallet displaying electronic tickets associated with successfully paid ticket orders.	High	3.17.5.7.1: #4
REQ-WC-ORD-005	Display electronic ticket usage status and a unique QR code for admission purposes.	High	3.17.5.7.1: #5
REQ-WC-ORD-006	Admission verification, QR code scanning, and redemption logic shall follow the Admission & Redemption scenario defined in Section 3.9.	Low	3.9

Table 3.17.5.7.2: Requirement List for Order Management & E-Ticket

3.17.5.8. Admission & Redemption

This scenario confirms that electronic credentials issued via the WeChat Mini Program can be validated and redeemed at WestK venues through the established admission control process.

Detailed admission workflows, validation logic, device usage, and operational procedures are defined in the Admission & Admission Control section and are not repeated here. Please refer to section 3.9.

Overview

Tickets, membership cards, and coupons issued through the WeChat Mini Program may be presented in the form of QR codes. These QR codes serve as official digital credentials that can be verified and redeemed by WestK operating staff at venue entry points or designated redemption locations.

The WeChat Mini Program supports the presentation of valid QR codes for admission or redemption purposes. Verification and redemption activities shall follow the operational rules, workflows, and control mechanisms defined in the Admission & Admission Control specifications.

3.17.5.8.1. Scope

No.	Component	Description
1	Ticket QR Code	Electronic tickets issued via the WeChat Mini Program can be verified and redeemed for venue admission.
2	Membership QR Code	Membership cards presented via the Mini Program can be verified for eligibility and access control.
3	Coupon QR Code	Eligible coupons issued via the Mini Program can be verified and redeemed according to defined usage rules.

Table 3.17.5.8.1: Description for the Scope of Admission & Redemption

3.17.5.8.2. Requirement List

RID	Description	Reference Process
REQ-WC-ADM-001	QR codes issued for tickets, membership cards, and coupons via the WeChat Mini Program shall be supported as valid credentials for admission and redemption.	3.9
REQ-WC-ADM-002	Validation, redemption, and exception handling of QR codes shall follow the established Admission & Admission Control processes.	3.9

Table 3.17.5.8.2: Requirement List of Admission & Redemption

3.17.5.9. Backend Integration & Data Governance

This section serves as a reference to clarify the system boundary between the WeChat Mini Program and backend ticketing and data governance systems. It does not introduce additional user or operational scenarios.

Overview

All transactional and operational data generated through the WeChat Mini Program, including ticket purchases, membership card purchases, merchandise orders, coupon usage, admission status, and order updates, are synchronized with backend systems in accordance with defined integration and data governance principles.

The WeChat Mini Program functions solely as a **front-end transaction and interaction channel**. It does not act as the system of record for ticketing configuration, inventory management, reporting, reconciliation, or data governance.

3.18. Douyin Mini Program

This section outlines the end-to-end business processes for the Douyin Mini Program integration. It covers high-level and detailed business requirements, the fulfilment approach, customer journey, and the functional specifications for both frontend and backend workflows.

3.18.1. Business Requirements

From the business perspective, WestK requires the Douyin Mini Program integration to support ticket sales for Mainland China users while ensuring operational efficiency, secure data handling, and compliance with platform policies.

The following business requirements have been identified.

3.18.1.1. Registration & Login

WestK requires the Douyin Mini Program to support seamless user authentication using Douyin's native login, with automatic account binding and secure session management to ensure identity consistency for order management and ticket ownership.

3.18.1.2. Douyin Ticket Sales Channel

WestK requires the ability to list exhibitions for M+/HKPM and shows for M+/PA on the Douyin Mini Program, enabling Mainland China users to browse and purchase tickets directly within the platform. In addition, WestK requires an intuitive home page that helps users easily discover and access events, featuring event listings, popular highlights, keyword search, and clear entry points to detailed event pages.

3.18.1.3. E-Ticket Wallet

WestK requires that users be able to enter the venue directly using electronic tickets generated through the Douyin Mini Program without exchanging physical tickets. Each ticket must contain a unique QR code for verification at the venue.

3.18.1.4. Order Traceability Across Systems

WestK requires that users be able to review their orders and related details within a centralized customer hub, including order status (such as pending payment, paid, cancelled, or refunded) and the associated ticket information. Furthermore, all orders created through the Douyin Mini Program must be consistently recorded both on the

Douyin platform—for end users to easily locate and review their orders—and within AnTank’s backend systems, where staff members can perform operational queries and reviews. Clear identifier mappings must be maintained across both systems to ensure complete cross-system traceability and to support financial and operational reconciliation processes.

3.18.1.5. Payment, Currency Handling & Settlement

WestK requires that currency conversion mechanisms be implemented to support internal financial reconciliation and reporting in Hong Kong Dollars (HKD), in accordance with WestK’s financial policies.

3.18.1.6. Ticket Verification & Admission Control

WestK requires that each E-Ticket contains a unique QR code and that admission verification updates ticket/order status in real time across Douyin and AnTank, preventing duplicate entries and ensuring operational traceability.

3.18.1.7. Integration with Ticketing System

WestK requires the Douyin Mini Program to integrate a centralized ticketing system which serves as the central system for ticket inventory management, pricing control, order processing, and ticket lifecycle management. Furthermore, WestK requires comprehensive access to ticket sales and operational data through a unified backend reporting portal, enabling financial reconciliation, operational oversight, and show analysis across all Douyin related transactions.

3.18.1.8. Privacy, Security, Scalability & Inclusive Accessibility

WestK requires compliant handling of personal data, including minimizing data collection, restricting use to authorized purposes, encrypting sensitive information, and enforcing strict access controls. The system must also support high concurrency during peak sales to ensure stable experience for all users. In line with WestK’s Diversity & Inclusion principles, wheelchair-accessible seating must be clearly identifiable and managed as a dedicated seating category to support accessibility needs, with full allocation logic planned for a future phase.

3.18.2. Business Requirement Analysis

This section provides further analysis and clarification of the business requirements described in Section 3.18.1. Each subsection elaborates on the business intent, operational considerations and functional expectations underpinning the Douyin Mini Program ticketing solution.

3.18.2.1. User Registration & Login

Map to Business Requirement 3.18.1.1 and 3.18.1.8.

- The Mini Program shall support user authentication via the Douyin Platform login mechanism, enabling seamless entry without requiring separate account creation.
- User accounts shall be created or bound automatically.
- Maintain user identity consistency for order management and ticket ownership.
- The system ensures secure session management and user data protection.

3.18.2.2. Home Page Display & Detailed Event Display

Map to Business Requirement 3.18.1.2.

The Douyin Mini Program shall provide intuitive event discovery capabilities:

- Home Page:
 - Event listings for M+/HKPM exhibitions and M+/PA shows.
 - Popular exhibitions and shows highlighting.
 - Keyword search functionality.
- Event Detail Page:
 - Event title, description, and poster.
 - Show dates, session times, and venue information.
 - Ticket tiers, pricing and seat availability and description.
 - Ticket availability status.

3.18.2.3. E-Ticket Wallet & Order Traceability

Map to Business Requirement 3.18.1.3 and 3.18.1.4.

- User shall be able to access a centralized hub – “My Account” page to:
 - View all orders (Pending Payment/Paid/Cancelled/Refunded).
 - View order details including Ticket information, QR code, and order status.
- A dedicated page shall display:
 - Unused E-Tickets.
 - Used E-Tickets.
 - Each E-Ticket shall contain a unique QR code for admission.
- Cross-System Traceability:
 - Order records will be aligned across Douyin Platform and AnTank Backend.
 - Mappings between Douyin Order ID, AnTank Order ID, Ticket Codes, and payment references must be maintained for financial and operational reconciliation.

3.18.2.4. Ticket Purchasing Process

Map to Business Requirement 3.18.1.2 and 3.18.1.5.

- The mini program should support two types of ticketing:
 - Admission ticket:
 - Ticket type selection
 - Quantity selection
 - Seat-selection ticket:
 - Interactive online seating plan
 - Seat selection
- An order confirmation page shall display including:
 - Selected ticket details including ticket types or seats information
 - Total payable amount
 - Payment Options
 - Real name verification is required during purchasing (per Douyin rules)

Platform constraints:

Due to Douyin platform requirements, the Mini Program must be operated by a Mainland China-registered entity and supports RMB (CNY) pricing only. According to Douyin's payment requirements, users must complete real-name verification before using Douyin Pay.

To meet WestK's business requirement for Hong Kong Dollar (HKD) pricing and financial settlement, ticket prices defined in HKD will be converted into RMB for listing and transaction purposes on the Douyin platform. Appropriate currency conversion and reconciliation mechanisms will be implemented to ensure alignment with WestK's financial policies.

In addition, due to Douyin's event category restrictions and approval process, only Admission tickets and seat-selection tickets will be launched on the platform.

3.18.2.5. Payment & Payment Result Handling

Map to Business Requirement 3.18.1.4 and 3.18.1.5.

- The Douyin Mini Program shall integrate with Douyin-supported payment methods.
 - Include WeChat Pay, Alipay China, Credit Card, Debit Card.
- During the payment flow:
 - Allow users to pay within a 15 min pending time window.
 - Allow users to cancel orders (before orders expiration).
- Upon a successfully Payment:
 - A payment success page shall be displayed.
 - E-Ticket containing QR code shall be automatically issued.

- AnTank backend system shall receive and process Douyin Payment callbacks to confirm payment status and successfully log an order in backend system.

3.18.2.6. Admission Control

Map to Business Requirement 3.18.1.6.

- Each e-Ticket shall generate a unique QR code for admission.
- Upon scanning:
 - Ticket status shall update to “Used” in real time.
 - The corresponding order status shall be synchronized to AnTank Backend system.
 - Verification data shall be recorded for audit purposes.

3.18.2.7. Backend Portal Requirements

Map to Business Requirement 3.18.1.7.

- A centralized backend portal to:
 - Create and modify the events displayed in Douyin Mini Program.
 - View and process orders sold in Douyin Mini Program.
 - Monitor Douyin Admission status synchronization.
 - Access sales reporting and operational analytics.

3.18.2.8. Douyin Mini Program Integration with AnTank Backend System

Map to Business Requirement 3.18.1.7.

- All ticket purchasing functions shall be implemented via the Douyin Mini Program.
- All Events and ticket master data will be sourced from the AnTank Backend system.
- Orders, payments, and ticket issuance can be tracked via the Backend System.
- Customers shall be able to enter the venue directly using the E-Ticket/QR code within Douyin Mini Program, without the need for physical ticket exchange.

3.18.2.9. System Show & High Concurrency

Map to Business Requirement 3.18.1.8.

- The solution should be designed to support high concurrent user access during peak ticket sales periods (e.g., major releases), ensuring system stability, transaction integrity, and consistent user experience.

- For the current release, the wheelchair seating allocation logic is not included in the implementation scope. The detailed handling approach will be clarified in the Approach section and addressed in a future phase.

3.18.3. Operating Model

This initiative uses a **three-party operating model** to deliver ticket sales through a Douyin Mini Program.

- **AnTank** delivers and operates the **ticketing technology layer** (provides ticketing system and Mini Program development).
 - Provides the **ticketing system** capabilities: event/session setup interfaces, inventory management, order lifecycle, ticket issuance (e-ticket/QR), order status queries, and reporting exports.
 - Develops and maintains the **Douyin Mini Program**: user journey, Douyin OAuth integration, UI/UX, basic operational dashboards.
 - Integrates with payment endpoints as required by the solution design (typically via Douyin Pay APIs).
 - Ensures functional correctness, system availability, logging/monitoring, and technical support within SLA.
- **Charisma** provides **Douyin operations**, acting as the operational interface for Douyin account opening and operation, payment configuration, monitoring, and **monthly settlement coordination** with WestK **Finance**.
 - Serves as the **operations bridge** between Douyin and WestK and oversees **Douyin Payment**.
 - Provide customer service, payment operations monitoring and incident triage (payment exceptions, callback anomalies, reconciliation support).
 - Owns the **monthly settlement process coordination** with WestK Finance team, e.g. preparing/collecting settlement statements, assisting reconciliation, tracking discrepancies to closure, ensuring monthly settlement completeness and audit readiness.
- **WestK** is the business owner, who
 - Owns the ticketing business (events, pricing, inventory policies, customer policies).
 - Owns customer relationships and defines after-sales policies (subject to agreed scope).
 - Receives settlement reporting and performs accounting recognition and reconciliation.
 - Acts as the primary finance counterparty for monthly settlement.

End-to-End Ticket Purchase Flow via Douyin Mini Program

The **Diagram 3.18.3** below is for illustrative purposes only. It is intended to present the roles involved and delineate the interactions across the business process flow. For the detailed user journey and end-to-end process specifications, please refer to **Section 3.18.4**.

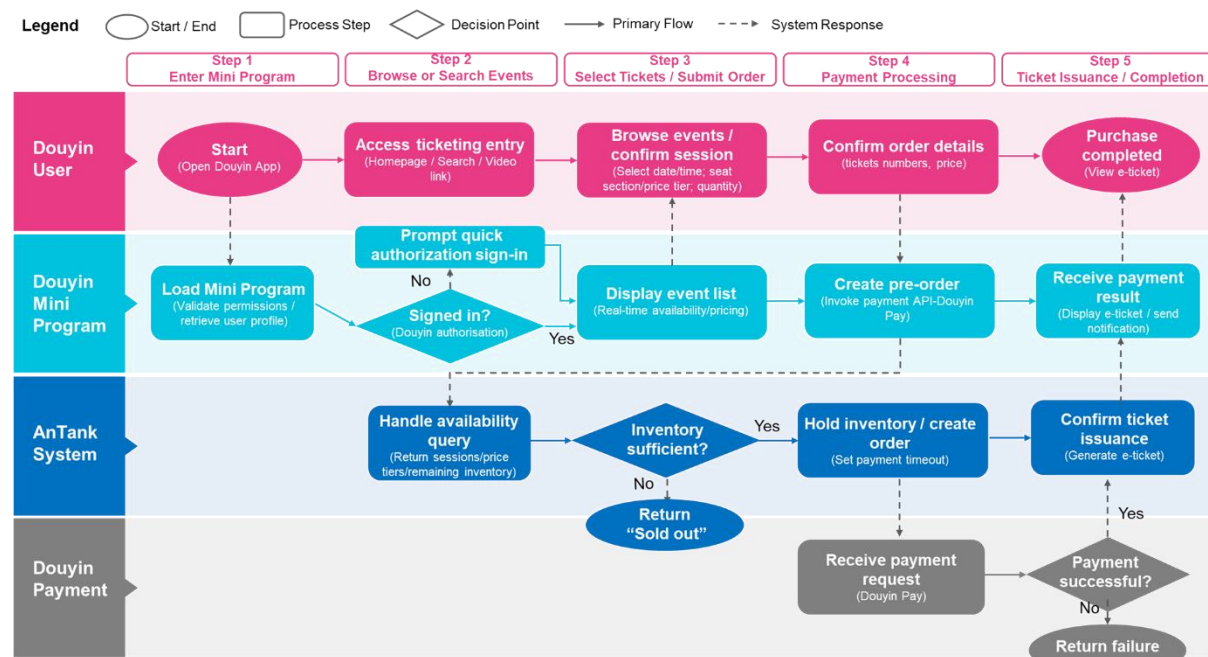


Diagram 3.18.3: Business Process Flow

3.18.4. Customer Journey

This section describes what users can do after entering the Douyin Mini Program.

The user story is used to describe the user's needs, goals, and expected outcomes, providing clarity on what the user wants to achieve and why the functionality is required. When users enter the Douyin Mini Program, they can complete the entire ticket purchasing and admission process without leaving the platform.

No.	Category	Role	User Story
1	Registration and Login	As a Douyin user	I want to log in with my phone number which binds with my Douyin account, so that I can access the Mini Program quickly and conveniently.
2	Event Discovery	As a potential customer	I want to view a homepage highlighting popular exhibitions and shows, so that I can easily discover trending events.
3	Event Discovery	As a potential customer	I want to use the keyword search functionality so that I can quickly find specific events I am interested in.
4	Event Discovery	As a potential customer	I want to view a detailed event page including the poster, dates, price, venue, etc., so that I have all the information needed to make a purchase decision.
5	Ticket Purchasing	As a customer	I want to select my desired ticket type and quantity for Admission events, so that I can

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			purchase the exact number of tickets my group needs.
6	Ticket Purchasing	As a customer	I want to use an interactive online seating plan when buying tickets for shows, so that I can choose my preferred seats quickly and easily.
7	Ticket Purchasing	As a customer	I want to see ticket prices listed in RMB (CNY), so that I can understand the cost in my local and platform-supported currency.
8	Ticket Purchasing	As a customer	I want to review an order confirmation page showing my selected tickets and total payable amount, so that I can verify my order details before proceeding with payment.
9	Checkout & Payment	As a customer	I want to pay using Douyin-supported methods (WeChat Pay, Alipay China, Credit/Debit Card), so that I can check out securely using my preferred payment method.
10	Checkout & Payment	As a customer	I want to see a payment success page to confirm my completed order and view my E-Ticket QR code for convenient admission.
11	Order Management	As a registered user	I want to access a "My Account" page to assess the status of all my orders, so that I can keep track of my purchases.
12	Order Management	As a registered user	I want to access an independent E-Ticket management page displaying my unused and used tickets with their QR codes, so that I can easily find and present my tickets on the day of the event.
13	Venue Admission	As an event attendee	I want to scan my E-Ticket QR code at the venue, so that I can enter directly without needing to queue for physical ticket exchange.
14	Venue Admission	As an event attendee	I want to complete real-name verification during admission (if required), so that I comply with the event's security and ticketing policies.
15	Backend Integration	As a WestK Staff	I want all event and ticket data to be synchronized from different BU's Backend to the Douyin's Backend, so that the event can be managed efficiently.

16	Backend Integration	As a WestK Staff	I want to use a centralized backend portal to create and modify the events displayed on the Douyin Mini Program, so that I can easily control the listed event.
17	Backend Ticket Report	As a WestK Staff	I want to access a centralized backend ticketing report so that I can view business-related ticketing data that is relevant to my team's operations.

Table 3.18.4: Description for the User Story

3.18.5. Business Scenarios

3.18.5.1. Registration and Login

This process defines how users access the Douyin Mini Program securely and in compliance with privacy regulations.

- **User Process:** When the system initiates an authentication request, it verifies whether the user is already logged in. If the user has an active session, they will be directed to the Home Page directly. If no valid session is detected, the system will prompt the user to complete authentication via the "One-Click Phone Login" process (which is a native login function supported by Douyin Platform) before accessing the Mini Program.
- **Privacy Policies:** Users register or log in using their Douyin registered phone number. To proceed, users must check and agree to the privacy terms before registration or login. Users can also click a "Privacy Policy" link, which redirects them to the full privacy policy page. The Privacy Policy content is issued by WestK.

3.18.5.1.1. User Process

The following table describes the user process of Registration and Login.

No.	Component	Description
1	Register and Login	Users can register or log in using their Douyin registered phone number via the "One-Click Phone Login" button, which is a native login function supported by Douyin Platform.
2	Privacy Protection Policy Checkbox	Users must agree with the Privacy Protection Policy before successful login or registration.
3	Privacy Policy Link	Clicking the "Privacy Policy" link redirects users to the full privacy policy page. The Privacy Policy content is issued by WestK.

Table 3.18.5.1.1: Description for the User Process of Registration and Login

3.18.5.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-DY-REG-001	Support one-click registration and login using the Douyin registered mobile phone number.	High	3.18.5.1.1: #1
REQ-DY-REG-002	Enforce mandatory checking of the privacy protection policy issued by WestK before logging in.	High	3.18.5.1.1: #2
REQ-DY-REG-003	Provide a clickable link to view the complete WestK privacy policy.	High	3.18.5.1.1: #3

Table 3.18.5.1.2: Requirement List of Registration and Login

3.18.5.2. Home Page

The Home Page serves as the central navigation page:

- **Global Search:** The page features a Global Search function allowing instant searches for specific exhibitions, shows.
- **Hero Banner:** A Hero Banner dynamically highlights the current significant exhibitions, shows, or the upcoming major events.
- **Ticketing Purchase Navigation:** Users can access the "Ticketing" page to purchase tickets or redirect to different venue pages to visit dedicated ticketing purchase pages for M+, HKPM, and PA.
- **Featured Exhibitions & Shows:** The page displays Featured Exhibitions and Featured Shows, allowing users to view event information or purchase tickets directly.
- **Customer Service:** The button redirects users to an online customer service portal with detailed contact information.

3.18.5.2.1. User Process

No.	Component	Description
1	Hero Banner	A banner dynamically highlighting the current significant shows, exhibitions, or upcoming major events.
2	Ticketing Button	Redirect to pages of purchasing tickets for exhibitions and shows. The Ticket Purchase page can filter the events by its type (exhibitions or shows) or by venue (HKPM, M+, and PA). The Ticket Purchase page also supports calendar mode, which allows users to check available events at different dates via a calendar.

3	Venue Redirection	Redirect to dedicated ticket purchase page for M+, HKPM, and PA. The ticket purchase page will filter the events by venue and only display the events held at designated venue.
4	Global Search	Allows instant searches for the name of specific exhibitions and shows. The search follows the inclusion rule, which means if an event name includes the searching content, the event will be displayed in the searching result.
5	Featured Exhibitions	List all current popular exhibitions in this section. Tapping any entry navigates to the details page for viewing event information or purchasing tickets directly.
6	Featured Shows	List all current popular Shows in this section. Tapping any entry navigates to the details page for viewing event information or purchasing tickets directly.
7	Online Customer Service	Redirect to an Online Customer Service Page for online customer support with detailed contact information. The user can directly use the online chat box to communicate with Online Customer Support.

Table 3.18.5.1.2: Description for the User Process of Home Page**3.18.5.2.2. Requirement List**

RID	Description	Priority	Related Process
REQ-DY-HOM-001	Support global search functionality for exhibitions and shows.	High	3.18.5.2.1: #4
REQ-DY-HOM-002	Display a Hero Banner linked to major event details.	High	3.18.5.2.1: #1
REQ-DY-HOM-003	Provide a redirection link to the ticket purchase page.	High	3.18.5.2.1: #2
REQ-DY-HOM-004	The ticket purchase page supports filter by event type and venue.	High	3.18.5.2.1: #2
REQ-DY-HOM-005	The ticket purchase page supports calendar mode to check event s at different date.	High	3.18.5.2.1: #2
REQ-DY-HOM-006	Provide direct redirection links to M+, HKPM, and PA ticket purchase page.	High	3.18.5.2.1: #3

REQ-DY-HOM-007	Provide a redirection button to the customer service page.	High	3.18.5.2.1: #7
REQ-DY-HOM-008	The customer service page provides an online chat box for the user to communicate with staff directly.	High	3.18.5.2.1: #7
REQ-DY-HOM-009	Display Featured Exhibitions list. Click the exhibition to redirect user to the event detail page.	High	3.18.5.2.1: #5
REQ-DY-HOM-010	Display Featured Shows list. Click the shows to redirect user to the event detail page.	High	3.18.5.2.1: #6

Table 3.18.5.2.2: Requirement List of Home Page

3.18.5.3. My Ticket Code

This section provides users with quick access to their purchased tickets for admission.

- **User Process:** Users enter the "My Ticket Code" page from the Home Page, where they can toggle between two tabs: "Tickets" (shown by default) and "Flex Tickets" (A Long-Term Ticket refers to a multi-day pass that can be used across more than one day).
- **Same-Day Tickets:** Displays all same-day tickets in one location, showing the ticket QR code.
- **Flex Tickets:** Displays Flex tickets in one location, showing the QR code.

3.18.5.3.1. User Process

The following table describes the user process of My Ticket Code.

No.	Component	Description
1	My Ticket Code Overview	Users can enter the My Ticket Code page from the home page. There are two tabs corresponding to "Tickets" and "Long Term Tickets". The "Tickets" tab is shown by default.
2	Same-Day Ticket	Users can view all the same-day tickets in one convenient location. After clicking the ticket, the ticket's QR code can be displayed.
3	Flex Ticket	Users can view all the Flex tickets in one convenient location. After clicking the ticket, the ticket's QR code can be displayed. A Long-Term Ticket refers to a multi-day pass that can be used across more than one day.

Table 3.18.5.3.1: Description for the User Process of My Ticket Code

3.18.5.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-DY-TIC-001	Users can enter the My Ticket Code page from Home Page.	High	3.18.5.3.1: #1
REQ-DY-TIC-002	Users can view their tickets on My Ticket Code pages.	High	3.18.5.3.1: #2, #3
REQ-DY-TIC-003	Users can select different types of ticket: Same-day Ticket or Flex Ticket.	High	3.18.5.3.1: #1
REQ-DY-TIC-004	Ticket QR codes can be displayed for valid tickets.	High	3.18.5.3.1: #2, #3

Table 3.18.5.3.2: Requirement List of My Ticket Code

3.18.5.4. My Account

The My Account page is a centralized dashboard for users to manage their personal information, orders, and system preferences.

- **Personalization Settings:** Users can access "Account Setting" by tapping their avatar to manage specific preferences, e.g.: update their profile, toggle Marketing Notifications, manage System Permissions, or delete their account. Users can also log out through this page.
- **Management Hubs:**
 - **Orders:** A centralized hub to manage all orders, with options to categorize by type and filter by payment status.
 - **Tickets:** A hub to manage all tickets, with options to categorize by ticket type.
- **Company Information:** The page includes an "Operation License" section showing company qualifications, and a "Contact Us" section with contact details for different WestK departments.

3.18.5.4.1. User Process

No.	Component	Description
1	My Account Page	Provide a landing page for accessing different personal information and related information, including account settings, orders, tickets, operation license, and contact us.
2	Tickets	A centralized hub for managing all tickets, providing a clear overview of all tickets. Tickets can be categorized by their types (all tickets, shows tickets, Flex tickets, and exhibition tickets).

		Click the ticket to display the order and the QR code.
3	Orders	A centralized hub for managing all orders across exhibitions and shows. You can categorize orders by type, and filter by payment status. Click the order to redirect to the order detail page.
4	Account Setting	Users can access the Account Setting page by tapping the avatar at My Account page, where you can manage specific account preferences.
5	Operation License	Users can view operation licenses and qualifications that the company possesses.
6	Contact Us	Users can view the contact information for WestK, including email addresses, telephone numbers, and contact addresses for different business units or departments.
7	Avatar & Nickname	Allows users to personalize your account identity by uploading or changing your profile avatar and updating your display nickname.
8	Marketing Notifications	Users can toggle settings to subscribe or unsubscribe from promotional updates.
9	System Permissions	Users can view and manage specific system-level permissions granted to the app (Service Message Management, Personal Information and Permission Usage Record, and Phone Number Management).
10	Security Setting	Users can delete their account on this page.
11	Log out	User can log out via the “Log Out” button at the Account Setting page.

Table 3.18.5.4.1: Description for the My Account Page User Process

3.18.5.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-DY-ACC-001	Provide a My Account page for users to access different account settings.	High	3.18.5.4.1: #1
REQ-DY-ACC-002	Allow users to view and categorize orders by type and filter by payment status.	High	3.18.5.4.1: #2

REQ-DY-ACC-003	Provide a centralized hub to manage and overview all tickets and filter by type.	High	3.18.5.4.1: #3
REQ-DY-ACC-004	Display company operation licenses and contact information for different departments.	High	3.18.5.4.1: #5
REQ-DY-ACC-005	Display contact information for different business units.	High	3.18.5.4.1: #6
REQ-DY-ACC-006	Support user profile personalization, including updating the avatar and display nickname.	High	3.18.5.4.1: #7
REQ-DY-ACC-007	Allow users to toggle settings to subscribe or unsubscribe from marketing notifications.	High	3.18.5.4.1: #8
REQ-DY-ACC-008	Allow users to view and manage specific system-level permissions granted to the app.	High	3.18.5.4.1: #9
REQ-DY-ACC-009	Provide a portal to delete the account.	High	3.18.5.4.1: #10
REQ-DY-ACC-010	Provide a button to log out the current account.	High	3.18.5.4.1: #11

Table 3.18.5.4.2: Requirement List of My Account

3.18.5.5. Admission Ticket Purchase

This process outlines the workflow for purchasing Admission tickets without seat selection.

- **Selection & Confirmation:** Users view "Event Detail" (introductions, dates, prices, venue). They select a Date and Session, followed by the Ticket Type & Quantity, which dynamically updates the total price.
- **Payment Processing:** Users can confirm the order and complete the payment, and an order number will be generated. Alternatively, payment can be cancelled on the confirmation page.
- **Order Pending State:** If unpaid, the order remains in a "Pending" state for 15 minutes, allowing the user to complete the transaction; it is automatically cancelled after 15 minutes if payment is not received.
- **Completion:** Upon success, users can view their ticket information and admission QR code.

3.18.5.5.1. User Process

The following table describes the user process of Admission Ticket Purchase.

No.	Component	Description
1	Event Detail	Users can check program introductions, date and time, price, venue location, and event content in the event detail page.

2	Date and Session Selection	Users can select different dates and sessions.
3	Select Ticket Type	Users can select tickets of different types. The ticket type selection page while dynamically updating the total prices.
4	Order Confirmation	Users can check the calculated subtotals and confirm the final payment.
5	Complete Payment	After the payment is completed, user can view the order in detail with an order number.
6	Cancel Payment	Payment can be cancelled at the order confirmation page by tapping the back button.
7	Order Pending	The unpaid order will be in the pending state for 15 minutes. The user will still be able to pay for the order within 15 minutes. The order will be automatically cancelled after 15 minutes.
8	View Tickets	Users can view detailed tickets for information on the order completion page. User can click “View Ticket Code” button to view the admission ticket code.

Table 3.18.5.5.1: Description for the User Process of Admission Ticket Purchase

3.18.5.5.2. Requirement List

RID	Description	Priority	Related Process
REQ-DY-UTP-001	Display comprehensive event information including introductions, schedules, and ticket availability on the details page.	High	3.18.5.5.1: #1
REQ-DY-UTP-002	Provide an interactive calendar interface for selecting dates and browsing available sessions.	High	3.18.5.5.1: #2
REQ-DY-UTP-003	Display real-time ticket availability, including labelling sold-out tickets.	High	3.18.5.5.1: #2
REQ-DY-UTP-004	Support the selection of specific ticket types and quantities while dynamically updating the total Price.	High	3.18.5.5.1: #3
REQ-DY-UTP-005	Require users to accept purchase terms before confirming order submission.	High	3.18.5.5.1: #4
REQ-DY-UTP-006	Support different payment methods.	High	3.18.5.5.1: #5

REQ-DY-UTP-007	Display a payment completion page confirming the transaction status, order number, and payment amount.	High	3.18.5.5.1: #5
REQ-DY-UTP-008	Order can be cancelled on the order confirmation page.	High	3.18.5.5.1: #6
REQ-DY-UTP-009	Order will be pending for 15 minutes if the user does not complete the payment.	High	3.18.5.5.1: #7
REQ-DY-UTP-010	The order will be automatically cancelled if the order is unpaid after 15 minutes.	High	3.18.5.5.1: #7
REQ-DY-UTP-011	Generate an order detail page providing individual admission QR codes and full transaction records.	High	3.18.5.5.1: #8

Table 3.18.5.5.2: Requirement List of Admission Ticket Purchase

3.18.5.6. Seated Ticket Purchase

This process outlines the workflow for purchasing tickets with seat selection.

- **Seats Selection:** Users view "Event Detail" (introductions, dates, prices, venue). They select a Date and Session, followed by the seat selection page which allows users to select their seats directly on the seating plan. After the seat selection, users will enter the order confirmation page to complete the payment.
- **Payment Processing:** Order number will be generated when users complete the payment. Alternatively, payment can be cancelled on the confirmation page.
- **Order Pending State:** If unpaid, the order remains in a "Pending" state for 15 minutes, allowing the user to complete the transaction; it is automatically cancelled after 15 minutes if payment is not received.
- **Completion:** Upon success, users can view their ticket information and admission QR code.

3.18.5.6.1. User Process

The following table describes the user process of Seated Ticket Purchase.

No.	Component	Description
1	Event Detail	Users can check program introductions, date and time, price, venue location, and event content in the event detail page.
2	Date and Session Selection	Users can select different dates and sessions.
3	Select Seats	Users can select seats with different location or price, while the total price will be calculated dynamically and displayed on the seat's selection page.

4	Order Confirmation	Users can check the calculated subtotals and confirm the final payment.
5	Complete Payment	After the payment is completed, users can view the order in detail with an order number.
6	Cancel Payment	Payment can be cancelled at the order confirmation page by tapping the back button.
7	Order Pending	The unpaid order will be in the pending state for 15 minutes. The user will still be able to pay for the order within 15 minutes. The order will be automatically cancelled after 15 minutes.
8	View Tickets	Users can view detailed tickets for information on the order completion page. User can click the “View Ticket Code” button to view the admission ticket code.

Table 3.18.5.6.1: Description for the User Process of Seated Ticket Purchase

3.18.5.6.2. Requirement List

RID	Description	Priority	Related Process
REQ-DY-STP-001	Display comprehensive event information including introductions, schedules, and ticket availability on the details page.	High	3.18.5.6.1: #1
REQ-DY-STP-002	Provide an interactive calendar interface for selecting dates and browsing available sessions.	High	3.18.5.6.1: #2
REQ-DY-STP-003	Display real-time ticket availability, including labelling sold-out tickets.	High	3.18.5.6.1: #2
REQ-DY-STP-004	Support the selection of specific seats while dynamically updating the total Price.	High	3.18.5.6.1: #3
REQ-DY-STP-005	Require users to accept purchase terms before confirming order submission.	High	3.18.5.6.1: #4
REQ-DY-STP-006	Support different payment methods.	High	3.18.5.6.1: #5
REQ-DY-STP-007	Display a payment completion page confirming the transaction status, order number, and payment amount.	High	3.18.5.6.1: #5
REQ-DY-STP-008	Order can be cancelled on the order confirmation page.	High	3.18.5.6.1: #6
REQ-DY-STP-009	Order will be pending for 15 minutes if the user does not complete the payment.	High	3.18.5.6.1: #7

REQ-DY-STP-010	The order will be automatically cancelled if the order is unpaid after 15 minutes.	High	3.18.5.6.1: #7
REQ-DY-STP-011	Generate an order detail page providing individual admission QR codes and full transaction records.	High	3.18.5.6.1: #8

Table 3.18.5.6.2: Requirement List of Seated Ticket Purchase

3.18.5.7. Douyin Account to Company Account Transfer

This section details the financial backend workflow of how funds move from user purchases to the company's bank account.

- **Fund Holding & Status:** When a user completes a purchase, funds are temporarily held on the Douyin platform. The order's admission status on Douyin is initiated to "Not Completed".
- **Admission Modes:** Administrators must set one of three admission modes before listing an event:
 - **Order Complete Mode:** Order Admission Status is labelled "Completed" immediately after purchase.
 - **Session Complete Mode:** Order Admission Status is labelled "Completed" automatically when the session finishes.
 - **Admission Complete Mode:** Order Admission Status is labelled "Completed" only after the ticket is successfully scanned by AnTank Admission App.
- **Fund Transfer:** Once the Admission Status of an order turns "Completed," funds will be available to move to the Douyin Transferable Account manually. Douyin deducts relevant fees (commission, advertisement) during the transfer. Finally, funds will be available to transfer to the designated Company Bank Account manually.

3.18.5.7.1. User Process

The following table describes the user process of Douyin Account to Company Account Transfer.

No.	Component	Description
1	Funds Hold	When the user completes the ticket purchase, the funds are temporarily held on the Douyin platform.
2	Douyin Admission Status	The Admission Status of the order on Douyin Platform will be initiated to "Not Completed".
3	Admission Mode Setting	Administrators can set up different admission modes for different event s before listing the event on Douyin Platform. One event can only be set to one admission mode.

4	Order Complete Mode	The order Admission Status is labelled "Completed" immediately after the user completes the purchase.
5	Session Complete Mode	All orders from Admission Status are labelled "Completed" automatically when the session finishes.
6	Admission Complete Mode	The order of Admission Status is labelled "Completed" only after the ticket is successfully scanned and admitted using a PDA machine.
7	Transferable Account	After the order Admission Status is labelled "Completed" on Douyin Platform, the funds can then be moved to the Douyin Transferable Account.
8	Fee Deduction	Douyin deducts relevant fees (e.g., commission and advertisement fees) from the total order amount. The amount depends on different situations.
9	Company Account Transfer	Funds can be transferred from the Douyin Transferable Account to the designated Company Bank Account. This process will take 4 days (determined by Douyin Platform).

Table 3.18.5.7.1: Description for the User Process of Douyin Account to Company Account Transfer

3.18.5.7.2. Requirement List

RID	Description	Priority	Related Process
REQ-DY-DCT-001	The funds are temporarily held in the Douyin platform upon user purchase completion.	High	3.18.5.7.1: #1
REQ-DY-DCT-002	The funds will not be able to transfer from Douyin Platform to Douyin Transferable Account until the order Admission Status is set to "Completed".	High	3.18.5.7.1: #2
REQ-DY-DCT-003	The system allows administrators to configure one of three mutually exclusive Admission modes per event: Session Complete, Order Complete, or Admission Complete.	High	3.18.5.7.1: #3
REQ-DY-DCT-004	Under Order Complete mode, the system automatically labels the order as "Admission Completed" status immediately upon a successful order.	High	3.18.5.7.1: #4

REQ-DY-DCT-005	Under Session Complete mode, the system automatically labels all relevant orders as "Admission Completed" status when the session is finished.	High	3.18.5.7.1: #5
REQ-DY-DCT-006	Under Admission Complete mode, the system automatically labels the order as "Admission Completed" status upon ticket is scanned via a PDA machine.	High	3.18.5.7.1: #6
REQ-DY-DCT-007	The funds can be moved to the Douyin Transferable Account once the "Admission" status is set to "Completed".	High	3.18.5.7.1: #7
REQ-DY-DCT-008	The commission fee will be deducted (commission, advertisement) by Douyin before funds move to the Douyin Transferable Account. The amount depends on different situations.	High	3.18.5.7.1: #8
REQ-DY-DCT-009	The funds in Transferable account can be transferred to the company bank account. This process will take 4 days (Determined by Douyin Platform).	High	3.18.5.7.1: #9

Table 3.18.5.7.2: Requirement List of Douyin Account to Company Account Transfer

3.18.5.8. Admission Control

Entry admission is an on-site access validation function provided by the ticketing system, available to all tickets or credential holders. The process begins when attendees present their e-ticket QR code. The system reads the information via AnTank Admission App installed on scanning devices and automatically verifies it based on multiple criteria, including ticket validity, timing, and redemption status.

The system performs automated triage based on verification results: if verification passes, tickets are automatically validated and entry is granted; if verification fails, specific reasons are immediately prompted and entry is denied. For cases requiring manual judgment (such as concession tickets, using electronic ticket codes after paper tickets have been collected, cross-seating areas entry, etc.), the system marks them as "requires secondary confirmation" and alerts on-site validation staff for final decision-making. The validation personnel can either grant entry or deny access based on the actual situation, thereby concluding the single verification process.

This process ensures the uniqueness and status accuracy of each ticket code by synchronizing in real time with the central ticketing inventory system, preventing duplicate admissions and invalid ticket verification. Meanwhile, verification results are recorded in real time, providing a full lifecycle flow for on-site crowd management, data statistics, and post- event audits.

3.18.5.8.1. User Process

The following table describes the user process of Admission Control.

No.	Component	Description
1	Start: Audience Presents Ticket Code	The process begins, and the audience presents their electronic or paper ticket code before verifying the device.
2	Admission Device Scan/Input	The verification device reads the ticket code information by scanning the QR code and uploading it to the backend system.
3	System Automatic Verification	The system automatically performs checks: QR code validity, ticket status, session validity, venue/event matching, and seat status.
4a	Verification Decision: Passed	The system determined the result as a pass based on predefined rules.
4b	Secondary confirmation	The system determines that secondary confirmation is required based on preset rules, prompting the device to display the reason (e.g., concession ticket entry, using an electronic ticket code after collecting a paper ticket, cross-seating areas entry), and waits for the verification personnel to select "Confirm" or "Cancel."
4d	Verification Decision: Not Passed	The system determines the result as a failure based on preset rules and prompts accordingly, such as invalid QR code, already redeemed ticket, or other conditions that do not meet the entry verification requirements.
5a	Result Execution: Display "Ticket Verification Successful" and Grant Passage.	For the scenario of 4a (Pass), the system automatically completes the verification: 1. Update ticket status: Mark the ticket as "Verified" in the database. 2. Device feedback: The verification device displays a "Check-in Successful" prompt, accompanied by a success of sound effect and vibration. 3. Execute passage.
5b	Secondary Judgment Gy Admission Staff	For scenarios requiring secondary confirmation (4b), control is transferred to the verification personnel: 1. Interface prompt: The device interface clearly displays the alarm reason (e.g., "Please confirm: senior ticket"), accompanied by a confirmation of sound and vibration, with "Confirm" and "Cancel" buttons provided.

		2. Manual verification: The verification personnel conduct a quick check based on the prompt. 3. Decision-making: The verification personnel click the corresponding button according to the on-site situation.
5c	Execution Result: Display "Ticket Verification Failed" and Deny Entry.	For cases of 4c (rejection), the system performs the rejection operation: 1. Device feedback: The validation device displays a "Ticket Verification Failed" prompt, clearly indicating the reason for failure (e.g., "Invalid Ticket"), accompanied by a failure of sound effect and vibration. 2. Entry is denied.
6a	Execution Result: Display "Ticket Verification Successful" and Grant Passage.	When the verification staff clicks "Confirm" in step 5b: 1. Record operation: The system logs the action in the verification log. 2. Update ticket status: Complete verification and mark the ticket as "Verified". 3. Execute release.
6b	Execution Result: Entry Denied	When the verification staff clicks "Cancel" in step 5b: 1. Process termination: The system will not verify the ticket, and the ticket status remains unchanged. 2. Entry denied.
7	End	The final status of a single verification transaction.

Table 3.18.5.8.1: Description for the User Process of Admission Control

3.18.5.8.2. User Process for Secondary Confirmation

The following table describes the verification scenarios that require manual intervention for secondary confirmation when ticket attributes trigger specific rules. These rules are typically configured at the ticket type or ticket level.

No.	Component	Description
1	Scan Code and Basic Verification	The verification personnel scan the ticket QR code, and the system performs basic validation to confirm that the ticket is valid.

2	Special Rules Trigger	The system determines whether the current ticket attributes match the predefined "requires secondary confirmation" rule set. Key rules include: 1. Ticket types requiring secondary confirmation: child tickets, senior tickets, and work-related tickets. 2. Ticket status is "physical ticket retrieved": when the user attempts to verify with an electronic QR code or ID card, but a physical ticket has already been redeemed. 3. Cross-seating areas entry: the area specified on the ticket does not match the current verification point area (only applicable to seat selection events).
3a	Process According to Standard Procedure	If no special rules are triggered, proceed to the admission verification process.
3b	Rule Type Judgment	If a rule is triggered, the system further determines the specific type of rule activated and proceeds to different processing branches.
4a	Trigger the "Ticket Type Confirmation" Prompt	When special ticket types (such as student tickets, senior tickets, etc.) are involved, the system displays a confirmation message on the device interface, requiring a secondary confirmation.
4b	Trigger "Paper Ticket Collected" Prompt	When the user has collected a paper ticket and uses an electronic ticket code for entry, the device prompts: "Please confirm! Paper tickets are already collected," requiring a secondary confirmation.
4c	Are Sessions Allowed to be Validated Across Regions?	When the event type is seat selection and cross-zone entry is involved, check the session configuration: "Refuse cross-zone ticket check" switch: If the switch is on (reject), then jump to node 5a. If the switch is turned off (allowed), then proceed to node 5b.
5a	Trigger "Cross-Seating Areas Entry" Prompt	When cross-zone ticket verification is allowed for the event, the device prompts: "Please confirm! No validation permission for the current zone," requiring secondary confirmation.
5b	Direct Refusal: Regional Independence	When the event is set to "Reject cross-seating areas ticket validation," the system will directly fail the verification.

6	Manual Secondary Confirmation	For all alerts requiring secondary confirmation (ticket type, ticket status, allowed cross-seating areas operations), the interface will be uniformly locked, with highlighted prompt messages, and only provide "Confirm" and "Cancel" buttons. The verification personnel will make decisions based on the current situation.
7a	Confirm Release	Click "Confirm". Device: Execute the admission and update the ticket status to "Used". The device indicates successful verification, proceed with release.
7b	Refused Entry	Click "Cancel". Device: Do not perform admission. Entry denied.
8	End	Complete a full special verification process.

Table 3.18.5.8.2: Description of Secondary Confirmation User Process**3.18.5.8.3. Requirement List**

RID	Description	Priority	Related Process
REQ-AC-M-001	The backend management system must provide an interface to configure permissions for each verification account, including the types of services, venues, facilities, and areas (area verification applies only to seat selection) they are authorized to verify.	High	Full process
REQ-AC-M-002	The system should record and enable traceability of admission data and admission logs.	High	Full process
REQ-AC-M-003	The admission app interface must clearly display the available verification points and functions for the logged-in user, while supporting both QR code scanning and manual ticket code input for admission.	High	3.18.5.8.1: #2
REQ-AC-M-004	The system must support rapid account switching on verification devices to accommodate shift changes or shared device scenarios, ensuring the switching process is simple and efficient.	High	Full process

REQ-AC-M-005	The system must support configuring differentiated ticket verification times for special ticket types (such as VIP tickets). For example, VIP ticket holders may enter early, with their verification validity window starting before the event 's commencement time.	High	3.18.5.8.1: #3
REQ-AC-M-006	When an invalid ticket (such as counterfeit, already used, or refunded) is detected, the system must trigger a clear audible alarm and text prompt on the verification device.	High	3.18.5.8.1: #4c-#7
REQ-AC-M-007	admission system provides multiple options for prompt tones (success, failure, secondary confirmation).	Low	3.18.5.8.1: #4b, #5a, #5c, #6a, #6b 3.7.8.2: #4b, #4b, #5a, #6, #7b, #7b
REQ-AC-M-008	When verifying ticket types that require secondary confirmation (such as senior tickets or student tickets), the system must trigger clear visual prompts on the device and require the verification personnel to perform a secondary confirmation.	High	3.18.5.8.2: #1-#8
REQ-AC-M-009	The system must support an event -level toggle for "prohibit cross-seating areas ticket validation." When the toggle is off, cross-seating areas tickets enter a manual review process; when the toggle is on, cross-seating areas tickets are directly invalidated.	Medium	3.18.5.8.2: #3b-#8

Table 3.18.5.8.3: Requirement List of Admission Control

3.18.5.9. Backend Management

This process defines how events are created and published to the Douyin Mini Program.

- **User Process:** The process begins by completing the event configuration through the ticketing system back-end.
- **Integration & Application:** The event is pushed to the Douyin via a System Integration portal and listed through the CMS System. Administrators then complete the Listing Application within the Douyin Listing Management System.
- **Approval:** The application awaits approval by the Douyin Platform. If rejected, administrators must re-apply. If approved, the listing is completed, and the event becomes available for sale on the Douyin Mini Program.

3.18.5.9.1. User Process

The following table describes the user process of Backend Management.

No.	Component	Description
1	Event Configuration	The Administrator should complete the Event Configuration on WestK Backend (Refer to Ticketing BRD document).
2	Event Push	The Administrator can push the event from WestK Backend to the Douyin Backend via the System integration portal.
3	CMS Listing	The Administrator can list the event on the Douyin Backend through the CMS System.
4	Douyin Listing Application	The Administrator can Complete the listing application on the Douyin Backend through the Douyin Listing Management System.
5	Application Approval	Wait for the application to be approved by the Douyin Platform. If the application is refused, the administrator should re-apply on the listing. If the application is approved, the event should automatically be available on Douyin Mini Program for sale.
6	Order Management	The Administrator can manage all customer orders generated from the Douyin Mini Program through WestK Backend, including viewing order details and checking order status.
7	Report Management	The Administrator can review various operational reports through WestK Backend for business analysis.
8	Refund Management	The Administrator can process order refund requests initiated via the Douyin Mini Program through WestK Backend, including reviewing refund reasons, approving/refusing requests, and checking refund status.

Table 3.18.5.9.1: Description for the User Process of Backend Listing Management

3.18.5.9.2. Requirement List

RID	Description	Priority	Related Process
REQ-DY-BLM-001	The system must support completing event configurations through the ticketing system back-end.	High	3.18.5.9.1: #1

REQ-DY-BLM-002	The system must be able to push events from the ticketing system back-end to the Douyin Backend via the system integration portal.	High	3.18.5.9.1: #2
REQ-DY-BLM-003	The system must allow listing the event on the Douyin Backend through the CMS System.	High	3.18.5.9.1: #3
REQ-DY-BLM-004	Administrators must be able to complete the listing application through the Douyin Listing Management System.	High	3.18.5.9.1: #4
REQ-DY-BLM-005	The system must track the application's approval status from the Douyin Platform.	High	3.18.5.9.1: #5
REQ-DY-BLM-006	The system must allow users to re-apply if the listing application is refused.	High	3.18.5.9.1: #5
REQ-DY-BLM-007	Upon completion, the event must automatically become available for sale on the Douyin Mini Program.	High	3.18.5.9.1: #5
REQ-DY-BLM-008	The system must allow authorized WestK staff to search, view, and filter orders generated through the Douyin Mini Program, including the ability to review order status, ticket details, payment results.	High	3.18.5.9.1: #6
REQ-DY-BLM-009	The system must provide WestK staff with access to centralized sales and operational reports, including sales show, inventory status, and admission analytics, with the ability to export reports for financial reconciliation and management review.	High	3.18.5.9.1: #7
REQ-DY-BLM-010	The system must allow authorized WestK staff to process refund requests and update the corresponding order and ticket statuses, with all refund actions logged for audit and reconciliation purposes.	High	3.18.5.9.1: #8

Table 3.18.5.9.2: Requirement List of Backend Management

3.19. Xiaohongshu Mini Program

This section outlines the end-to-end business processes for the Xiaohongshu Mini Program integration. It covers the business problem statement, background and objectives, high-level and detailed business requirements, the fulfilment approach, customer journey, and the functional specifications for both frontend and backend workflows.

3.19.1. Business Requirements

From the business perspective, WestK requires the Xiaohongshu Mini Program integration to support ticket sales for Mainland China users while ensuring operational efficiency, secure data handling, and compliance with platform policies.

The following business requirements have been identified.

3.19.1.1. Registration & Login

WestK requires the Xiaohongshu Mini Program to support seamless user authentication using Xiaohongshu's native login, with automatic account binding and secure session management to ensure identity consistency for order management and ticket ownership.

3.19.1.2. Xiaohongshu Ticket Sales Channel

WestK requires the ability to list exhibitions for M+/HKPM and shows for M+/PA on the Xiaohongshu Mini Program, enabling Mainland China users to browse and purchase tickets directly within the platform. In addition, WestK requires an intuitive home page that helps users easily discover and access events, featuring event listings, popular highlights, keyword search, and clear entry points to detailed event pages.

3.19.1.3. E-Ticket Wallet

WestK requires that users be able to enter the venue directly using electronic tickets generated through the Xiaohongshu Mini Program without exchanging physical tickets. Each ticket must contain a unique QR code for verification at the venue.

3.19.1.4. Order Traceability Across Systems

WestK requires that users be able to review their orders and related details within a centralized customer hub, including order status (such as pending payment, paid, cancelled, or refunded) and the associated ticket information. Furthermore, all orders created through the Xiaohongshu Mini Program must be consistently recorded both on the Xiaohongshu platform—for end users to easily locate and review their orders—and within AnTank's backend systems, where staff members can perform operational queries and reviews. Clear identifier mappings must be maintained across both systems to ensure complete cross-system traceability and to support financial and operational reconciliation processes.

3.19.1.5. Payment, Currency Handling & Settlement

WestK requires that currency conversion mechanisms be implemented to support internal financial reconciliation and reporting in Hong Kong Dollars (HKD), in accordance with WestK's financial policies.

3.19.1.6. Ticket Verification & Admission Control

WestK requires that each E-Ticket contains a unique QR code and that admission verification updates ticket/order status in real time across Xiaohongshu and AnTank, preventing duplicate entries and ensuring operational traceability.

3.19.1.7. Integration with Ticketing System

WestK requires the Xiaohongshu Mini Program to integrate a centralized ticketing system which serves as the central system for ticket inventory management, pricing control, order processing, and ticket lifecycle management. Furthermore, WestK requires comprehensive access to ticket sales and operational data through a unified backend reporting portal, enabling financial reconciliation, operational oversight, and show analysis across all Xiaohongshu related transactions.

3.19.1.8. Privacy, Security, Scalability & Inclusive Accessibility

WestK requires compliant handling of personal data, including minimizing data collection, restricting use to authorized purposes, encrypting sensitive information, and enforcing strict access controls. The system must also support high concurrency during peak sales to ensure stable experience for all users. In line with WestK's Diversity & Inclusion principles, wheelchair-accessible seating must be clearly identifiable and managed as a dedicated seating category to support accessibility needs, with full allocation logic planned for a future phase.

3.19.2. Business Requirement Analysis

This section provides further analysis and clarification of the business requirements described in Section 3.18.1. Each subsection elaborates on the business intent, operational considerations and functional expectations underpinning the Xiaohongshu Mini Program ticketing solution.

3.19.2.1. User Registration & Login

Map to Business Requirement 3.19.1.1 and 3.19.1.8.

- The Mini Program shall support user authentication via the Xiaohongshu Platform login mechanism, enabling seamless entry without requiring separate account creation.
- User accounts shall be created or bound automatically.
- Maintain user identity consistency for order management and ticket ownership.
- The system ensures secure session management and user data protection.

3.19.2.2. Home Page Display & Detailed Event Display

Map to Business Requirement 3.19.1.2.

The Xiaohongshu Mini Program shall provide intuitive event discovery capabilities:

- Home Page:
 - Event listings for M+/HKPM exhibitions and M+/PA shows.
 - Popular exhibitions and shows highlighting.
 - Keyword search functionality.
- Event Detail Page:
 - Event title, description, and poster.
 - Show dates, session times, and venue information.
 - Ticket tiers, pricing and seat availability and description.
 - Ticket availability status.

3.19.2.3. E-Ticket Wallet & Order Traceability

Map to Business Requirement 3.19.1.3 and 3.19.1.4.

- User shall be able to access a centralized hub – “My Account” page to:
 - View all orders (Pending Payment/Paid/Cancelled/Refunded).
 - View order details including Ticket information, QR code, and order status.
- A dedicated page shall display:
 - Unused E-Tickets.
 - Used E-Tickets.
 - Each E-Ticket shall contain a unique QR code for admission.
- Cross-System Traceability:
 - Order records will be aligned across Xiaohongshu Platform and AnTank Backend.
 - Mappings between Xiaohongshu Order ID, AnTank Order ID, Ticket Codes, and payment references must be maintained for financial and operational reconciliation.

3.19.2.4. Ticket Purchasing Process

Map to Business Requirement 3.19.1.2 and 3.19.1.5.

- The mini program should support two types of ticketing:
 - Admission ticket:
 - Ticket type selection.
 - Quantity selection.
 - Seat-selection ticket:
 - Interactive online seating plan.
 - Seat selection.
- An order confirmation page shall display including:
 - Selected ticket details including ticket types or seats information.

- Total payable amount.
- Payment Options.
- Real name verification is required during purchasing (per Xiaohongshu rules).

Platform constraints:

Due to the Xiaohongshu platform restriction, different types of tickets with multiple prices cannot be purchased together in one order.

Due to Xiaohongshu platform requirements, the Mini Program must be operated by a Mainland China-registered entity and supports RMB (CNY) pricing only.

To meet WestK's business requirement for Hong Kong Dollar (HKD) pricing and financial settlement, ticket prices defined in HKD will be converted into RMB for listing and transaction purposes on the Xiaohongshu platform. Appropriate currency conversion and reconciliation mechanisms will be implemented to ensure alignment with WestK's financial policies.

In addition, due to Xiaohongshu's event category restrictions and approval process, only Admission tickets and seat-selection tickets will be launched on the platform.

3.19.2.5. Payment & Payment Result Handling

Map to Business Requirement 3.19.1.4 and 3.19.1.5.

- The Xiaohongshu Mini Program shall integrate with Xiaohongshu-supported payment methods.
 - Include Alipay China only.
- During the payment flow:
 - Allow users to pay within a 15 min pending time window.
 - Allow users to cancel orders (before orders expiration).
- Upon a successfully Payment:
 - A payment success page shall be displayed.
 - E-Ticket containing QR code shall be automatically issued.
- AnTank backend system shall receive and process Xiaohongshu Payment callbacks to confirm payment status and successfully log an order in backend system.

3.19.2.6. Admission Control

Map to Business Requirement 3.19.1.6.

- Each e-Ticket shall generate a unique QR code for admission.
- Upon scanning:
 - Ticket status shall update to "Used" in real time.
 - The corresponding order status shall be synchronized to AnTank Backend system.
 - Verification data shall be recorded for audit purposes.

3.19.2.7. Backend Portal Requirements

Map to Business Requirement 3.19.1.7.

- A centralized backend portal to:
 - Create and modify the events displayed in Xiaohongshu Mini Program.
 - View and process orders sold in Xiaohongshu Mini Program.
 - Monitor Xiaohongshu Admission status synchronization.
 - Access sales reporting and operational analytics.

3.19.2.8. Xiaohongshu Mini Program Integration with AnTank Backend System

Map to Business Requirement 3.19.1.7.

- All ticket purchasing functions shall be implemented via the Xiaohongshu Mini Program.
- All Events and ticket master data will be sourced from the AnTank Backend system.
- Orders, payments, and ticket issuance can be tracked via the Backend System.
- Customers shall be able to enter the venue directly using the E-Ticket/QR code within Xiaohongshu Mini Program, without the need for physical ticket exchange.

3.19.2.9. System Show & High Concurrency

Map to Business Requirement 3.19.1.8.

- The solution should be designed to support high concurrent user access during peak ticket sales periods (e.g. major releases), ensuring system stability, transaction integrity, and consistent user experience.
- For the current release, the wheelchair seating allocation logic is not included in the implementation scope. The detailed handling approach will be clarified in the Approach section and addressed in a future phase.

3.19.3. Operating Model

This initiative uses a **three-party operating model** to deliver ticket sales through a Xiaohongshu Mini Program.

- **AnTank** delivers and operates the **ticketing technology layer** (provides ticketing system and Mini Program development).
 - Provides the **ticketing system** capabilities: event/session setup interfaces, inventory management, order lifecycle, ticket issuance (e-ticket/QR), order status queries, and reporting exports.
 - Develops and maintains the **Xiaohongshu Mini Program**: user journey, Xiaohongshu OAuth integration, UI/UX, basic operational dashboards.

- Integrates with payment endpoints as required by the solution design (typically via Xiaohongshu Pay APIs).
 - Ensures functional correctness, system availability, logging/monitoring, and technical support within SLA.
- **Charisma** provides **Xiaohongshu operations**, acting as the operational interface for Xiaohongshu account opening and operation, payment configuration, monitoring, and **monthly settlement coordination** with **WestK Finance**.
 - Serves as the **operations bridge** between Xiaohongshu and WestK and oversees **Xiaohongshu Payment**.
 - Provide customer service, payment operations monitoring and incident triage (payment exceptions, callback anomalies, reconciliation support).
 - Owns the **monthly settlement process coordination** with WestK Finance team, e.g. preparing/collecting settlement statements, assisting reconciliation, tracking discrepancies to closure, ensuring monthly settlement completeness and audit readiness.
- **WestK** is the business owner, who
 - Owns the ticketing business (events, pricing, inventory policies, customer policies).
 - Owns customer relationships and defines after-sales policies (subject to agreed scope).
 - Receives settlement reporting and performs accounting recognition and reconciliation.
 - Acts as the primary finance counterparty for monthly settlement.

End-to-End Ticket Purchase Flow via Xiaohongshu Mini Program

The **Diagram 3.19.3** below is for illustrative purposes only. It is intended to present the roles involved and delineate the interactions across the business process flow. For the detailed user journey and end-to-end process specifications, please refer to **Section 3.19.4**.

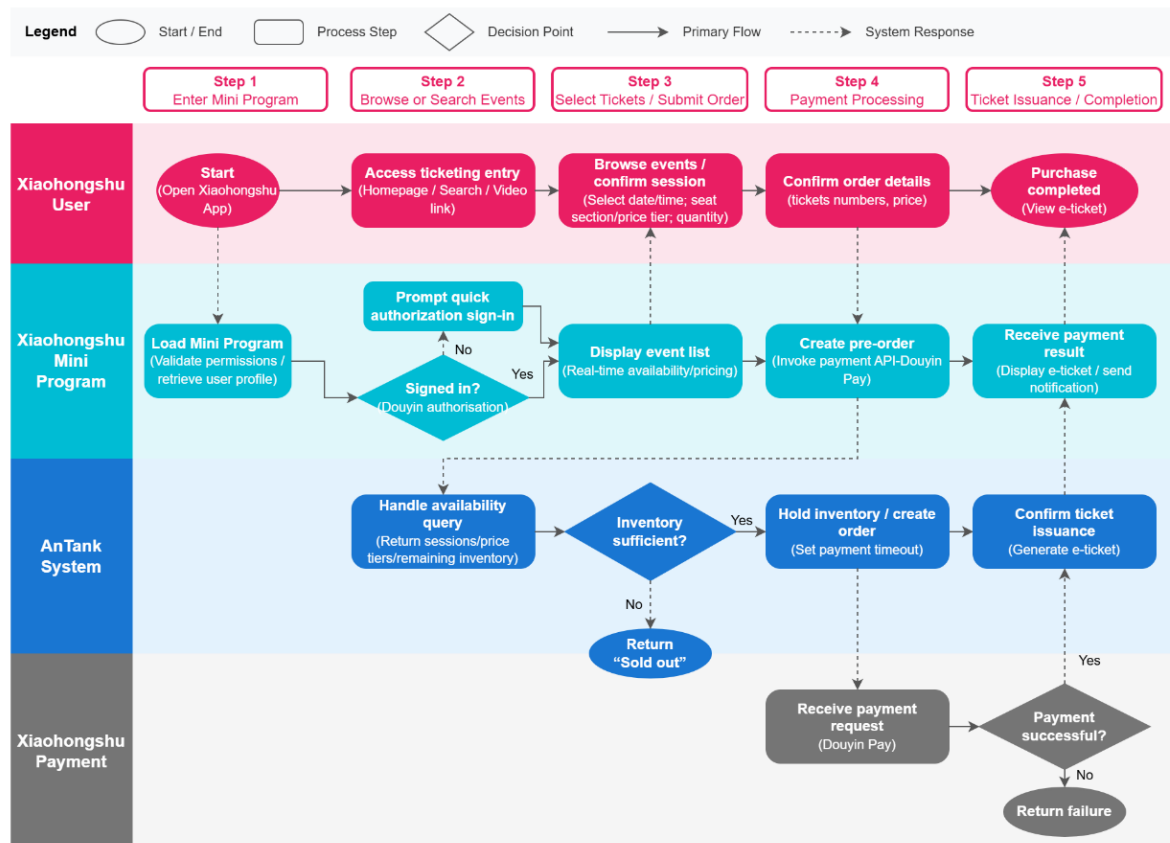


Diagram 3.19.3: Business Process Flow

3.19.4. Customer Journey

This section describes what users can do after entering the Xiaohongshu Mini Program.

The user story is used to describe the user's needs, goals, and expected outcomes, providing clarity on what the user wants to achieve and why the functionality is required. When users enter the Xiaohongshu Mini Program, they can complete the entire ticket purchasing and admission process without leaving the platform.

No.	Category	Role	User Story
1	Registration and Login	As a Xiaohongshu user	I want to log in with my phone number which binds with my Xiaohongshu account, so that I can access the Mini Program quickly and conveniently.
2	Event Discovery	As a potential customer	I want to view a homepage highlighting popular exhibitions and shows, so that I can easily discover trending events.
3	Event Discovery	As a potential customer	I want to use the keyword search functionality so that I can quickly find specific events I am interested in.

Technology Partnership Package 2
Requirement Document

4	Event Discovery	As a potential customer	I want to view a detailed event page including the poster, dates, price, venue, etc., so that I have all the information needed to make a purchase decision.
5	Ticket Purchasing	As a customer	I want to select my desired ticket type and quantity for Admission events, so that I can purchase the exact number of tickets my group needs.
6	Ticket Purchasing	As a customer	I want to use an interactive online seating plan when buying tickets for shows, so that I can choose my preferred seats quickly and easily.
7	Ticket Purchasing	As a customer	I want to see ticket prices listed in RMB (CNY), so that I can understand the cost in my local and platform-supported currency.
8	Ticket Purchasing	As a customer	I want to review an order confirmation page showing my selected tickets and total payable amount, so that I can verify my order details before proceeding with payment.
9	Checkout & Payment	As a customer	I want to pay using Xiaohongshu-supported methods (WeChat Pay, Alipay China, Credit/Debit Card), so that I can check out securely using my preferred payment method.
10	Checkout & Payment	As a customer	I want to see a payment success page to confirm my completed order and view my E-Ticket QR code for convenient admission.
11	Order Management	As a registered user	I want to access a "My Account" page to assess the status of all my orders, so that I can keep track of my purchases.
12	Order Management	As a registered user	I want to access an independent E-Ticket management page displaying my unused and used tickets with their QR codes, so that I can easily find and present my tickets on the day of the event.
13	Venue Admission	As an event attendee	I want to scan my E-Ticket QR code at the venue, so that I can enter directly without needing to queue for physical ticket exchange.
14	Venue Admission	As an event attendee	I want to complete real-name verification during admission (if required), so that I comply with the event's security and ticketing policies.

15	Backend Integration	As a WestK Staff	I want all event and ticket data to be synchronized from different BU's Backend to the Xiaohongshu's Backend, so that the event can be managed efficiently.
16	Backend Integration	As a WestK Staff	I want to use a centralized backend portal to create and modify the events displayed on the Xiaohongshu Mini Program, so that I can easily control the listed event.
17	Backend Ticket Report	As a WestK Staff	I want to access a centralized backend ticketing report so that I can view business-related ticketing data that is relevant to my team's operations.

Table 3.19.4: Description for the User Story

3.19.5. Business Scenarios

3.19.5.1. Registration and Login

This process defines how users access the Xiaohongshu Mini Program securely and in compliance with privacy regulations.

- **User Process:** When the system initiates an authentication request, it verifies whether the user is already logged in. If the user has an active session, they will be directed to the Home Page directly. If no valid session is detected, the system will prompt the user to complete authentication via the "One-Click Phone Login" process (which is a native login function supported by Xiaohongshu Platform) before accessing the Mini Program.
- **Privacy Policies:** Users register or log in using their Xiaohongshu registered phone number. To proceed, users must check and agree to the privacy terms before registration or login. Users can also click a "Privacy Policy" link, which redirects them to the full privacy policy page. The Privacy Policy content is issued by WestK.

3.19.5.1.1. User Process

The following table describes the user process of Registration and Login.

No.	Component	Description
1	Register and Login	Users can register or log in using their Xiaohongshu registered phone number via the "One-Click Phone Login" button, which is a native login function supported by Xiaohongshu Platform.
2	Privacy Protection Policy Checkbox	Users must agree with the Privacy Protection Policy before successful login or registration.

3	Privacy Policy Link	Clicking the "Privacy Policy" link redirects users to the full privacy policy page. The Privacy Policy content is issued by WestK.
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Table 3.19.5.1.1: Description for the User Process of Registration and Login

3.19.5.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-XHS-REG-001	Support one-click registration and login using the Xiaohongshu registered mobile phone number.	High	3.19.5.1.1: #1
REQ-XHS-REG-002	Enforce mandatory checking of the privacy protection policy issued by WestK before logging in.	High	3.19.5.1.1: #2
REQ-XHS-REG-003	Provide a clickable link to view the complete WestK privacy policy.	High	3.19.5.1.1: #3

Table 3.19.5.1.2: Requirement List of Registration and Login

3.19.5.2. Home Page

The Home Page serves as the central navigation page:

- **Global Search:** The page features a Global Search function allowing instant searches for specific exhibitions, shows.
- **Hero Banner:** A Hero Banner dynamically highlights the current significant exhibitions, shows, or the upcoming major events.
- **Ticketing Purchase Navigation:** Users can access the "Ticketing" page to purchase tickets or redirect to different venue pages to visit dedicated ticketing purchase pages for M+, HKPM, and PA.
- **Featured Exhibitions & Shows:** The page displays Featured Exhibitions and Featured Shows, allowing users to view event information or purchase tickets directly.
- **Customer Service:** The button redirects users to an online customer service portal with detailed contact information.

3.19.5.2.1. User Process

No.	Component	Description
1	Hero Banner	A banner dynamically highlighting the current significant shows, exhibitions, or upcoming major events.

2	Ticketing Button	Redirect to pages of purchasing tickets for exhibitions and shows. The Ticket Purchase page can filter the events by its type (exhibitions or shows) or by venue (HKPM, M+, and PA). The Ticket Purchase page also supports calendar mode, which allows users to check available events at different dates via a calendar.
3	Venue Redirection	Redirect to dedicated ticket purchase page for M+, HKPM, and PA. The ticket purchase page will filter the events by venue and only display the events held at designated venue.
4	Global Search	Allows instant searches for the name of specific exhibitions and shows. The search follows the inclusion rule, which means if an event name includes the searching content, the event will be displayed in the searching result.
5	Featured Exhibitions	List all current popular exhibitions in this section. Tapping any entry navigates to the details page for viewing event information or purchasing tickets directly.
6	Featured Shows	List all current popular Shows in this section. Tapping any entry navigates to the details page for viewing event information or purchasing tickets directly.
7	Online Customer Service	Redirect to an Online Customer Service Page for online customer support with detailed contact information. The user can directly use the online chat box to communicate with Online Customer Support.
8	Belt Banner	A smaller banner displayed in the middle of the Home Page dynamically highlighting the current significant shows, exhibitions, or upcoming major events.

Table 3.19.5.2.1: Description for the Home Page User Process

3.19.5.2.2. Requirement List

RID	Description	Priority	Related Process
REQ-XHS-HOM-001	Support global search functionality for exhibitions and shows.	High	3.19.5.2.1: #4
REQ-XHS-HOM-002	Display a Hero Banner linked to major event details.	High	3.19.5.2.1: #1

REQ-XHS-HOM-003	Provide a redirection link to the ticket purchase page.	High	3.19.5.2.1: #2
REQ-XHS-HOM-004	The ticket purchase page supports filter by event type and venue.	High	3.19.5.2.1: #2
REQ-XHS-HOM-005	The ticket purchase page supports calendar mode to check events at different date.	High	3.19.5.2.1: #2
REQ-XHS-HOM-006	Provide direct redirection links to M+, HKPM, and PA ticket purchase page.	High	3.19.5.2.1: #3
REQ-XHS-HOM-007	Provide a redirection button to the customer service page.	High	3.19.5.2.1: #7
REQ-XHS-HOM-008	The customer service page provides an online chat box for the user to communicate with staff directly.	High	3.19.5.2.1: #7
REQ-XHS-HOM-009	Display Featured Exhibitions list. Click the exhibition to redirect user to the event detail page.	High	3.19.5.2.1: #5
REQ-XHS-HOM-010	Display Featured Shows list. Click the shows to redirect user to the event detail page.	High	3.19.5.2.1: #6
REQ-XHS-HOM-010	Display a Belt Banner linked to major event details.	High	3.19.5.2.1: #8

Table 3.19.5.2.2: Requirement List of Home Page

3.19.5.3. My Ticket Code

This section provides users with quick access to their purchased tickets for admission.

- **User Process:** Users enter the "My Ticket Code" page from the Home Page, where they can toggle between two tabs: "Tickets" (shown by default) and "Flex Tickets" (A Long-Term Ticket refers to a multi-day pass that can be used across more than one day).
- **Same-Day Tickets:** Displays all same-day tickets in one location, showing the ticket QR code.
- **Flex Tickets:** Displays Flex tickets in one location, showing the QR code.

3.19.5.3.1. User Process

The following table describes the user process of My Ticket Code.

No.	Component	Description
1	My Ticket Code Overview	Users can enter the My Ticket Code page from the home page. There are two tabs corresponding to "Tickets" and "Long Term Tickets". The "Tickets" tab is shown by default.
2	Same-Day Ticket	Users can view all the same-day tickets in one convenient location. After clicking the ticket, the ticket's QR code can be displayed.
3	Flex Ticket	Users can view all the Flex tickets in one convenient location. After clicking the ticket, the ticket's QR code can be displayed. A Long-Term Ticket refers to a multi-day pass that can be used across more than one day.

Table 3.19.5.3.1: Description for the User Process of My Ticket Code

3.19.5.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-XHS-TIC-001	Users can enter the My Ticket Code page from Home Page.	High	3.19.5.3.1: #1
REQ-XHS-TIC-002	Users can view their tickets on My Ticket Code pages.	High	3.19.5.3.1: #2, #3
REQ-XHS-TIC-003	Users can select different types of ticket: Same-day Ticket or Flex Ticket.	High	3.19.5.3.1: #1
REQ-XHS-TIC-004	Ticket QR codes can be displayed for valid tickets.	High	3.19.5.3.1: #2, #3

Table 3.19.5.3.2: Requirement List of My Ticket Code

3.19.5.4. My Account

The My Account page is a centralized dashboard for users to manage their personal information, orders, and system preferences.

- **Personalization Settings:** Users can access "Account Setting" by tapping their avatar to manage specific preferences, e.g.: update their profile, toggle Marketing Notifications, manage System Permissions, or delete their account. Users can also log out through this page.
- **Management Hubs:**
 - **Orders:** A centralized hub to manage all orders, with options to categorize by type and filter by payment status.

- **Tickets:** A hub to manage all tickets, with options to categorize by ticket type.
- **Company Information:** The page includes an "Operation License" section showing company qualifications, and a "Contact Us" section with contact details for different WestK departments.

3.19.5.4.1. User Process

No.	Component	Description
1	My Account Page	Provide a landing page for accessing different personal information and related information, including account settings, orders, tickets, operation license, and contact us.
2	Tickets	A centralized hub for managing all tickets, providing a clear overview of all tickets. Tickets can be categorized by their types (all tickets, shows tickets, Flex tickets, and exhibition tickets). Click the ticket to display the order and the QR code.
3	Orders	A centralized hub for managing all orders across exhibitions and shows. You can categorize orders by type, and filter by payment status. Click the order to redirect to the order detail page.
4	Account Setting	Users can access the Account Setting page by tapping the avatar at My Account page, where you can manage specific account preferences.
5	Operation License	Users can view operation licenses and qualifications that the company possesses.
6	Contact Us	Users can view the contact information for WestK, including email addresses, telephone numbers, and contact addresses for different business units or departments.
7	Avatar & Nickname	Allows users to personalize your account identity by uploading or changing your profile avatar and updating your display nickname.
8	Marketing Notifications	Users can toggle settings to subscribe or unsubscribe from promotional updates.
9	System Permissions	Users can view and manage specific system-level permissions granted to the app (Service Message Management, Personal Information and Permission Usage Record, and Phone Number Management).

10	Security Setting	Users can delete their account on this page.
11	Log out	User can log out via the “Log Out” button at the Account Setting page.
12	My Favourite	User can check their favourite shows and exhibitions in the My Favourite page.

Table 3.19.5.4.1: Description for the My Account Page User Process

3.19.5.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-XHS-ACC-001	Provide a My Account page for users to access different account settings.	High	3.19.5.4.1: #1
REQ-XHS-ACC-002	Allow users to view and categorize orders by type and filter by payment status.	High	3.19.5.4.1: #2
REQ-XHS-ACC-003	Provide a centralized hub to manage and overview all tickets and filter by type.	High	3.19.5.4.1: #3
REQ-XHS-ACC-004	Display company operation licenses and contact information for different departments.	High	3.19.5.4.1: #5
REQ-XHS-ACC-005	Display contact information for different business units.	High	3.19.5.4.1: #6
REQ-XHS-ACC-006	Support user profile personalization, including updating the avatar and display nickname.	High	3.19.5.4.1: #7
REQ-XHS-ACC-007	Allow users to toggle settings to subscribe or unsubscribe from marketing notifications.	High	3.19.5.4.1: #8
REQ-XHS-ACC-008	Allow users to view and manage specific system-level permissions granted to the app.	High	3.19.5.4.1: #9
REQ-XHS-ACC-009	Provide a portal to delete the account.	High	3.19.5.4.1: #10

REQ-XHS-ACC-010	Provide a button to log out the current account.	High	3.19.5.4.1: #11
REQ-XHS-ACC-011	Provide a My Favourite page to display all shows and exhibitions marked Favourite.	High	3.19.5.4.1: #12

Table 3.19.5.4.2: Requirement List of My Account

3.19.5.5. Admission Ticket Purchase

This process outlines the workflow for purchasing Admission tickets without seat selection.

- **Selection & Confirmation:** Users view " event Detail" (introductions, dates, prices, venue). They select a Date and Session, followed by the Ticket Type & Quantity, which dynamically updates the total price.
- **Payment Processing:** Users can confirm the order and complete the payment, and an order number will be generated. Alternatively, payment can be cancelled on the confirmation page.
- **Order Pending State:** If unpaid, the order remains in a "Pending" state for 15 minutes, allowing the user to complete the transaction; it is automatically cancelled after 15 minutes if payment is not received.
- **Completion:** Upon success, users can view their ticket information and admission QR code.

3.19.5.5.1. User Process

The following table describes the user process of Admission Ticket Purchase.

No.	Component	Description
1	Event Detail	Users can check program introductions, date and time, price, venue location, and event content in the event detail page. Users can also add the event to their favourite list.
2	Date and Session Selection	Users can select different dates and sessions.
3	Select Ticket Type	Users can select tickets of different types. The ticket type selection page while dynamically updating the total prices. Due to the Xiaohongshu platform restriction, different types of tickets with multiple prices cannot be purchased together in one order.
4	Order Confirmation	Users can check the calculated subtotals and confirm the final payment.

5	Complete Payment	After the payment is completed, user can view the order in detail with an order number.
6	Cancel Payment	Payment can be cancelled at the order confirmation page by tapping the back button.
7	Order Pending	The unpaid order will be in the pending state for 15 minutes. The user will still be able to pay for the order within 15 minutes. The order will be automatically cancelled after 15 minutes.
8	View Tickets	Users can view detailed tickets for information on the order completion page. User can click “View Ticket Code” button to view the admission ticket code.

Table 3.19.5.5.1: Description for the User Process of Admission Ticket Purchase

3.19.5.5.2. Requirement List

RID	Description	Priority	Related Process
REQ-XHS-UTP-001	Display comprehensive event information including introductions, schedules, and ticket availability on the details page.	High	3.19.5.5.1: #1
REQ-XHS-UTP-002	Provide an interactive calendar interface for selecting dates and browsing available sessions.	High	3.19.5.5.1: #2
REQ-XHS-UTP-003	Display real-time ticket availability, including labelling sold-out tickets.	High	3.19.5.5.1: #2
REQ-XHS-UTP-004	Support the selection of specific ticket types and quantities while dynamically updating the total Price.	High	3.19.5.5.1: #3
REQ-XHS-UTP-005	Require users to accept purchase terms before confirming order submission.	High	3.19.5.5.1: #4
REQ-XHS-UTP-006	Support different payment methods.	High	3.19.5.5.1: #5
REQ-XHS-UTP-007	Display a payment completion page confirming the transaction status, order number, and payment amount.	High	3.19.5.5.1: #5
REQ-XHS-UTP-008	Order can be cancelled on the order confirmation page.	High	3.19.5.5.1: #6

REQ-XHS-UTP-09	Order will be pending for 15 minutes if the user does not complete the payment.	High	3.19.5.5.1: #7
REQ-XHS-UTP-010	The order will be automatically cancelled if the order is unpaid after 15 minutes.	High	3.19.5.5.1: #7
REQ-XHS-UTP-011	Generate an order detail page providing individual admission QR codes and full transaction records.	High	3.19.5.5.1: #8
REQ-XHS-UTP-012	Provide a button that allow users to add current event to their favourite list.	High	3.19.5.5.1: #1

Table 3.19.5.5.2: Requirement List of Admission Ticket Purchase

3.19.5.6. Seated Ticket Purchase

This process outlines the workflow for purchasing tickets with seat selection.

- **Seats Selection:** Users view " event Detail" (introductions, dates, prices, venue). They select a Date and Session, followed by the seat selection page which allows users to select their seats directly on the seating plan. After the seat selection, users will enter the order confirmation page to complete the payment.
- **Payment Processing:** Order number will be generated when users complete the payment. Alternatively, payment can be cancelled on the confirmation page.
- **Order Pending State:** If unpaid, the order remains in a "Pending" state for 15 minutes, allowing the user to complete the transaction; it is automatically cancelled after 15 minutes if payment is not received.
- **Completion:** Upon success, users can view their ticket information and admission QR code.

3.19.5.6.1. User Process

The following table describes the user process of Seated Ticket Purchase.

No.	Component	Description
1	Event Detail	Users can check program introductions, date and time, price, venue location, and event content in the event detail page. Users can also add the event to their favourite list.
2	Date and Session Selection	Users can select different dates and sessions.
3	Select Seats	Users can select seats with different location or price, while the total price will be calculated dynamically and displayed on the seat's selection page.

		Due to the Xiaohongshu platform restriction, different types of tickets with multiple prices cannot be purchased together in one order.
4	Order Confirmation	Users can check the calculated subtotals and confirm the final payment.
5	Complete Payment	After the payment is completed, users can view the order in detail with an order number.
6	Cancel Payment	Payment can be cancelled at the order confirmation page by tapping the back button.
7	Order Pending	The unpaid order will be in the pending state for 15 minutes. The user will still be able to pay for the order within 15 minutes. The order will be automatically cancelled after 15 minutes.
8	View Tickets	Users can view detailed tickets for information on the order completion page. User can click the “View Ticket Code” button to view the admission ticket code.

Table 3.19.5.6.1: Description for the User Process of Seated Ticket Purchase**3.19.5.6.2. Requirement List**

RID	Description	Priority	Related Process
REQ-XHS-STP-001	Display comprehensive event information including introductions, schedules, and ticket availability on the details page.	High	3.19.5.6.1: #1
REQ-XHS-STP-002	Provide an interactive calendar interface for selecting dates and browsing available sessions.	High	3.19.5.6.1: #2
REQ-XHS-STP-003	Display real-time ticket availability, including labelling sold-out tickets.	High	3.19.5.6.1: #2
REQ-XHS-STP-004	Support the selection of specific seats while dynamically updating the total Price.	High	3.19.5.6.1: #3
REQ-XHS-STP-005	Require users to accept purchase terms before confirming order submission.	High	3.19.5.6.1: #4
REQ-XHS-STP-006	Support different payment methods.	High	3.19.5.6.1: #5

REQ-XHS-STP-007	Display a payment completion page confirming the transaction status, order number, and payment amount.	High	3.19.5.6.1: #5
REQ-XHS-STP-008	Order can be cancelled on the order confirmation page.	High	3.19.5.6.1: #6
REQ-XHS-STP-009	Order will be pending for 15 minutes if the user does not complete the payment.	High	3.19.5.6.1: #7
REQ-XHS-STP-010	The order will be automatically cancelled if the order is unpaid after 15 minutes.	High	3.19.5.6.1: #7
REQ-XHS-STP-011	Generate an order detail page providing individual admission QR codes and full transaction records.	High	3.19.5.6.1: #8
REQ-XHS-UTP-012	Provide a button that allow users to add current event to their favourite list.	High	3.19.5.6.1: #1

Table 3.19.5.6.2: Requirement List of Seated Ticket Purchase

3.19.5.7. Xiaohongshu Account to Company Account Transfer

This section details the financial backend workflow of how funds move from user purchases to the company's bank account.

- **Fund Holding & Status:** When a user completes a purchase, funds are temporarily held on the Xiaohongshu platform. The order's admission status on Xiaohongshu is initiated to "Not Completed".
- **Admission Modes:** Administrators must set one of three admission modes before listing an event:
 - **Session Complete Mode:** Order Admission Status is labelled "Completed" automatically when the session finishes.
 - **Admission Complete Mode:** Order Admission Status is labelled "Completed" only after the ticket is successfully scanned by AnTank Admission App.
- **Fund Transfer:** Once the Admission Status of an order turns "Completed," funds will be available to move to the Xiaohongshu Transferable Account manually. Xiaohongshu deducts relevant fees (commission, advertisement) during the transfer. Finally, funds will be available to transfer to the designated Company Bank Account manually.

3.19.5.7.1. User Process

The following table describes the user process of Xiaohongshu Account to Company Account Transfer.

No.	Component	Description
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1	Funds Hold	When the user completes the ticket purchase, the funds are temporarily held on the Xiaohongshu platform.
2	Xiaohongshu Admission Status	The Admission Status of the order on Xiaohongshu Platform will be initiated to “Not Completed”.
3	Admission Mode Setting	Administrators can set up different admission modes for different events before listing the event on Xiaohongshu Platform. One event can only be set to one admission mode.
4	Session Complete Mode	All orders from Admission Status are labelled "Completed" automatically when the session finishes.
5	Admission Complete Mode	The order of Admission Status is labelled "Completed" only after the ticket is successfully scanned and admitted using a PDA machine.
6	Transferable Account	After the order Admission Status is labelled “Completed” on Xiaohongshu Platform, the funds can then be moved to the Xiaohongshu Transferable Account.
7	Fee Deduction	Xiaohongshu deducts relevant fees (e.g., commission and advertisement fees) from the total order amount. The amount depends on different situations.
8	Company Account Transfer	Funds can be transferred from the Xiaohongshu Transferable Account to the designated Company Bank Account. This process will take 4 days (determined by Xiaohongshu Platform).

Table 3.19.5.7.1: Description for the User Process of Xiaohongshu Account to Company Account Transfer

3.19.5.7.2. Requirement List

RID	Description	Priority	Related Process
REQ-XHS-DCT-001	The funds are temporarily held in the Xiaohongshu platform upon user purchase completion.	High	3.19.5.7.1: #1
REQ-XHS-DCT-002	The funds will not be able to transfer from Xiaohongshu Platform to Xiaohongshu Transferable Account until the order Admission Status is set to “Completed”.	High	3.19.5.7.1: #2
REQ-XHS-DCT-003	The system allows administrators to configure one of three mutually exclusive Admission modes per event: Session	High	3.19.5.7.1: #3

	Complete, Order Complete, or Admission Complete.		
REQ-XHS-DCT-004	Under Session Complete mode, the system automatically labels all relevant orders as "Admission Completed" status when the session is finished.	High	3.19.5.7.1: #4
REQ-XHS-DCT-005	Under Admission Complete mode, the system automatically labels the order as "Admission Completed" status upon ticket is scanned via a PDA machine.	High	3.19.5.7.1: #5
REQ-XHS-DCT-006	The funds can be moved to the Xiaohongshu Transferable Account once the "Admission" status is set to "Completed".	High	3.19.5.7.1: #6
REQ-XHS-DCT-007	The commission fee will be deducted (commission, advertisement) by Xiaohongshu before funds move to the Xiaohongshu Transferable Account. The amount depends on different situations.	High	3.19.5.7.1: #7
REQ-XHS-DCT-008	The funds in Transferable account can be transferred to the company bank account. This process will take 4 days (Determined by Xiaohongshu Platform).	High	3.19.5.7.1: #8

Table 3.19.5.7.2: Requirement List of Xiaohongshu Account to Company Account Transfer

3.19.5.8. Admission Control

Entry admission is an on-site access validation function provided by the ticketing system, available to all tickets or credential holders. The process begins when attendees present their e-ticket QR code. The system reads the information via AnTank Admission App installed on scanning devices and automatically verifies it based on multiple criteria, including ticket validity, timing, and redemption status.

The system performs automated triage based on verification results: if verification passes, tickets are automatically validated and entry is granted; if verification fails, specific reasons are immediately prompted and entry is denied. For cases requiring manual judgment (such as concession tickets, using electronic ticket codes after paper tickets have been collected, cross-seating areas entry, etc.), the system marks them as "requires secondary confirmation" and alerts on-site validation staff for final decision-making. The validation personnel can either grant entry or deny access based on the actual situation, thereby concluding the single verification process.

This process ensures the uniqueness and status accuracy of each ticket code by synchronizing in real time with the central ticketing inventory system, preventing duplicate admissions and invalid ticket verification. Meanwhile, verification results are recorded in real time, providing a full lifecycle flow for on-site crowd management, data statistics, and post-event audits.

3.19.5.8.1. User Process

The following table describes the user process of Admission Control.

No.	Component	Description
1	Start: Audience Presents Ticket Code	The process begins, and the audience presents their electronic or paper ticket code before verifying the device.
2	Admission Device Scan/Input	The verification device reads the ticket code information by scanning the QR code and uploading it to the backend system.
3	System Automatic Verification	The system automatically performs checks: QR code validity, ticket status, session validity, venue/event matching, and seat status.
4a	Verification Decision: Passed	The system determined the result as a pass based on predefined rules.
4b	Secondary Confirmation	The system determines that secondary confirmation is required based on preset rules, prompting the device to display the reason (e.g., concession ticket entry, using an electronic ticket code after collecting a paper ticket, cross-seating areas entry), and waits for the verification personnel to select "Confirm" or "Cancel."
4d	Verification Decision: Not Passed	The system determines the result as a failure based on preset rules and prompts accordingly, such as invalid QR code, already redeemed ticket, or other conditions that do not meet the entry verification requirements.
5a	Result Execution: Display "Ticket Verification Successful" and Grant Passage.	For the scenario of 4a (Pass), the system automatically completes the verification: 1. Update ticket status: Mark the ticket as "Verified" in the database. 2. Device feedback: The verification device displays a "Check-in Successful" prompt, accompanied by a success of sound effect and vibration. 3. Execute passage.
5b	Secondary Judgment By Admission Staff	For scenarios requiring secondary confirmation (4b), control is transferred to the verification personnel: 1. Interface prompt: The device interface clearly displays the alarm reason (e.g., "Please confirm: senior ticket"), accompanied by a confirmation of sound and vibration, with "Confirm" and "Cancel" buttons provided.

		2. Manual verification: The verification personnel conduct a quick check based on the prompt. 3. Decision-making: The verification personnel click the corresponding button according to the on-site situation.
5c	Execution Result: Display "Ticket Verification Failed" and Deny Entry.	For cases of 4c (rejection), the system performs the rejection operation: 1. Device feedback: The validation device displays a "Ticket Verification Failed" prompt, clearly indicating the reason for failure (e.g., "Invalid Ticket"), accompanied by a failure of sound effect and vibration. 2. Entry is denied.
6a	Execution Result: Display "Ticket Verification Successful" and Grant Passage.	When the verification staff clicks "Confirm" in step 5b: 1. Record operation: The system logs the action in the verification log. 2. Update ticket status: Complete verification and mark the ticket as "Verified". 3. Execute release.
6b	Execution Result: Entry Denied	When the verification staff clicks "Cancel" in step 5b: 1. Process termination: The system will not verify the ticket, and the ticket status remains unchanged. 2. Entry denied.
7	End	The final status of a single verification transaction.

Table 3.19.5.8.1: Description for the User Process of Admission Control

3.19.5.8.2. User Process for Secondary Confirmation

The following table describes the verification scenarios that require manual intervention for secondary confirmation when ticket attributes trigger specific rules. These rules are typically configured at the ticket type or ticket level.

No.	Component	Description
1	Scan Code and Basic Verification	The verification personnel scan the ticket QR code, and the system performs basic validation to confirm that the ticket is valid.

2	Special Rules Trigger	The system determines whether the current ticket attributes match the predefined "requires secondary confirmation" rule set. Key rules include: 1. Ticket types requiring secondary confirmation: child tickets, senior tickets, and work-related tickets. 2. Ticket status is "physical ticket retrieved": when the user attempts to verify with an electronic QR code or ID card, but a physical ticket has already been redeemed. 3. Cross-seating areas entry: the area specified on the ticket does not match the current verification point area (only applicable to seat selection events).
3a	Process According to Standard Procedure	If no special rules are triggered, proceed to the 3.1.1.1 Admission verification process.
3b	Rule Type Judgment	If a rule is triggered, the system further determines the specific type of rule activated and proceeds to different processing branches.
4a	Trigger the "Ticket Type Confirmation" Prompt	When special ticket types (such as student tickets, senior tickets, etc.) are involved, the system displays a confirmation message on the device interface, requiring a secondary confirmation.
4b	Trigger "Paper Ticket Collected" Prompt	When the user has collected a paper ticket and uses an electronic ticket code for entry, the device prompts: "Please confirm! Paper tickets are already collected," requiring a secondary confirmation.
4c	Are Sessions Allowed to be Validated Across Regions?	When the event type is seat selection and cross-zone entry is involved, check the session configuration: "Refuse cross-zone ticket check" switch: If the switch is on (reject), then jump to node 5a. If the switch is turned off (allowed), then proceed to node 5b.
5a	Trigger "Cross-Seating Areas Entry" Prompt	When cross-zone ticket verification is allowed for the event, the device prompts: "Please confirm! No validation permission for the current zone," requiring secondary confirmation.
5b	Direct Refusal: Regional Independence	When the event is set to "Reject cross-seating areas ticket validation," the system will directly fail the verification.
6	Manual Secondary Confirmation	For all alerts requiring secondary confirmation (ticket type, ticket status, allowed cross-seating areas operations), the interface will be uniformly

		locked, with highlighted prompt messages, and only provide "Confirm" and "Cancel" buttons. The verification personnel will make decisions based on the current situation.
7a	Confirm Release	Click "Confirm". Device: Execute the admission and update the ticket status to "Used". The device indicates successful verification, proceed with release.
7b	Refused Entry	Click "Cancel". Device: Do not perform admission. Entry denied.
8	End	Complete a full special verification process.

Table 3.19.5.8.2: Description of Secondary Confirmation User Process

3.19.5.8.3. Requirement List

RID	Description	Priority	Related Process
REQ-AC-M-001	The backend management system must provide an interface to configure permissions for each verification account, including the types of services, venues, facilities, and areas (area verification applies only to seat selection) they are authorized to verify.	High	Full process
REQ-AC-M-002	The system should record and enable traceability of admission data and admission logs.	High	Full process
REQ-AC-M-003	The admission app interface must clearly display the available verification points and functions for the logged-in user, while supporting both QR code scanning and manual ticket code input for admission.	High	3.19.5.8.1: #2
REQ-AC-M-004	The system must support rapid account switching on verification devices to accommodate shift changes or shared device scenarios, ensuring the switching process is simple and efficient.	High	Full process
REQ-AC-M-005	The system must support configuring differentiated ticket verification times for special ticket types (such as VIP tickets). For example, VIP ticket holders may enter early, with their verification validity window	High	3.19.5.8.1: #3

	starting before the event's commencement time.		
REQ-AC-M-006	When an invalid ticket (such as counterfeit, already used, or refunded) is detected, the system must trigger a clear audible alarm and text prompt on the verification device.	High	3.19.5.8.1: #4c-#7
REQ-AC-M-007	admission system provides multiple options for prompt tones (success, failure, secondary confirmation).	Low	3.19.5.8.1: #4b, #5a, #5c, #6a, #6b 3.7.8.2: #4b, #4b, #5a, #6, #7b, #7b
REQ-AC-M-008	When verifying ticket types that require secondary confirmation (such as senior tickets or student tickets), the system must trigger clear visual prompts on the device and require the verification personnel to perform a secondary confirmation.	High	3.19.5.8.2: #1-#8
REQ-AC-M-009	The system must support an event-level toggle for "prohibit cross-seating areas ticket validation." When the toggle is off, cross-seating areas tickets enter a manual review process; when the toggle is on, cross-seating areas tickets are directly invalidated.	Medium	3.19.5.8.2: #3b-#8

Table 3.19.5.8.3: Requirement List of Admission Control

3.19.5.9. Backend Management

This process defines how events are created and published to the Xiaohongshu Mini Program.

- **User Process:** The process begins by completing the event configuration through the ticketing system back-end.
- **Integration & Application:** The event is pushed to the Xiaohongshu via a System Integration portal and listed through the CMS System. Administrators then complete the Listing Application within the Xiaohongshu Listing Management System.
- **Approval:** The application awaits approval by the Xiaohongshu Platform. If rejected, administrators must re-apply. If approved, the listing is completed, and the event becomes available for sale on the Xiaohongshu Mini Program.

3.19.5.9.1. User Process

The following table describes the user process of Backend Management.

No.	Component	Description
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1	Event Configuration	The Administrator should complete the Event Configuration on WestK Backend (Refer to Ticketing BRD document).
2	Event Push	The Administrator can push the event from WestK Backend to the Xiaohongshu Backend via the System integration portal.
3	CMS Listing	The Administrator can list the event on the Xiaohongshu Backend through the CMS System.
4	Xiaohongshu Listing Application	The Administrator can Complete the listing application on the Xiaohongshu Backend through the Xiaohongshu Listing Management System.
5	Application Approval	Wait for the application to be approved by the Xiaohongshu Platform. If the application is refused, the administrator should re-apply on the listing. If the application is approved, the event should automatically be available on Xiaohongshu Mini Program for sale.
6	Order Management	The Administrator can manage all customer orders generated from the Xiaohongshu Mini Program through WestK Backend, including viewing order details and checking order status.
7	Report Management	The Administrator can review various operational reports through WestK Backend for business analysis.
8	Refund Management	The Administrator can process order refund requests initiated via the Xiaohongshu Mini Program through WestK Backend, including reviewing refund reasons, approving/refusing requests, and checking refund status.

Table 3.19.5.9.1: Description for the User Process of Backend Listing Management

3.19.5.9.2. Requirement List

RID	Description	Priority	Related Process
REQ-XHS-BLM-001	The system must support completing event configurations through the ticketing system back-end.	High	3.19.5.9.1: #1
REQ-XHS-BLM-002	The system must be able to push events from the ticketing system back-end to the Xiaohongshu Backend via the system integration portal.	High	3.19.5.9.1: #2

REQ-XHS-BLM-003	The system must allow listing the event on the Xiaohongshu Backend through the CMS System.	High	3.19.5.9.1: #3
REQ-XHS-BLM-004	Administrators must be able to complete the listing application through the Xiaohongshu Listing Management System.	High	3.19.5.9.1: #4
REQ-XHS-BLM-005	The system must track the application's approval status from the Xiaohongshu Platform.	High	3.19.5.9.1: #5
REQ-XHS-BLM-006	The system must allow users to re-apply if the listing application is refused.	High	3.19.5.9.1: #5
REQ-XHS-BLM-007	Upon completion, the event must automatically become available for sale on the Xiaohongshu Mini Program.	High	3.19.5.9.1: #5
REQ-XHS-BLM-008	The system must allow authorized WestK staff to search, view, and filter orders generated through the Xiaohongshu Mini Program, including the ability to review order status, ticket details, payment results.	High	3.19.5.9.1: #6
REQ-XHS-BLM-009	The system must provide WestK staff with access to centralized sales and operational reports, including sales show, inventory status, and admission analytics, with the ability to export reports for financial reconciliation and management review.	High	3.19.5.9.1: #7
REQ-XHS-BLM-010	The system must allow authorized WestK staff to process refund requests and update the corresponding order and ticket statuses, with all refund actions logged for audit and reconciliation purposes.	High	3.19.5.9.1: #8

Table 3.19.5.9.2: Requirement List of Backend Management

4. Technical Requirement Definition

4.1. Customer Data Platform

4.1.1. Technical Requirement

The following requirements summarize the main business requirements in this document. Each requirement can subsequently establish further mapping relationships with detailed business requirement documents, functional specification documents, and technical design documents.

4.1.1.1. Unified Event Scope and Standards

- A unified event classification and naming principle should be established to ensure that data from different business domains can be horizontally compared and aggregated.
- The scope of event tracking at the page, object, and action levels should be clearly defined to avoid omissions and duplicate records.
- Specific events and attributes should be clearly defined item by item in the subsequent detailed business requirements document.

4.1.1.2. Object-Oriented Tracking Coverage

- Membership cards, tickets, special events, and product information should each be recorded as independent Business Objects.
- Each type of object should support the recording of key links from exposure, browsing to interaction, and conversion.
- If there is a jump or linkage relationship between objects, it should support identifying their upstream and downstream associations.

4.1.1.3. Data Governance and Boundaries

- Event tracking should follow the principle of minimum necessity, and the use of event tracking data is managed by WestK's system.
- Event tracking data is transmitted via the MQTT protocol, supporting TLS/SSL encryption (similar to HTTPS).
- Detailed requirements related to specific event fields, attribute dictionaries, and calibre definitions should be further confirmed in subsequent special documents.

4.1.2. Technical Requirement Analysis

4.1.2.1. Front-end and Back-end Collaborative Event Tracking Mechanism

To ensure the accuracy, completeness, and consistency of event tracking data, the system should support the mechanism of front-end + back-end collaborative event tracking, clearly define the responsibility boundaries and data reporting division of labour for front-end and back-end event tracking, ensure that all key events and

states can be effectively recorded, and that data from both ends can be associated and integrated.

- **Frontend event tracking:** Responsible for reporting behavioural events that are visible to the user, actively triggered by the user, or directly perceptible by the frontend, with Data Acquisition cantered around user operations and page display, including but not limited to page exposure, product/membership card/event exposure, page click, stay, jump, search, share, favourite, authorization, and other behaviours.
- **Server-side event tracking:** Responsible for reporting events and data that cannot be reliably obtained from the front end, require system verification and confirmation, or represent the final state of the business, with Data Acquisition cantered around system business logic and state results, including but not limited to order creation, payment results, order status changes, the final state of membership card collection/activation/use, user membership information synchronization, product transaction conversion results, etc.
- Event tracking at both ends should have unified correlation identifiers (such as user unique ID, order ID, behaviour trace ID, etc.) to ensure that front-end behaviours and server-side states within the same business link can achieve precise matching and traceability analysis.

4.1.2.2. Data Reporting Frequency and Collection Rules

The system should clearly define the reporting frequency and collection trigger rules for different types of events tracking data, considering both the real-time requirements of the data and system Show optimization, to avoid over-reporting or reporting delays affecting the effectiveness of operational analysis. The reporting rules for all types of data need to match the requirements of business analysis.

4.1.2.2.1. Immediate reporting

Trigger Conditions: User core behaviours, key business status changes, and operational indicator-related data that requires real-time monitoring need to be immediately triggered for reporting after the event occurs/status changes to ensure that data enters the analysis pipeline without delay.

Covered Data Types:

1. Front-end: High-value user behaviours (such as payment clicks, membership card collection/activation operations, event participation submissions, sharing/authorization behaviours), exposure and conversion click of key pages.
2. Server: Core business states such as order placement, payment success/failure, order status change (cancellation/ completion), membership card benefit usage, product transaction results, user information addition/update, etc.

4.1.2.2.2. Batch/Delayed Reporting

Trigger conditions: user behaviour data with high frequency, low value, and no real-time analysis requirements. To reduce system interaction pressure, batch collection

or fixed delay can be used for reporting. The reporting delay should be within an acceptable range (such as ≤ 30 minutes).

Coverage data types:

1. Front-end: High-frequency, low-value behaviours such as regular browsing of ordinary pages, clicking on non-core buttons, Time Stay on pages, exposure of ordinary products/content (not key monitoring targets for operations), etc.
2. Non-core user behaviour trajectory aggregation data, behaviour logs without real-time operational monitoring requirements.

4.1.2.2.3. Continuous Collection/Timed Synchronization

Trigger Condition: For basic data and statistical data that need to be continuously recorded or regularly updated, use the method of continuous collection + scheduled synchronization to ensure the freshness and integrity of the data.

Covered Data Types:

1. Regular synchronization of Mini Program user basic information (authorized) and basic data of user segmentation tags.
2. Continuous collection and regular update of statistical summary indicators for various event tracking data (such as total exposure and total clicks within a time period).
3. Continuous collection and scheduled integration of behavioural trajectory data across business domains.
 - The reporting frequency and collection rules for all data should support flexible configuration, allowing adjustment of the reporting methods for various events based on subsequent operational analysis requirements and changes in business priorities.
 - The data reporting process should have a failure retransmission mechanism to avoid data loss caused by temporary network or system failures and ensure the integrity of event tracking data.

4.1.3. Technical Requirement List

RID	Description	Priority	Related Process
REQ-CDP-FRO-001	Front-end event tracking.	High	4.1.2.1
REQ-CDP-BCK-001	Server-side event tracking.	High	4.1.2.1
REQ-CDP-UID-001	Both ends of the event tracking should have a unified associated identifier.	Medium	4.1.2.1

REQ- CDP-UID- 002	Immediate reporting.	Medium	4.1.2.1
REQ- CDP-UID- 003	Batch/Delayed Reporting.	Medium	4.1.2.1

Table 4.1.3: Requirement List of Kiosk Terminal Monitoring

4.2. Reporting

4.2.1. Technical Requirements

WestK requires the system to inherit the existing user permission configuration of AnTank to ensure the unity of data and permissions. The system needs to complete lightweight integration with the AnTank business system, mainly through API interfaces, to achieve data update frequency, as well as the functions of filtering, displaying, exporting, and sharing based on various dimensional conditions.

4.2.2. Technical Requirements Analysis

4.2.2.1. Permissions and Management Functions

- Support role permission configuration: Control permissions at the navigation menu level, assign data viewing and export permissions by role to achieve data isolation.
- Support data calibre management: unified maintenance of indicator calibres to avoid calibre confusion.

4.2.2.2. Report Data Independence

- Report data and production data remain relatively independent to avoid mutual influence of business logic.
- can effectively isolate query pressure and ensure the stability of the production database operation.

4.2.2.3. Data Export

- Support for report export: Supports export in Excel and PDF formats.
- Exported data includes indicator names, values, and statistical time periods.

4.2.2.4. Data Privacy Protection

- User sensitive information involved in the report data shall be included in the scope of privacy protection management.
- Take certain desensitization measures for relevant sensitive fields to avoid the leakage of sensitive data.

4.2.2.5. Audit Log

The following operations will be recorded in the audit log:

- User access: User access dashboard name, access device, referrer, access type (view report, data export).
- Report Export: Report export records are logged into the audit log, including the report name, export conditions, and the number of exported records.

4.2.3. Technical Requirement List

RID	Description	Priority	Related Process
REQ-INT-SYS-001	Permission control by navigation menu level.	High	4.2.2.1
REQ-INT-SYS-003	Data Export.	Medium	4.2.2.3
REQ-INT-SYS-004	Audit Log Storage	Medium	4.2.2.5

Table 4.2.3: Requirement List of Reporting

4.3. Ticketing Operation Centre

4.3.1. Technical Requirements

WestK requires the system to inherit the existing user permission configuration of AnTank to ensure the unity of data and permissions. The system needs to complete a lightweight integration with the AnTank business system, mainly through API interfaces, to achieve data synchronization frequencies according to different business requirements, as well as functions for filtering, displaying, exporting, and sharing data based on various dimensional conditions.

4.3.2. Requirements Analysis

4.3.2.1. Permissions and Management Functions

- Supports role-based permission configuration performs permission control at the navigation menu level, assigns data viewing and export permissions by role, and achieves data isolation.
- Supports data calibre management: unified maintenance of indicator calibres to avoid calibre confusion.

4.3.2.2. BI Dashboard Data Independence

- BI Dashboard data remains relatively independent from production data to avoid mutual influence of business logic.
- can effectively isolate query pressure and ensure the stability of the production database operation.

4.3.2.3. Data Interface

- Ticketing Operation Centre and BI tools need to implement data synchronization through API interface docking. The docking must meet network interconnection requirements, ensure data security through encrypted transmission and desensitized storage of sensitive data, and at the same time strictly verify data to ensure data consistency and integrity.
- It is necessary to implement automatic updates of datasets and dashboards, while considering system show stability, full-process monitoring and operation and maintenance, ensuring that the docking process is efficient and compliant, and fully adapting to the data-driven and decision support requirements of enterprise digital operations.

4.3.2.4. Data display function

Supports visual charts: line chart, bar chart, pie chart, radar chart, heat map, and table, adapting to different analysis scenarios.

4.3.2.5. Data Export

Support for Dashboard export: Supports export in Excel, PDF, and image formats, with exported data including indicator names, values, etc.

4.3.2.6. Data Privacy Protection

User sensitive information involved in the Dashboard shall be included in the scope of privacy protection management.

Take certain desensitization measures for relevant sensitive fields to avoid the leakage of sensitive data.

4.3.2.7. Audit Log

The following operations will be recorded in the audit log:

User access: User accesses dashboard, dashboard name, access device, referrer, access type (viewing dashboard, data export).

4.3.3. Technical Requirement List

RID	Description	Priority	Related Process
REQ-DSB-SYS-001	Perform permission control by navigation menu level.	High	4.3.2.1

REQ- DSB-SYS- 002	Data Export.	Medium	4.3.2.5
REQ- DSB-SYS- 004	Audit Log Storage.	Medium	4.3.2.7

Table 4.3.3: *Technical Requirement List for Ticketing Operation Centre*

4.4. Monitoring

4.4.1. Technical Requirement

The following technical requirements have been identified.

4.4.1.1. Kiosk Terminal Monitoring

As the core carrier of on-site self-service ticketing services, I need the system to comprehensively monitor the status of Kiosk terminals, known in real time whether each Kiosk is online and whether its heartbeat is normal, and be able to detect any device offline or heartbeat interruption immediately; at the same time, I need to view the running status list of all Kiosk terminals to clearly understand the current status of each device, facilitating unified management by operation and maintenance personnel.

4.4.1.2. Octopus Device Monitoring

The Octopus card reader is the core device for on-site Octopus payment and card reading, directly affecting ticketing transactions. I need the system to monitor in real time whether the Octopus devices on each Kiosk are online and whether the connections are normal, and be able to accurately identify device abnormalities, disconnections, and other situations; once a device malfunctions, the system must immediately trigger an alarm notification to enable operation and maintenance personnel to handle it promptly, avoiding any impact on on-site Octopus-related ticketing services.

4.4.1.3. Apollo Device Monitoring

Apollo devices are key equipment that support ticketing payment and transaction communication. Device anomalies can lead to transaction failures or delays. I need the system to monitor the service status, network connection status, and service response status of Apollo devices in real time, so as to promptly detect issues such as device service unavailability, network interruption, and response timeout; when a device experiences a service anomaly, the system must immediately trigger an alarm to ensure the smooth communication link of ticketing transactions.

4.4.1.4. Printer Monitoring

Printers are important devices for on-site ticket issuance. I need the system to monitor in real-time the network connectivity status and remaining ticket paper

quantity of the printers paired with Kiosks, so as to promptly detect situations such as printer offline and insufficient ticket paper; when a printer malfunctions or the remaining ticket paper is below the threshold, the system shall push alarm information via email to remind operation and maintenance personnel to handle it promptly, thus preventing tourists from being unable to obtain tickets after purchasing them.

4.4.1.5. Uninterruptible Power Supply (UPS) Monitoring

UPS devices provide emergency power supply for Kiosk terminals and their supporting peripherals, ensuring the normal operation of the equipment during sudden power outages. I need the system to monitor the network connectivity and health status of UPS devices in real time; if the UPS device supports interface queries, it also needs to monitor the battery level. When the UPS device is offline, its health status is abnormal, or the battery level is below the threshold, the system can issue an alarm in a timely manner to ensure stable power supply for the equipment.

4.4.1.6. Server hardware and Software Environment Monitoring

The basic hardware environment is the foundation for the stable operation of devices. I need the system to monitor in real time the hardware show indicators such as CPU, memory, and disk of Kiosk terminals to keep track of the usage of hardware resources. When the usage of hardware resources exceeds the preset threshold, the system can trigger an alarm notification to avoid issues such as device freezing and laggy operation caused by resource exhaustion.

4.4.1.7. Microservice Monitoring

Ticketing-related microservices are the core support for ticketing business processes. I need the system to monitor the running status of all ticketing-related microservices in real time to confirm whether the microservices are properly started and providing services. When a microservice experiences downtime, abnormal response, or other issues, the system should push alarm information via email to enable development and operations personnel to intervene promptly, ensuring the normal operation of ticketing business processes.

4.4.2. Technical Requirement Analysis

This chapter conducts a detailed analysis of the user requirements in Section 4.4.1, clarifying the business intent, monitoring indicators, functional requirements, and implementation key points of each requirement, providing detailed basis for system design.

4.4.2.1. Kiosk Terminal Monitoring

Map to Technical Requirement 4.4.1.1.

The Kiosk terminal is the physical carrier of on-site ticketing services, and its online status and heartbeat are the foundation for device availability. The monitoring of this module needs to achieve the dual capabilities of basic status monitoring + resource auxiliary monitoring, with the specific requirements as follows:

Core Status Monitoring

Monitoring indicators: device online status, device heartbeat (heartbeat frequency $\leq$ 30 seconds per time).

Anomaly determination: If the heartbeat is interrupted for more than 60 seconds, it is determined that the device is offline, and a serious alarm is immediately triggered.

Functional Requirements: Provide a list of the operating status of all Kiosk terminals, support filtering by dimensions such as device number, location, status, etc., and the list is refreshed in real time.

Basic Resource Monitoring (Auxiliary Operation and Maintenance)

Monitoring metrics: CPU utilization, memory utilization, disk utilization, network utilization.

Anomaly determination: When resource utilization exceeds 95% (extremely high threshold), a low-priority alarm is triggered.

Functional Requirements: Support viewing the resource usage trend curve of a single device, with the time dimension being selectable (hour/day/week).

4.4.2.2. Octopus Device Monitoring

Map to Technical Requirement 4.4.1.2.

Octopus devices are core business peripherals, and their availability directly determines whether Octopus-related ticketing transactions can be carried out. This module is the core business monitoring module, and the specific requirements are as follows:

Monitoring indicators: device online status, USB connection status (identified by matching the USB serial port signature CH34x), and device card reading function status.

Abnormality determination: If the system fails to recognize the USB connection of the Octopus device or the card reading function does not respond, both are determined as device abnormalities, immediately triggering a serious alarm.

Functional Requirements: Support viewing Octopus device status by associating with Kiosk terminal numbers, record the occurrence time and recovery time of device anomalies, and form a device fault ledger.

4.4.2.3. Apollo Device Monitoring

Map to Technical Requirement 4.4.1.3.

Apollo devices serve as the communication core for ticketing transactions, and their service and network status directly affect the success rate and efficiency of transactions. This module is the core business monitoring module, with the following specific requirements:

Monitoring indicators: device service online status, network connection status, HTTP connectivity, service response time.

Abnormality Judgment

Severe Alarm: HTTP connectivity to the Apollo backend service is interrupted, and the service is unresponsive.

Warning Alarm: Apollo service response time > 300ms (preset threshold), not interrupted but affecting transaction efficiency.

Functional Requirements: Read the Apollo service URL from the Kiosk configuration file for connectivity testing, with a testing frequency ≤ 30 seconds per test; support viewing the trend curve of service response time.

4.4.2.4. Printer Monitoring

Map to Technical Requirement 4.4.1.4.

The printer is a key device for on-site ticket issuance, and its status directly affects the ticketing experience of tourists. The monitoring requirements for this module are as follows:

Monitoring indicators: printer network status, remaining ticket paper quantity (measured in sheets/percentages), and print task execution status.

Abnormality Judgment

Severe Alarm: Printer offline, print job execution failed.

Warning Alarm: The remaining number of tickets is below the preset threshold (e.g., 100 sheets).

Functional Requirements: Support viewing printer status associated with Kiosk terminal number, and ticket paper remaining data needs to be synchronized in real-time with the printer hardware interface; in case of an anomaly, push alarm information via email, including device location and anomaly type.

4.4.2.5. UPS (Uninterruptible Power Supply) Monitoring

Map to Technical Requirement 4.4.1.5.

PS equipment provides emergency power supply support for ticketing equipment, and its health status directly affects the operation of the equipment during sudden power outages. The monitoring requirements for this module are as follows:

Monitoring indicators: UPS device network connectivity status, device health status (returned by the UPS hardware interface), battery level (if the device supports interface query).

Abnormality Judgment

Severe Alarm: UPS device is offline, and its health status is abnormal.

Warning Alarm: Battery level is below the preset threshold (e.g., 20%).

Functional Requirements: Only interface with UPS devices that support standard data interfaces; for devices without interfaces, only monitor network connectivity status; trigger an alarm when an anomaly occurs, and support viewing the trend curve of battery capacity.

(If the UPS supports reading battery level, this function is supported; if it does not support reading computer information, this function is unavailable).

4.4.2.6. Server Hardware Environment Monitoring

Map to Technical Requirement 4.4.1.6.

The basic hardware environment is the guarantee for the stable operation of the device. This module is the basic monitoring module, and the specific requirements are as follows:

Monitoring metrics: CPU utilization, memory utilization, disk utilization (partition level), network upload/download rate, network latency.

Anomaly Detection: Alarms are triggered when various indicators exceed the preset thresholds (configurable), with thresholds distinguishing between two levels: warning (e.g., CPU usage 80%) and critical (e.g., CPU usage 95%).

Functional Requirements: Support comparative analysis of hardware indicators for single/multiple devices, and support setting custom threshold parameters.

4.4.2.7. Microservice Monitoring

Map to Technical Requirement 4.4.1.7.

The ticketing-related microservices are the software core of the ticketing business, and their operational status directly determines the smoothness of the ticketing process. The monitoring requirements for this module are as follows:

Monitoring indicators: microservice instance startup status, service interface call success rate, service response time, service concurrency.

Abnormality Judgment

Severe Alarm: Microservice instance downtime, service interface call success rate below 90%.

Warning Alert: Service response time > 500ms, concurrency exceeds the preset threshold.

Functional Requirements: Support filtering monitoring data by microservice name and instance number, record the occurrence and recovery time of service exceptions; in case of an exception, push alarm information via email, including the microservice name, exception type, and scope of impact.

4.4.2.8. Monitoring and Alarming

Map to Business Requirement 4.4.1.1 to 4.4.1.7.

Monitoring and alerting are one of the core capabilities of this system, which requires the implementation of hierarchical, multi-channel, and traceable alert management. The specific requirements are as follows:

Alarm Classification: Alarms are classified into three levels, namely critical alarms, warning alarms, and low-priority alarms, based on the impact level of faults. The classification criteria refer to Appendix D.

Alarm types: cover all abnormal types in monitoring scenarios such as device offline, service exception, network exception, resource exception, insufficient consumables, etc.

Alarm Notification

Notification Channels: Email Notification, Operation and Maintenance System Notification (Integrated with the existing WestK Operation and Maintenance System).

Notification content: includes key information such as alarm level, monitored object, fault location, anomaly type, occurrence time, handling suggestions, etc.

Notification Rules: Severe alarms are immediately pushed (within 5 minutes), warning alarms are delayed for pushing (within 30 minutes, and not pushed if the fault is resolved), and low-priority alarms are only displayed on the monitoring dashboard.

Alarm Management: Supports full-process operations for alarms, including confirmation, ticket assignment, handling, and closure, and records all logs of alarm handling; supports custom configuration of alarm rules (thresholds, notification channels, and notified personnel).

4.4.2.9. System Integration

Map to Business Requirement 4.4.1.1 to 4.4.1.7

This system needs to complete a lightweight integration with the existing systems of WestK to ensure data interoperability and unified permissions. The specific requirements are as follows:

Core Integrated Object: AnTank Business System (WestK Core Ticketing System).

Integration Method: Data interaction is implemented based on standard RESTful API interfaces, which use encrypted transmission to ensure data security.

Integrated Content

Data Transmission Service: Synchronize basic Master Data such as Kiosk terminals, device points, and ticketing business scope from the AnTank system, and push device anomaly alarm data to the AnTank system.

Permission Integration: Connect to the existing unified permission system of WestK to implement SSO single sign-on, ensuring that the operation permissions of maintenance personnel remain consistent with those of the existing system.

Integration Requirements: The interface shall provide detailed interface documentation, support high-concurrency calls, and have an interface response time $\leq 100\text{ms}$; the system shall reserve expansion interfaces to support subsequent integration with other systems.

4.4.3. Operating Model

This initiative uses a three-party operating model to describe the monitoring features.

The user usage process of this system **revolves around "monitoring and viewing - abnormal alarm - fault handling - status recovery - data traceability", with the core users being the operation and maintenance personnel and operation management personnel** of WestK. The usage processes and operation permissions vary for different roles, and the specific operation model is as follows.

4.4.3.1. Overall Usage Process

Device status collection → Real-time monitoring and viewing → Anomaly detection and graded alarm → Fault reception and location → Fault handling and feedback → Status recovery and alarm closure → Data recording and statistical analysis

1. **Device Status Collection:** The system automatically collects the operational status data of all monitored objects (Kiosk, Octopus, Apollo, etc.) at a preset frequency (≤ 30 seconds per collection), achieving real-time data synchronization.
2. **Real-time monitoring and viewing**
 - Operation and maintenance personnel: After logging into the system, view the real-time status of devices and services in the responsible area on the **Unified Monitoring Dashboard**, which supports filtering by dimensions such as location, device type, and status.
 - Operation managers: **View aggregated indicators such as the overall equipment operation rate, failure occurrence rate, and alarm handling efficiency on the operation monitoring dashboard;** understand the overall operation status of ticketing equipment.
3. **Anomaly Detection and Graded Alerts:** The system automatically detects device/service anomalies based on preset thresholds and rules, classifies them according to the impact level, and pushes alert information via email/operation and maintenance system.
4. **Fault Reception and Location:** After receiving the alarm information, the operation and maintenance personnel can view the detailed information of the fault (location, type, occurrence time) through the system and quickly locate the fault site and the cause of the fault.
5. **Fault handling and feedback:** Operation and maintenance personnel go to the site or remotely handle the fault and record the **handling actions and handling time** in the system during the handling process; if the fault cannot be repaired immediately, a work order can be initiated in the system to notify relevant technical personnel to intervene.
6. **Status Recovery and Alarm Closure:** After the fault handling is completed, the system detects in real time that the device/service status has returned to normal and automatically triggers **recovery notification**; the operation and maintenance personnel confirm the fault recovery in the system and close the alarm.
7. **Data Recording and Statistical Analysis :** The system automatically records all data related to equipment operation, abnormal alarms, and fault handling, forming a ledger; operation and maintenance/operation management personnel can view data such as equipment operation trends, fault type distribution, and alarm handling efficiency through the system's **statistical analysis module** , providing a basis for equipment maintenance and operation optimization.

4.4.3.2. Role Permission Division

1. **Operation and Maintenance Personnel:** Have operation permissions such as monitoring and viewing, alarm confirmation, fault handling, alarm closure, and fault recording, and can only view device data in the responsible area.
2. **Operations and Maintenance Supervisor:** Has all the permissions of operations and maintenance personnel, and can also perform operations such as alarm rule configuration, fault ticket assignment, permission allocation, data export, etc., and can view device data across the entire region.

- 3. Operations Management Personnel:** Have **read-only permissions** and can only view the summary data and statistical analysis reports on the monitoring dashboard but cannot perform fault handling or system configuration operations.
- 4. System Administrator:** Has the highest authority in the system and can perform all operations such as system parameter configuration, interface management, user management, and log management.

4.4.4. Customer Journey

This section describes what users can do after entering the Douyin Mini Program.

The user story is used to describe the user's needs, goals, and expected outcomes, providing clarity on what the user wants to achieve and why the functionality is required.

No.	Category	Role	User Story
1	Real-time monitoring	As a frontline operation and maintenance staff member in WestK.	After logging into the system, I need to be able to view the real-time status of devices such as Kiosk, Octopus, and Apollo in the responsible area, support filtering by device location, so that I can quickly grasp the operating conditions of devices within the jurisdiction without the need for on-site manual inspection.
2	Real-time Monitoring	As WestK Operations and Maintenance Supervisor.	I need a unified monitoring dashboard that can display the operating status of all devices and services across the entire region, while presenting aggregated metrics such as device online rate and failure incidence rate, enabling me to have an overall grasp of the overall operation of ticketing equipment.
3	Abnormal Alarm	As a frontline operation and maintenance personnel in WestK.	I need the system to promptly push alarm information via email and the operation and maintenance system when an abnormality occurs in the equipment/service, and the alarm information should include the fault location and abnormality type, enabling me to quickly locate the fault and arrive at the site for handling in a timely manner.
4	Fault Handling	As a frontline operation and maintenance personnel in WestK.	I need to record the troubleshooting process and results in the system, while supporting the fault dispatch function. When I am unable to resolve the fault, I can quickly notify technicians to

			intervene, ensuring that faults can be handled efficiently.
5	Statistical Analysis	As an operational management staff member of WestK.	I need the system to provide statistical data such as equipment operation rate, failure occurrence rate, and alarm handling efficiency, support filtering by time, region, and equipment type, and be able to export data reports, so that I can analyse equipment operation conditions and provide a basis for operational decision-making.
6	System Configuration	As WestK Operations and Maintenance Supervisor.	I need the system to support custom configuration of alarm thresholds, notification channels, and notification personnel, while also being able to assign permissions to operation and maintenance personnel, enabling the system to adapt to on-site operation and maintenance workflows and management requirements.
7	Data Traceability	As WestK Operations and Maintenance Supervisor.	I need the system to record the operation logs, alarm logs, and fault handling logs of all devices, support queries by dimensions such as time, device number, and alarm type, so that I can trace the root cause of faults and analyse the patterns of device failures.
8	System Integration	As a WestK IT administrator.	I need this system to integrate with the existing AnTank business system and unified authorization system, support SSO single sign-on, and synchronize basic Master Data, so that operation and maintenance personnel do not need to log in to the system multiple times, ensuring data consistency.

4.4.5. Business Scenarios

This chapter sorts out the core business scenarios of this system, assigns a unique RID to each scenario, and clearly defines the requirement description, priority, whether it is specified in the tender, type of technical investment, requirement type, and associated business process in tabular form, thereby achieving standardized management of requirements.

This section organizes the user usage process and corresponding requirement list according to business scenarios.

4.4.5.1. Kiosk Terminal Monitoring

4.4.5.1.1. User Process

The user enters the monitoring system to view the Kiosk terminal status list.

The system displays the online status, heartbeat status, last reported time, and basic resource status of the terminal.

Users can identify offline terminals, terminals that have not reported for a long time, or terminals with abnormal resources, and further arrange for handling.

4.4.5.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-KSK-001	The system should support monitoring the online status, heartbeat status, and last reported time of Kiosk terminals, and be able to identify terminal offline status.	High	4.4.5.1.1
REQ-MON-KSK-002	The system should support the collection of basic operating indicators of Kiosk terminals, including CPU, memory, disk, and network, etc., and support the display of status lists.	Medium	4.4.5.1.1

Table 4.4.5.1.2: Requirement List of Kiosk Terminal Monitoring

4.4.5.2. Octopus Device Monitoring

4.4.5.2.1. User Process

Users view the status of Octopus devices or enter the details page after an abnormal alarm.

The system displays whether the device is online, whether the connection is normal, whether there are identification anomalies or disconnections.

When the device is abnormal, the system generates an alarm and notifies relevant personnel to handle it.

4.4.5.2.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-OCT-001	The system shall support monitoring the online status, connection status, and identification status of Octopus devices, and promptly detect disconnection or unidentifiable situations.	High	4.4.5.2.1

REQ-MON-OCT-002	The system should support triggering alarm notifications when an Octopus device malfunctions and displaying the cause of the malfunction and the scope of impact on the monitoring interface.	High	4.4.5.2.1
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Table 4.4.5.2.2: Requirements List of Octopus Device Monitoring

4.4.5.3. Apollo Device Monitoring

4.4.5.3.1. User Process

The user checks the status of Apollo devices or payment communication services.

The system checks its connectivity, availability, and response status, and identifies issues such as timeouts, failures, or unreachability.

When an exception occurs, the system pushes an alarm to assist operation and maintenance personnel in location and recovery.

4.4.5.3.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-APO-001	The system shall support monitoring the connectivity status, service availability, and response of Apollo devices or related payment services.	High	4.4.5.3.1
REQ-MON-APO-002	The system should support triggering an alarm when Apollo is abnormal and assist operation and maintenance personnel in quickly locating transaction communication issues.	High	4.4.5.3.1

Table 4.4.5.3.2: Requirements List of Apollo Device Monitoring

4.4.5.4. Printer Monitoring

4.4.5.4.1. User Process

The user views the printer status list or enters the details of the specified device after an alarm.

The system displays information such as the printer's network status, online status, working status, and remaining ticket paper.

When the printer is abnormal, out of paper, or unavailable, the system sends an alarm and prompts the user to handle it.

4.4.5.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-PRN-001	The system shall support monitoring the network connectivity status, online status, and working status of printers.	High	4.4.5.4.1
REQ-MON-PRN-002	The system shall support monitoring the remaining amount of printer ticket paper or the status of equivalent consumables and sending email alerts when an anomaly occurs.	High	4.4.5.4.1

Table 4.4.5.4.2: Requirements List of Printer Monitoring

4.4.5.5. UPS (Uninterruptible Power Supply) Monitoring

4.4.5.5.1. User Process

User checks the status of UPS devices.

The system displays whether the UPS is networked, as well as battery level, health status, or equivalent collectible metrics.

When the UPS status is abnormal, the system records the event and notifies relevant personnel to conduct troubleshooting.

4.4.5.5.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-UPS-001	The system should support monitoring the network connectivity status of UPS and whether the device is normally reachable.	Medium	4.4.5.5.1
REQ-MON-UPS-002	The system should support monitoring UPS battery charge, health status, or equivalent power status indicators (subject to device capabilities).	Medium	4.4.5.5.1

Table 4.4.5.5.2: Requirements List of UPS Monitoring

4.4.5.6. Hardware Environment Monitoring

4.4.5.6.1. User Process

The user views the monitoring data of the production line's hardware environment.

The system displays metrics such as CPU, memory, and disk, and identifies high load or abnormal states based on thresholds.

When resource usage is abnormal, the system sends an early warning to prompt the user to take corrective measures.

4.4.5.6.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-HWM-001	The system should support monitoring show indicators such as CPU, memory, and disk of the production line's hardware environment.	Medium	4.4.5.6.1
REQ-MON-HWM-002	The system shall support setting thresholds for hardware environment indicators and issuing an alarm when the thresholds are exceeded.	Medium	4.4.5.6.1

Table 4.4.5.6.2: *Requirements List of Hardware Environment Monitoring*

4.4.5.7. Microservice Monitoring

4.4.5.7.1. User Process

The user views the running status or health check results of critical microservices.

The system identifies states such as normal, abnormal, and unavailable, and records abnormal events.

When a microservice is abnormal, the system sends an alert to relevant users and supports subsequent queries.

4.4.5.7.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-MSV-001	The system should support monitoring the running status, health check results, and basic availability of key microservices.	High	4.4.5.7.1
REQ-MON-MSV-002	The system should support sending alerts via email or other means when a microservice encounters an exception and retain exception records.	High	4.4.5.7.1

Table 4.4.5.7.2: *Requirements List of Microservice Monitoring*

4.4.5.8. Monitoring Alerts and Notifications

4.4.5.8.1. User Process

After the system identifies an abnormality in a device or service, it generates a corresponding alarm record.

The system sends notifications such as emails according to the alarm level and notification rules and updates the status after recovery.

Users can query historical alarm records and track the handling and recovery status of issues.

4.4.5.8.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-ALT-001	The system shall support a unified alarm notification mechanism, at least support email alarms, and support differentiating notification content by object and severity.	High	4.4.5.8.1
REQ-MON-ALT-002	The system shall support alarm classification, recovery notification, and historical alarm record query to facilitate closed-loop management of issues.	High	4.4.5.8.1

Table 4.4.5.8.2: Requirements List of Monitoring Alerts and Notifications

4.4.5.9. Monitoring Overview, Recording, and Integration

4.4.5.9.1. User Process

Users can view the overall monitoring status through the unified dashboard and query historical events and processing records when necessary.

The system interfaces with the AnTank business system or subsequent interfaces to aggregate device, service, and alarm data.

The system supports recording monitoring events and their handling results, providing a basis for statistical analysis and review.

4.4.5.9.2. Requirement List

RID	Description	Priority	Related Process
REQ-MON-DSH-001	The system shall support providing a unified monitoring dashboard and a list of device/service statuses to facilitate users in viewing the current overall operation status.	Medium	4.4.5.9.1
REQ-MON-RPT-001	The system shall support recording monitoring events, alarm information, and processing results for subsequent statistical analysis and operation and maintenance review.	Medium	4.4.5.9.1

REQ-MON-INT-001	The system shall support integration with the AnTank business system and subsequent monitoring interfaces to obtain device, service, and alarm-related data.	Low	4.4.5.9.1
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Table 4.4.5.9.2: *Requirements List of Monitoring Overview, Recording, and Integration*

4.5. Permission

4.5.1. Technical Requirements

This section outlines the IAM (User and Permissions) business process. It covers the business problem statement, background and objectives, high-level and detailed business requirements, and implementation methods.

4.5.1.1. Data permission management requirements based on user BU attributes

WestK has multiple different business units (BUs). The system needs to be able to construct a data permission management mechanism based on BUs for the data that users from different BUs can operate on when they are operating the same function.

The system shall ensure that users' data permissions take effect within the scope of their affiliated Business Unit (BU) to achieve data isolation between different business units.

4.5.1.2. Position-based functional authority management system

The system shall support using positions as the main body for permission configuration, uniformly configuring menu, page, and operation permissions, and allowing users to inherit permissions through their affiliated positions.

- The system should support configuring functional permissions at the position level, including menu permissions and operation permissions (such as add, edit, delete, etc.).
- The system should support using positions as the configuration entity for functional permissions to centrally manage system functional permissions.
- The system should support users inheriting functional permissions through their assigned positions, eliminating the need to configure permissions for individual users one by one.
- The system should support flexible adjustment of job permissions, and user permissions should be updated synchronously after permission changes.

4.5.1.3. Data permission management requirements based on user groups

4.5.1.3.1. Data management requirements during business operations

Data access rights not only manifest in the scope of data display but also need to be integrated throughout the user's actual business operation process.

- In scenarios involving data selection, the system shall restrict users to only select Business Objects within their authorized scope.
- During the data creation process, the system shall automatically determine data ownership and ensure it falls within the scope of user authorization.
- During data editing and operation, the system shall verify data ownership and only allow operations on data within the authorized scope.

4.5.1.3.2. Data management requirements for orders and reports

The system shall perform permission control on order data based on the user's data permission level:

1. Super: Return the data of the BU to which the user belongs and the sales data of the BU where the user is located.
2. Group Level: Only return data belonging to the user group the user belongs to (including cross-BU data).
3. Personal Level: Returns order data generated by the user himself (including cross-BU).

Report Data Permission Management Requirements.

The system shall perform permission control on report data based on the user's data permission level, and the statistical results shall be consistent with the scope of order data accessible to the user.

4.5.1.3.3. Implementation of User Permissions

The user's final permissions are jointly determined by "Position (Functional Permissions)" and "User Group/User Attribute (Data Permissions)". The system needs to support configuring a position and a user group for a user when creating them and specifying the scope of data permissions.

4.5.2. Technical Requirement Analysis

4.5.2.1. Data permission management requirements based on user BU attributes

- User BU Attribute Binding: The system shall support binding each user to a specific Business Unit (BU).
- Data Access Permission Control: The system shall automatically filter data such as orders, reports, user group verification permissions, and all promotional activities that can set items, sessions, ticket types across BUs, etc., based on the user's BU affiliation and data permission level.

4.5.2.2. Function Permission Management Based on Position

- Business Unit Position Management: The system shall support the creation and management of business units as the basis for data isolation.
- Position Management: As the configuration entity for functional permissions, positions require the system to support their creation, editing, and deletion, and each position can be bound to a set of functional permissions.

4.5.2.3. Data permission management requirements based on user groups

- Creation and Management of User Groups: The system shall support creating, configuring, and managing multiple user groups under the BU.
- Binding of user groups and users: However, only users with the same BU attribute can be bound under the same user group.
- User Group and Data Permission Binding: The system should support binding user groups to specific cross-BU data entities (such as events, sessions, ticket prices, etc.) to achieve unified data permission allocation and cross-BU data permission allocation.
- The system should support configuring the accessible venue scope at the user group level and controlling access to relevant business data (including items, sessions, and ticket types) based on venue promotion activities, ticketing types, payment methods, associated available fare printing methods, SMS templates, associated membership cards, and other permission-related controls.

4.5.2.4. Based on the user's permission management requirements

- User Management: As the ultimate bearer of permissions, users inherit functional permissions through their affiliated positions and can be associated with specific user groups.
- User Data Permission Management: The system should support configuring data permission levels (super, group-level, personal-level) for each user.

4.5.3. Business Scenario

4.5.3.1. Data Permission Management Based on User BU Attributes

4.5.3.1.1. User Process

No.	Component	Description
1	Data Permission Management for BU Dimension	The system shall support data permission management by BU, and all data permission configurations shall be performed within the scope of the corresponding BU.
2	Data Access Scope Control	The system shall control the data access scope based on the permission configuration of the user group under a certain BU to which the user belongs, and unauthorized data shall not be displayed or returned.
3	Users can cross BU to use discounts	The system assigns the scope of use for cross-BU events, sessions, and ticket prices to membership card discounts, coupons, membership levels, and promotional discounts, so that discounts can be used across BUs.

4	Verification Permission Settings	The event authorization verification can be verified by the verification user group across BUs, enabling cross-BU verification.
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Table 4.4.5.1.1: Description of Data Permission Management Functional Button Based on User BU Attribute

4.5.3.1.2. Requirement List

RID	Description	Priority	Related Process
REQ-BU-DATA-001	The system should support associating and binding business data (such as events, sessions, ticket types, etc.) with BU 's to ensure clear data ownership.	High	4.4.5.1.1: #1
REQ-BU-DATA-002	The system shall support access control of data based on the BU to which the user belongs, and users can only access data within the scope of their authorized BU.	High	4.4.5.1.1: #2
REQ-BU-DATA-003	The system supports setting membership card discounts for use across BU events, sessions, and ticket prices.	High	4.4.5.1.1: #3
REQ-BU-DATA-004	The system supports setting coupon discounts to be used across BU events, sessions, and ticket prices.	High	4.4.5.1.1: #3
REQ-BU-DATA-005	The system supports setting member level discounts and granting the use of cross - BU events, sessions, and ticket prices.	High	4.4.5.1.1: #3
REQ-BU-DATA-006	The system supports setting up promotional discounts to be used across BU events, sessions, and ticket prices.	High	4.4.5.1.1: #3
REQ-BU-DATA-007	The event authorization verification can be carried out by verification user groups with cross-BU verification permissions, enabling cross-BU verification.	High	4.4.5.1.1: #4
REQ-BU-DATA-008	Membership card authorization verification can cross BU verification user groups for verification permissions, enabling cross-BU verification.	High	4.4.5.1.1: #4

Table 4.4.5.1.2: Requirements Description of Data Permission Management Based on User BU Attributes

4.5.3.2. Function and Menu Permission Management Based on Position

4.5.3.2.1. User Process

No.	Component	Description
1	BU Management	BU Management Function refers to the system's support for creating and configuring multiple Business Units (BUs), ensuring data isolation and access control.
2	User Group Management	The User Group Management and BU Binding Function refers to the system's support for creating and managing user groups under a BU, thereby controlling user groups' access rights to data.
3	Position Management	The Position Management and Permission Binding function refers to the system's support for creating and managing different positions under the BU and assigning corresponding permissions to each position to ensure that the permission configuration of the position can control the user's functional visibility scope and operation permissions within the system.
4	User Management	The User Management Function refers to the system's support for creating, configuring, and managing user accounts under the BU, and assigning corresponding positions, permissions, and user groups to each user to ensure that the user's operation permissions and data access scope comply with their responsibilities and business requirements.

Table 4.4.5.2.1: Functional Button Description of Position-based Function and Menu Permission Management

4.5.3.2.2. Requirement List.

RID	Description	Priority	Related Process
REQ-ROLE-001	Add/Edit BU.	High	4.4.5.2.1: #1
REQ-ROLE-002	Supports the addition of user groups.	High	4.4.5.2.1: #2

REQ- ROLE- 003	Supports User Group Modification.	High	4.4.5.2.1: #2
REQ- ROLE- 004	Supports user groups to view associated users.	High	4.4.5.2.1: #2
REQ- ROLE- 005	Supports resetting the effective status of user groups.	High	4.4.5.2.1: #2
REQ- ROLE- 006	New Support Positions Added.	High	4.4.5.2.1: #3
REQ- ROLE- 007	Support position duplication.	High	4.4.5.2.1: #3
REQ- ROLE- 008	Supports setting job permissions.	High	4.4.5.2.1: #3
REQ- ROLE- 009	Support staff can view associated users.	High	4.4.5.2.1: #3
REQ- ROLE- 010	Supports resetting the effective status of positions.	High	4.4.5.2.1: #3
REQ- ROLE- 011	Supports adding new system users.	High	4.4.5.2.1: #4
REQ- ROLE- 012	Supports editing system users.	High	4.4.5.2.1: #4
REQ- ROLE- 013	Supports viewing user list.	High	4.4.5.2.1: #4
REQ- ROLE- 014	Supports forced user logout.	High	4.4.5.2.1: #4
REQ- ROLE- 015	Supports resetting user passwords.	High	4.4.5.2.1: #4

REQ-ROLE-016	Supports clearing the number of login limit attempts.	High	4.4.5.2.1: #4
REQ-ROLE-017	Supports resetting user active status.	High	4.4.5.2.1: #4

Table 4.4.5.2.2: Requirements Description Based on Job Function and Menu Permission Management

4.5.3.3. Data permission management requirements based on user groups

4.5.3.3.1. User Process

No.	Component	Description
1	Data Access and Business Scope Control	The system shall support controlling data access scope and business operation scope based on user groups, and different user groups can be configured with different data visibility scopes and operable objects.
2	User Group Associated Venue Data Permissions	The system should support configuring the accessible venue scope at the user group level and controlling the access scope of relevant business data (including events, sessions, and ticket types) based on venue permissions.
3	User group associated with promotional activities	The system shall support user group-based promotional event data permission management, where users can only view and operate promotional event data within their authorized scope.
4	User Group Associated Available Ticket Types	The system shall support configuring available ticketing types based on user groups, which is used to control the range of ticketing types that can be selected when placing an order at the box office, and users can only use the ticketing types within their authorized scope.
5	User Group Associated Available Payment Methods	The system shall support configuring available payment methods based on user groups to control the range of payment methods that users can choose during transactions, and users can only use payment methods within their authorized scope.
6	User Group Associated Available Fare Printing Method	The system shall support configuring available ticket price printing methods based on user groups, which is used to control the drop-down selection range of ticket price printing methods when placing orders at offline box offices. Users can only select printing methods within their authorized scope.

7	User Group Associated with Available SMS Templates	The system shall support configuring available SMS templates based on user groups to control the scope of SMS templates that users can select during business operations, and users can only use SMS templates within their authorized scope.
8	User Group Association Associated Membership Card	The system shall support configuring the permissions to purchase membership cards based on user groups, which are used to control the scope of membership card types that users can purchase. Users can only purchase membership cards within their authorized scope.

Table 4.4.5.3.1: Functional Button Description of Data Permission Management Requirements Based on User Groups

4.5.3.3.2. Requirement List.

RID	Description	Priority	Related Process
REQ-GRP-001	The system shall support configuring the saleable scope based on user groups, which is used to control the items, sessions, and ticket types that can be sold by different user groups or channels.	High	4.4.5.3.1: #1
REQ-GRP-002	The system should support user groups to associate with events, sessions, and ticket price available ticket groups, enabling offline, OTA, and online ticketing permissions for users in specified user groups.	High	4.4.5.3.1: #1
REQ-GRP-003	The system shall ensure that user group permissions take effect within the scope of the authorized BU, and the data access and saleable scope configured for user groups shall not exceed the data boundaries of their authorized BU.	High	4.4.5.3.1: #1
REQ-GRP-004	The system shall support controlling venue data permissions based on user groups, which is used to limit the scope of venues accessible to users and restrict the access scope of relevant event, session, and ticket type data based on venues.	High	4.4.5.3.1: #2
REQ-GRP-005	The system should support controlling promotional event data permissions based on user groups to limit the scope of promotional activities that users can view and operate on.	High	4.4.5.3.1: #3

REQ-GRP-006	The system should support controlling available ticketing types based on user groups, which is used to limit the range of ticketing types that can be selected when placing offline box office orders.	High	4.4.5.3.1: #4
REQ-GRP-007	The system should support controlling available payment methods based on user groups to limit the range of payment methods that users can choose during transactions.	High	4.4.5.3.1: #5
REQ-GRP-008	The system should support controlling the ticket price printing method based on user groups, which is used to limit the drop-down selection range of the ticket price printing method when placing orders at offline box offices.	High	4.4.5.3.1: #6
REQ-GRP-009	The system shall support controlling available SMS templates based on user groups to limit the scope of SMS templates that users can select during business operations.	High	4.4.5.3.1: #7
REQ-GRP-010	The system should support controlling the scope of membership cards based on user groups to limit the types of membership cards that users can use and purchase.	High	4.4.5.3.1: #8

Table 4.4.5.3.2: Requirements Description for Data Permission Management
Requirements Based on User Groups

4.5.3.4. User-based Permission Management

4.5.3.4.1. User Process

No.	Component	Description
1	Primary Administrator User	The system shall support users with all permissions.
2	Users need support for inheriting permissions	The system not only directly assigns permissions to users but also supports integrating the permissions of the user group to which they belong.
3	User Interface OTA Interface Permissions	The system needs to support OTA user interface permission restrictions.
4	User Functional Permissions	The system should support users inheriting the functional permissions configured for a position through the associated position.

Table 4.4.5.4.1: *Functional Button Description of User-Based Permission Management*

4.5.3.4.2. Requirement List

RID	Description	Priority	Related Process
REQ-USER-001	When the user data permission is primary administrator, return all data related to orders and reports belonging to the BU and sold by the BU.	High	4.4.5.4.1: #1
REQ-USER-002	Users should inherit permissions through associated user groups, and the system should automatically assign all permissions under the user group.	High	4.4.5.4.1: #2
REQ-USER-003	The system needs to support the activation of OTA users and the configuration of OTA access interface permissions for these users.	High	4.4.5.4.1: #3
REQ-USER-004	The system should support users inheriting the functional permissions configured for a position through the associated position.	High	4.4.5.4.1: #4

Table 4.4.5.4.2: *Requirements Description Based on User Permission Management*

4.6. Real-Name Authentication

4.6.1. Technical Requirements

This section outlines the technical requirements, business analysis, operating models, user journeys, and business scenarios related to real-name authentication, with a focus on the simplified real-name authentication solution to meet the real-name management requirements under different events, different customer groups, and different compliance scenarios.

4.6.1.1. Manual verification of name

As a ticket purchasing user, I hope that in events requiring real-name management, I only need to fill in partial name information as required, or directly have my name displayed on the ticket, so that staff can manually verify my identity through my ID card or passport at the time of entry, without having to complete a complex online authentication process.

4.6.1.2. Hong Kong iAM Smart Certification

As a local user in Hong Kong, I hope to be able to directly use iAM Smart for real-name authentication in applicable scenarios, so as to quickly complete identity verification through a familiar and official local authentication method.

4.6.1.3. Strong Real-name Authentication for Mainland Chinese ID Cards

As an inland tourist, I hope that in applicable mini-program scenarios, I can use my inland resident ID card to complete strong real-name authentication, so that I can smoothly purchase tickets and enter venues when real-name system is required for inland tours or related events.

4.6.2. Technical Requirement Analysis

This section expands on and analyses the user requirements listed in 4.6.1, explaining the business applicable scenarios, usage principles, and management requirements of various real-name authentication methods.

4.6.2.1. Manual verification of name

This approach is suitable for scenarios where only the establishment of a basic real-name correspondence relationship is required, but the system is not required to complete strong identity verification.

Users can enter partial name information according to business rules, or the system can display the registered name on the ticket surface for on-site staff to verify.

Upon entry, staff will manually verify whether the ticket registration information matches the name on the user's ID card or passport presented.

This approach focuses on lowering the threshold for users' online operations while retaining basic real-name management capabilities.

4.6.2.2. Hong Kong iAM Smart Certification

This method is suitable for scenarios where the proportion of local Hong Kong users is relatively high and where there is a desire to adopt local official identity verification capabilities.

The system shall support integration with iAM Smart to obtain authentication results or necessary identity information.

This approach can reduce users' repeated input of information and enhance the credibility and completion efficiency of authentication.

The specific scope of activation shall be configured by the event according to business needs.

4.6.2.3. Strong Real-name Authentication for Mainland Chinese ID Cards

This approach is applicable to scenarios related to mainland tourists, especially in mainland tour events or other events with explicit requirements for strong real-name authentication.

The system shall support enabling a strong real-name authentication process based on the Mainland Resident Identity Card on the Mini Program platform.

This method is only enabled when business needs require it and is not used as the default authentication method for all events.

Once enabled, event-level configuration should be used to control whether users are required to complete this authentication before purchasing tickets or entering the venue.

4.6.3. Operating Model

The real-name authentication capability is driven by event configuration. Different events can choose to enable one or more applicable methods, such as manual name verification, Hong Kong iAM Smart authentication, or strong real-name authentication of Mainland Chinese identity cards, based on ticketing policies, venue requirements, and target customer groups.

For scenarios where manual verification of names is used, users fill in their name information during the ticket purchase or registration process, and the system writes the relevant real-name information into the order and ticket information for on-site staff to conduct manual comparison upon entry.

For scenarios using Hong Kong iAM Smart, after the user initiates authentication and completes authorization on the front end, the system records the authentication result and uses it for subsequent ticket purchase eligibility judgment, real-name ticket display, or entry verification reference.

For scenarios that adopt strong real-name authentication using Mainland Chinese ID cards, users are required to complete real-name authentication with a Mainland Chinese ID card as required within the mini-program. Only after the authentication is approved can the system allow them to continue to complete the relevant ticketing or usage process.

At the operational level, it is necessary to support the definition of real-name strategies by event and ensure that the front-end display, order data, ticket data, and on-site verification criteria remain consistent.

4.6.4. Customer Journey

This section describes User Stories based on roles, reflecting the usage requirements of different users under different real-name authentication methods.

No.	Category	Role	User Story
1	Manually verify the name	Ticketing User	As a ticketing user, I hope to only need to fill in partial name information or have my name displayed on the ticket, so that I can complete the ticketing process more quickly and cooperate with the staff for manual verification upon entry.
2	Manually verify the name	On-site Staff	As a field staff member, I hope to be able to see the registered name in the ticket or order information so that I can manually compare it with the user's ID card or passport presented.

3	iAM Smart Certification	Hong Kong local users	As a local Hong Kong user, I hope to directly use iAM Smart to complete real-name authentication, so as to reduce the need for repeatedly filling in information and improve the success rate of authentication.
4	Strong real-name authentication for Mainland Chinese ID cards	Mainland tourists	As an inland tourist, I hope to use my inland ID card to complete real-name authentication in events that require strong real-name verification, so as to smoothly complete ticket purchase and subsequent entry.
5	Event Operators	WestK Operations Staff	As a operator, I hope to be able to configure the real-name authentication method according to the event, so as to flexibly enable the appropriate real-name rules based on event requirements.
6	Ticketing Management Staff	WestK Ticketing Staff	As a ticketing administrator, I hope that relevant real-name information or authentication results can be reflected in order and ticket data to support customer service handling, on-site verification, and subsequent traceability.

Table 4.6.4: User Story Description

4.6.5. Business Scenarios

4.6.5.1. Manual verification of name

This scenario applies to events that adopt basic real-name registration and on-site manual comparison.

RID	Description	Priority	Related Process
BR-RNA-001	The system supports users to fill in partial name information in specified scenarios for real-name registration.	High	Manual verification of name
BR-RNA-002	The system supports displaying the registered name on tickets, order details, or relevant ticketing vouchers for on-site manual verification.	High	Manual verification of name
BR-RNA-003	The system supports on-site staff to manually compare the identity documents or passports presented by users with the registered names.	High	Manual verification of name

BR-RNA-004	The system supports event-level configuration to determine whether to enable the manual verification of names.	High	Manual verification of name
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Table 4.6.5.1: Requirements List of Manual verification of name

4.6.5.2. Hong Kong iAM Smart Certification

This scenario applies to events where Hong Kong local users can complete real-name authentication through Smart Easy.

RID	Description	Priority	Related Process
BR-RNA-005	The system supports integration with Hong Kong iAM Smart for user identity authentication.	High	iAM Smart Certification
BR-RNA-006	The system supports guiding users to initiate the iAM Smart authentication process in applicable events.	High	iAM Smart Certification
BR-RNA-007	The system supports recording the user's authentication results and using these results in subsequent business processes.	High	iAM Smart Certification
BR-RNA-008	The system supports event-level configuration to enable or disable the iAM Smart authentication method.	High	iAM Smart Certification

Table 4.6.5.2: Requirements List of Hong Kong iAM Smart Certification

4.6.5.3. Strong Real-name Authentication for Mainland Chinese ID Cards

This scenario applies to scenarios for mainland tourists using mini-programs, as well as events that require strong real-name authentication, such as mainland tour events.

RID	Description	Priority	Related Process
BR-RNA-009	The system supports enabling the strong real-name authentication process for Mainland Chinese residents' ID cards on the mini-program platform.	High	Strong real-name authentication for Mainland Chinese ID cards
BR-RNA-010	The system supports requiring users to complete strong real-name authentication with Mainland Chinese ID cards in the specified event before they can continue purchasing tickets.	High	Strong real-name authentication for Mainland

			Chinese ID cards
BR-RNA-011	The system supports enabling strong real-name authentication for Mainland Chinese identity cards only when business requires it and does not make it a default requirement for all events.	High	Strong real-name authentication for Mainland Chinese ID cards
BR-RNA-012	The system supports event-level configuration to determine whether to enable the strong real-name authentication method for Mainland Chinese identity cards.	High	Strong real-name authentication for Mainland Chinese ID cards

Table 4.6.5.3: Requirements List

4.7. Technical Requirements

4.7.1. General

This section defines the technical requirements, covering the technical architecture, infrastructure, and technology standards required to implement the system, ensuring that the system maintains high reliability, show, and security, etc.

4.7.1.1. Technical Requirements List

RID	Name	Priority	Category	Description
TR-PER-001	High Concurrency Handling	High	Show	The system must support high-concurrency scenarios during peak ticket sales.
TR-SEC-001	Data Encryption	High	Security	Sensitive data must be protected through encryption mechanisms.
TR-SEC-002	Data Privacy Compliance	High	Security	The system must comply with applicable data protection regulations including: • GDPR • PDPO • PIPL
TR-SEC-003	Secure API Signature Mechanism	High	Security	A secure API signature mechanism must be provided to ensure the security of API access.

TR-DRR-001	Backend System Disaster Recovery	High	Disaster Recovery	Disaster recovery requirements including Recovery Time Objective (RTO)/ Recovery Point Objective (RPO) targets and service coverage.
TR-SBR-001	Recovery Procedures	High	System Backup and Recovery	The system should design a recovery procedure under various system failures.
TR-SBR-002	Backup Procedures	High	System Backup and Recovery	The system shall have backup arrangements for systems involved.
TR-TSR-001	Software Support Requirement	High	Technical Support Requirement	The support levels and test/diagnostics should be provided for the system.
TR-TST-001	Full Lifecycle Testing Process	High	Testing	Testing procedures designed for each stage during the whole software development lifecycle.

Table 4.7.1.1: Technical Requirement List Table**4.7.1.2. TR-PER-001 High Concurrency Handling****Technical requirements description:**

Item	Description
Requirement ID	TR-PER-001
Requirement Name	High Concurrency Handling
Priority	High
Category	Show
Technical requirement description	The system shall support high-concurrency traffic during peak ticket-sale periods. The system shall handle at least 20,000 concurrent API requests per second with a maximum p95 response time of 500 ms and an error rate below 1%. This show target shall be validated through load and stress testing prior to production deployment.

Table 4.7.1.2: Technical Requirement Description of High Concurrency Handling**4.7.1.3. TR-SEC-001 Data Encryption****Technical requirements description:**

Item	Description
Requirement ID	TR-SEC-001
Requirement Name	Data Encryption
Priority	Medium
Category	Security
Technical requirement description	Sensitive data must be protected through encryption mechanisms. All data in transit and data at rest should be encrypted.

Table 4.7.1.3: Technical Requirement Description of Data Encryption

4.7.1.4. TR-SEC-002 Data Privacy Compliance

Technical requirements description:

Item	Description
Requirement ID	TR-SEC-002
Requirement Name	Data Privacy Compliance
Priority	High
Category	Security
Technical requirement description	The system must comply with applicable data protection regulations including: GDPR, PDPO, and PIPL.

Table 4.7.1.4: Technical Requirement Description of Data Privacy Compliance

4.7.1.5. TR-SEC-003 Secure API Signature Mechanism

Technical requirements description:

Item	Description
Requirement ID	TR-SEC-003
Requirement Name	Secure API Signature Mechanism
Priority	High
Category	Security
Technical requirement description	AnTank TAS system integrates with OTA via a standard REST Application Programming Interface (API) over HTTPS, adopting a dual-mechanism authentication solution: RSA for request signature (ensuring data integrity and anti-

	tampering) and the appkey/partnerKey as the unique merchant identifier (for identity verification).
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Table 4.7.1.5: Technical Requirement Description of Data Privacy Compliance

4.7.1.6. TR-DRR-001 Backend System Disaster Recovery

Technical requirements description:

Item	Description
Requirement ID	TR-DRR-001
Requirement Name	Backend System Disaster Recovery
Priority	High
Category	Disaster Recovery Requirements
Technical requirement description	1. Disaster recovery scope: Covers all core backend components including AnTank backend services. 2. Disaster recovery tiers and objectives: - Level 1 (Intra-region failure): Multi-AZ failure within the same Alibaba Cloud region, with Recovery Time Objective (RTO) ≤ 30 minutes and RPO ≤ 5 minutes. - Level 2 (Cross-region failure): Overall failure of the Alibaba Cloud Hong Kong region, switching to the Singapore standby region, with RTO ≤ 2 hours and RPO ≤ 30 minutes. 3. Service coverage: During disaster recovery, core services such as ticket purchase, order inquiry, payment processing, and ticket verification must remain available, while non-core services (e.g., marketing push, historical data statistics) can be degraded. 4. Disaster recovery drills: Conduct cross-region disaster recovery drills semi-annually and intra-region failure recovery drills quarterly, record drill results, and optimize procedures.

Table 4.7.1.6: Technical Requirement Description of Backend System Disaster Recovery

4.7.1.7. TR-SBR-001 Recovery Procedures

Technical requirements description:

Item	Description
Requirement ID	TR-SBR-001
Requirement Name	Recovery Procedures

Priority	High
Category	System Backup and Recovery
Technical requirement description	The system should design a recovery procedure under various system failures. This includes: 1. Database failure recovery: automatic failover to standby instances with RTO ≤ 5 minutes. 2. Application service recovery: auto-scaling to replace failed nodes within 10 minutes. 3. A documented recovery playbook with clear steps, roles, and escalation paths.

Table 4.7.1.7: *Technical Requirement Description of Recovery Procedures*

4.7.1.8. TR-SBR-002 Backup Procedures

Technical requirements description:

Item	Description
Requirement ID	TR-SBR-002
Requirement Name	Backup Procedures
Priority	High
Category	System Backup and Recovery
Technical requirement description	The system shall have backup arrangements for systems involved. This includes: 1. Database backups: daily full backups and 6-hour incremental backups, retained for 7 days. 2. Configuration backups: real-time backups on change, with 30-day version history.

Table 4.7.1.8: *Technical Requirement Description of Backup Procedures*

4.7.1.9. TR-TSR-001 Software Support Requirement

Technical requirements description:

Item	Description
Requirement ID	TR-TSR-001
Requirement Name	Software Support Requirement
Priority	High
Category	Technical Support Requirement

Technical requirement description	The system shall define support severity levels (e.g. Severity 1–4) with clearly documented impact descriptions, and each severity level shall have a corresponding SLA specifying response time, workaround time, and resolution time.
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Table 4.7.1.9: *Technical Requirement Description of Software Support Requirement***4.7.1.10. TR-TST-001 Full Lifecycle Testing Process****Technical requirements description:**

Item	Description
Requirement ID	TR-TST-001
Requirement Name	Full Lifecycle Testing Process
Priority	High
Category	Testing
Technical requirement description	1. Requirements Analysis Phase The system shall have a Requirement Validation Test Plan that includes test cases covering all functional and non-functional requirements 2. Design Phase The technical design shall undergo formal Architecture Review Testing and Interface Compatibility Testing. Review results, gaps, and approval records shall be documented in the Design Review Report. 3. Development Phase The development team shall perform Unit Testing (≥80% code coverage unless justified otherwise) and Integration Testing. Test results shall be documented and stored in the event repository. 4. Testing Phase The testing team shall conduct System Testing (functional, show, security, compatibility) followed by User Acceptance Testing (UAT). All defects shall be logged, triaged, and resolved according to the defect severity Service Level Agreement (SLA). 5. Operation Phase The system shall undergo regular Regression Testing and Disaster Recovery Testing. A full regression test shall be executed at least once per quarter, with documented test reports and remediation actions.

Table 4.7.1.10: *Technical Requirement Description of Full Lifecycle Testing Process*

4.7.2. Log System

This chapter defines the core content required for the implementation of WestK Log System, including technical architecture, framework selection, format specifications, show indicators, security requirements, etc., ensuring that the log system has **standardization, integrity, readability, high show, and compliance**, and can be seamlessly integrated into Douyin Mini Programs and core business systems, providing reliable log support for system operation and maintenance and business operations.

4.7.2.1. Technical Requirements List

RID	Name	Priority	Classification	Description
TR-LOG-001	Standardization of Logging Framework	High	Framework Selection	Unify the technical stack logging framework, adopt the facade pattern for decoupling, and clarify dependency management and version specifications.
TR-LOG-002	Log Collection Covers all Dimensions	High	Collection Management	Covers all types of logs including system, business, interface, exception, audit, etc., achieving full-scenario and comprehensive collection.
TR-LOG-003	Structured Specification for Log Format	High	Format Definition	Develop a unified structured log format, clarify mandatory fields and exclusive fields, and standardize output rules.
TR-LOG-004	Standardized Use of Log Levels	High	Level Management	Define the usage scenarios and specifications for log levels, prohibit the abuse of levels across scenarios, and implement environmentalized level control.
TR-LOG-005	Refined recording of exception logs	High	Exception handling	Standardize the elements of exception log recording, requiring it to include on-site information and stack information, and prohibiting swallowing exceptions or missing records.
TR-LOG-006	Log Storage and Retention Compliance	High	Storage Management	Meets regulatory requirements such as GDPR, PDPO, and PIPL,

				sets retention periods in a graded manner, and achieves secure storage and backup.
TR-LOG-007	Log Show and Efficiency Optimization	High	Show optimization	Prohibit string concatenation, implement asynchronous output, control the frequency of log output, and avoid affecting the show of the business system.
TR-LOG-008	Log Security and sensitive information management and control	High	Security Management	Prohibit logging of sensitive information, mask necessary user information, and implement control over log access permissions.
TR-LOG-009	Log Retrieval and Link Tracing	High	Retrieval and Tracing	Supports efficient multi-condition retrieval, realizes whole-link log association of globally unique identifiers, and meets the requirement for rapid problem location.
TR-LOG-010	Log File Management Specification	Medium	File Management	Standardize the naming, classification, and rolling strategies of log files to avoid duplicate printing and disk overflow.
TR-LOG-011	Full Lifecycle Testing of Logging System	High	Test Verification	Design a full lifecycle testing process for the logging system to ensure that functions such as log output, collection, storage, and retrieval meet the standards.

Table 4.7.2.1: Technical Requirement List Table

4.7.2.2. TR-LOG-001 Log Framework Standardisation

Item	Description
Requirement ID	TR-LOG-001
Requirement Name	Log Framework Standardisation

Priority	High
Category	Framework Selection
Technical Requirement Description	1. Technology Stack Framework Selection: The Java technology stack uniformly adopts the Simple Logging Facade for Java (SLF4J) + Logback combination as the core logging framework, uses the facade pattern to decouple the logging interface from its implementation, and prohibits direct use of low-level logging framework APIs such as Log4j and JUL; 2. Simplify development specifications: It is recommended to use Lombok's @Slf4j annotation to simplify the definition of Logger objects. In non-Lombok scenarios, Logger should be defined as a private static final constant to avoid repeated creation. 3. Dependency Management Specification: Logging framework dependencies must use a unified version, be globally managed via Maven/Gradle, set implementation layer dependencies to the runtime scope to avoid compile-time dependency coupling, and set optional=true to prevent dependency transitivity. 4. Bridging Compatibility Specification: If there is a conflict in the logging framework when introducing third-party components, compatibility must be achieved through SLF4J bridging packages (such as log4j-over-slf4j, jul-to-slf4j) to ensure that all logs are uniformly output to the SLF4J+Logback framework. 5. Multilingual Adaptation: Non-Java technology stack components need to follow the design principles of the facade pattern, select a standardized logging framework corresponding to the technology stack, and ensure that the logging output rules are consistent with the overall system.

Table 4.7.2.2: Technical Requirement Description of Log Framework Standardisation

4.7.2.3. TR-LOG-002 Log Collection with Full-Dimensional Coverage

Item	Description
Requirement ID	TR-LOG-002
Requirement Name	Log Collection with Full-Dimensional Coverage
Priority	High
Category	Collection Management

Technical Requirement Description	1. Log type coverage: It is necessary to collect six major categories of logs, including system logs, business logs, interface logs, exception logs, audit logs, and monitoring logs. Among them, interface logs need to cover all external interfaces such as TAS/B2B/Douyin callbacks, and audit logs need to cover all sensitive operations such as merchant operations, order changes, and configuration modifications. 2. Collection scenario coverage: Covers all scenarios including normal business processes, high-concurrency peaks, system failures, interface call failures, permission changes, data anomalies, etc., with no blind spots in collection. 3. Core Node Collection: All core methods (entry/exit), critical branch logic, third-party service calls, database operations, cache operations, and other nodes must add log records to ensure the traceability of business processes. 4. Requirements for collected content: Interface logs must record request/response parameters, caller information, elapsed time, and status codes; business logs must record the operator, operation object, operation content, and operation result; system logs must record the running status and key events of servers/containers/databases. 5. Collection Real-time Show: Core business and interface logs achieve second-level real-time collection, non-core logs achieve minute-level collection, and the collection process has a CPU utilization $\leq 5\%$ and memory utilization $\leq 10\%$ of the business system, with no show impact.
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Table 4.7.2.3: Technical Requirement Description of Log Collection with Full-Dimensional Coverage

4.7.2.4. TR-LOG-003 Log Format Structuring Specification

Item	Description
Requirement ID	TR-LOG-003
Requirement Name	Log Format Structuring Specification
Priority	High
Category	Format Definition
Technical Requirement Description	1. Overall Format: All logs shall be uniformly output in JSON structured format, and unstructured plain text logs are prohibited to ensure that logs are parsable, analysable, and aggregable. 2. Required common fields: All logs must include log_id (unique identifier), timestamp (timestamp), service_name (service name), module_name (module name), log_level (log

	level), thread_name (thread name), logger_name (logger name), with the timestamp format being UTC+8 dd-MM-yyyy HH:mm:ss.SSS, accurate to milliseconds; 3. Exclusive Extension Fields: Different types of logs supplement exclusive fields. Interface logs must include request_id (global link ID), partnerKey, requestUrl, status_code, and cost_time; audit logs must include merchant_id, operator, operate_type, old_status, and new_status; exception logs must include error_code and error_msg. 4. Log Message Specification: The log message body should be concise, clear, and semantically unambiguous, using parameterized placeholders {} to concatenate variables, and string concatenation using + is prohibited. The message body must include key business identifiers (such as order ID, user ID, merchant ID). 5. Multithreaded output specification: A single business log must be output on the same line to avoid log serialization and confusion in a multithreaded environment, ensuring the integrity of the log.
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Table 4.7.2.4: Technical Requirement Description of Log Format Structuring Specification

4.7.2.5. TR-LOG-004 Standardised Use of Log Levels

Item	Description
Requirement ID	TR-LOG-004
Requirement Name	Standardised Use of Log Levels
Priority	High
Category	Level Management
Technical Requirement Description	1. Level Definitions and Scenarios: DEBUG, INFO, WARN, and ERROR levels of logs are uniformly defined, with the usage scenarios of each level strictly distinguished, and cross-scenario abuse is prohibited: ① DEBUG: Used during the development/testing phase, records detailed debugging information, variable values, and execution processes, and must be turned off in the production environment. ② INFO: Records key information about the normal operation of the system, including service startup/stop, core business operation results, successful interface call status, completion of configuration loading, etc., and only retains valuable key information.

	③ WARN: Records non-severe exceptions within expectations, including invalid parameters, empty data, minor exceptions in third-party services, business process branches deviating from expectations but not affecting execution, etc., which do not require urgent handling but need attention. ④ ERROR: Records serious and unrecoverable exceptions, including system failures, interface call failures, database operation exceptions, business process interruptions, etc., which require triggering an alarm and prompt handling. 2. Environmental Level Control: The Development Environment enables DEBUG level, the Test Environment enables INFO level and disables DEBUG, the Production Environment only enables INFO/WARN/ERROR levels, and DEBUG log output is prohibited in the Production Environment. 3. Level judgment specification: For DEBUG/INFO level logs, if the log content contains complex method calls or object conversions, level judgment should first be performed via <code>log.isDebugEnabled()/log.isInfoEnabled()</code> to avoid unnecessary calculations. 4. Usage taboos for levels: It is prohibited to use the WARN/ERROR levels for normal business process logging, and it is prohibited to use the INFO level to log exception information, so as to avoid false triggering of alarms and difficulties in problem location.
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Table 4.7.2.5: Technical Requirement Description of Standardized Use of Log Levels

4.7.2.6. TR-LOG-005 Refined Recording of Exception Logs

Item	Description
Requirement ID	TR-LOG-005
Requirement Name	Refined Recording of Exception Logs
Priority	High
Category	Exception handling
Technical Requirement Description	1. Elements of exception records: Exception logs must simultaneously include information about the scene of the incident and exception stack information. The scene information should include the key parameters that triggered the exception, business identifiers, and execution context. The stack information must fully print the exception chain, and it is prohibited to print only the exception message. 2. Exception handling specification: Exceptions must be caught and subdivided by specific types. Directly catching

	Exception/Throwable for global handling is prohibited. RuntimeExceptions that can be avoided through pre-checks (such as NullPointerException) are prohibited from being handled via catch. 3. Log output specification: Exception logs should be uniformly output at the ERROR level, with the output format being <code>log.error("Exception description: {}, Business ID: {}", description information, business identifier, exception object)</code>. The exception object must be passed as the last parameter to ensure the integrity of the stack information. 4. Prohibit abnormal behaviour: Prohibit swallowing exceptions (capturing without logging or rethrowing), prohibit only printing <code>e.getMessage()</code>, prohibit nested exception capturing, and prohibit rethrowing after logging an exception (to avoid secondary stack printing). 5. Custom Exception Specification: Custom exceptions shall include an error code, error message, and user prompt message. The error code shall follow the 5-digit string specification (A/B/C + four digits). Exception logs shall record the complete information of custom exceptions, including the error code and business context. 6. Third-party exception handling: For exceptions that occur when calling third-party services/interfaces, the third-party error code, original exception information must be recorded in the log, and error code escaping must be performed (e.g., converting third-party Class C errors to Class B errors in this system).
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Table 4.7.2.6: Technical Requirement Description of Refined Recording of Exception Logs

4.7.2.7. TR-LOG-006 Log Storage and Retention Compliance

Item	Description
Requirement ID	TR-LOG-006
Requirement Name	Log Storage and Retention Compliance
Priority	High
Category	Storage Management
Technical Requirement Description	1. Storage Architecture: Adopt a distributed log storage system (Alibaba Cloud SLS) as the core storage, which supports efficient storage, expansion, and retrieval of massive logs. The storage system must implement multi-Available Zone deployment to avoid single point of failure. 2. Data Encryption: In compliance with the requirements of TR-SEC-001, log data during transmission (HTTPS/TLS) and at rest are both encrypted, with encryption keys managed

	independently, and sensitive fields requiring separate encryption. 3. Retention Period: Set retention periods in a graded manner according to regulations and business requirements to meet data protection regulations such as GDPR, PDPO, and PIPL: ① Audit logs, exception logs, core business/interface logs: Retain for ≥ 1 year. ② System operation logs and INFO-level ordinary logs: retained for ≥ 15 days. ③ Network operation status, security-related logs, and user sensitive operation logs: Retain for ≥ 6 months and perform off-site multi-machine backup. 4. Data Backup: In accordance with the requirements of TR-SBR-002, log data is fully backed up daily and incrementally backed up every 6 hours, with backup data retained for ≥ 3 months. Backup files are stored in off-site nodes to ensure that log data is not lost. 5. Compliance Requirements: Support compliant export and deletion of log data, meet the query and access requests of data subjects, prohibit tampering and forgery during log storage, and ensure that all log operations are traceable.
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Table 4.7.2.7: Technical Requirement Description of Log Storage and Retention Compliance

4.7.2.8. TR-LOG-007 Log Show and Efficiency Optimization

Item	Description
Requirement ID	TR-LOG-007
Requirement Name	Log Show and Efficiency Optimization
Priority	High
Category	Show optimization
Technical Requirement Description	1. Output method optimization: The core service log implements asynchronous output, configures an asynchronous queue through Logback's AsyncAppender, and sets the queue size reasonably (recommended ≥ 512) to avoid synchronous output blocking business threads. 2. Optimization of concatenation method: Strictly prohibit using + for log string concatenation, uniformly use parameterized placeholders {}, reduce StringBuilder creation and memory usage, and improve log output efficiency. 3. Output frequency control: Prohibit outputting a large number of logs in loops or high-frequency calling methods.

	The log output frequency in core business loops must be ≤ 1 time per 100 loops to avoid log flushing to disk occupying a large amount of IO resources. 4. Object output optimization: When outputting an object in the log, ensure that the object overrides the toString() method, prohibit directly using JSON tools to convert the object to a string (to avoid exceptions thrown by get methods affecting business operations), and only print business-related attributes. 5. Resource Usage Control: The operating resources (CPU / memory / disk IO) of the logging system need to be rate limited. Log flushing to disk adopts a batch write strategy to avoid frequent IO operations caused by single-record writes, ensuring that the resource usage of the logging system for the business system accounts for $\leq 10\%$. 6. Avoid invalid logs: Prohibit the output of meaningless and valueless logs. Log content must meet the requirements of "being traceable, analysable, and auditable" to avoid wasting disk space and reducing log retrieval efficiency.
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Table 4.7.2.8: Technical Requirement Description of Log Show and Efficiency Optimization

4.7.2.9. TR-LOG-008 Log Security and Sensitive Information Management and Control

Item	Description
Requirement ID	TR-LOG-008
Requirement Name	Log Security and Sensitive Information Management and Control
Priority	High
Category	Security Management
Technical Requirement Description	1. Prohibition on Recording Sensitive Information: Any sensitive information, including user passwords, Tokens, bank card numbers, mobile phone numbers, ID card numbers, merchant keys, etc., is prohibited from being recorded in logs. Directly outputting sensitive fields in request headers/request bodies is also prohibited. 2. Sensitive Information Masking: If the business needs to record user/merchant identification information (such as mobile phone numbers, email addresses), masking and desensitization must be performed. The rules are as follows: for mobile phone numbers, keep the first 3 and last 4 digits (1381234); for email addresses, keep the first 2 characters of

	the prefix and the domain name (ab@xxx.com); for ID numbers, keep the first 6 and last 4 digits; 3. Log access permission management and control: The log storage system implements fine-grained access permission control, assigns log query, export, and view permissions by role, only authorized operation and maintenance/development personnel can access logs, and all log access operations are recorded in audit logs. 4. Audit Log Security: Audit logs are physically stored separately from ordinary logs, ensuring that once written, they cannot be tampered with or deleted. Timestamp/blockchain technology is used for anti-tampering verification, and all modification/deletion operations on audit logs are recorded in the super log. 5. Log Transmission Security: The transmission of logs between services, between the collection end and the storage end, must use encrypted transmission protocols (HTTPS/TLS), and plaintext transmission is prohibited to prevent log data from being stolen or tampered with during transmission. 6. Log export security: Log export requires permission verification and operation records, exported files need to be encrypted, exported log data needs to be desensitized, and exporting un-desensitized raw logs is prohibited.
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Table 4.7.2.9: *Technical Requirement Description of Log Security and Sensitive Information Management and Control*

4.7.2.10. TR-LOG-009 Log Retrieval and Link Tracing

Item	Description
Requirement ID	TR-LOG-009
Requirement Name	Log Retrieval and Link Tracing
Priority	High
Category	Retrieval and Tracing
Technical Requirement Description	1. Global Link Identifier: All requests across services and modules generate a globally unique request_id, which is passed into all downstream calls, and all associated logs must carry this ID to achieve whole-link log association for a single request. 2. Search Capability: Supports efficient multi-condition combined search, allowing searches by fields such as log_id, request_id, service_name, module_name, log_level, timestamp, business_id (order/user/merchant ID),

	error_code, etc., and supports fuzzy search, range search, and precise search. 3. Retrieval Show: Achieves second-level response (≤ 3 seconds) for single-condition retrieval, response time ≤ 10 seconds for multi-condition complex retrieval, supports efficient retrieval of billions of logs, and the retrieval process does not affect log storage and collection. 4. Link Tracing Capability: Through the request_id, whole-link log tracing can be achieved, allowing one-click association of all upstream and downstream logs including request entry, interface call, business processing, database operation, cache operation, exception feedback, etc. Logs are sorted and displayed by timestamp, clearly presenting the request execution process. 5. Exception Location Capability: Exception logs need to be correlated with relevant business logs and interface logs. Clicking on an exception log allows direct viewing of upstream and downstream correlated logs, enabling rapid location of the root cause of the exception. 6. Search Result Processing: Search results support multi-dimensional filtering, sorting, and pagination, support one-click viewing of log details including complete fields, exception stack, request/response parameters, and support encrypted export of search results.
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Table 4.7.2.10: Technical Requirement Description of Log Retrieval and Link Tracing

4.7.2.11. TR-LOG-010 Log File Management Specification

Item	Description
Requirement ID	TR-LOG-010
Requirement Name	Log File Management Specification
Priority	Medium
Category	File Management
Technical Requirement Description	1. File Naming Convention: Log files shall uniformly adopt the naming rule of appName_logType_logName.log, where appName is the service name, logType is the log type (stats/monitor/access/error/audit), and logName is the log description. Example: wkcd_miniprogram_access_api.log. 2. Log classification and storage: Physically classify and store logs by log type, separately storing error logs, audit logs, interface logs, and ordinary business logs. Logs of different services are stored in independent directories

	named after the service names, facilitating log search and management. 3. Rolling Strategy: Adopt a hybrid rolling strategy based on time + size, generating log files daily (with the file suffix yyyy-MM-dd), and automatically splitting the log file when its size exceeds 1GB to prevent the retrieval efficiency from decreasing due to an excessively large single log file. 4. Avoid duplicate printing: Set additivity=false in the logging framework configuration to prevent child loggers from inheriting the output sources of parent loggers, avoid duplicate printing of the same log in multiple files, and save disk space. 5. Disk Capacity Monitoring: The log storage directory must implement disk capacity monitoring. When the disk usage reaches or exceeds 80%, an alarm will be triggered, and expired log files will be automatically cleaned up to prevent business system failures caused by disk overflow. 6. Log file encoding: All log files uniformly use UTF-8 encoding to ensure that log content is free of garbled characters and support the normal display of multilingual log content.
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Table 4.7.2.11: *Technical Requirement Description of Log File Management Specification*

4.7.2.12. TR-LOG-011 Lifecycle Testing of Logging System

Item	Description
Requirement ID	TR-LOG-011
Requirement Name	Full Lifecycle Testing of Logging System
Priority	High
Category	Test Verification
Technical Requirement Description	1. Requirements Analysis Phase: Develop a log requirements verification test plan, sort out the functional and non-functional requirements of all log scenarios, and ensure that all log requirements are testable and traceable. 2. Design Phase: Conduct formal architecture review tests on the design of the logging framework, format specifications, and storage architecture to verify the rationality of framework selection, the feasibility of format specifications, and the reliability of the storage architecture. The review results are recorded in the "Design Review Report". 3. Development Phase: The development team conducts unit tests on the log output code, with code coverage $\geq 80\%$, verifies whether the log output format, level, and content

	comply with the specifications, and simultaneously conducts integration tests to verify the link identification and correlation capabilities of cross-service logs. 4. Testing Phase: The testing team conducts system testing + user acceptance testing (UAT). System testing includes dimensions such as log collection completeness, format correctness, exception record accuracy, retrieval show, and storage compliance. UAT verifies the traceability and auditability of logs in actual business scenarios. All defects are tracked and fixed according to the severity level SLA. 5. Pre-launch Phase: Conduct log show testing in the stress testing environment, verify the resource utilization, output efficiency, and storage stability of the logging system under high-concurrency scenarios, and ensure that the logging system does not affect core business during peak business periods. 6. Operation Phase: Conduct regular regression tests and compliance tests on the log system, perform a full regression test once a quarter to verify the implementation of log specifications, conduct a compliance test once every six months to verify whether log storage and retention meet regulatory requirements, record the test results, and output corrective measures.
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Table 4.7.2.12: *Technical Requirement Description of Lifecycle Testing of Logging System*

5. Organisation Chart

To ensure the successful delivery of the Douyin Mini Program, the structure of the WKCD Project team and AnTank team is as follows:

Project Steering Committee (PjSC):

WKCD		Finsoft Antank	
Chief Financial Officer (CFO)	Carmen LEE (Chairperson)	Finsoft:	
Technology Partnership (TP)	Rebecca YANG	Wun Yan Kelly TIN	
M+ Museum (M+)	David TSUI	Kwan Nam Gabriel YU	
Hong Kong Palace Museum (HKPM)	Brian YUEN	Antank:	
Performing Arts (PA)	Joycelyn CHOI	Yong LIU, Sam	
Development Department (DD)	Diana TSUI, Anca COMAN-BAL	Weiji CHEN, Ricky	
External Affairs (EA)	Kenneth NG	Songwei GUO	
Finance Department	Elaine TY Chung		
Innovation & Technology Department (ITD)	Margod YAU		

Project Management Team (PjMT):

WKCD		Finsoft Antank	
TP	Howard CHENG (Co-Chairperson)	Finsoft:	
IT	Wilson Yeung (Co-Chairperson)	Thomas LI	
M+	Sewon BARERA, Jackie Chan	Antank:	
HKPM	Amelie CHENG	Troy SHU (Project Director),	
PA	Jonas CHAU	Jason WU (Project Manager)	
DD	Carmen Man (M+), Wing Yun CHAN (HKPM), Anna CHUENG (PA), Michael NG (District)		
EA	Allan YU, Diamond CHENG (HKPM)		
Digital	Keith CO LAM		
Ticketing	Kamen LAM		
Finance	Samanda CHEUNG		
ITD	Benny LAI, Alan LAM, Vivienne LAM (Project Manager)		

Project Groups:

WKCD Business User Group		WKCD Support Team		Implementation Team	
TP	Rex WONG	Data & Analytics	Franco HO	Alibaba Group	
M+	Sewon BARERA, Sandy Sze, Jackie Chan	ITD (CRM)	Chris KWAN, Shirley LEE	Alicloud Consulting	Jonathan PENG
HKPM	Amelie CHENG	ITD (Ticketing)	Fiona CHAU	Finsoft Antank	
PA	Jonas CHAU	ITD (ERP)	Stephen WONG	Finsoft:	
Central Digital	Keith CO LAM	Enterprise Architecture	Stanley LAW, Hydro CHOY, Angelo MAO	Technical Manager	Percy CHIU
EA	Mandy CHEUNG, Kiko YAU (HKPM), Ivan POON (PA), Shirley PU (District)	Cloud Infrastructure and Network	Radek CHAN, Dicky WAN	Antank:	
DD	Carmen MAN (M+), Kiko YAU (HKPM), Ivan POON (PA), Shirley PU (District)	Cybersecurity	Jeffrey CHEUNG	Back-End Solution Architect	Feng WANG
Ticketing	Kamen LAM			Front-End Solution Architect	Erpeng ZHAO
Central CRM	Kenni WONG, Kay NG			Hardware Manager	David FU
Finance	Sevil YIP			Quality Assurance Manager	James XUE

6. RACI Chart and Roles & Responsibilities

6.1. Roles & Responsibilities and Detail Job Descriptions

Name	Role	Responsibility
Wun Yan Kelly Tin	PMO/Business Director (Finsoft)	Oversee project progress from Finsoft side; secure resources and support commercial and settlement matters.
Kwan Nam Gabriel Yu	PMO/Risk Control (Finsoft)	Focuses on project risk control and compliance; monitors major project risks and supports mitigation and governance activities.
Ricky Chen	PMO/Business Director	Overall business owner: co-leads PMO and aligns delivery with WKCD business objectives.
Songwei Guo	PMO/Project Director	Co-leads PMO oversees overall planning, governance and cross-workstream coordination.
Troy Shu	Project Director	Maintains master timeline; organizes meetings and prepares project progress reports. Drives project execution; coordinates development/implementation tasks and dependencies.
Jason Wu	Project Manager	Leads SIT/UAT execution; prepares test progress/result reports and tracks resolution of testing issues.
Feng Wang	Development Lead/Back-End Solution Architect	Owns back-end architecture and integration design leads to development.
Erpeng Zhao	Front-End Solution Architect	Owns front-end architecture and UI components across web, kiosk, and other channels.
James Xue	Quality Assurance Manager	Defines test strategy and plans; oversees test quality and non-functional testing.
Jade Zhang	Business Analyst	Gathers requirements, documents processes, and prepares UAT scenarios and training inputs.

Siviya Gong	Business Analyst	Supports requirements analysis, process documentation, and UAT preparation with business users.
David Fu	Hardware Manager	Manages kiosk/device scope and infrastructure readiness with TAS-IT and venue operations.

6.2. RACI Chart

Responsible (R) – Who is assigned to do the work.

Accountable (A) – Who makes the final decision and has the ownership.

Consulted (C) – Who is consulted before a decision or action is taken.

Informed (I) – Who is informed of a decision or action?

Process/Deliverable	AnTank Team	WKCD A PjMT	WKCD A PjSC	WKCD A Business Users Group	WKCD A Support Team
Project Initiation	R	A	C	C	C
Proof of concept (POC)	R	A	I	C	C
System Analysis and Design (SA&D) Stage	R	A	I	C	C
Implementation Phase 1	R	A	I	C	C
Implementation Phase 2	R	A	I	C	C
Defects Liability Period (DLP)	R	A	I	C	C
Elective and Optional Items - Phase 3	R	A	C	C	C
Elective and Optional Items - Phase 4	R	A	C	C	C

Appendix

Appendix A – Abbreviations and Definitions

Abbreviation	Definition
API	Application Programming Interface
APK	Android Application Package
B2B	Business-to-Business
B2C	Business-to-Consumer
BI	Business Intelligence
BIN	Bank Identification Number
BO	Box Office
BRD	Business Requirement Document
BSDD	Business Solution Design Document
BU	Business Unit
CAPTCHA	Completely Automated Public Turing Test to Tell Computers and Humans Apart
CDP	Customer Data Platform
CMS	Content Management System
CRM	Customer Relationship Management
ECR	Electronic Cash Register
EDM	Electronic Direct Mail
ERP	Enterprise Resource Planning
ETL	Extract-Transform-Load
FPS	Faster Payment System
FSD	Functional Specification Document
GDPR	General Data Protection Regulation
H5	H5 refers to HTML5-based web pages that provide lightweight, browser-accessible user experiences without requiring application installation.
HKPM	Hong Kong Palace Museum
HTML	Hyper Text Markup Language
HTTPS	Hyper Text Transfer Protocol Secure
IAM	Identity and Access Management
IT	Innovation and Technology
M+	M+ Museum
NGO	Non-Governmental Organization

Technology Partnership Package 2
Requirement Document

OAuth	Open Authorisation
OOTB	Out of the Box
OTA	Online Travel Agency
PA	Performing Arts
PDA	Personal Digital Assistant
PDPO	Personal Data (Privacy) Ordinance
PII	Personally Identifiable Information
PIN	Personal Identification Number
PIPL	Personal Information Protection Law
PjMT	Project Management Team
PjSC	Project Steering Committee
PMO	Project Management Office
POS	Point of Sales
PRD	Product Requirements Document
QR	Quick Response
RD	Requirement Document
REQ	Requirement
RID	Requirement Identification Number
RTM	Requirements Traceability Matrix
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SA&D	Solution Architecture & Design
SLA	Service Level Agreement
SIP	System Integration Platform
SISTIC	Singapore Ticketing Information & Cultural Services
SIT	System Integration Testing
SMS	Short Message Service
SLF4J	Simple Logging Facade for Java
SLS	Simple Log Service
SSL	Secure Sockets Layer
SSO	Single Sign-On
T&C	Terms and Conditions
TAS	Ticketing and Admission System
TLS	Transport Layer Security
TOC	Ticketing Operation Center
TP2	Technology Partnership Package 2

TRD	Technical Requirement Document
UAT	User Acceptance Testing
UI	User Interface
UPS	Uninterruptible Power Supply
UX	User Experience
VIP	Very Important Person
WCAG	Web Content Accessibility Guidelines (WCAG) 2.1 AA
WestK	West Kowloon
WKCD	West Kowloon Cultural District
WKCDA	West Kowloon Cultural District Authority

Appendix B – OTA Platform List

No.	OTA Name
1	Damai
2	Trip.com
3	Maoyan
4	Fliggy
5	UUTIX
6	Popticket
7	Tongcheng
8	Meituan
9	Digital Travel
10	KKDay
11	Klook
12	Art-mate
13	GetYourGuide
14	Cityline
15	CTSHK

Appendix C – OTA Specific Requirements

No.	OTA Name	Support Admission Ticket Projects	Support Marked-seat	Specific Requirements	Requirement Not Applicability Rules
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Technology Partnership Package 2
Requirement Document

			Ticket Projects		
1	Damai	Yes	Yes	Damai—apply strict character limits (e.g., 40 characters) to project name fields.	All Requirement Applicable
2	Trip.com	Yes	No	Trip.com displays a frontend order number to customers, which must also be searchable within TAS for customer service, operational verification, and financial reconciliation.	REQ-OTA-007; REQ-OTA-008
3	Maoyan	No	Yes		REQ-OTA-005; REQ-OTA-006
4	Fliggy	Yes	No	Fliggy can only support event name of project information automatically updated from AnTank interface. Other information, such as project images, project description can only update manually.	REQ-OTA-007; REQ-OTA-008
5	UUTIX	Yes	No		REQ-OTA-007; REQ-OTA-008
6	Popticket	Yes	Yes		All Requirement Applicable
7	Tongcheng	Yes	No		REQ-OTA-007; REQ-OTA-008
8	Meituan	Yes	No		REQ-OTA-007; REQ-OTA-008
9	Digital Travel	Yes	No		REQ-OTA-007; REQ-OTA-008

10	KKDay	Yes	No	KKday does not support seating ticket and Flex ticket.	REQ-OTA-007; REQ-OTA-008
11	Klook	Yes	No		REQ-OTA-007; REQ-OTA-008
12	Art-mate	Yes	Yes		All Requirement Applicable
13	GetYourGuide	Yes	No		REQ-OTA-007; REQ-OTA-008
14	Cityline	Yes	Yes		All Requirement Applicable
15	CTSHK	Yes	No		REQ-OTA-007; REQ-OTA-008

Appendix D – Definitions of Technical Key Terms

Term	Definition
Severe Alarm	Device anomalies that directly lead to the interruption of ticketing transactions (e.g., Octopus offline, Apollo network interruption).
Warning Alarm	Does not directly interrupt the transaction but may affect the experience or indicate potential failures (such as Apollo response timeout).
Failure Recovery Time	Total duration from alarm triggering to device status returning to normal.